Whisper tiny 15-language performance analysis

Completed seed-42 study · Original recording splits · Self-contained report with all plots
Generated 2026-10-25 01:09 UTC from saved results; no models were refitted.

1. Main findings

> **Test-set gains do not transfer uniformly to the external recordings.** Fine-tuning raises combined held-out accuracy from 96.07% to 97.59%, while complete external accuracy moves from 76.95% to 63.85%.

Both models complete the 15-language task and are exported. Fine-tuning makes 198 fewer combined-test errors, mostly through better Pacific recognition. On external recordings, its biggest losses are Tongan and the three added Niuean sources; Samoan improves. These observations describe this fixed experiment, not a deployment winner chosen from external labels.

Both runs selected epoch 8 and stopped after epoch 13. All 15 outputs stay enabled. The original-language accuracy cost relative to each corresponding Whisper12 control is about 0.44 percentage points. Hindi/Urdu remain the main weakness on FLEURS. High aggregate accuracy therefore hides meaningful differences between languages and domains.

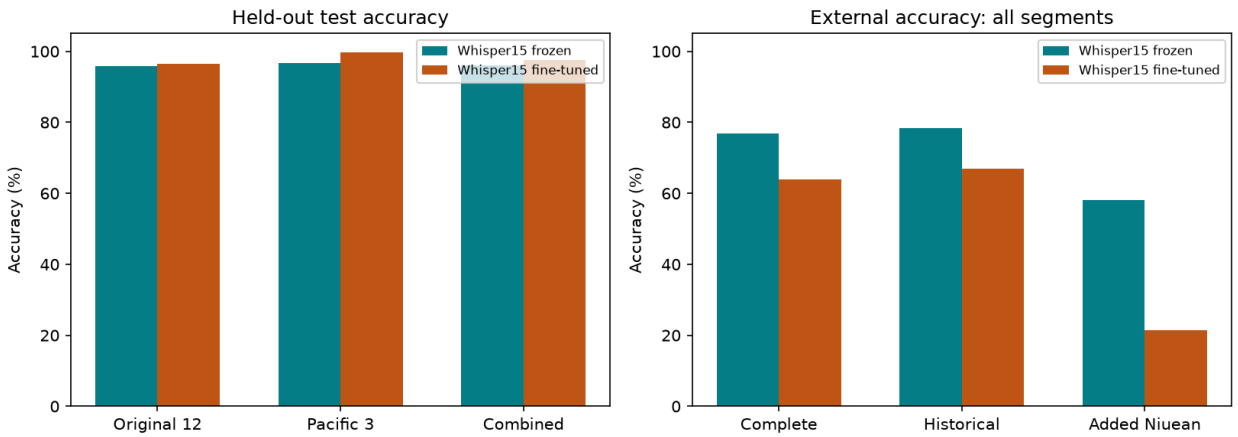


Figure 1. Matched held-out and external cohorts. External references are provisional folder labels; the added cohort contains 93 Niuean segments from three recordings.

2. Data, training and comparison protocol

The original twelve-label order is preserved and Samoan (sam), Niuean (niu) and Tongan (ton) are appended. Both variants start from the selected frozen-encoder Whisper12 classifier. Hidden layers and

original output rows are copied exactly; the new rows share seed-42 initialization. Frozen training changes the classifier only. Fine-tuning updates the full audio encoder and classifier. SpeechBrain partial15 updates its encoder tail, so it is a different adaptation policy.

Training replays the original VoxLingua recording splits and uses the unchanged Pacific manifests. Both variants use identical class-balanced sampled orders: 32,768 draws per epoch, effective batch 32, AdamW, classifier LR 1e-4, encoder LR 1e-5, weight decay 1e-4, at most 20 epochs and patience 5. Checkpoints maximize combined selection macro F1, breaking ties with class-balanced NLL.

Cohort	Examples	Role
vox_train	204,234	Original-language replay
new_train	35,668	Pacific training
vox_validation	23,607	Checkpoint selection
new_selection	2,325	Checkpoint selection
fleurs_validation	4,172	Calibration only
new_calibration	2,284	Calibration only
fleurs_test	8,519	Original-language test
new_test	4,545	Recording-held-out Pacific test
external	1,336	External evaluation only

Preprocessing remains mono 16 kHz, maximum ten-second center crop, native thirty-second Whisper padding, and mean/std pooling over real audio tokens. The external corpus uses one frozen Silero VAD snapshot with common segment IDs and boundaries for every model. Its 24 sources yield 1,336 segments: 1,243 historical plus 93 added Niuean segments. Historical PCM samples and timestamps match v3 exactly; float-WAV creation timestamps explain differing container hashes.

One paired seed is available. Recording separation does not establish speaker independence. Correlated cuts from a recording are not independent trials; no segment-level significance claims or confidence intervals are inferred from these counts.

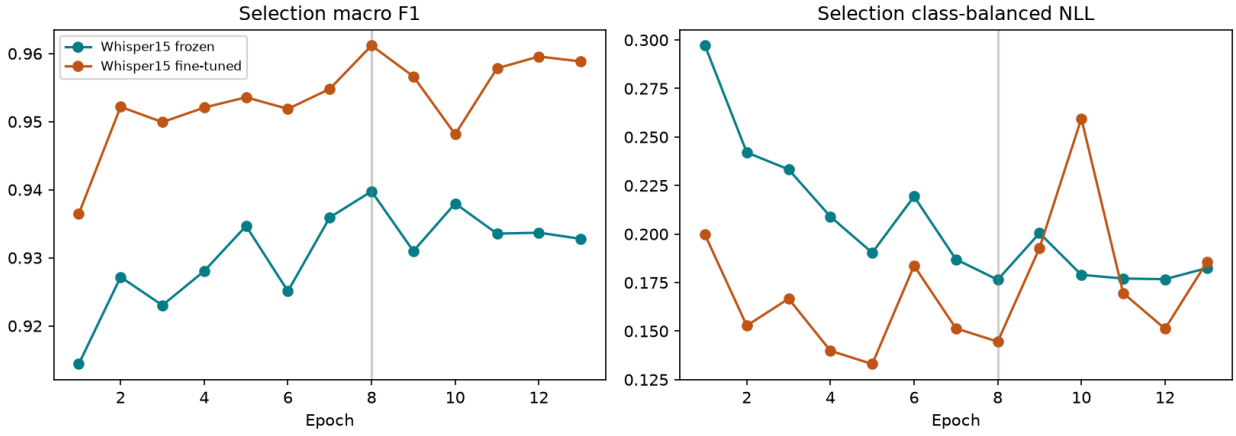


Figure 2. Checkpoint selection uses these curves only; the selected epoch is 8 for both models. Neither test nor external performance enters selection.

3. Held-out test performance

Model	Test	N	Accuracy	Precision	Recall	Macro F1	NLL	Brier	ECE
Whisper15 frozen	fleurs_test	8,519	95.79%	93.26%	92.38%	92.68%	0.1094	0.0595	0.69%
Whisper15 frozen	pacific_test	4,545	96.61%	97.60%	96.64%	97.11%	0.1081	0.0528	1.67%
Whisper15 frozen	combined_test	13,064	96.07%	93.57%	93.23%	93.29%	0.1090	0.0572	0.41%
Whisper15 fine-tuned	fleurs_test	8,519	96.44%	94.07%	93.25%	93.41%	0.0928	0.0520	0.59%
Whisper15 fine-tuned	pacific_test	4,545	99.74%	99.83%	99.74%	99.78%	0.0096	0.0038	0.40%
Whisper15 fine-tuned	combined_test	13,064	97.59%	95.08%	94.55%	94.62%	0.0638	0.0352	0.30%

Joint calibration is used in this table. Combined accuracy improves by 1.52 percentage points and macro F1 by 1.33 points with fine-tuning. Pacific error counts fall from 154 to 12. The combined test mixes FLEURS speech with Pacific radio cuts and should be read alongside both constituent results.

Macro metrics average represented reference languages: twelve for FLEURS, three for Pacific and fifteen for the combined test. Confusions into any of the other enabled outputs remain errors. Class-specific precision in a Pacific-only cohort can differ from precision on the combined test because false positives from old-language examples are absent.

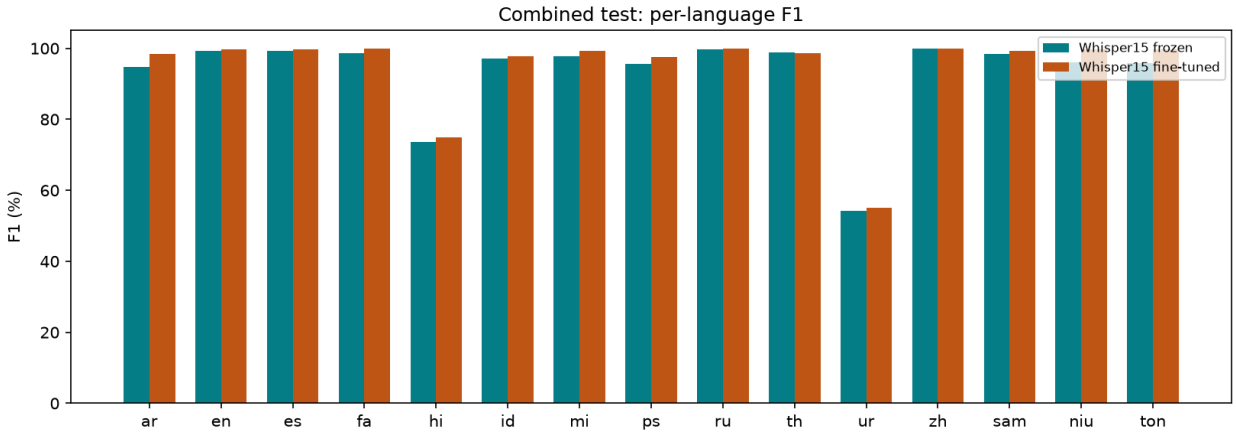
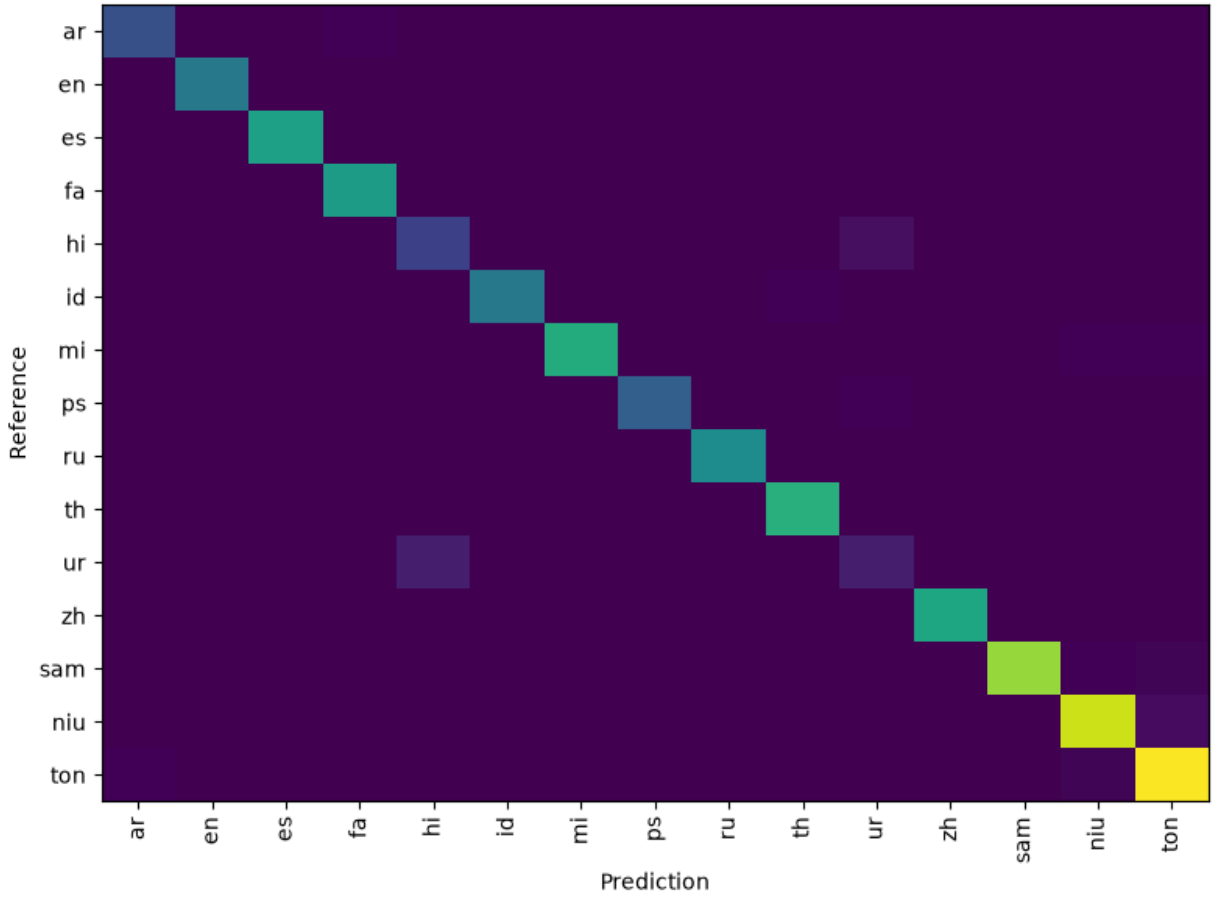
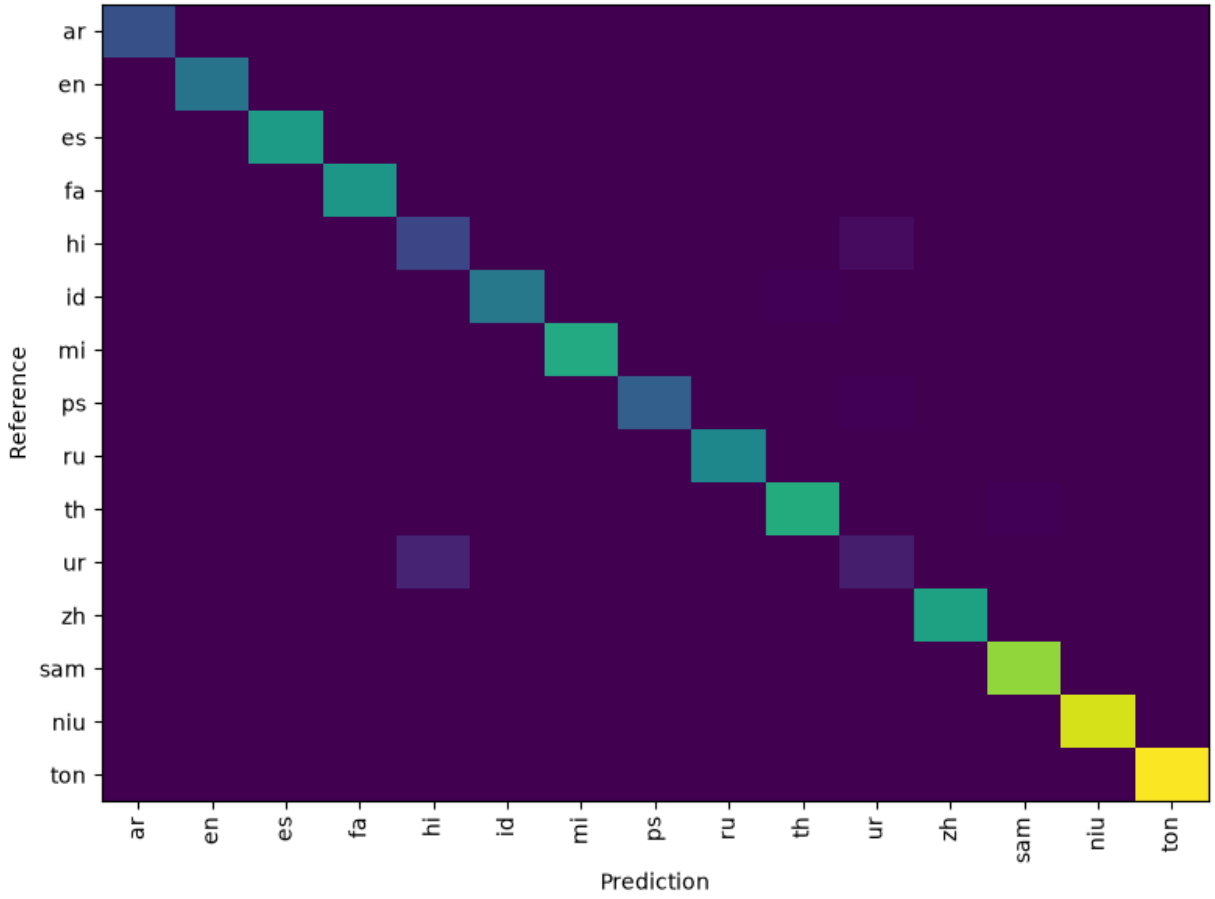


Figure 3. Hindi and Urdu constrain the macro score despite near-perfect performance on several other languages.



Combined-test confusion counts: Whisper15 frozen. Rows are references and columns predictions; all 15 outputs remain enabled.



Combined-test confusion counts: Whisper15 fine-tuned. Rows are references and columns predictions; all 15 outputs remain enabled.

Model	Language	N	Precision	Recall / accuracy	F1	NLL	Brier	ECE
Whisper15 frozen	ar — Arabic	428	94.85%	94.63%	94.74%	0.1612	0.0746	1.94%
Whisper15 frozen	en — English	647	98.93%	99.85%	99.38%	0.0051	0.0027	0.30%
Whisper15 frozen	es — Spanish	908	99.02%	99.67%	99.34%	0.0092	0.0042	0.21%
Whisper15 frozen	fa — Persian	871	97.42%	99.89%	98.64%	0.0130	0.0043	0.96%
Whisper15 frozen	hi — Hindi	418	68.30%	79.90%	73.65%	0.5279	0.2880	3.42%
Whisper15 frozen	id — Indonesian	687	98.36%	95.92%	97.13%	0.1298	0.0611	1.30%
Whisper15 frozen	mi — Maori	1008	98.79%	96.83%	97.80%	0.1098	0.0509	3.03%
Whisper15 frozen	ps — Pushto	512	96.61%	94.73%	95.66%	0.1530	0.0730	2.80%
Whisper15 frozen	ru — Russian	775	99.61%	100.00%	99.81%	0.0014	0.0007	0.12%
Whisper15 frozen	th — Thai	1021	97.79%	99.90%	98.84%	0.0024	0.0009	0.19%
Whisper15 frozen	ur — Urdu	299	63.11%	47.49%	54.20%	1.1117	0.7022	27.39%
Whisper15 frozen	zh — Mandarin Chinese	945	100.00%	99.79%	99.89%	0.0057	0.0036	0.26%
Whisper15 frozen	sam — Samoan	1374	99.33%	97.53%	98.42%	0.0792	0.0379	1.23%
Whisper15 frozen	niu — Niuean	1538	96.77%	95.38%	96.07%	0.1435	0.0724	1.76%

Model	Language	N	Precision	Recall / accuracy	F1	NLL	Brier	ECE
Whisper15 frozen	ton — Tongan	1633	94.62%	97.00%	95.80%	0.0991	0.0467	2.30%
Whisper15 fine-tuned	ar — Arabic	428	98.82%	98.13%	98.48%	0.0590	0.0241	1.88%
Whisper15 fine-tuned	en — English	647	99.69%	99.85%	99.77%	0.0146	0.0030	1.31%
Whisper15 fine-tuned	es — Spanish	908	99.78%	99.67%	99.72%	0.0133	0.0060	0.52%
Whisper15 fine-tuned	fa — Persian	871	99.66%	100.00%	99.83%	0.0074	0.0016	0.68%
Whisper15 fine-tuned	hi — Hindi	418	67.43%	84.21%	74.89%	0.4189	0.2345	3.17%
Whisper15 fine-tuned	id — Indonesian	687	99.40%	96.22%	97.78%	0.1027	0.0554	1.04%
Whisper15 fine-tuned	mi — Maori	1008	99.50%	99.01%	99.25%	0.0339	0.0154	1.30%
Whisper15 fine-tuned	ps — Pushto	512	98.60%	96.48%	97.53%	0.1430	0.0622	3.21%
Whisper15 fine-tuned	ru — Russian	775	99.74%	100.00%	99.87%	0.0011	0.0002	0.11%
Whisper15 fine-tuned	th — Thai	1021	98.06%	99.12%	98.59%	0.0307	0.0150	1.14%
Whisper15 fine-tuned	ur — Urdu	299	67.48%	46.49%	55.05%	1.1502	0.7415	32.33%
Whisper15 fine-tuned	zh — Mandarin Chinese	945	100.00%	99.79%	99.89%	0.0083	0.0031	0.56%
Whisper15 fine-tuned	sam — Samoan	1374	98.78%	99.85%	99.31%	0.0040	0.0014	0.28%
Whisper15 fine-tuned	niu — Niuean	1538	99.80%	99.61%	99.71%	0.0111	0.0044	0.40%
Whisper15 fine-tuned	ton — Tongan	1633	99.51%	99.76%	99.63%	0.0128	0.0051	0.66%

4. Original-language regression and SpeechBrain controls

Each Whisper15 variant is compared with its corresponding Whisper12 variant using FLEURS-only temperature fits for both; all 15 extension outputs remain active. Accuracy falls by 0.43 points for the frozen variant and 0.45 points for the fine-tuned variant. Fine-tuning's largest per-language F1 regression is Urdu; the frozen variant's largest is Hindi. Aggregate accuracy alone masks these language-specific costs.

Model	Δ accuracy (pp)	Δ macro F1 (pp)	Δ NLL	Δ Brier	Δ ECE (pp)
frozen	-0.43	-0.19	+0.0113	+0.0055	+0.03
finetuned	-0.45	-0.48	+0.0116	+0.0070	-0.00

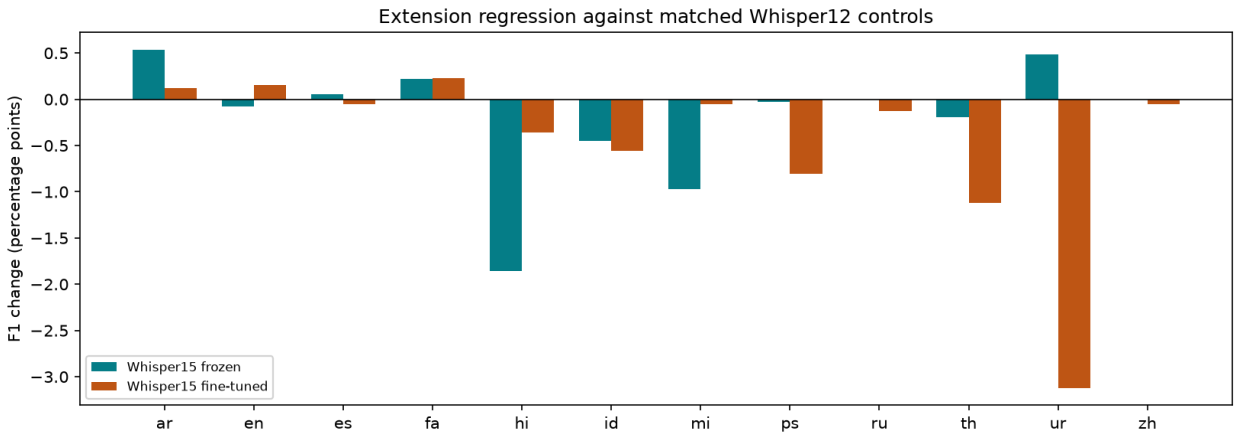


Figure 4. Differences are extension minus corresponding twelve-language control on the same FLEURS test recordings.

Model	Group	N	Accuracy	Macro F1	NLL	Brier	ECE
speechbrain15_frozen	old12	8519	96.74%	95.20%	0.0936	0.0473	0.50%
speechbrain15_frozen	new3	4545	99.71%	99.74%	0.0213	0.0073	1.39%
speechbrain15_frozen	all15	13064	97.77%	95.91%	0.0685	0.0334	0.34%
speechbrain15_partial	old12	8519	97.73%	96.47%	0.0700	0.0350	0.56%
speechbrain15_partial	new3	4545	99.78%	99.82%	0.0146	0.0051	0.86%
speechbrain15_partial	all15	13064	98.45%	97.02%	0.0508	0.0246	0.34%
whisper15_frozen	all15	13064	96.07%	93.29%	0.1090	0.0572	0.41%
whisper15_frozen	old12	8519	95.79%	92.68%	0.1094	0.0595	0.69%
whisper15_frozen	new3	4545	96.61%	97.11%	0.1081	0.0528	1.67%
whisper15_finetuned	all15	13064	97.59%	94.62%	0.0638	0.0352	0.30%
whisper15_finetuned	old12	8519	96.44%	93.41%	0.0928	0.0520	0.59%
whisper15_finetuned	new3	4545	99.74%	99.78%	0.0096	0.0038	0.40%

SpeechBrain partial15 has higher combined-test accuracy (98.45%) than Whisper15 fine-tuning (97.59%) in this study. SpeechBrain's frozen encoder already handles Pacific test cuts extremely well (99.71%), whereas Whisper's frozen encoder reaches 96.61%. The backbone pretraining and encoder update policies differ; these are system comparisons, not an isolated causal comparison of update depth.

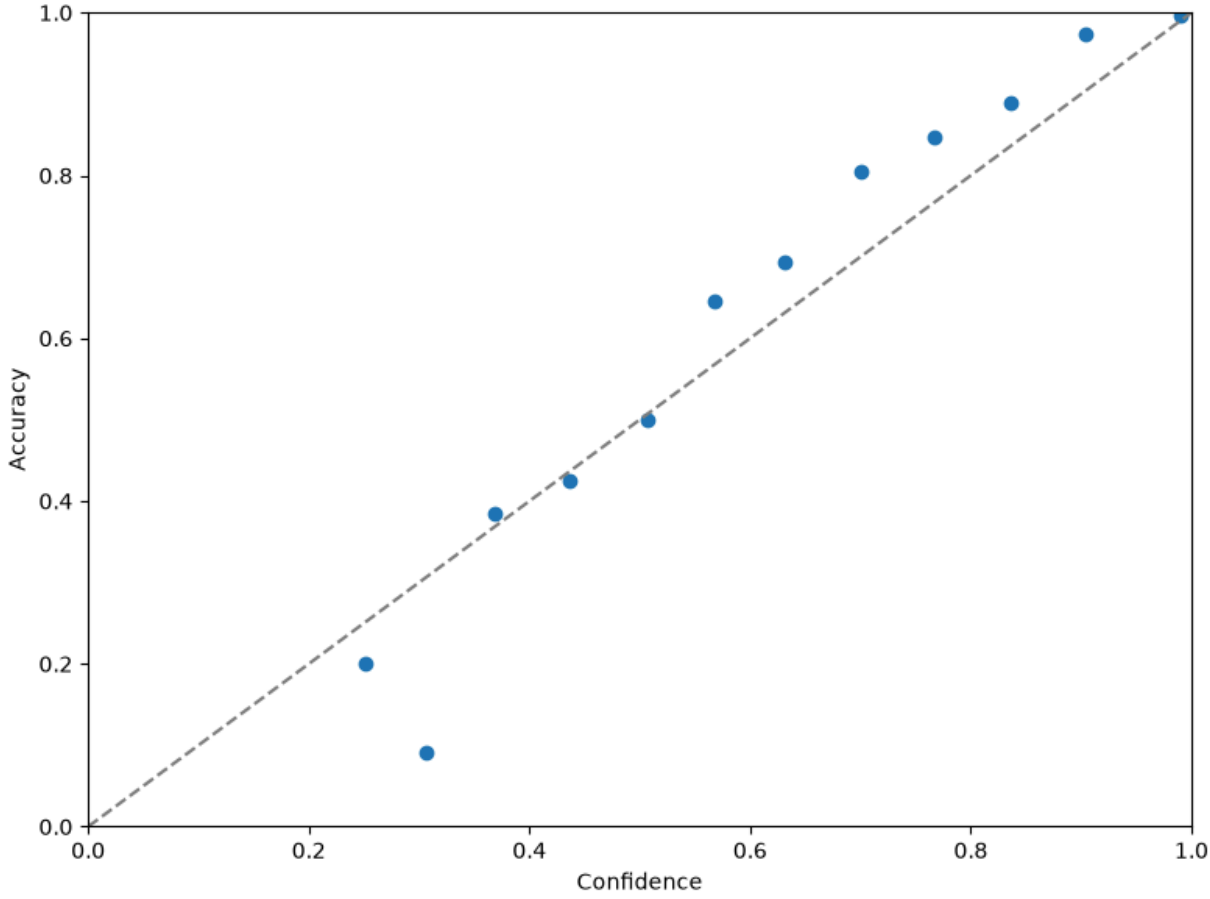
5. Calibration and confidence

Joint temperature fitting uses 4,172 FLEURS validation recordings and 2,284 Pacific calibration cuts. The old-only comparison fit uses FLEURS validation alone. No test or external audio fits either temperature. Calibration rescales confidence and does not change top-1 predictions.

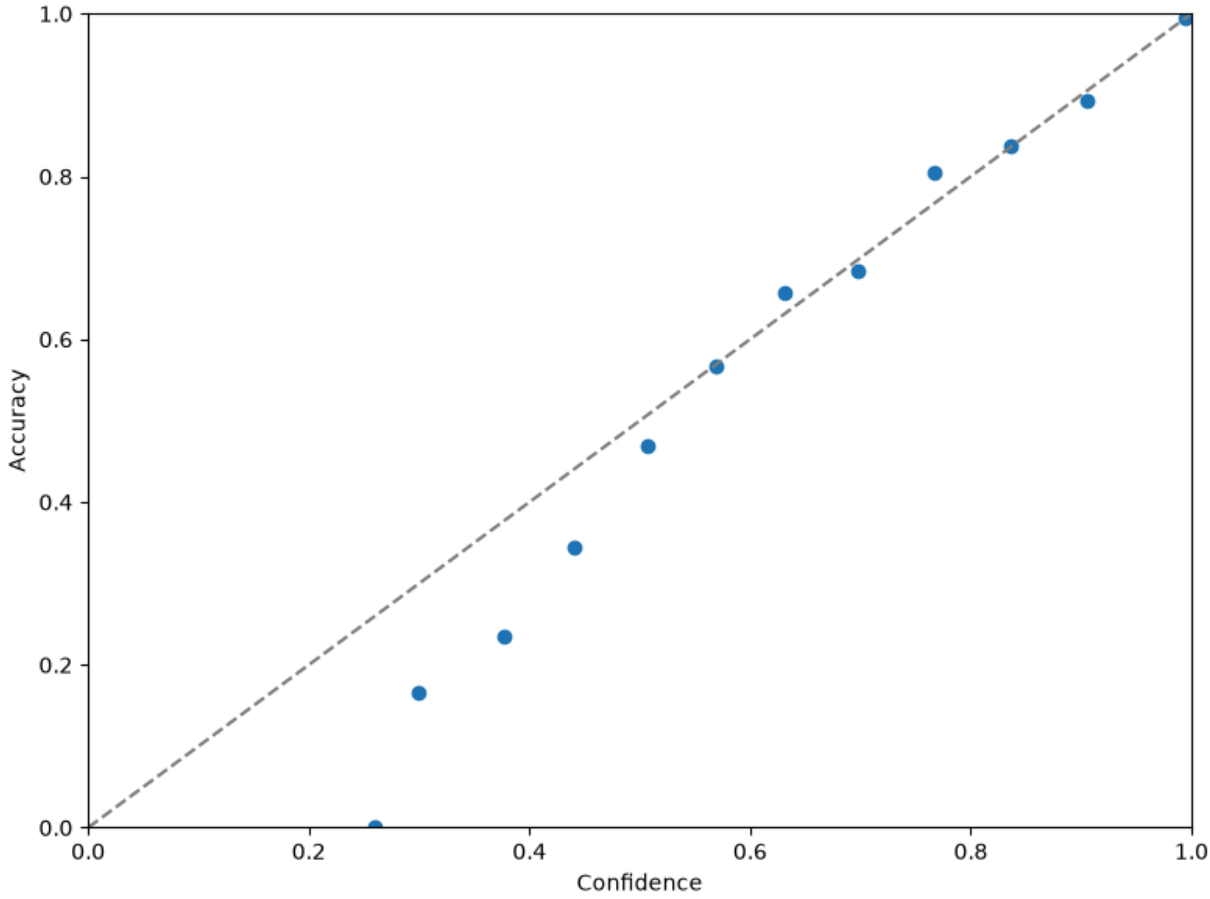
Model	Joint temperature	FLEURS-only temperature
Whisper15 frozen	0.7722	0.8775
Whisper15 fine-tuned	1.6710	1.7350

The frozen temperature below 1 sharpens scores; the fine-tuned temperature above 1 softens them. Joint calibration reduces combined-test ECE for both models. It does not improve every subgroup: fine-tuned Pacific raw ECE is 0.07%, versus 0.40% after joint calibration, and Pacific NLL increases. A single fitted temperature trades off the pooled calibration domains.

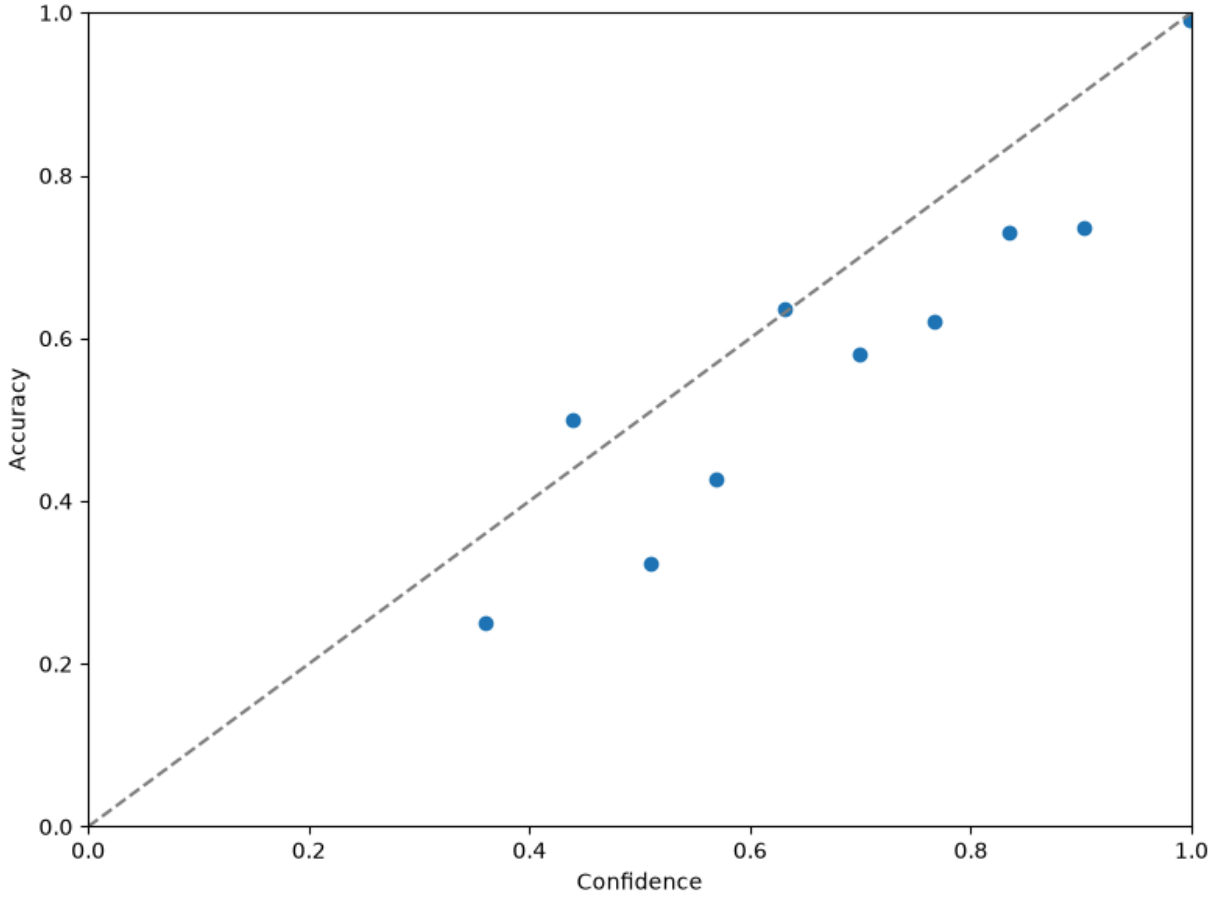
Model	Test	Variant	Temperature	NLL	Brier	ECE
Whisper15 frozen	fleurs_test	raw	1.0000	0.1103	0.0594	0.87%
Whisper15 frozen	fleurs_test	joint	0.7722	0.1094	0.0595	0.69%
Whisper15 frozen	fleurs_test	old_only	0.8775	0.1085	0.0592	0.47%
Whisper15 frozen	pacific_test	raw	1.0000	0.1274	0.0577	4.19%
Whisper15 frozen	pacific_test	joint	0.7722	0.1081	0.0528	1.67%
Whisper15 frozen	pacific_test	old_only	0.8775	0.1153	0.0546	2.77%
Whisper15 frozen	combined_test	raw	1.0000	0.1162	0.0588	1.90%
Whisper15 frozen	combined_test	joint	0.7722	0.1090	0.0572	0.41%
Whisper15 frozen	combined_test	old_only	0.8775	0.1109	0.0576	1.04%
Whisper15 fine-tuned	fleurs_test	raw	1.0000	0.1122	0.0554	1.95%
Whisper15 fine-tuned	fleurs_test	joint	1.6710	0.0928	0.0520	0.59%
Whisper15 fine-tuned	fleurs_test	old_only	1.7350	0.0928	0.0519	0.51%
Whisper15 fine-tuned	pacific_test	raw	1.0000	0.0067	0.0034	0.07%
Whisper15 fine-tuned	pacific_test	joint	1.6710	0.0096	0.0038	0.40%
Whisper15 fine-tuned	pacific_test	old_only	1.7350	0.0103	0.0038	0.46%
Whisper15 fine-tuned	combined_test	raw	1.0000	0.0755	0.0373	1.28%
Whisper15 fine-tuned	combined_test	joint	1.6710	0.0638	0.0352	0.30%
Whisper15 fine-tuned	combined_test	old_only	1.7350	0.0641	0.0352	0.32%



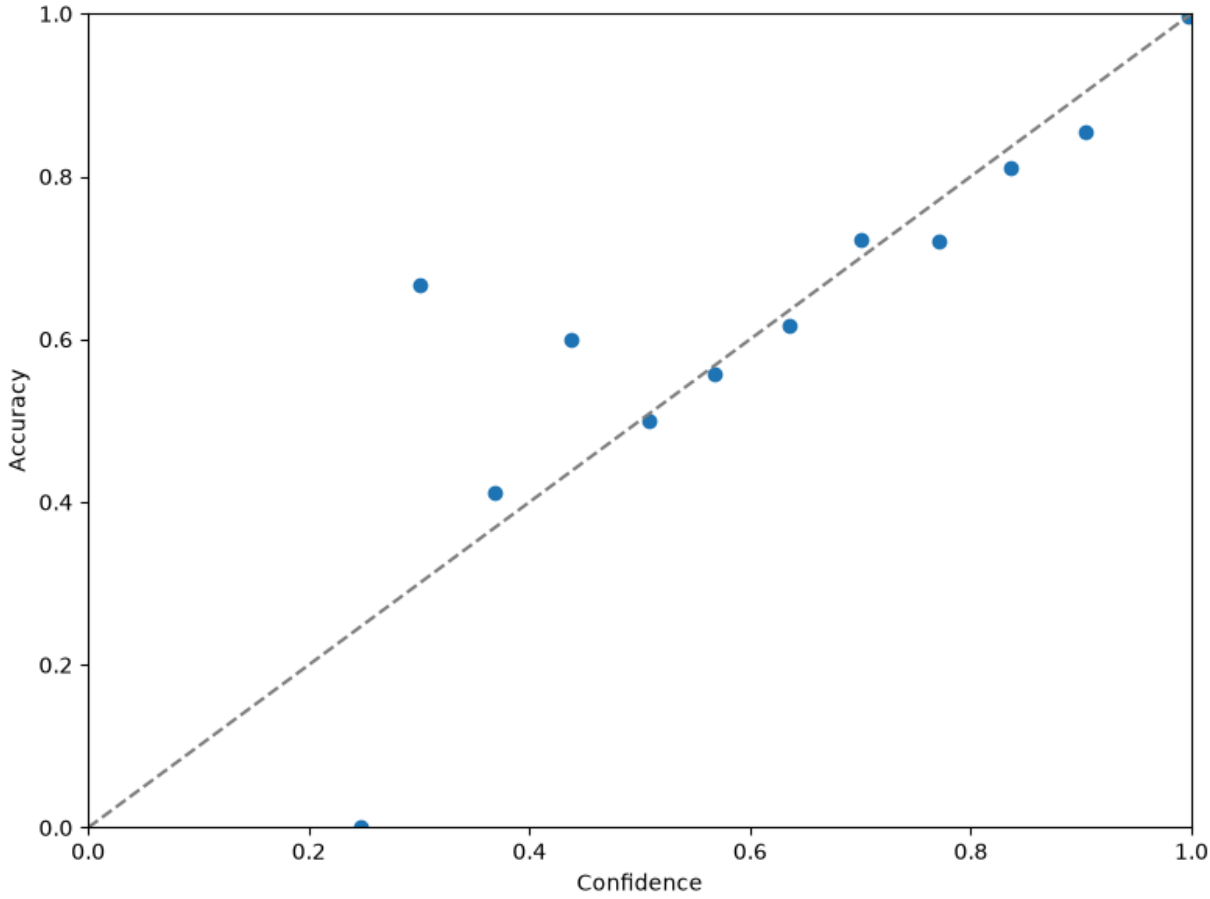
Combined-test reliability: Whisper15 frozen, raw. Empty bins are omitted; ECE uses fifteen confidence bins.



Combined-test reliability: Whisper15 frozen, joint. Empty bins are omitted; ECE uses fifteen confidence bins.



Combined-test reliability: Whisper15 fine-tuned, raw. Empty bins are omitted; ECE uses fifteen confidence bins.



Combined-test reliability: Whisper15 fine-tuned, joint. Empty bins are omitted; ECE uses fifteen confidence bins.

6. External recordings: transfer, exclusions and source balance

External labels are provisional directory labels, not segment-by-segment human annotations. Predictions of English can reflect real code-switching. All-segment accuracy counts every mismatch; English-excluded accuracy omits each model's English predictions. Shared cohorts exclude any segment predicted English by any of the four extension models, plus the compared model's own English exclusions. For these four extension models the shared cohort is identical: 1,266 segments (94.76% coverage).

Model	Policy	Scored	Coverage	Accuracy	Macro F1	ECE	Equal-source accuracy
Whisper15 frozen	all	1336	100.00%	76.95%	79.36%	9.97%	67.30%
Whisper15 frozen	english_excluded	1280	95.81%	80.31%	82.32%	6.54%	69.46%
Whisper15 frozen	shared	1266	94.76%	80.88%	82.84%	6.20%	70.03%
Whisper15 fine-tuned	all	1336	100.00%	63.85%	74.55%	19.12%	55.50%
Whisper15 fine-tuned	english_excluded	1282	95.96%	66.54%	77.29%	16.07%	57.63%
Whisper15 fine-tuned	shared	1266	94.76%	66.90%	77.62%	15.83%	57.84%
SpeechBrain15 frozen	all	1336	100.00%	73.73%	76.73%	11.42%	60.49%
SpeechBrain15 frozen	english_excluded	1275	95.43%	77.25%	80.38%	7.50%	62.88%
SpeechBrain15 frozen	shared	1266	94.76%	77.41%	80.52%	7.56%	62.95%
SpeechBrain15 partial	all	1336	100.00%	76.57%	78.92%	10.59%	63.27%
SpeechBrain15 partial	english_excluded	1274	95.36%	80.30%	82.59%	6.76%	65.66%
SpeechBrain15 partial	shared	1266	94.76%	80.49%	82.77%	6.65%	65.89%

The fine-tuning deficit persists on the shared cohort (66.90% versus 80.88%), so unequal English exclusions do not explain it. Source-balanced accuracy is lower than segment-weighted accuracy for both Whisper variants: performance varies substantially by recording and longer/easier sources affect the pooled score.

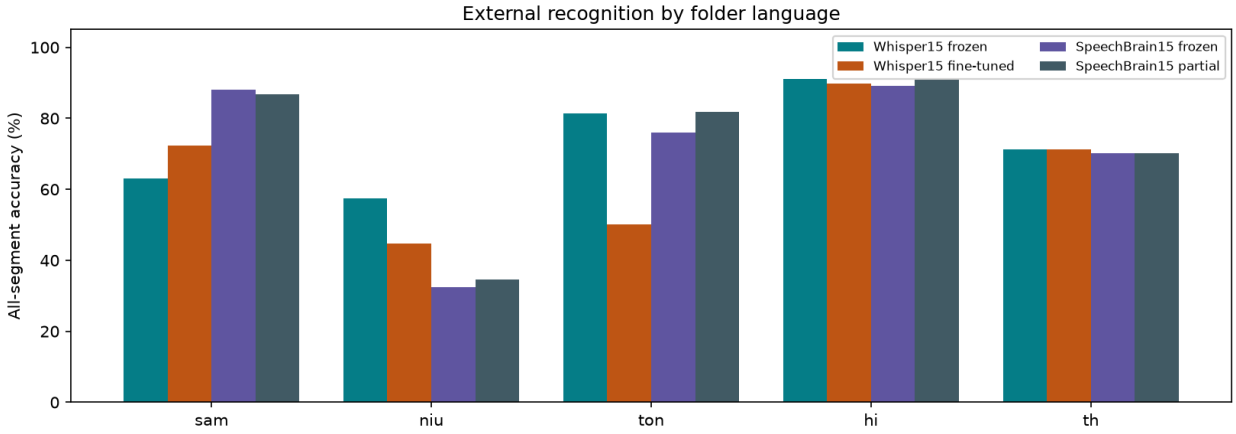


Figure 5. All 1,336 segments are included. Twelve-language models are omitted from Pacific recognition comparisons because their label sets cannot represent these languages.

Fine-tuning improves external Samoan from 63.16% to 72.37%, but Niuean falls from 57.54% to 44.69% and Tongan from 81.35% to 50.19%. The Tongan loss is dominated by Māori predictions: 155/520 fine-tuned

predictions versus 21/520 frozen predictions. Niuean-to-Tongan confusion decreases (53/179 to 9/179), yet total Niuean accuracy falls because Māori errors grow (12/179 to 58/179). Reducing one directional confusion therefore does not establish better recognition.

Model	Direction	Count	Reference support	Rate
Whisper15 frozen	niu → ton	53	179	29.61%
Whisper15 frozen	ton → niu	35	520	6.73%
Whisper15 frozen	niu → mi	12	179	6.70%
Whisper15 frozen	ton → mi	21	520	4.04%
Whisper15 fine-tuned	niu → ton	9	179	5.03%
Whisper15 fine-tuned	ton → niu	38	520	7.31%
Whisper15 fine-tuned	niu → mi	58	179	32.40%
Whisper15 fine-tuned	ton → mi	155	520	29.81%

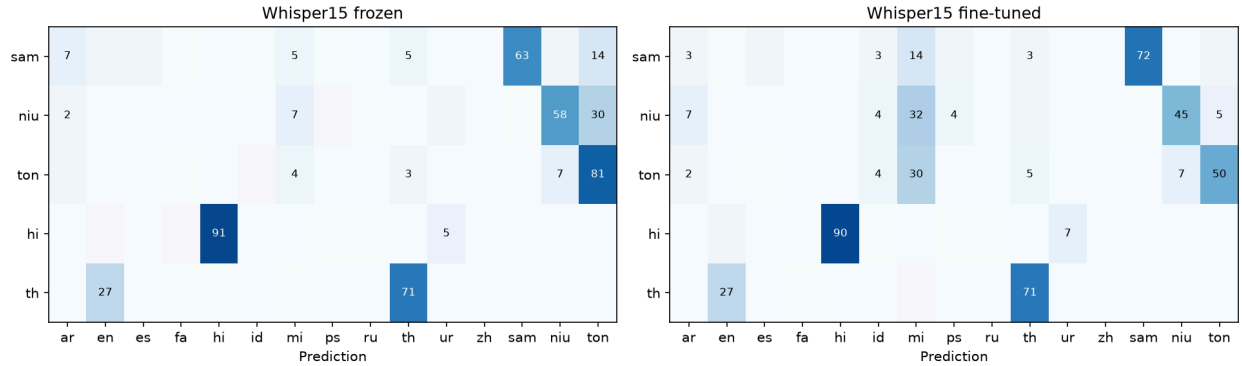


Figure 6. External confusion percentages normalized within each reference language; all 15 predicted labels are visible.

Model	External confidence	NLL	Brier	ECE
Whisper15 frozen	raw	0.9882	0.3450	6.32%
Whisper15 frozen	joint	1.1536	0.3537	9.97%
Whisper15 fine-tuned	raw	2.0394	0.5992	26.48%
Whisper15 fine-tuned	joint	1.3723	0.5381	19.12%

Fine-tuned external ECE is 19.12% after joint calibration, compared with 0.30% on the combined test. This gap is consistent with a domain/calibration-transfer problem; it does not prove a specific cause such as channel, speaker, genre or volume. External confidence was not recalibrated to remove it.

Model	Cohort	N	Accuracy	Equal-source accuracy
Whisper15 frozen	historical	1243	78.36%	68.64%
Whisper15 frozen	added	93	58.06%	57.87%
Whisper15 fine-tuned	historical	1243	67.02%	60.31%

Model	Cohort	N	Accuracy	Equal-source accuracy
Whisper15 fine-tuned	added	93	21.51%	21.78%
SpeechBrain15 frozen	historical	1243	77.80%	66.37%
SpeechBrain15 frozen	added	93	19.35%	19.33%
SpeechBrain15 partial	historical	1243	80.61%	69.08%
SpeechBrain15 partial	added	93	22.58%	22.57%

Added-source accuracy is 58.06% frozen and 21.51% fine-tuned; the older Niuean cohort is 56.98% and 69.77%, respectively. The apparent Niuean ordering changes with the recording cohort. The report keeps both cohorts separate to make that sensitivity explicit.

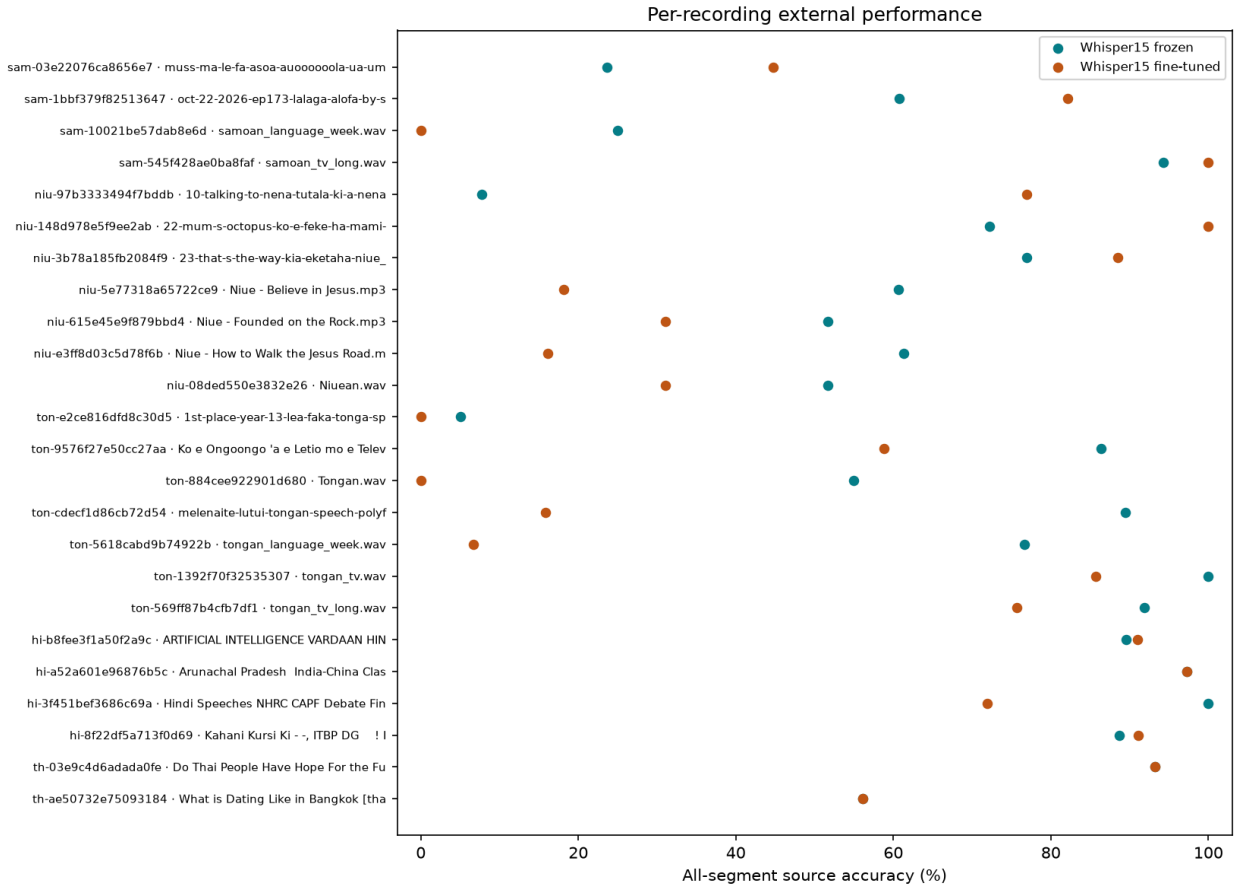


Figure 7. One point per original source; differences across recordings limit interpretations based on pooled cuts.

7. Duration and volume diagnostics

Duration diagnostics use nested 3/5/10-second center crops of identical ten-second VAD speech windows, shared across models. They are a subset of the external corpus, not the complete segmentation. There are 134 Hindi, 76 Thai, 10 Samoan and 124 Tongan windows per duration. **No Niuean VAD region qualified for a ten-second window, so no matched Niuean duration estimate exists.** Directional Niuean/Tongan rates on the complete corpus are reported above instead.

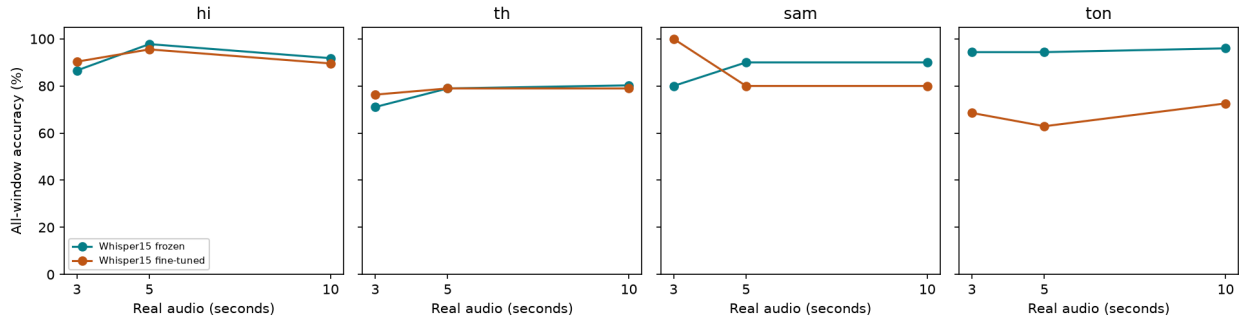


Figure 8. Longer excerpts do not improve all cohorts monotonically. Samoan has only ten windows, so one changed prediction is ten percentage points.

Volume probes use twelve checkpoint-selection cuts per language, never external recordings. Positive gain is capped before clipping. For Samoan the mean applied positive gain is 10.38 dB rather than 12 dB. Frozen Samoan changes one prediction under −12 dB; fine-tuning changes one under positive gain. These small probes do not establish general volume invariance or explain the external Tongan/Niuean loss.

Model	Requested dB	Mean applied dB	N	Changed	Accuracy before	Accuracy after	Mean |Δ confidence|
whisper15_frozen	-12	-12.00	12	1	91.67%	100.00%	0.0059
whisper15_frozen	12	10.38	12	0	91.67%	91.67%	0.0162
whisper15_finetuned	-12	-12.00	12	0	100.00%	100.00%	0.0093
whisper15_finetuned	12	10.38	12	1	100.00%	91.67%	0.0137

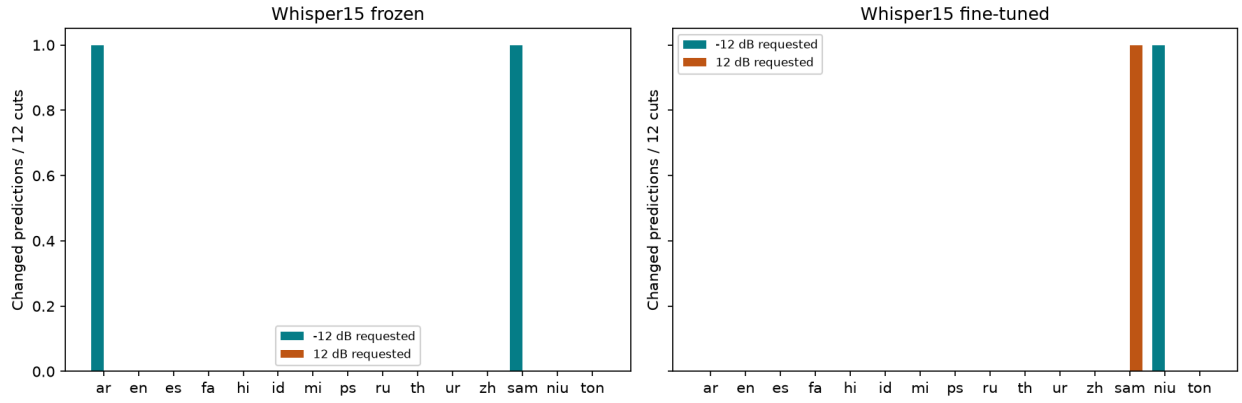


Figure 9. Selection-only gain sensitivity. Positive applied gain varies by recording because clipping is prevented.

8. Deployment artifacts and validation

Model	Archive	Size	Verified examples	Top-1 differences	Max |Δ log p|
Whisper15 frozen	artifacts/deployment/whisper_tiny15_frozen.tar.gz	19.65 MiB	14400	0	1.14e-05
Whisper15 fine-tuned	artifacts/deployment/whisper_tiny15_finetuned.tar.gz	29.64 MiB	14400	0	2.38e-05

Both decoder-free bundles support SpeechBrain `EncoderClassifier.from_hparams(...)`, contain fifteen labels and a custom Whisper backend, and have calibration folded exactly once into the classifier. Keep the complete extracted folder together. Each bundle passed relocated offline checks on 8,519 FLEURS test recordings, 4,545 Pacific test cuts and 1,336 external segments, plus CPU/file and padded-batch checks. Archive hashes and the hashes of archived files were rechecked while building this report.

```
from speechbrain.inference.classifiers import EncoderClassifier

model = EncoderClassifier.from_hparams(
    source="/path/to/whisper_tiny15_finetuned",
    savedir="/tmp/whisper15_loaded",
    run_opts={"device": "cpu"},  # "cuda" is also supported
)
log_probs, score, index, labels = model.classify_file("audio.wav")
probabilities = log_probs.exp()
```

Use the frozen folder name to load the other variant. VAD and unknown-language rejection are not included. These are language-ID exports; native Whisper transcription loaders expect a different checkpoint.

Archive	SHA-256
artifacts/deployment/whisper_tiny15_frozen.tar.gz	030af3d7284eb9e90e9f90e1984b47d33f6d174377485b0876e518fd2f55d1b5
artifacts/deployment/whisper_tiny15_finetuned.tar.gz	68d047b2e466f9bca01e685a039b5bc4d34da9350d34eadc34928eafd1af598a

9. Interpretation and limits

The held-out tests support improved Pacific recognition from full Whisper encoder fine-tuning under these recording splits. The external results show that this gain is not a reliable predictor across the present collection of channels and genres. Frozen Whisper and partial SpeechBrain have similar pooled external accuracy, but their language-specific strengths differ. No universal deployment winner is established by this one paired seed and provisional external reference set.

Original-language regression should be reviewed per language, especially Urdu and Hindi. English exclusion is a reporting policy, not ground-truth annotation or an unknown-language detector. Equal-source averages reduce domination by long recordings, but the corpus still contains only 24 sources with uneven language coverage. Manual timestamped review queues are saved for follow-up annotation; no labels were revised to improve these metrics.

The study enforced a 4 GiB host-memory ceiling, zero swap, a 200% CPU quota and a single GPU job below 7 GiB. Training completion records report peak reserved VRAM of 0.24 GiB frozen and 0.50 GiB fine-tuned; these are allocator measurements, not total GPU memory or latency benchmarks. Some stage resource files reflect the final resumed verification run, so they should not be interpreted as a continuous memory trace of the entire experiment.

10. Corpus inventory and numerical appendices

Source ID	Language	File	Cohort
sam-03e22076ca8656e7	sam	muss-ma-le-fa-asoa-auooooooola-ua-uma-upu-ae-taunuu-mustheman78_7zLrpyjK.mp3	historical
sam-1bbf379f82513647	sam	oct-22-2026-ep173-lalaga-alofa-by-su-a-andrew-leota-leuluaialii-muavaa_NZ0qkOn4.mp3	historical
sam-10021be57dab8e6d	sam	samoan_language_week.wav	historical
sam-545f428ae0ba8faf	sam	samoan_tv_long.wav	historical
niu-97b3333494f7bddb	niu	10-talking-to-nena-tutala-ki-a-nena-niue_Jf3qRCTf.mp3	historical
niu-148d978e5f9ee2ab	niu	22-mum-s-octopus-ko-e-feke-ha-mami-niue_RgwdRyES.mp3	historical
niu-3b78a185fb2084f9	niu	23-that-s-the-way-kia-eketaha-niue_EMIhO9ef.mp3	historical
niu-5e77318a65722ce9	niu	Niue - Believe in Jesus.mp3	added
niu-615e45e9f879bbd4	niu	Niue - Founded on the Rock.mp3	added
niu-e3ff8d03c5d78f6b	niu	Niue - How to Walk the Jesus Road.mp3	added
niu-08ded550e3832e26	niu	Niuean.wav	historical
ton-e2ce816dfd8c30d5	ton	1st-place-year-13-lea-faka-tonga-speech-polyfest-2013_yVpmU9mO.mp3	historical
ton-9576f27e50cc27aa	ton	Ko e Ongoongo 'a e Letio mo e Television Tonga.mp3	historical
ton-884cee922901d680	ton	Tongan.wav	historical
ton-cdecf1d86cb72d54	ton	melenaite-lutui-tongan-speech-polyfest-2026_jWBDXS5X.mp3	historical
ton-5618cabd9b74922b	ton	tongan_language_week.wav	historical
ton-1392f70f32535307	ton	tongan_tv.wav	historical
ton-569ff87b4cfb7df1	ton	tongan_tv_long.wav	historical
hi-b8fee3f1a50f2a9c	hi	ARTIFICIAL INTELLIGENCE VARDAAN HINDI DEBATE 2024.mp3	historical
hi-a52a601e96876b5c	hi	Arunachal Pradesh में India-China Clash की बात किसने छिपाई, ITBP Chief ने क्यों चेताया LAC[hindi].mp3	historical
hi-3f451bef3686c69a	hi	Hindi Speeches NHRC CAPF Debate Final Competition (हिंदी भाषण).mp3	historical
hi-8f22df5a713f0d69	hi	Kahani Kursi Ki भारत-चीन आमने-सामने, ITBP DG का बड़ा स्टेटमेंट ! India China Border Conflict [hindi].mp3	historical
th-03e9c4d6adada0fe	th	Do Thai People Have Hope For the Future of Thailand.mp3	historical
th-ae50732e75093184	th	What is Dating Like in Bangkok [thai].mp3	historical

▶ **artifacts/whisper_tiny15/speechbrain_comparison.csv — 32 rows**

▶ **artifacts/whisper_tiny15/original_language_per_class_regression.csv — 24 rows**

▶ **artifacts/whisper_tiny15/diagnostics/duration.csv — 132 rows**

▶ **artifacts/whisper_tiny15/diagnostics/gain.csv — 330 rows**

▶ **artifacts/youtube_validation_v4/matched_summary.csv — 144 rows**

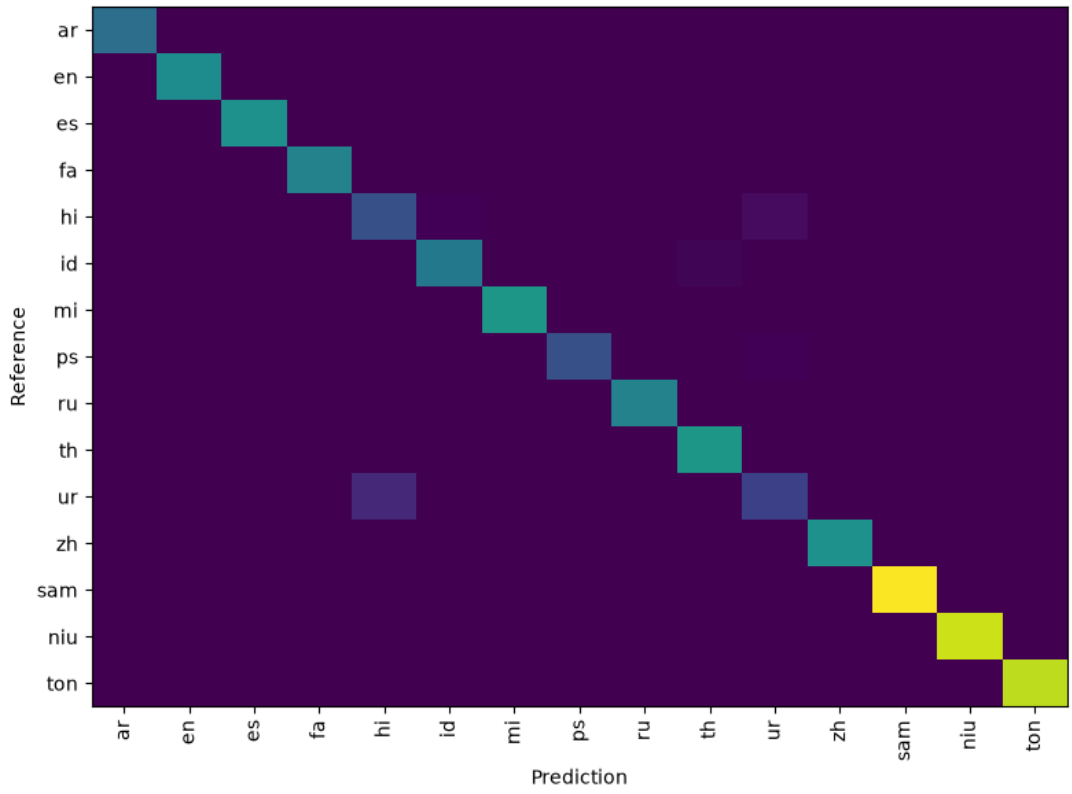
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▶ **artifacts/youtube_validation_v4/matched_per_source.csv — 1,944 rows**

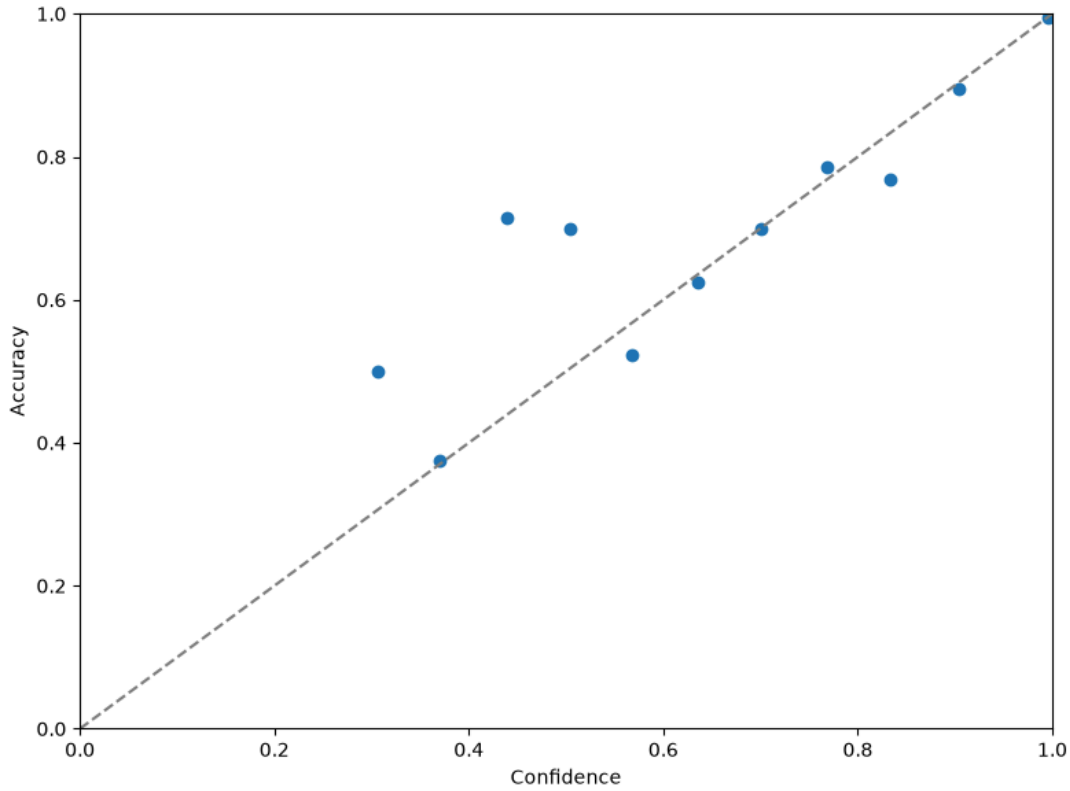
Download the complete numerical metrics embedded in this report.

11. Complete plot appendix (204 original plots)

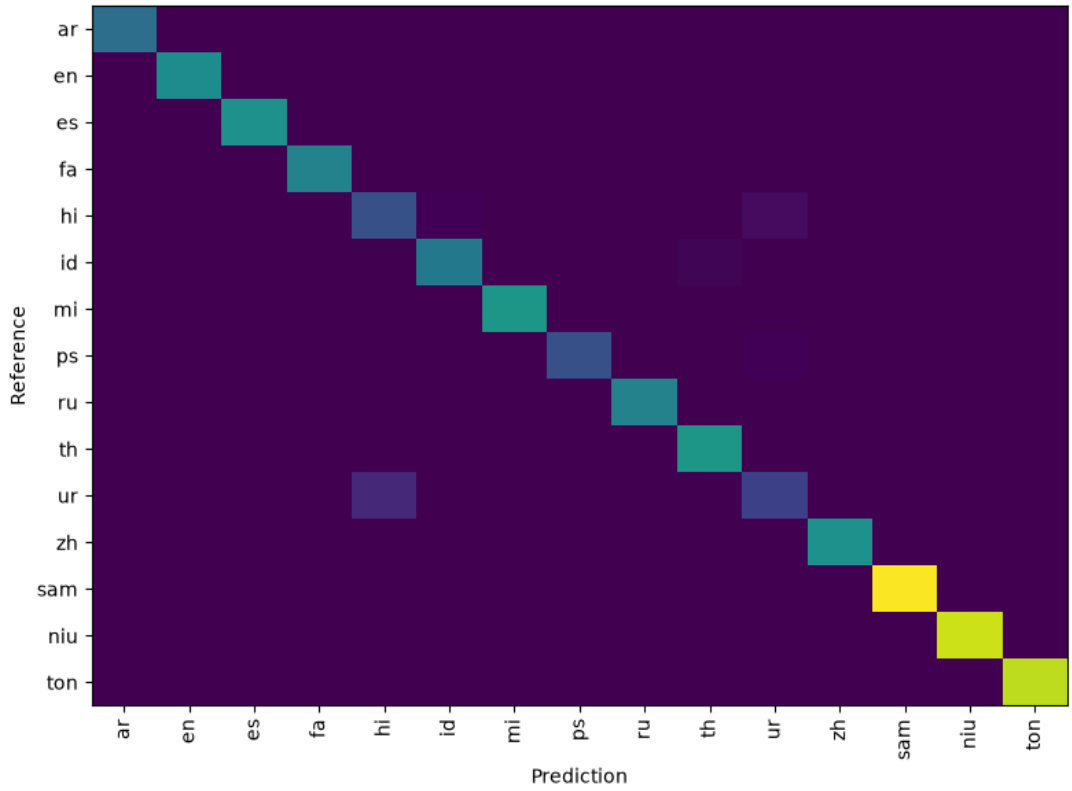
Every original study and refreshed-validation plot is embedded below, including selection/calibration cohorts, raw/joint/old-only calibration variants, and all supported external baseline/cohort/exclusion comparisons. Selection and calibration panels are development diagnostics, not independent test evidence. The main analysis above adds training, regression, domain-transfer, per-language, per-source, duration and gain figures.



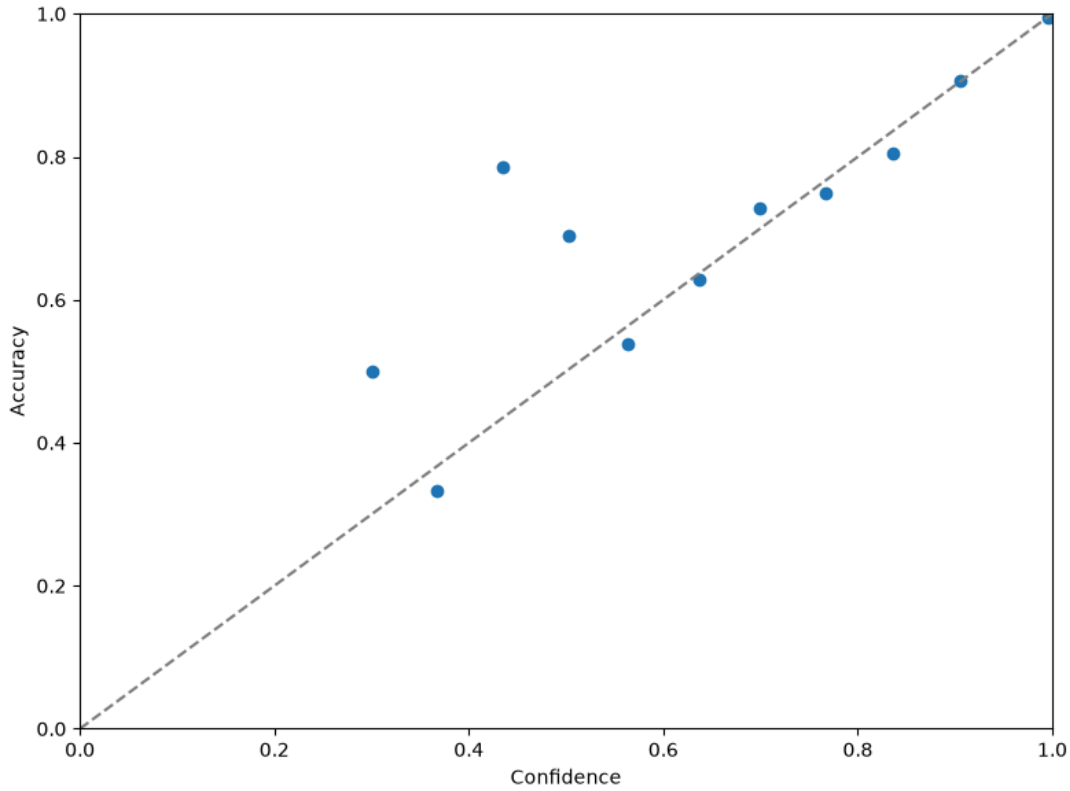
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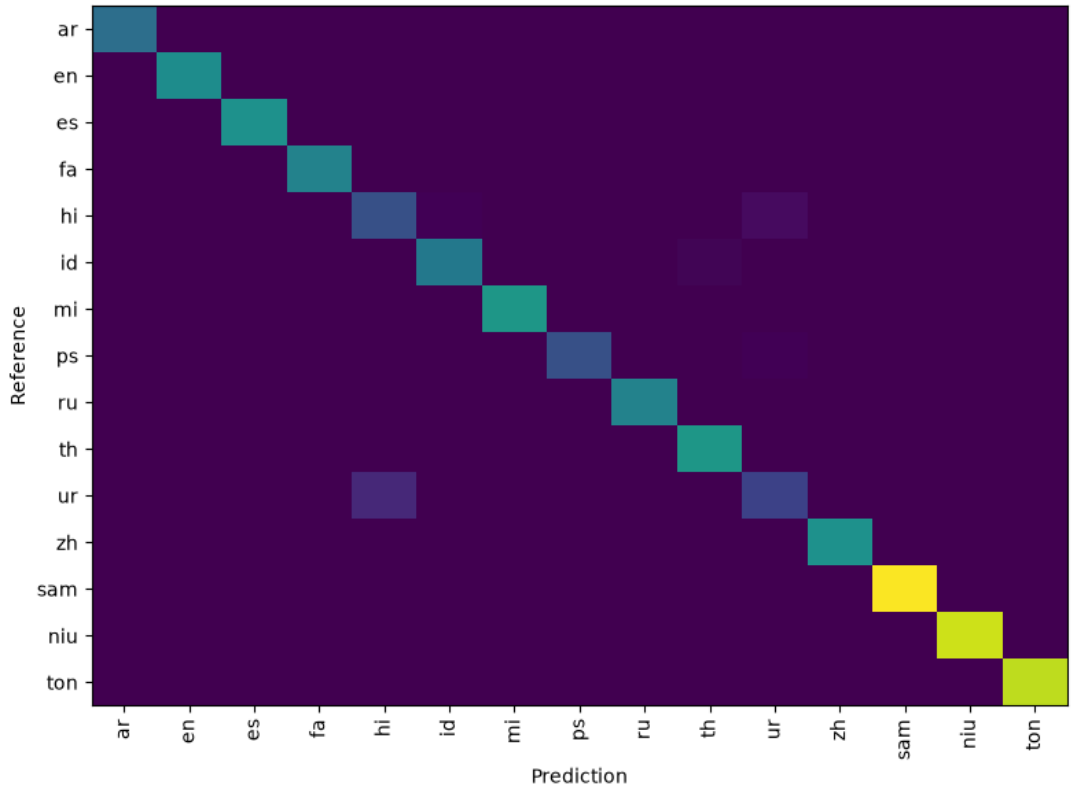
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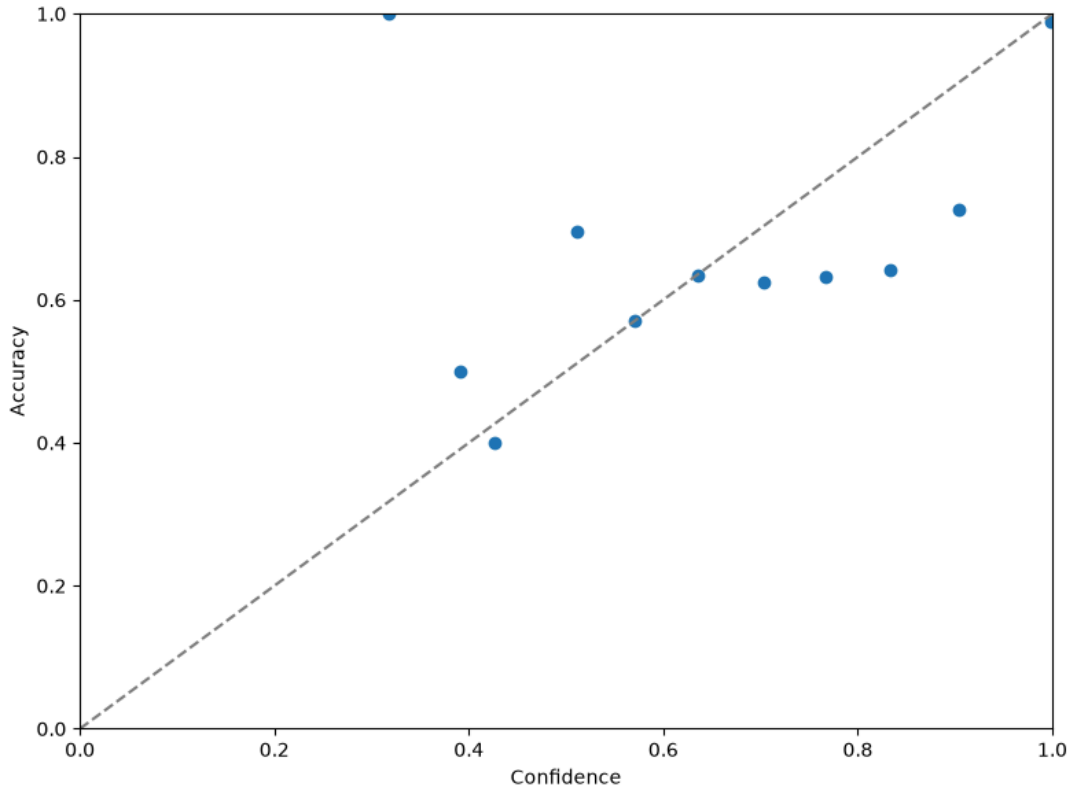
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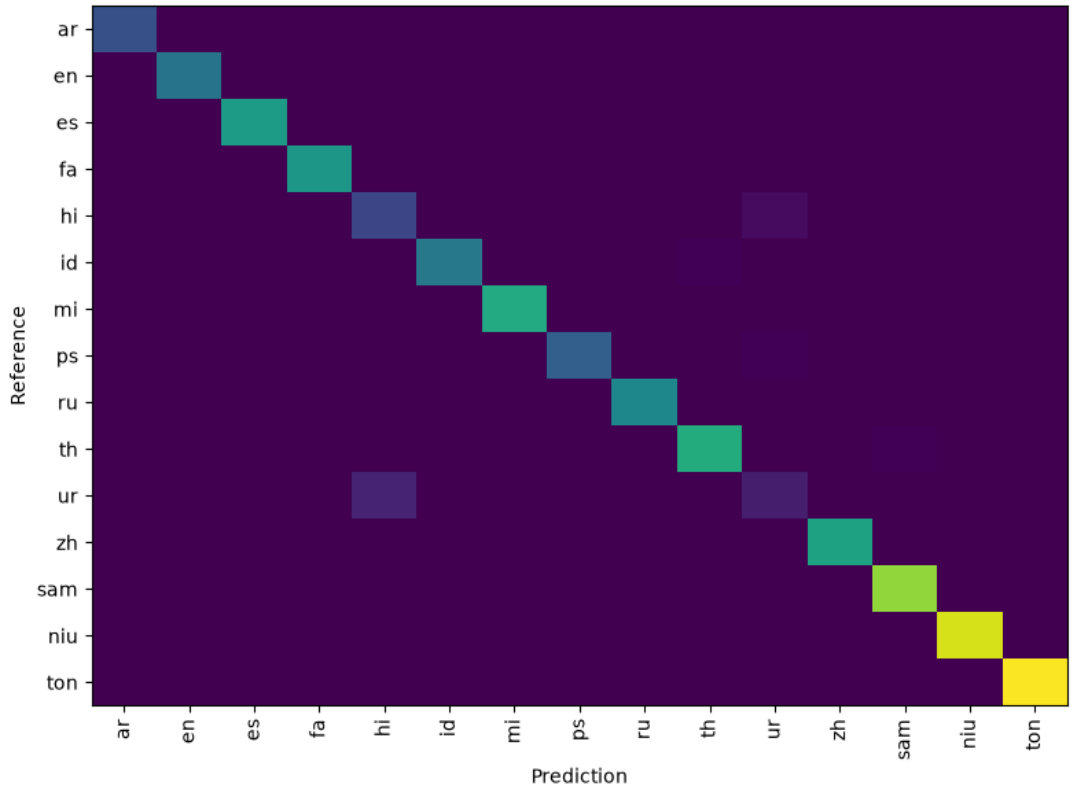
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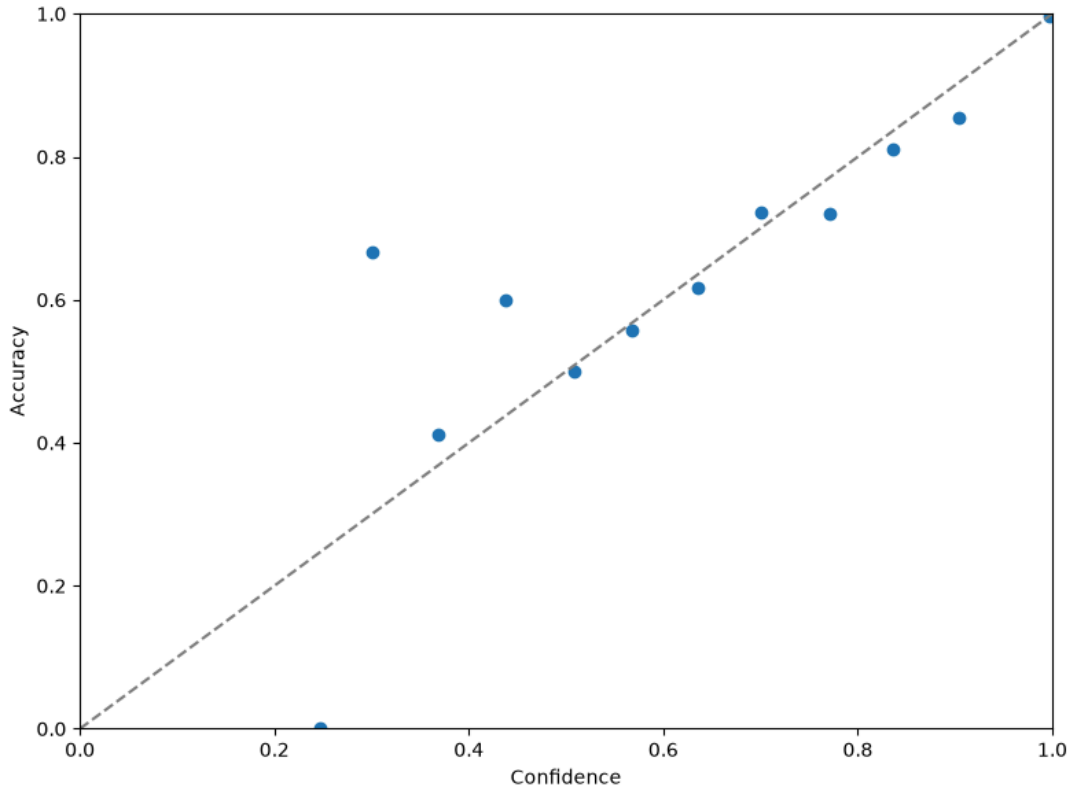
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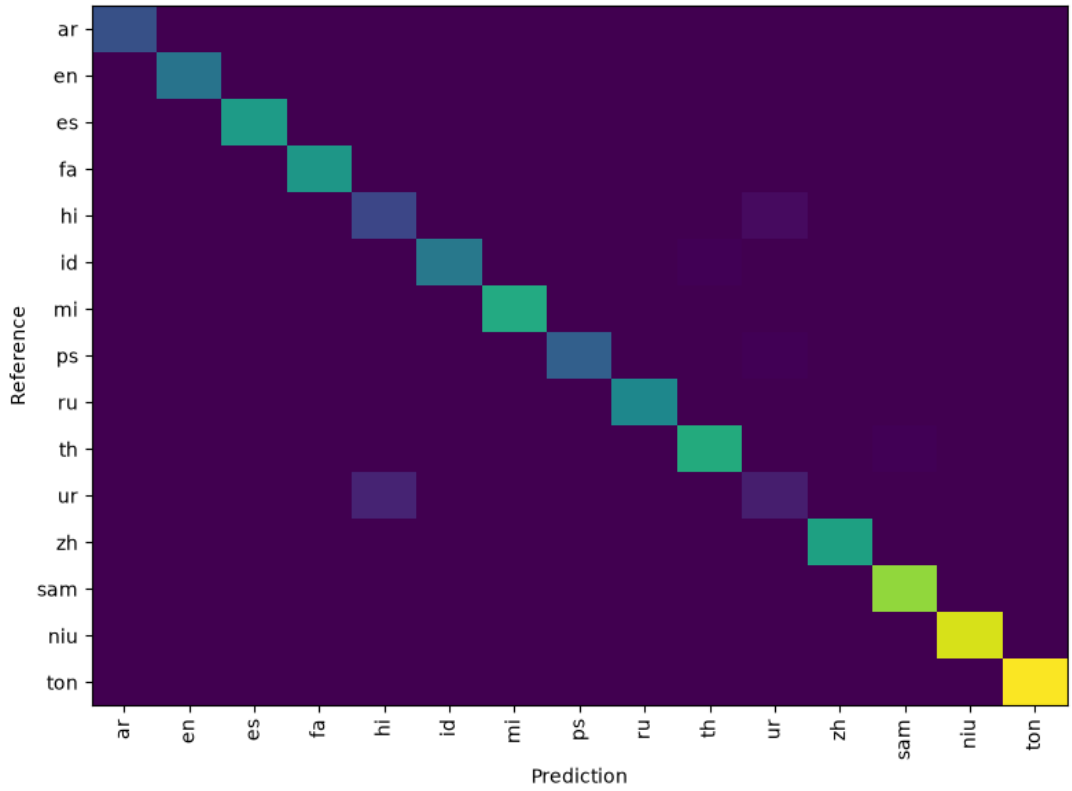
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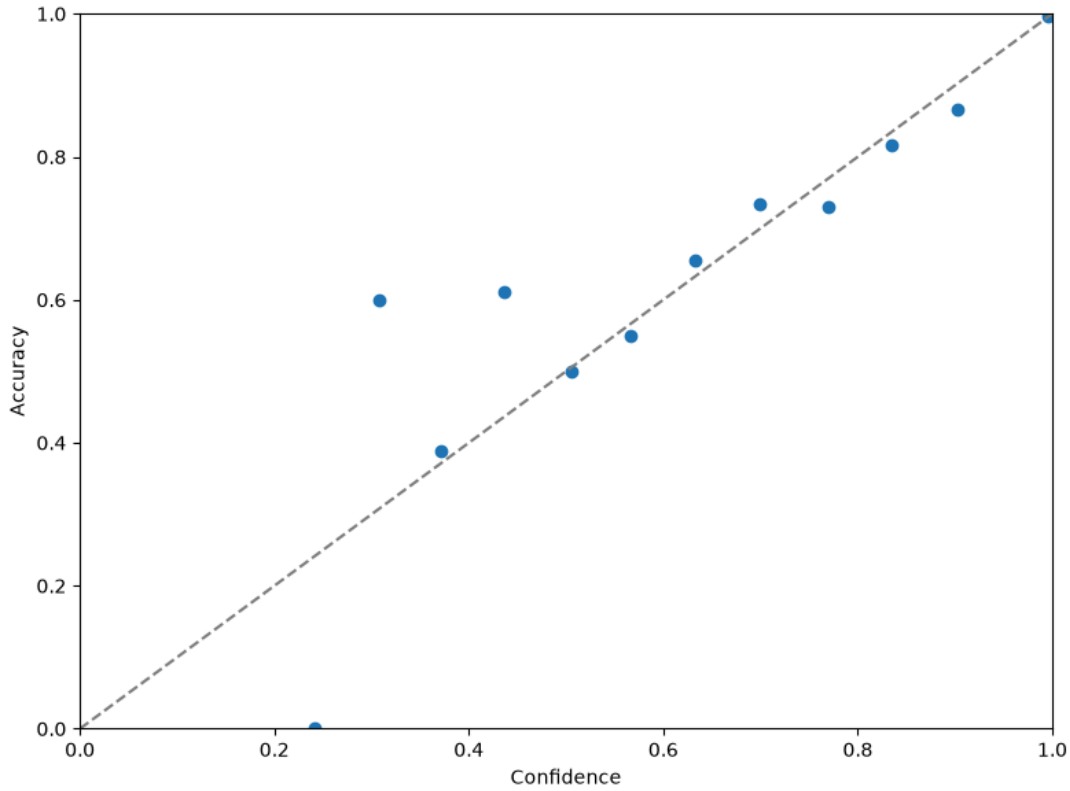
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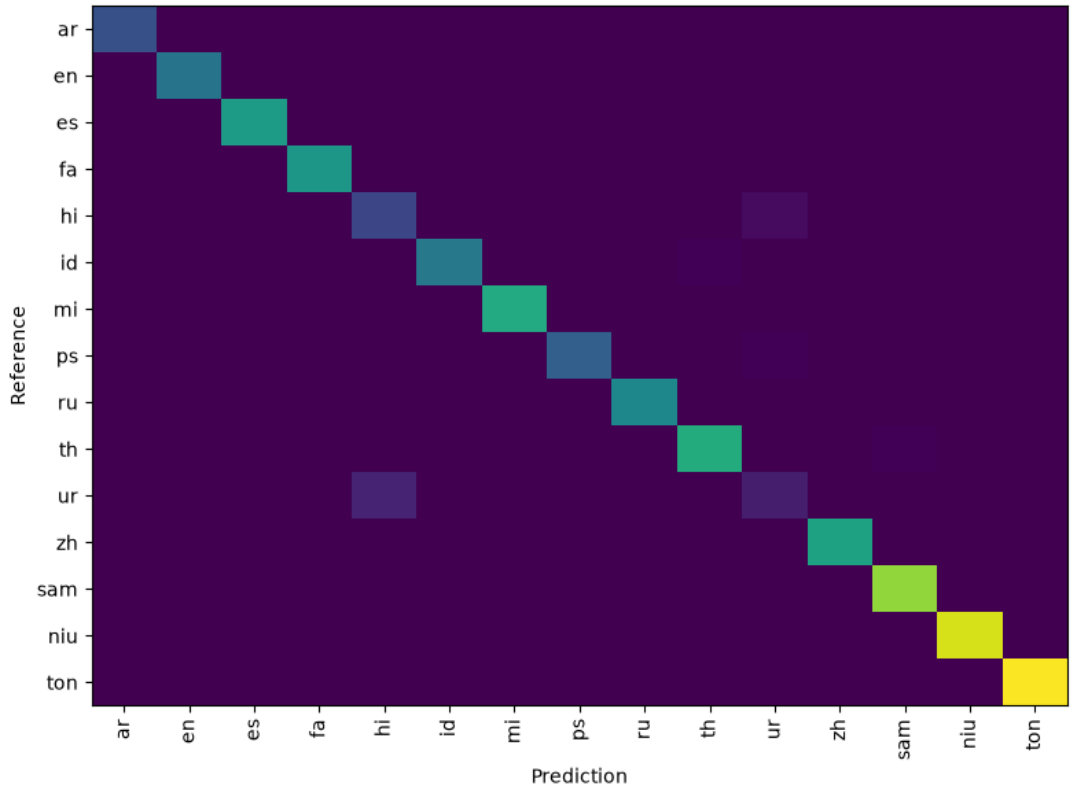
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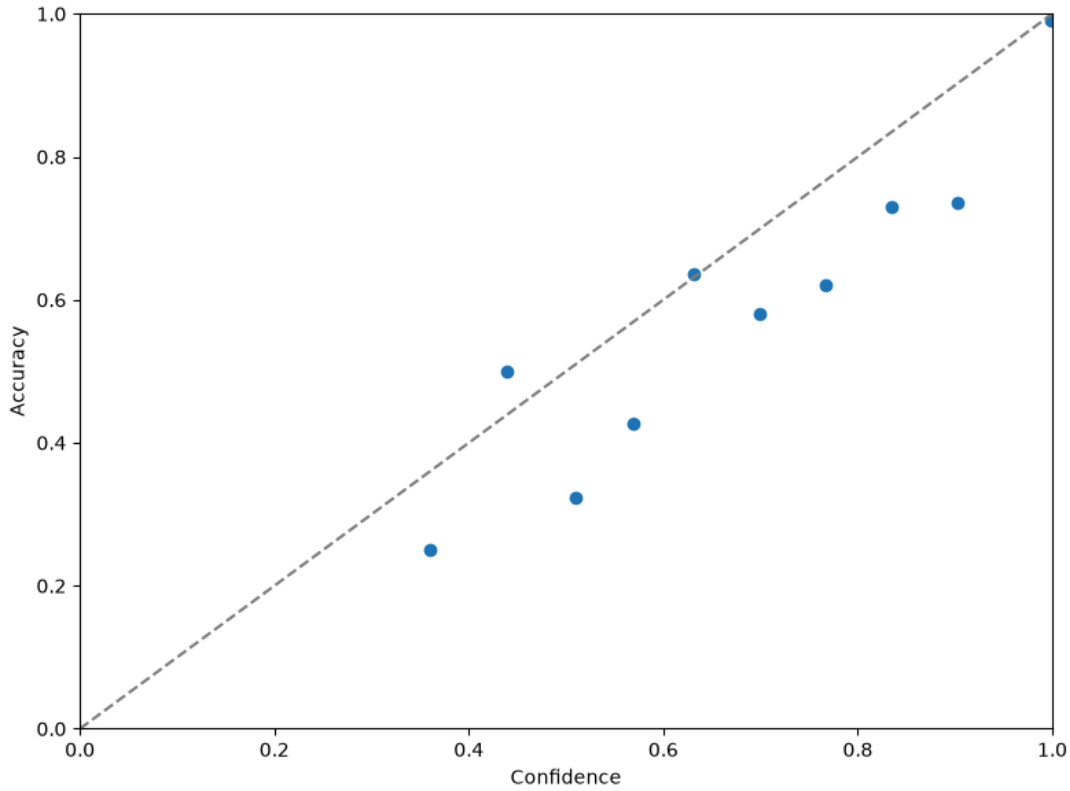
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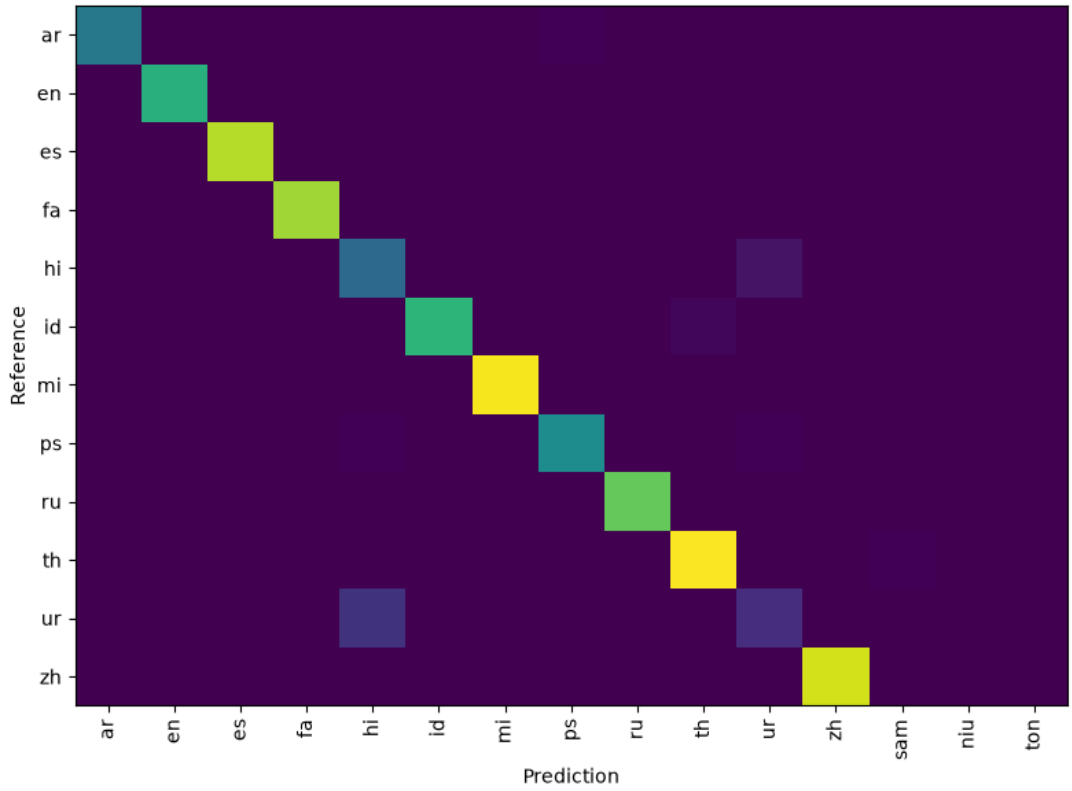
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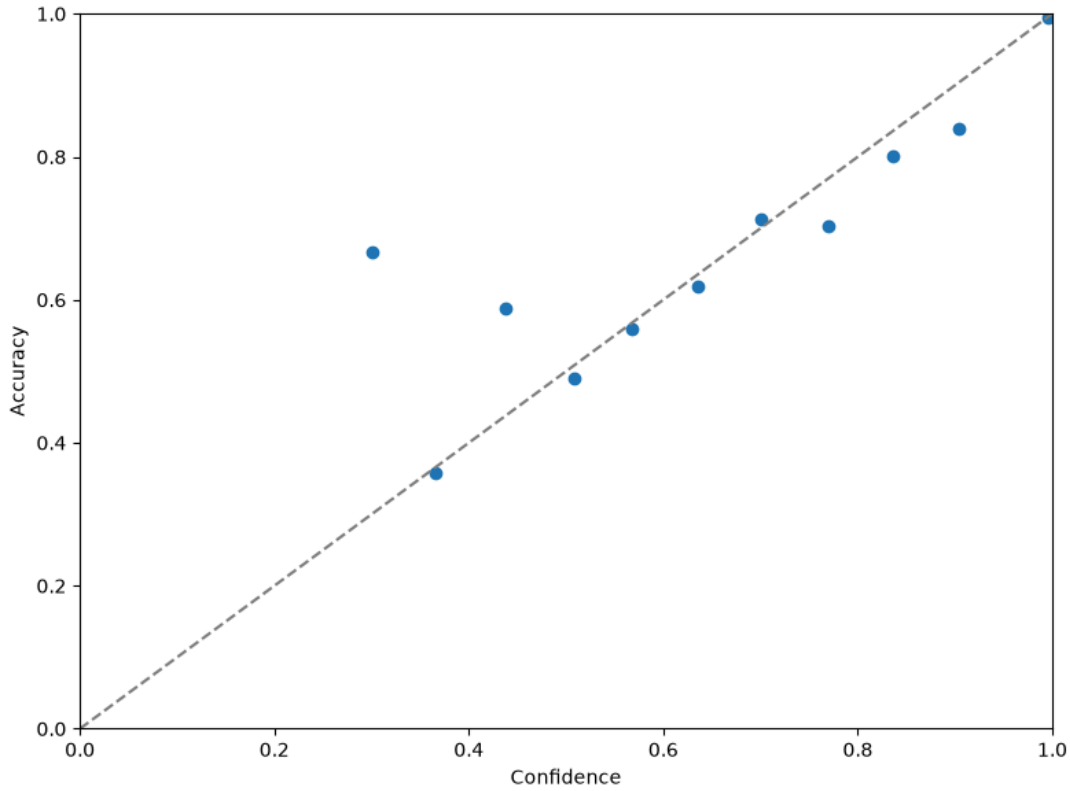
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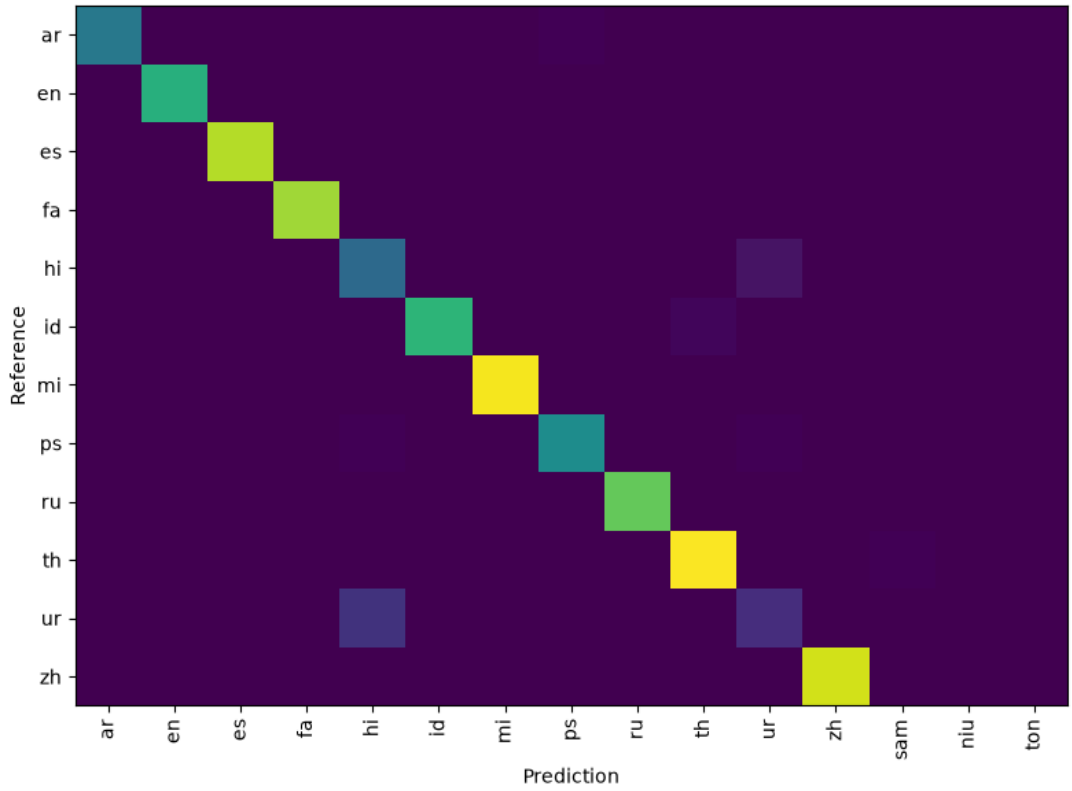
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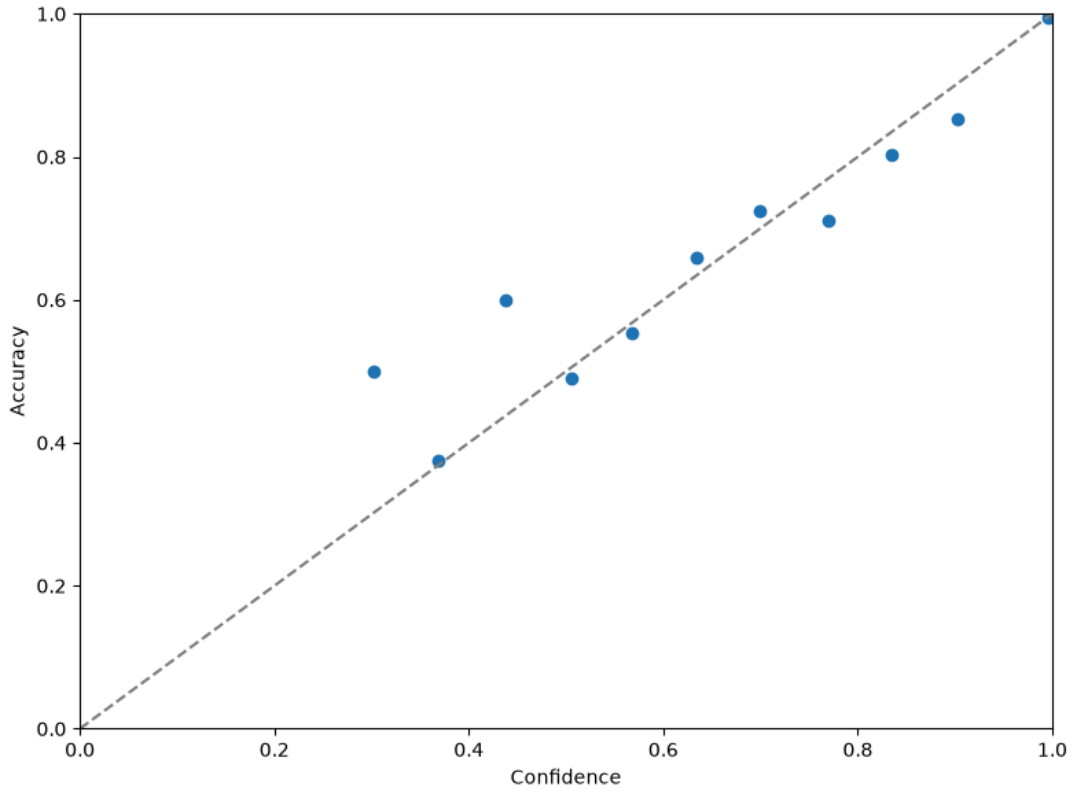
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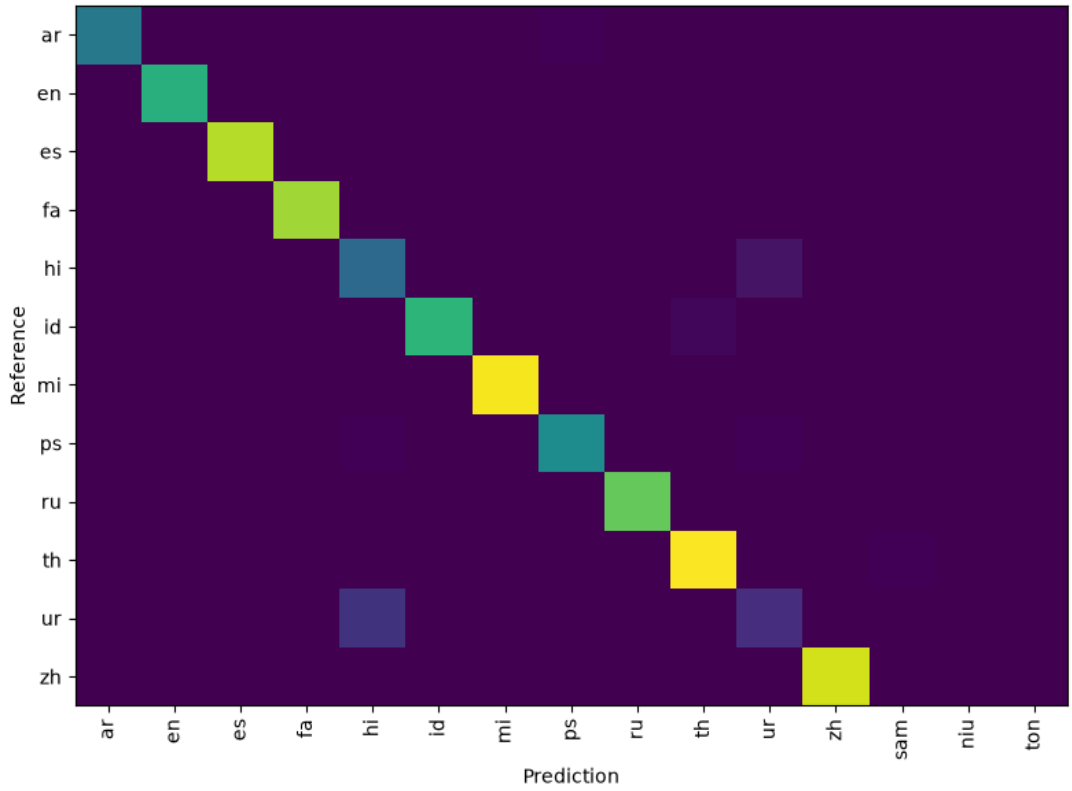
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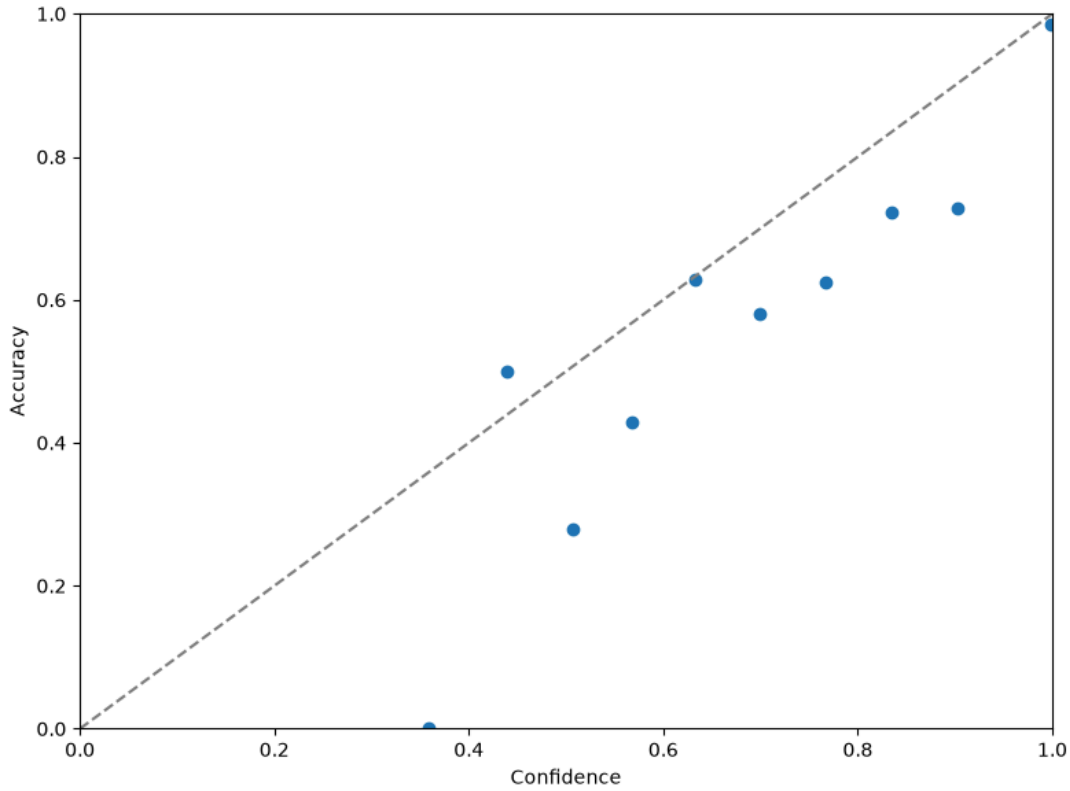
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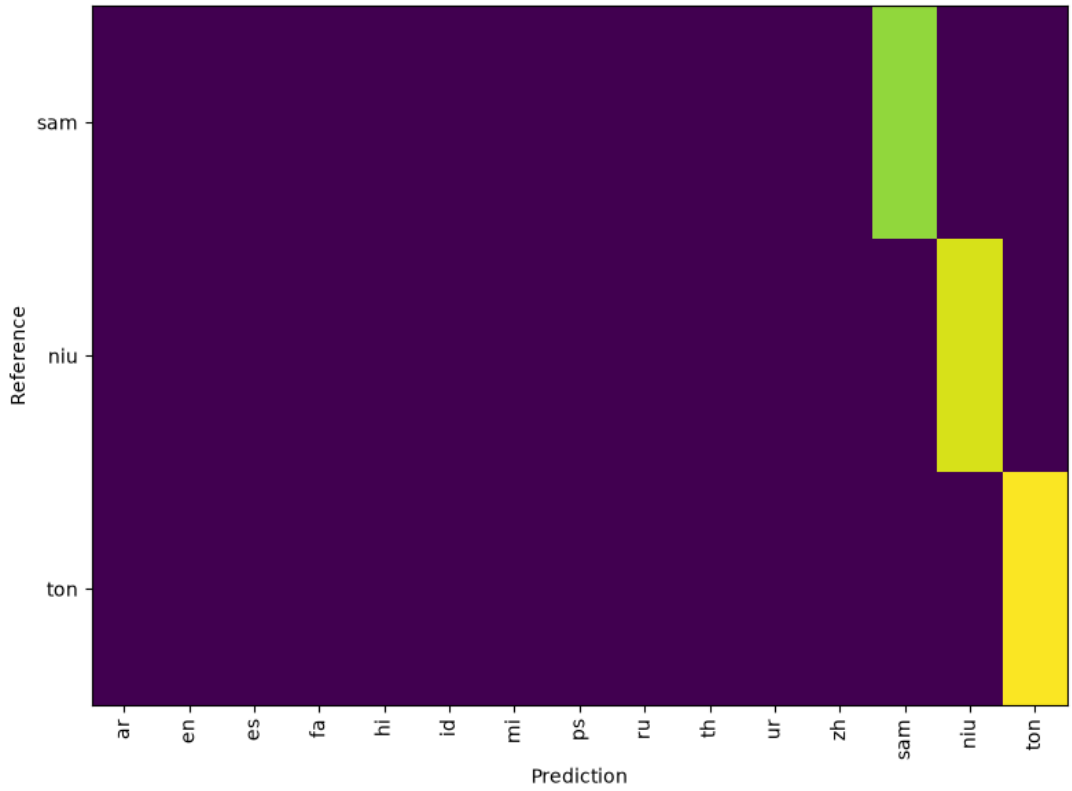
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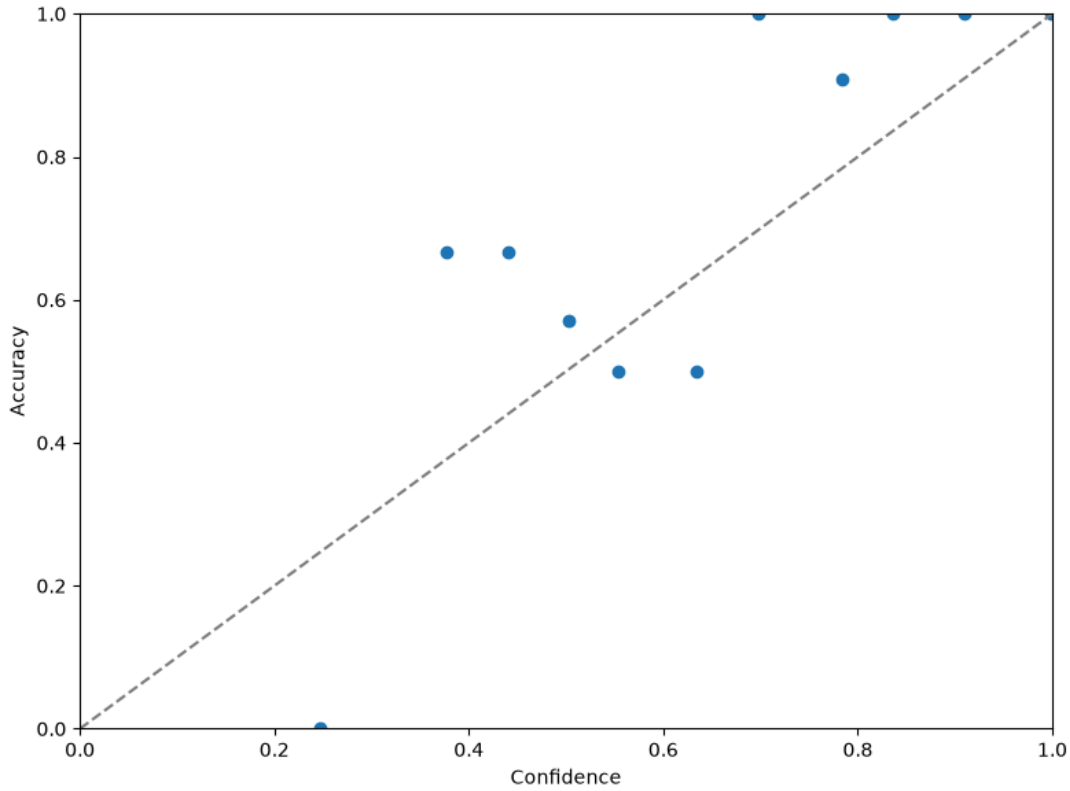
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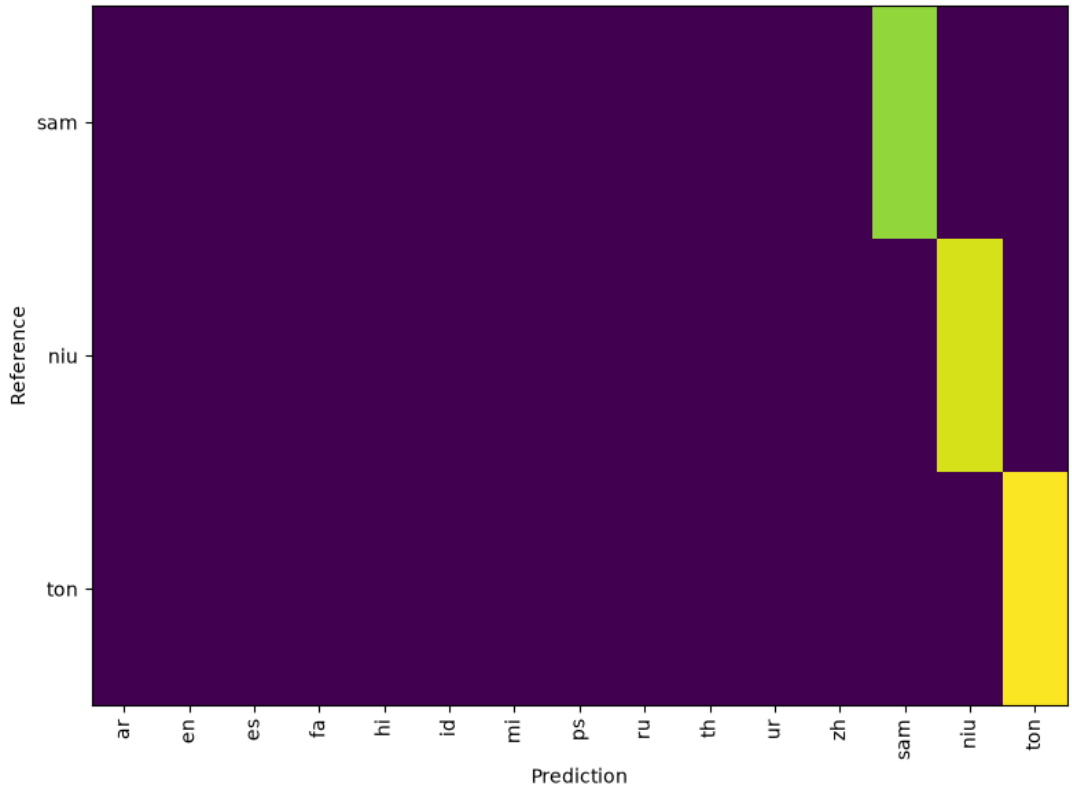
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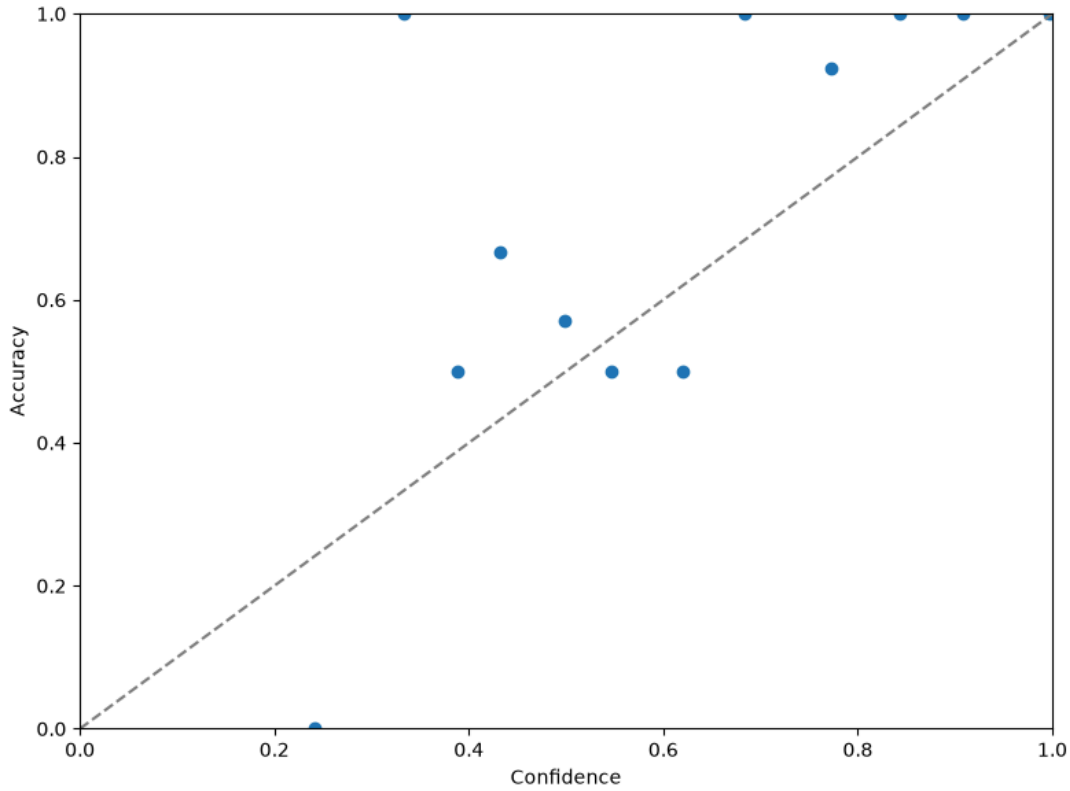
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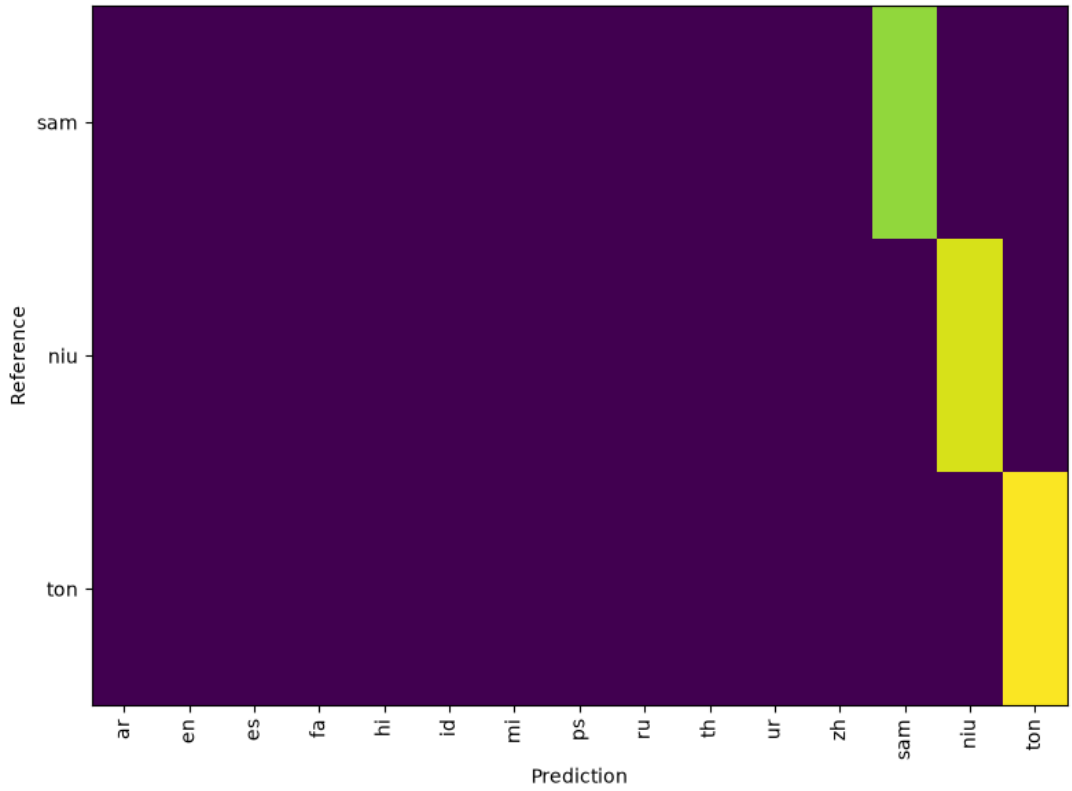
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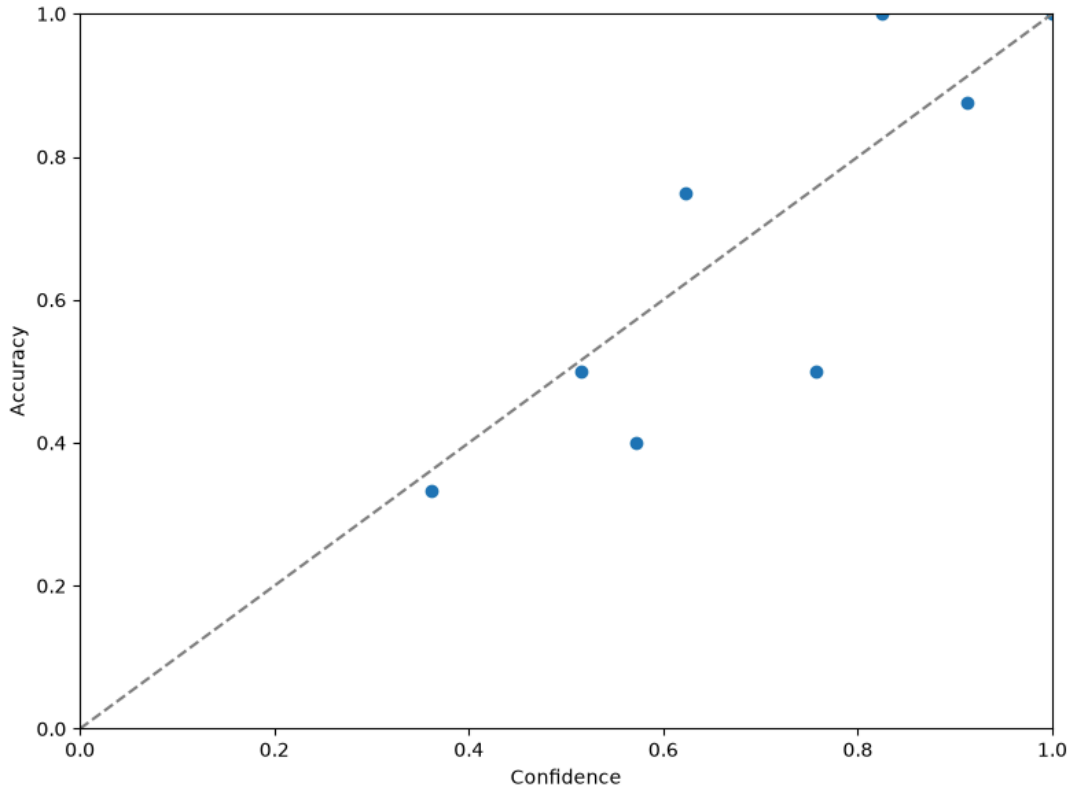
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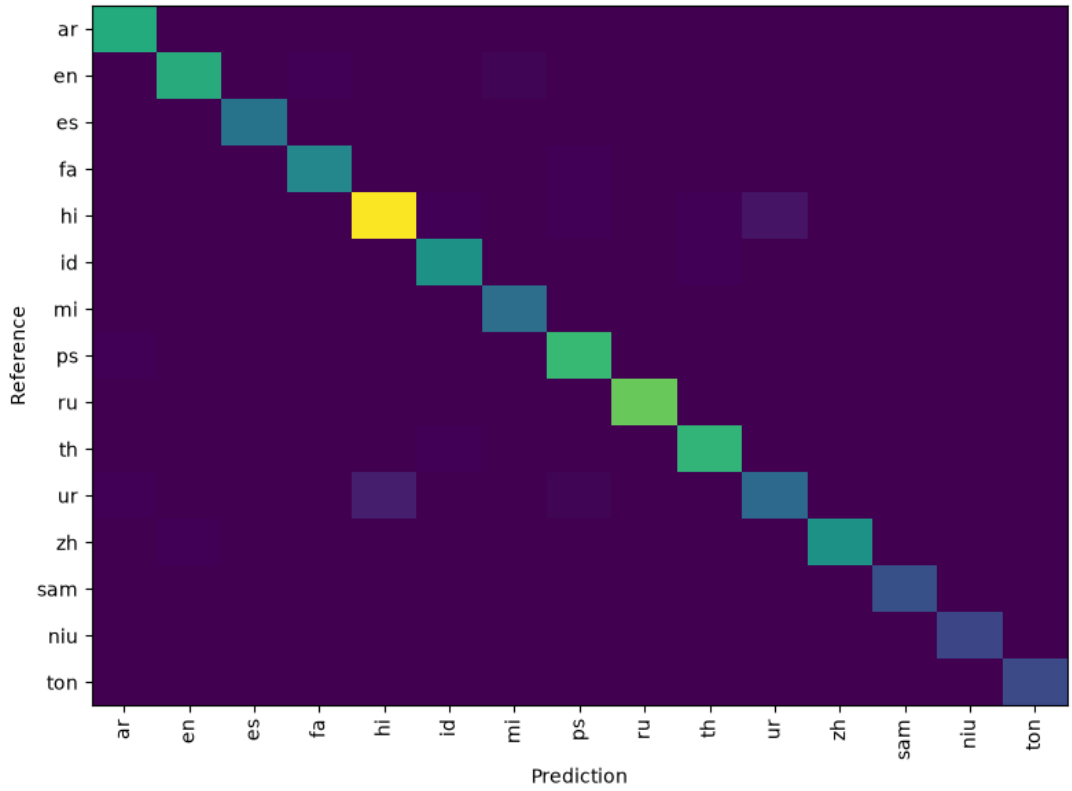
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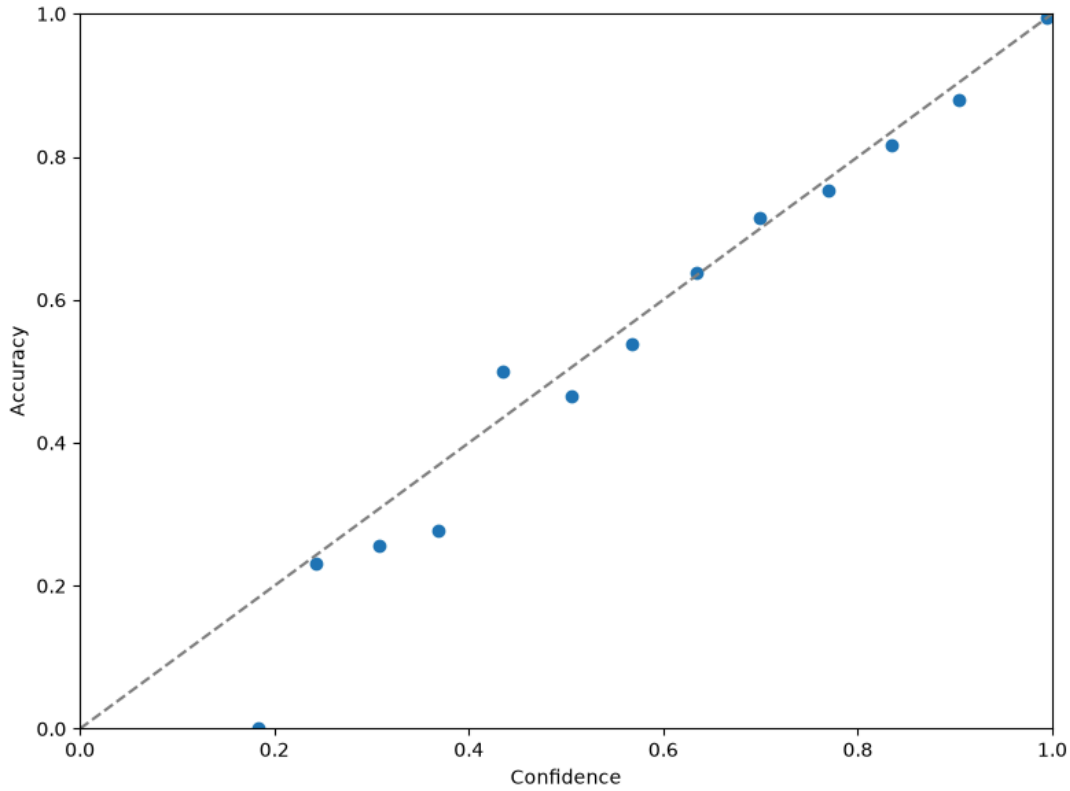
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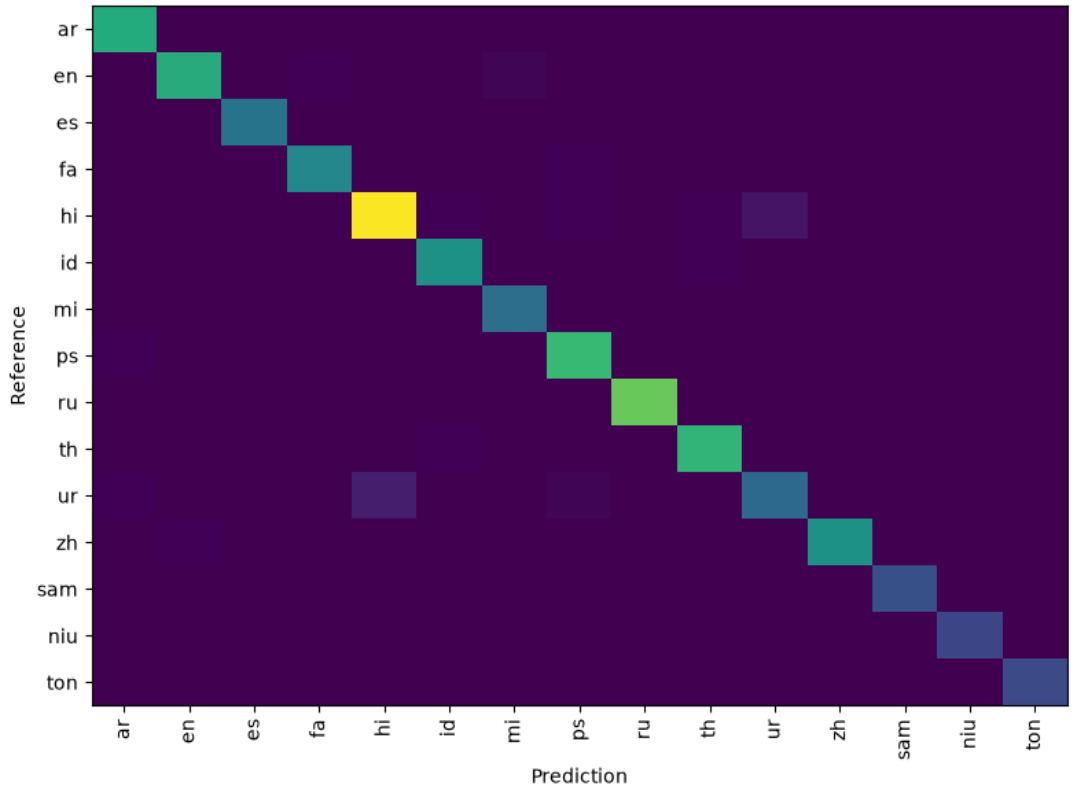
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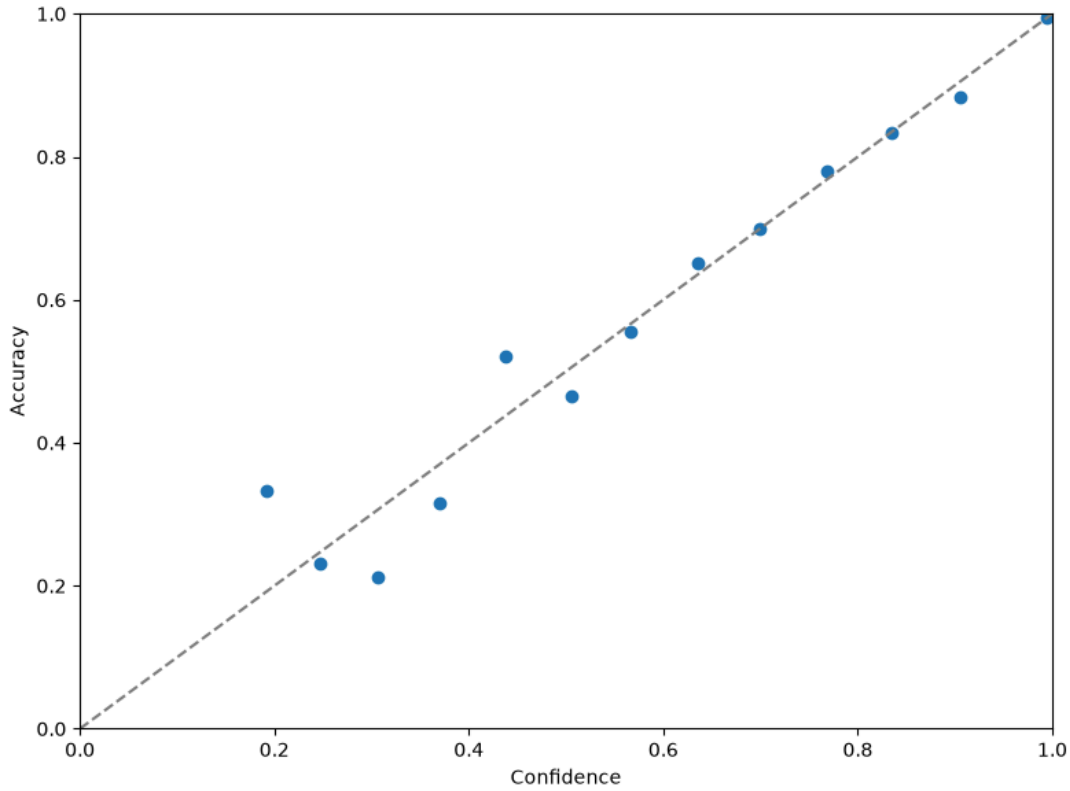
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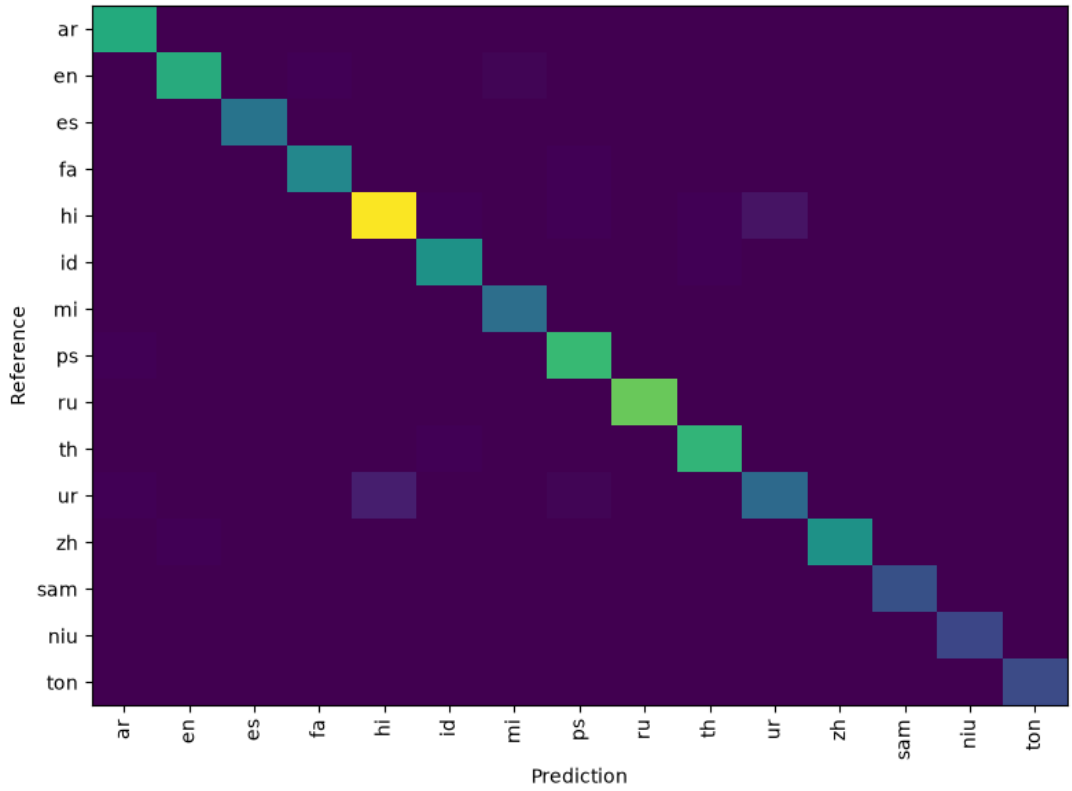
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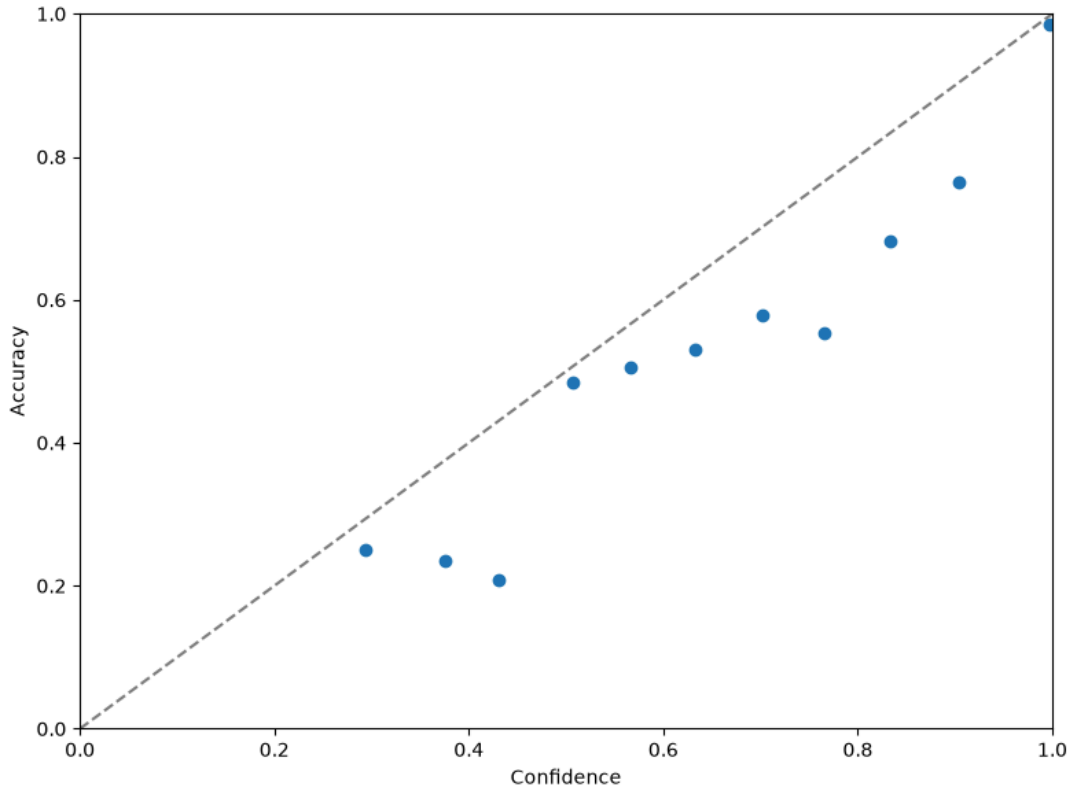
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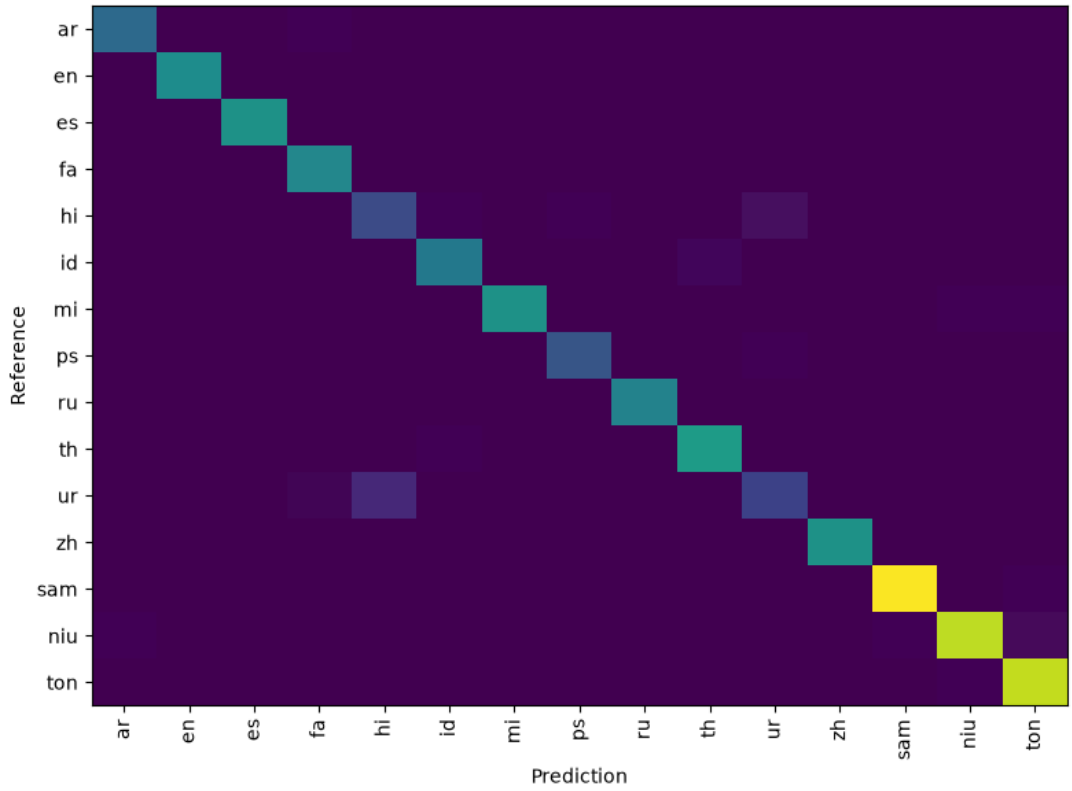
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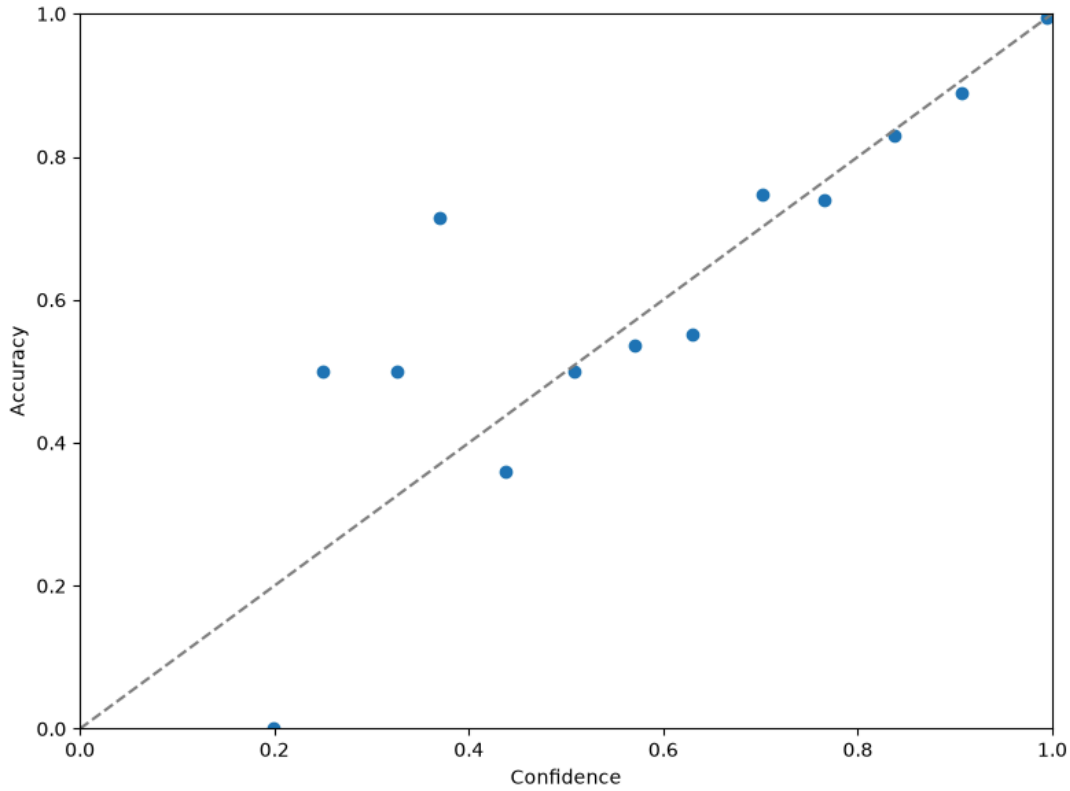
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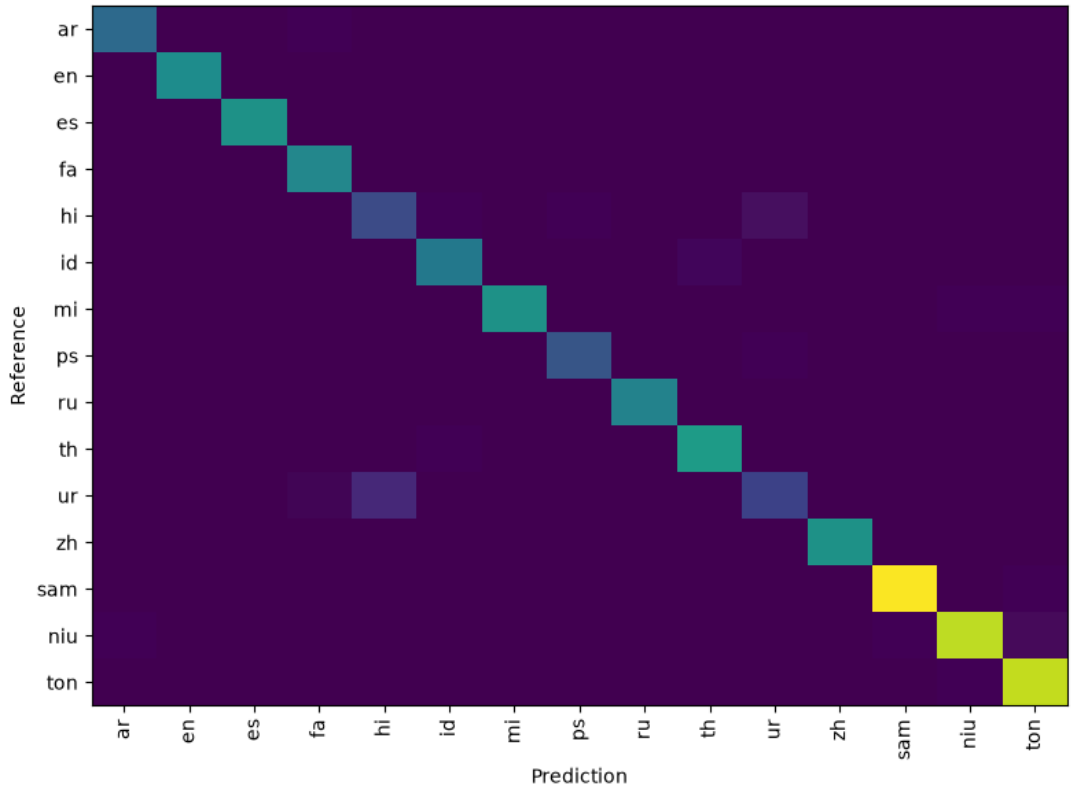
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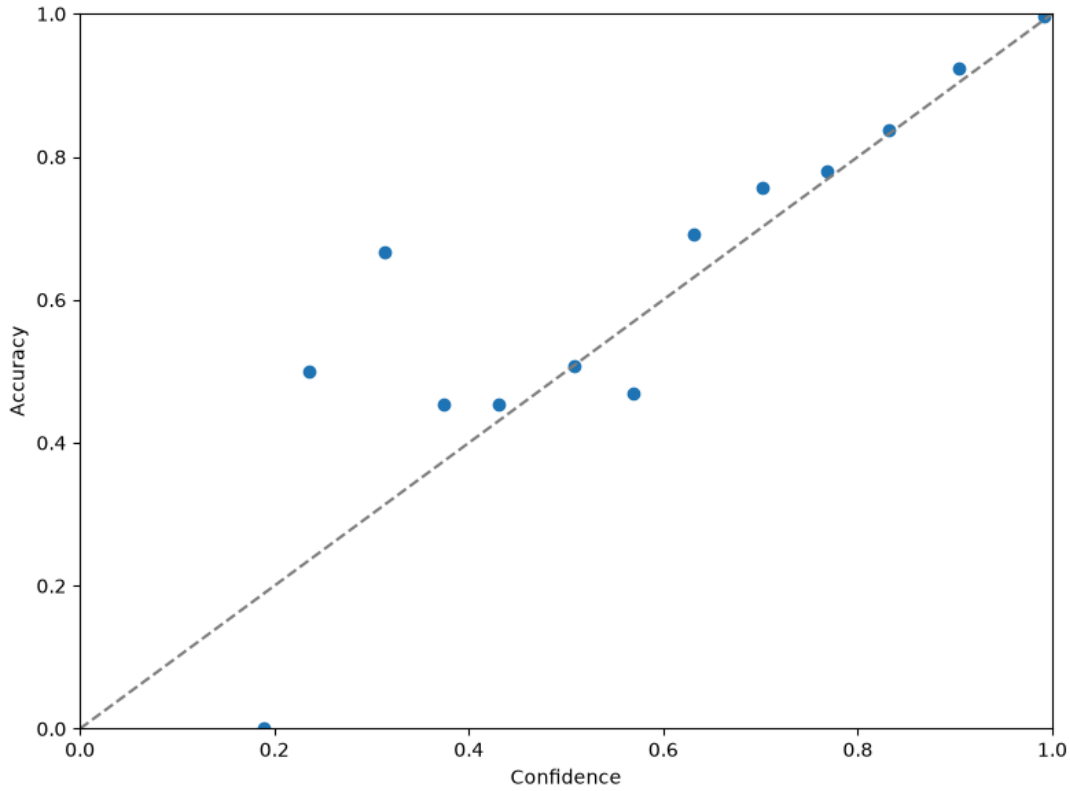
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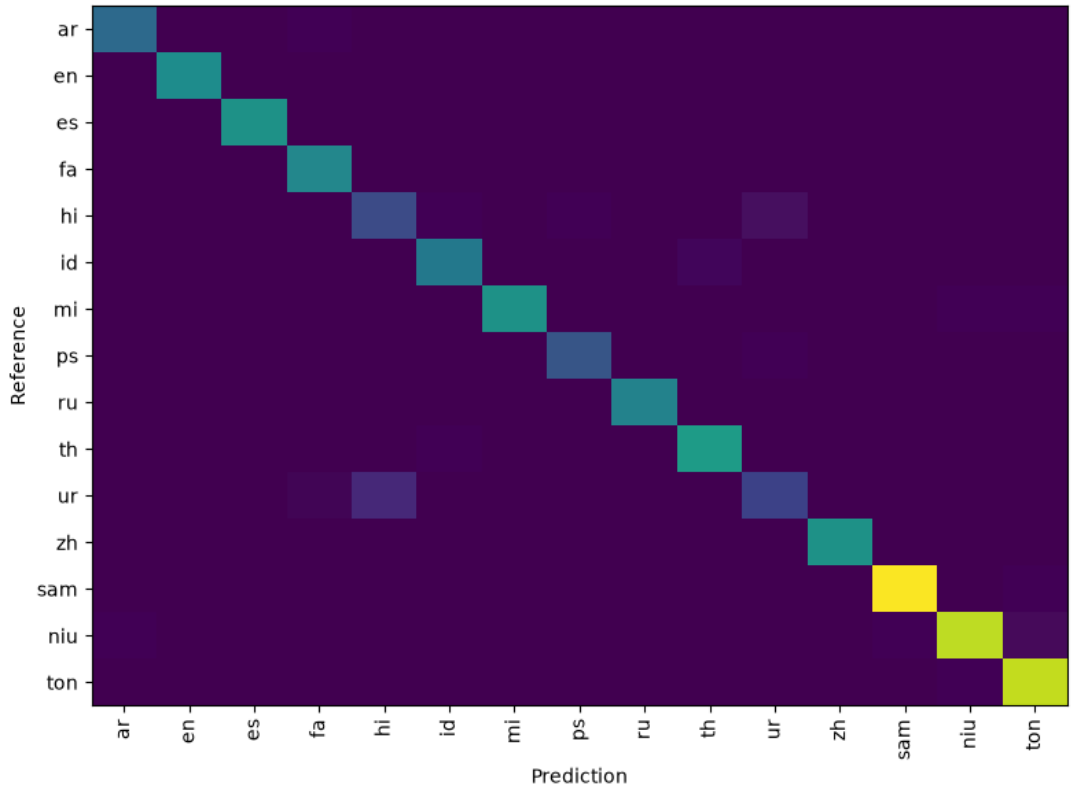
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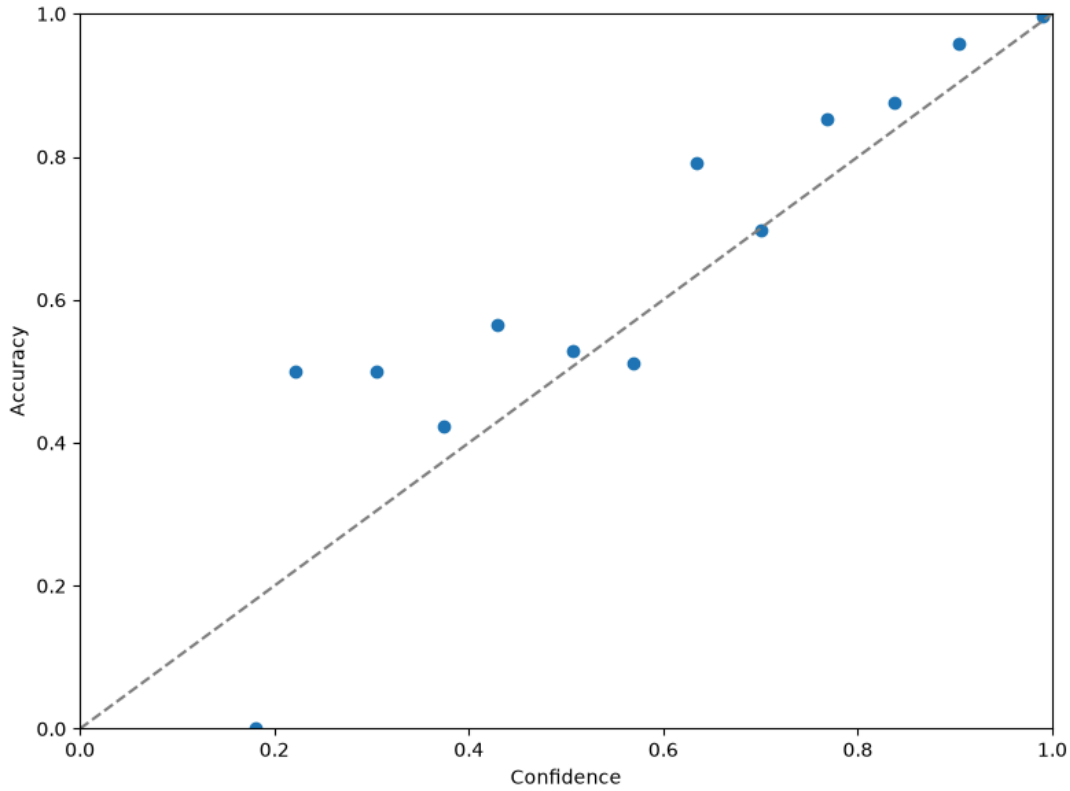
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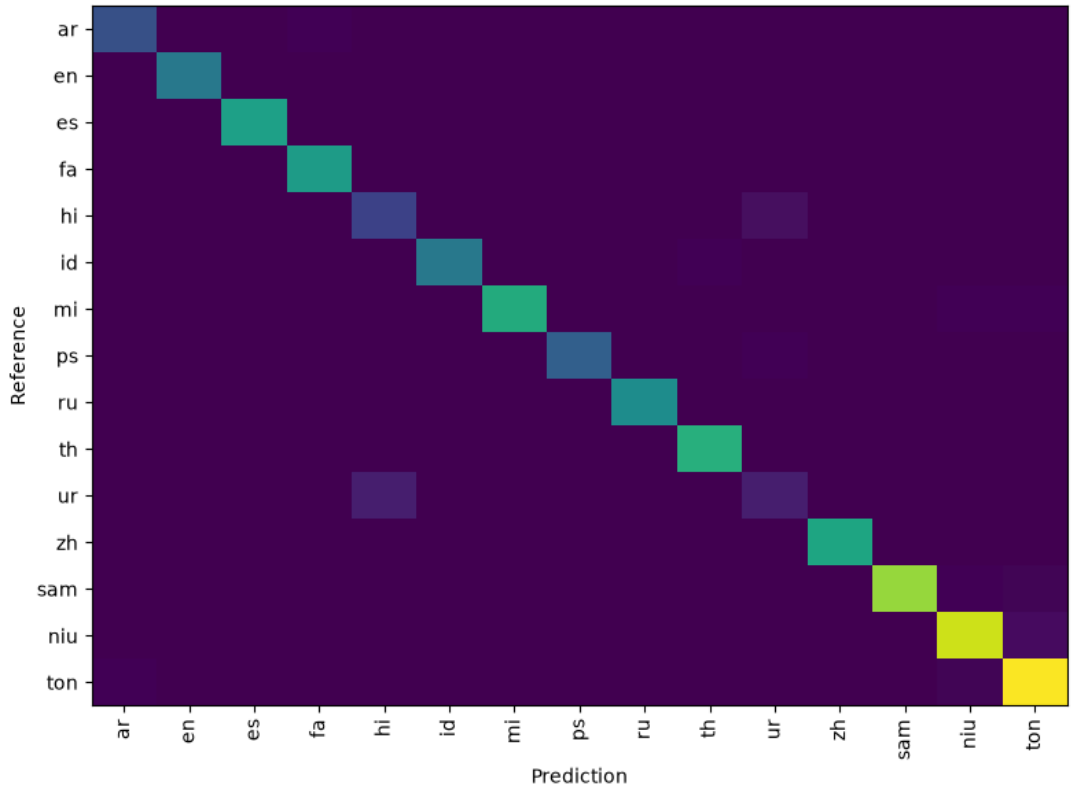
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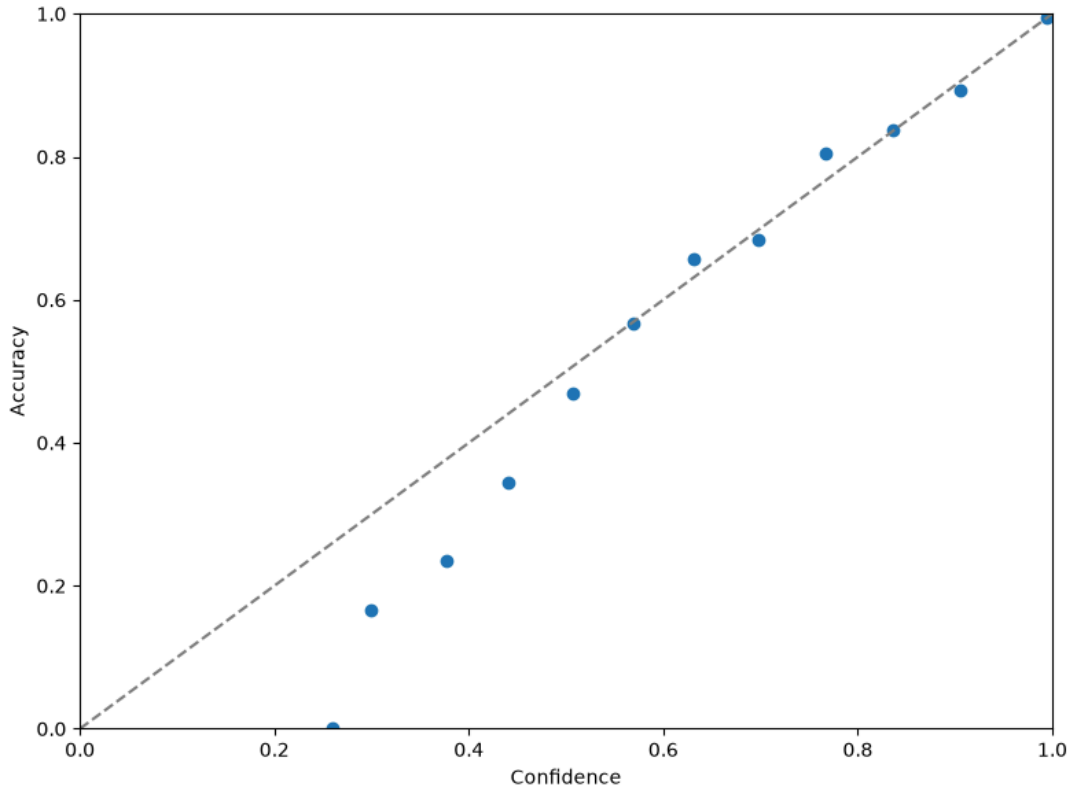
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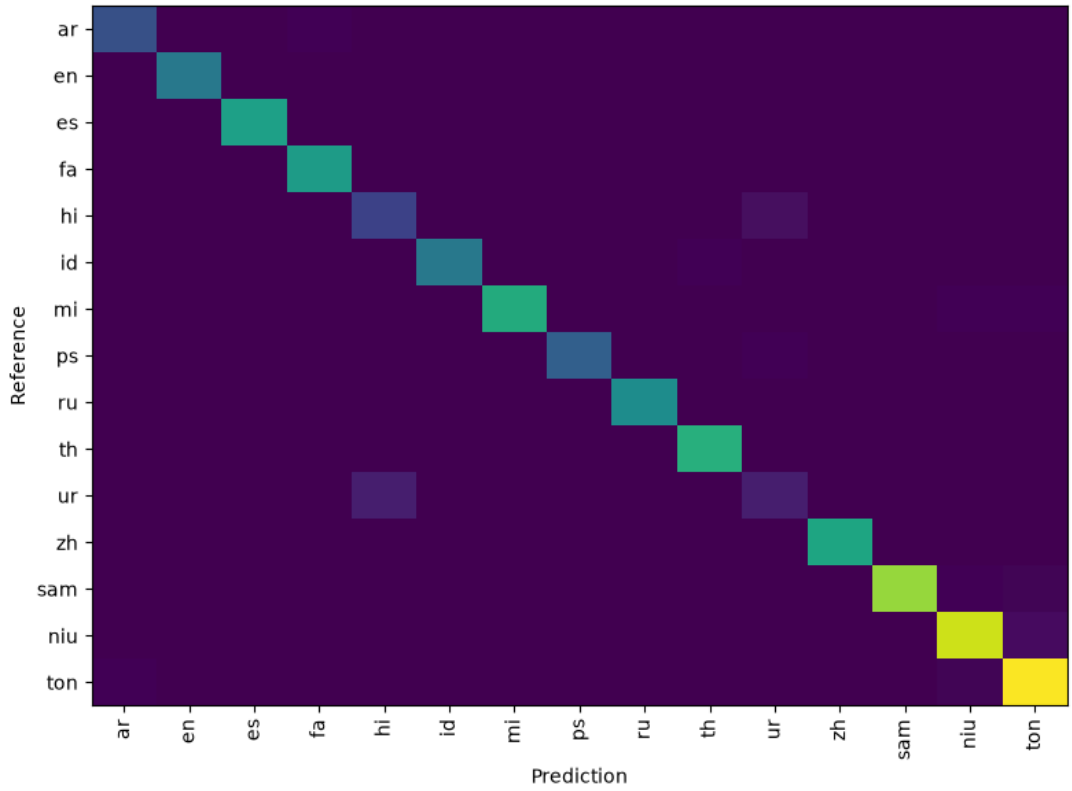
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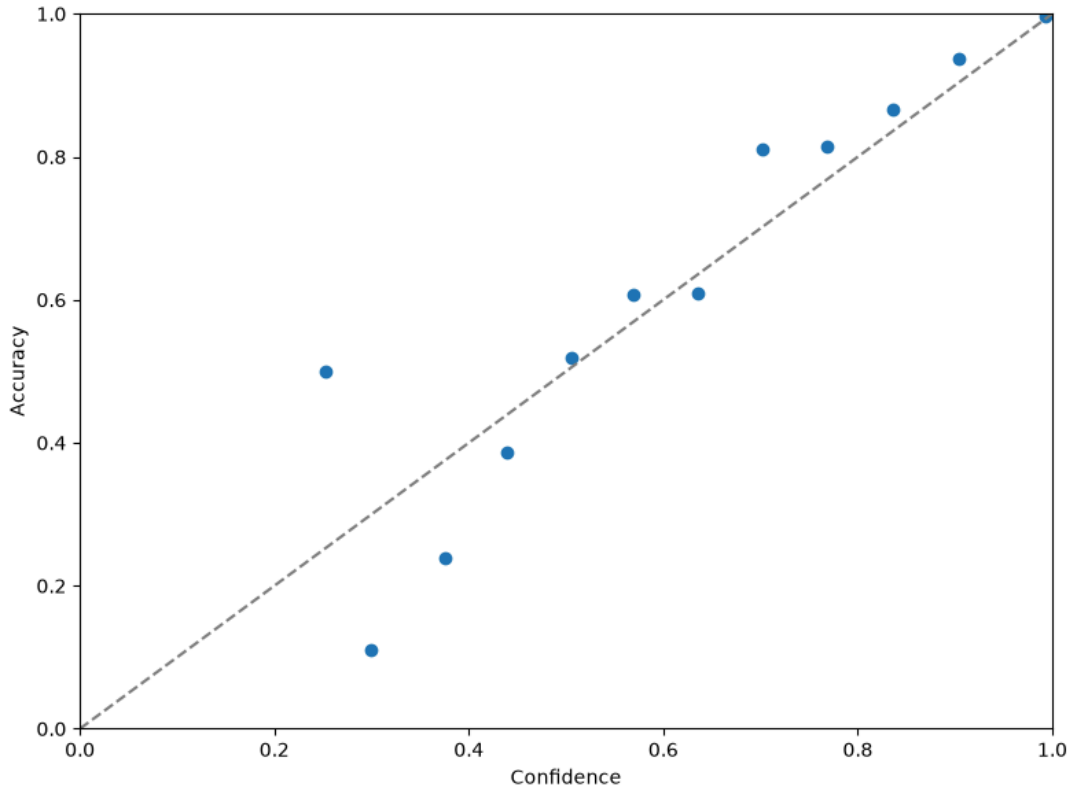
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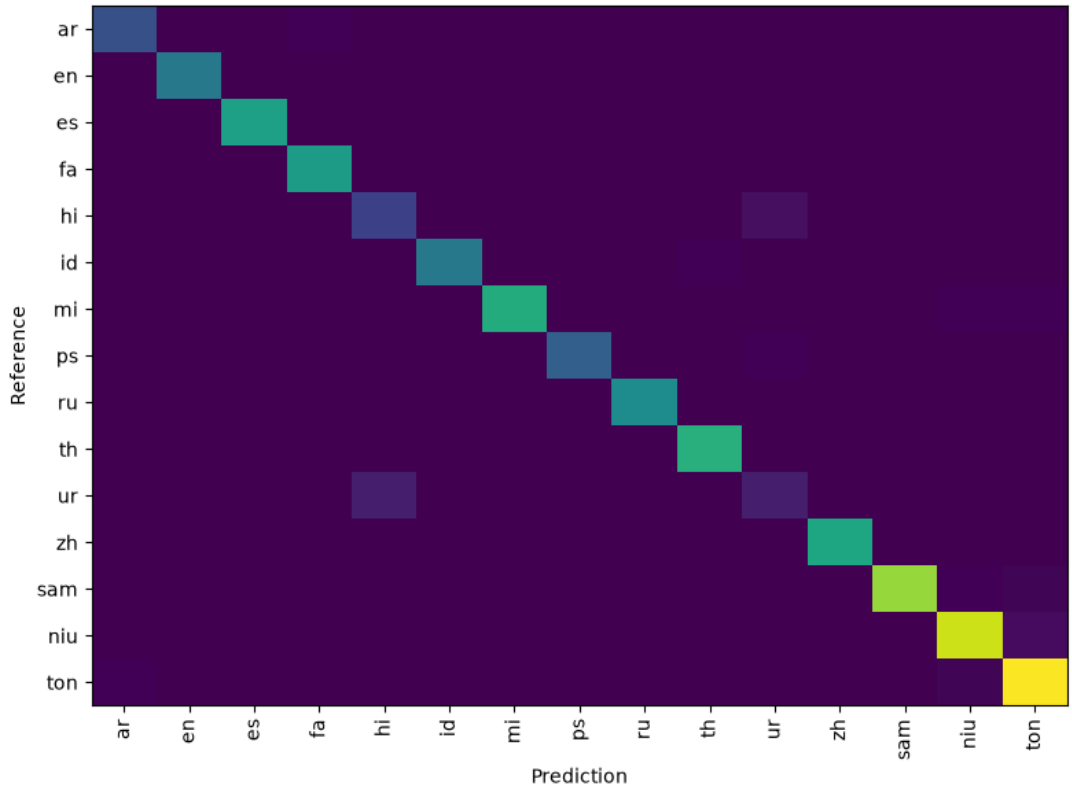
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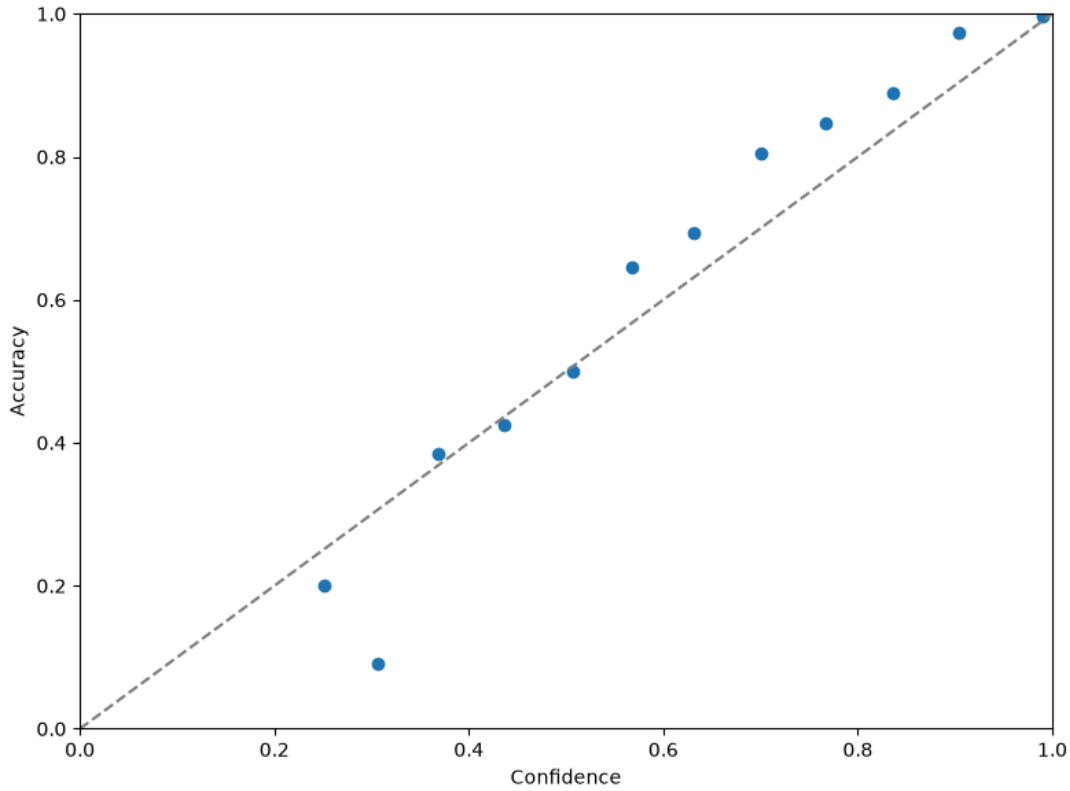
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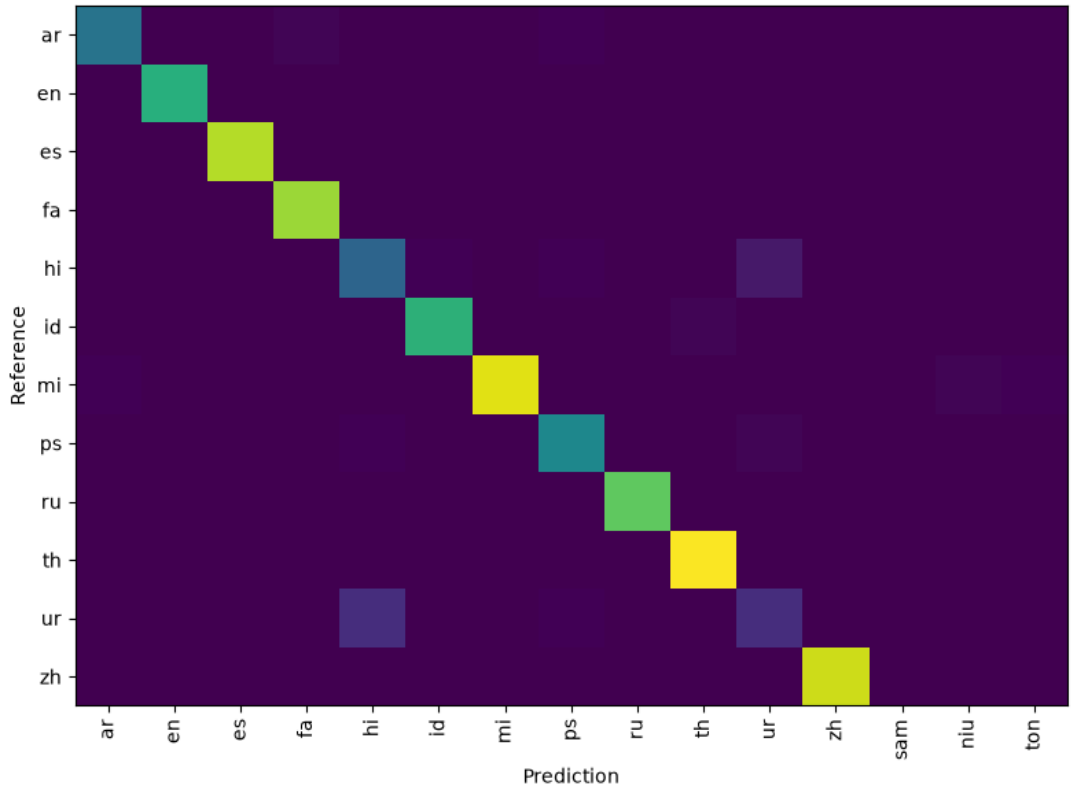
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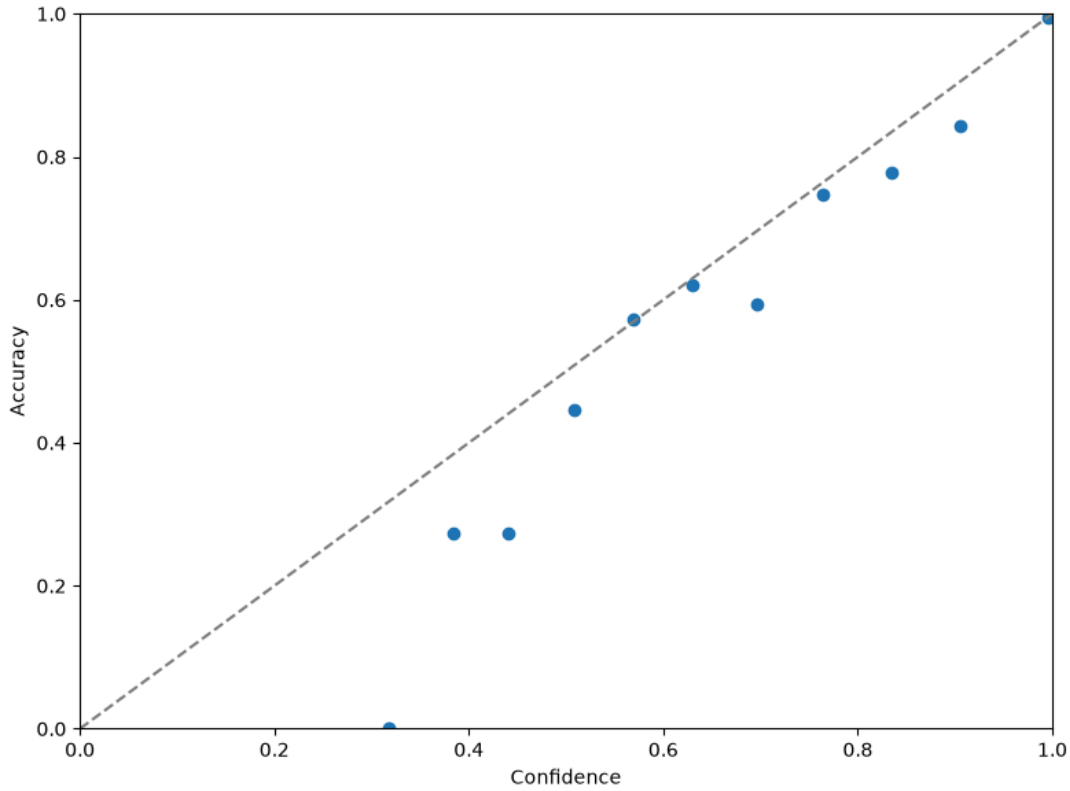
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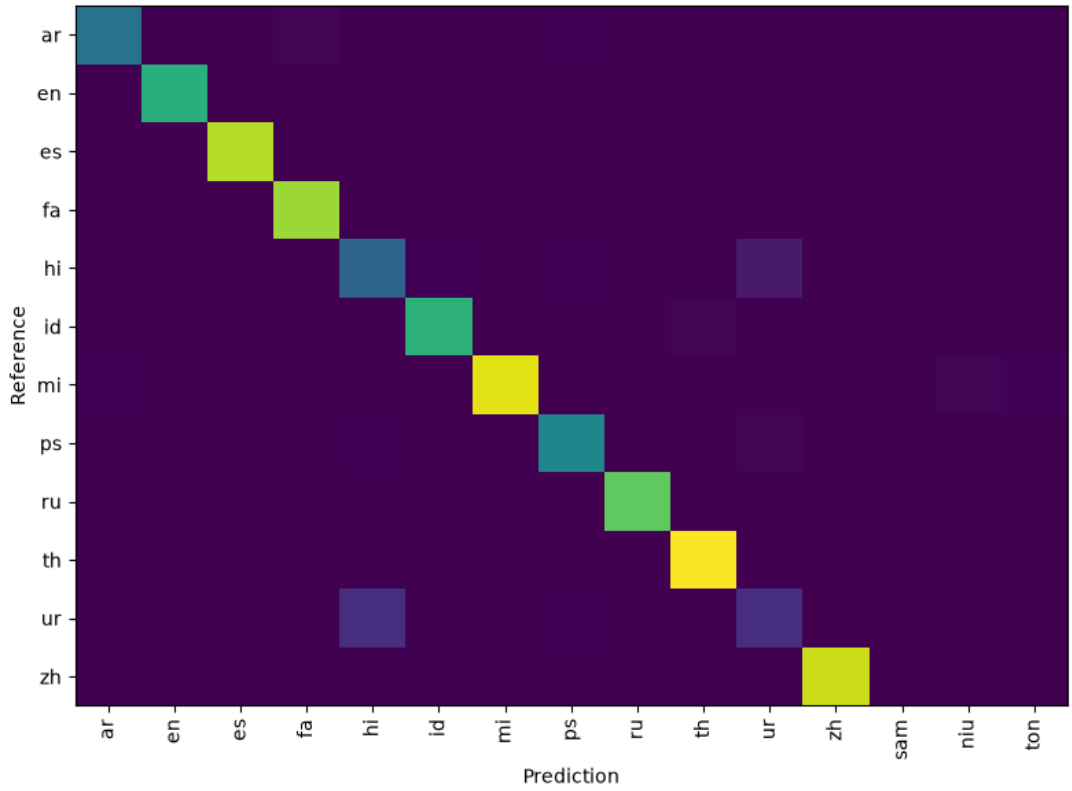
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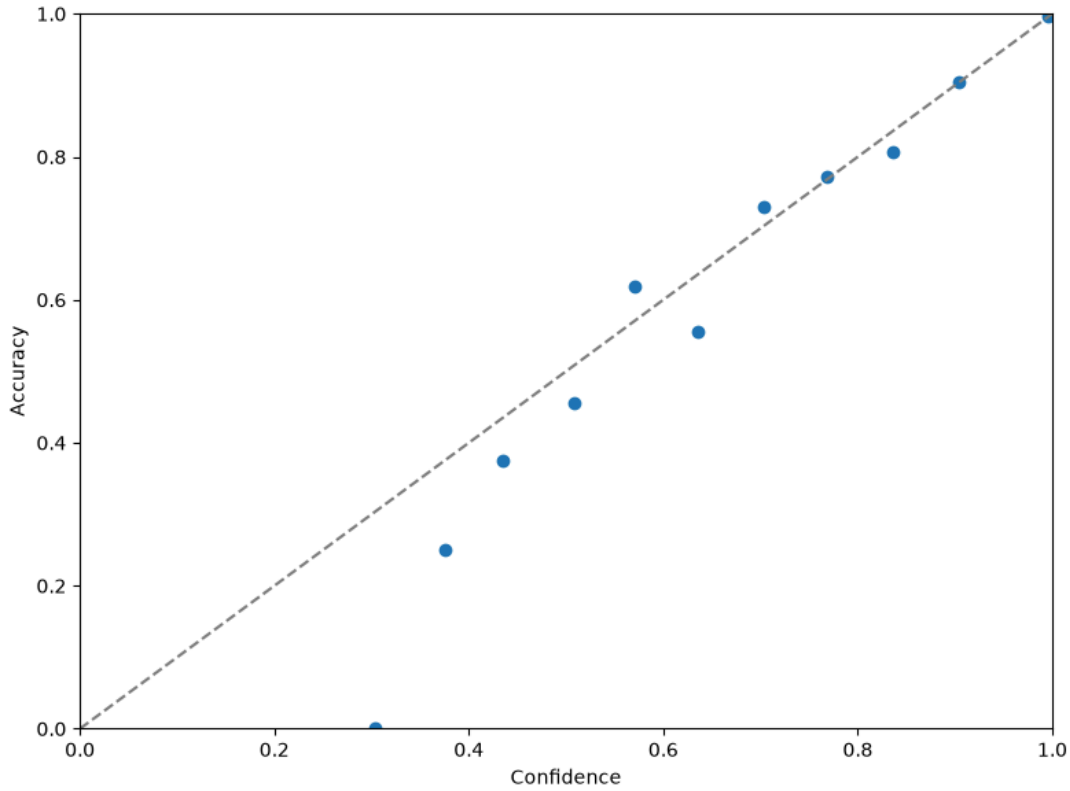
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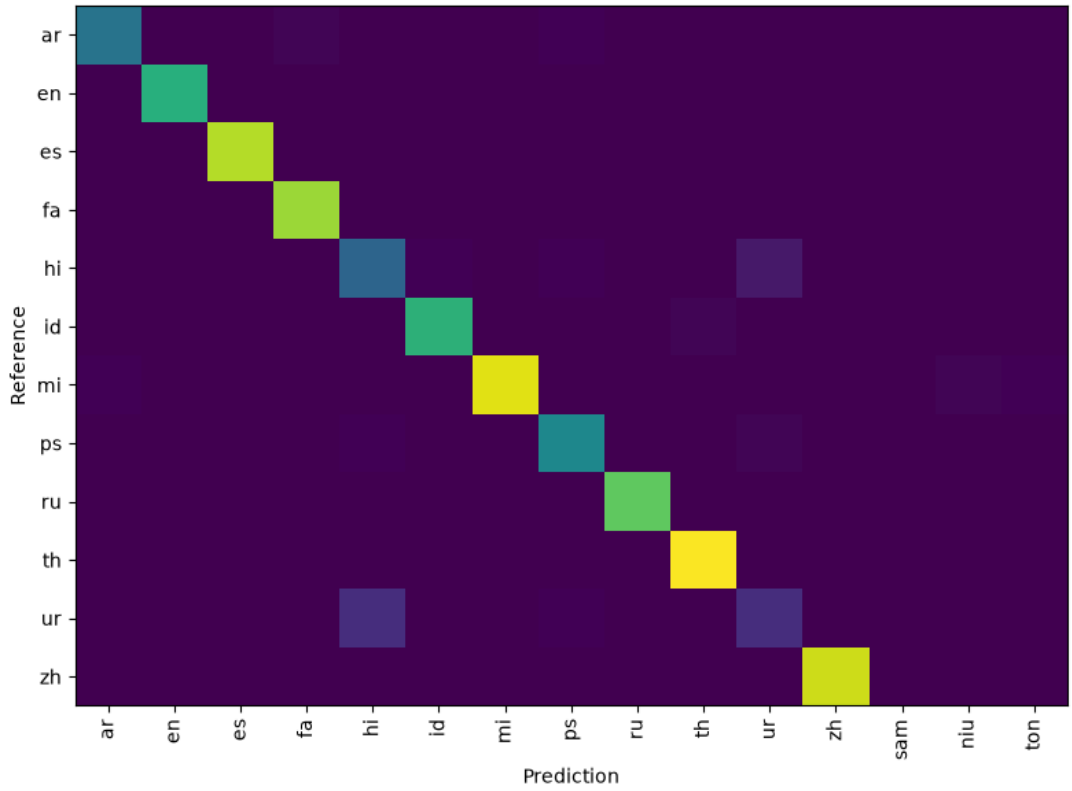
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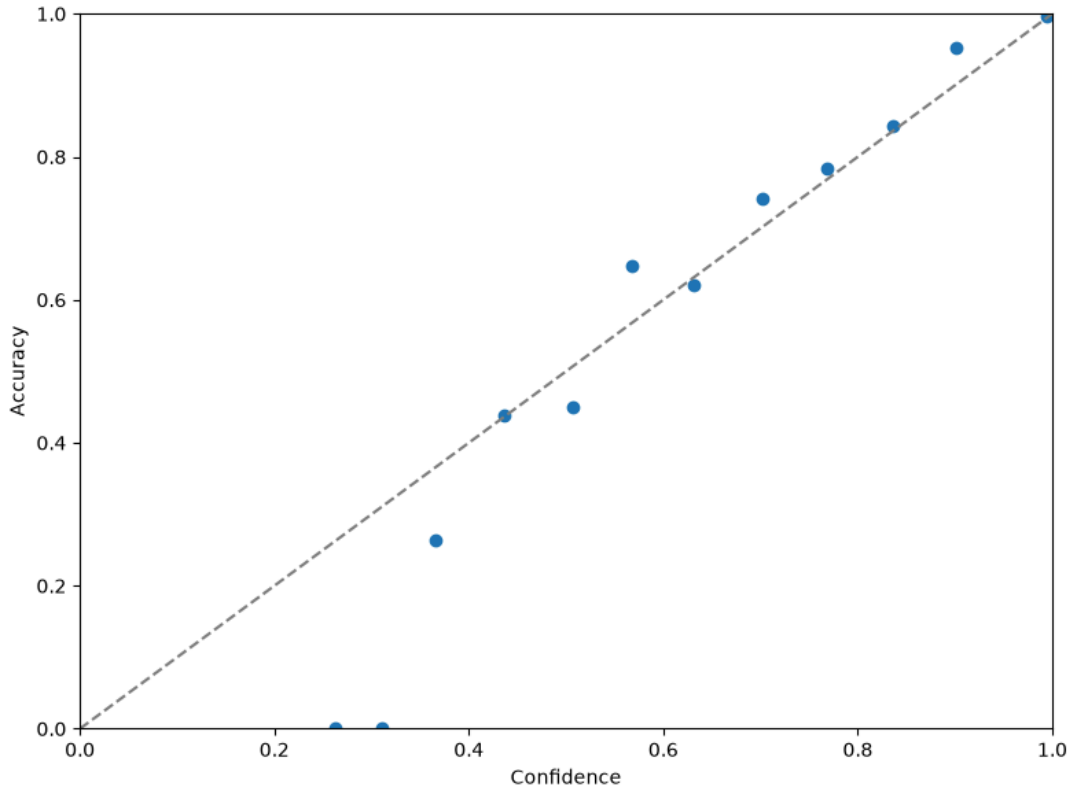
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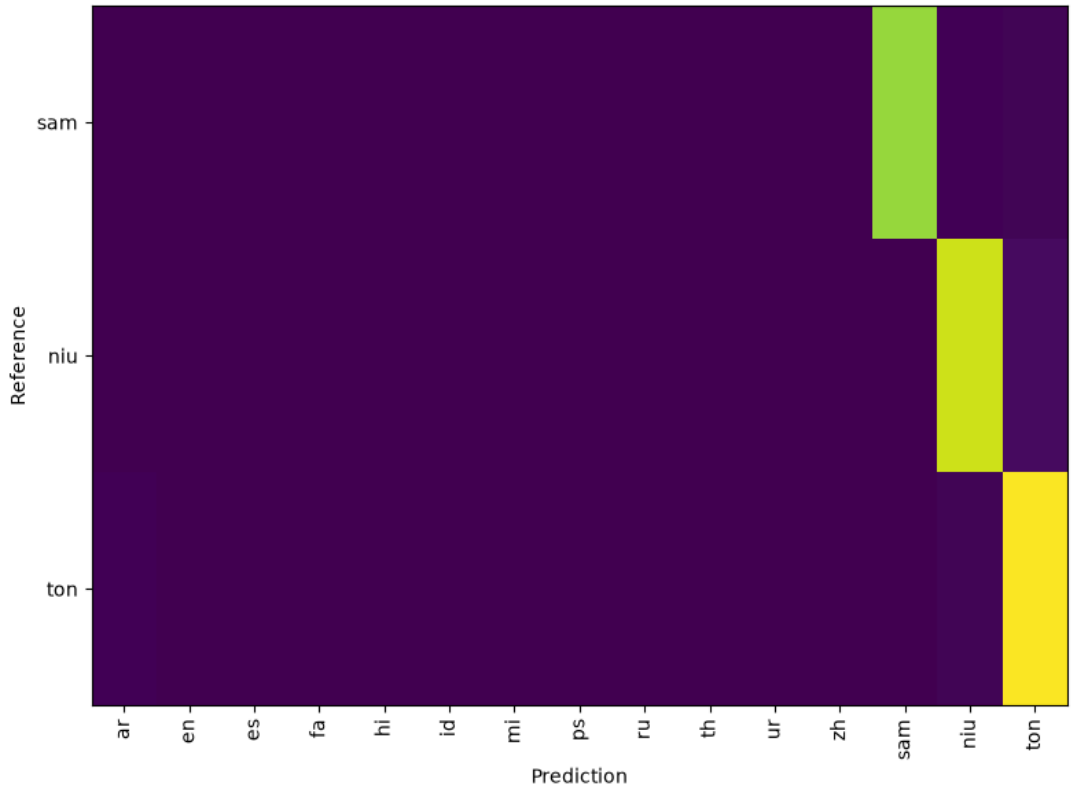
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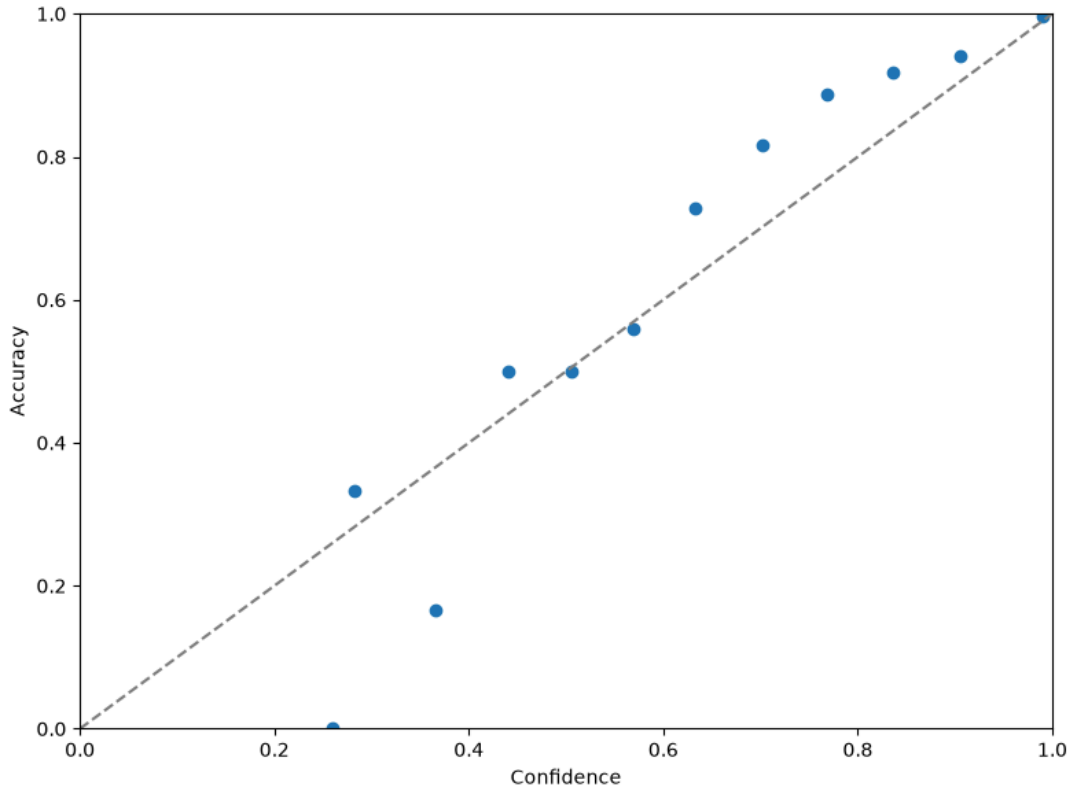
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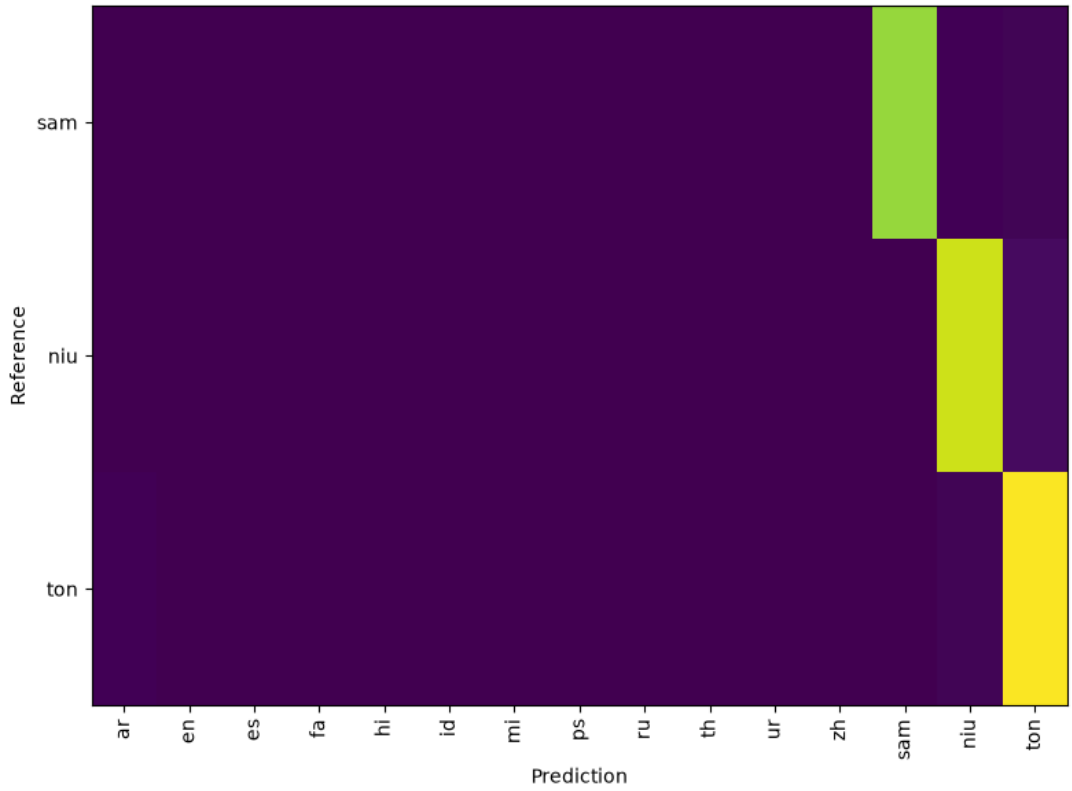
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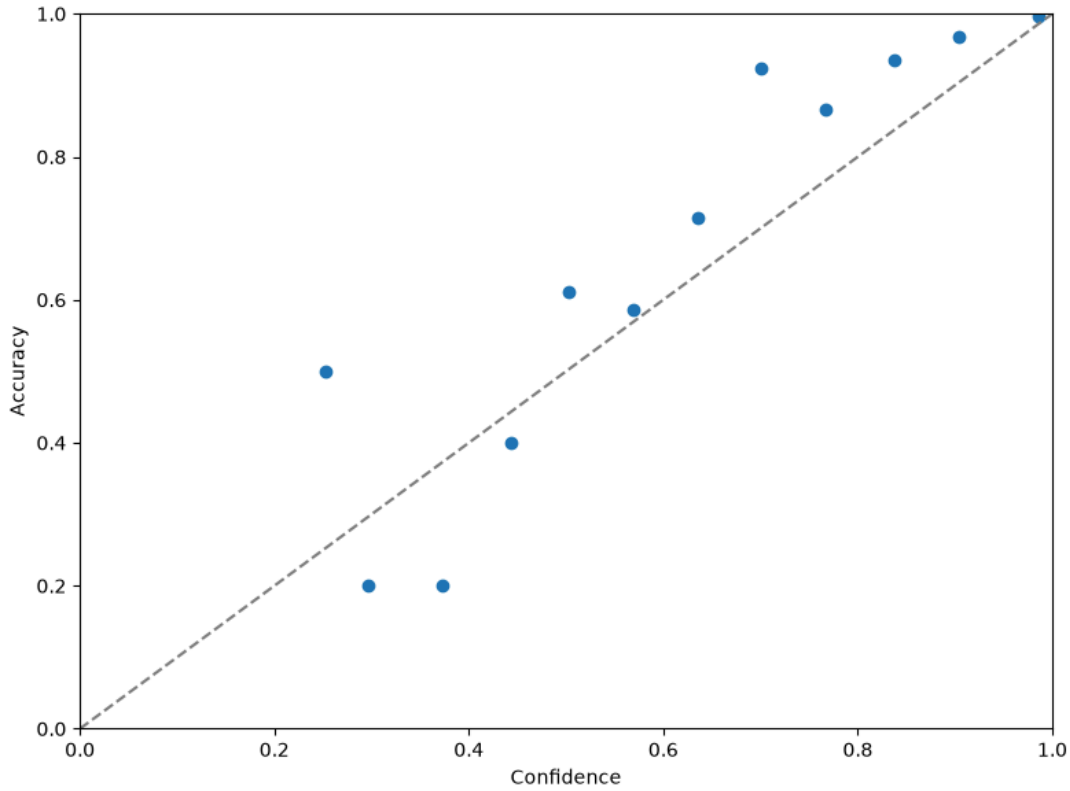
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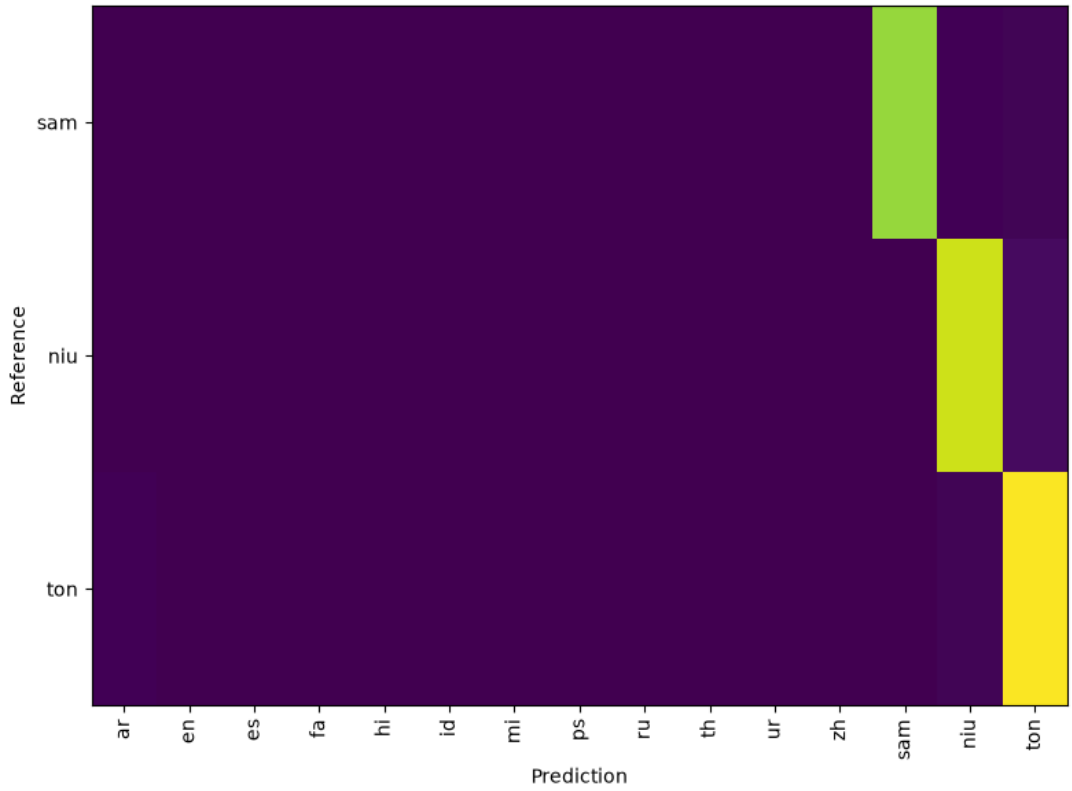
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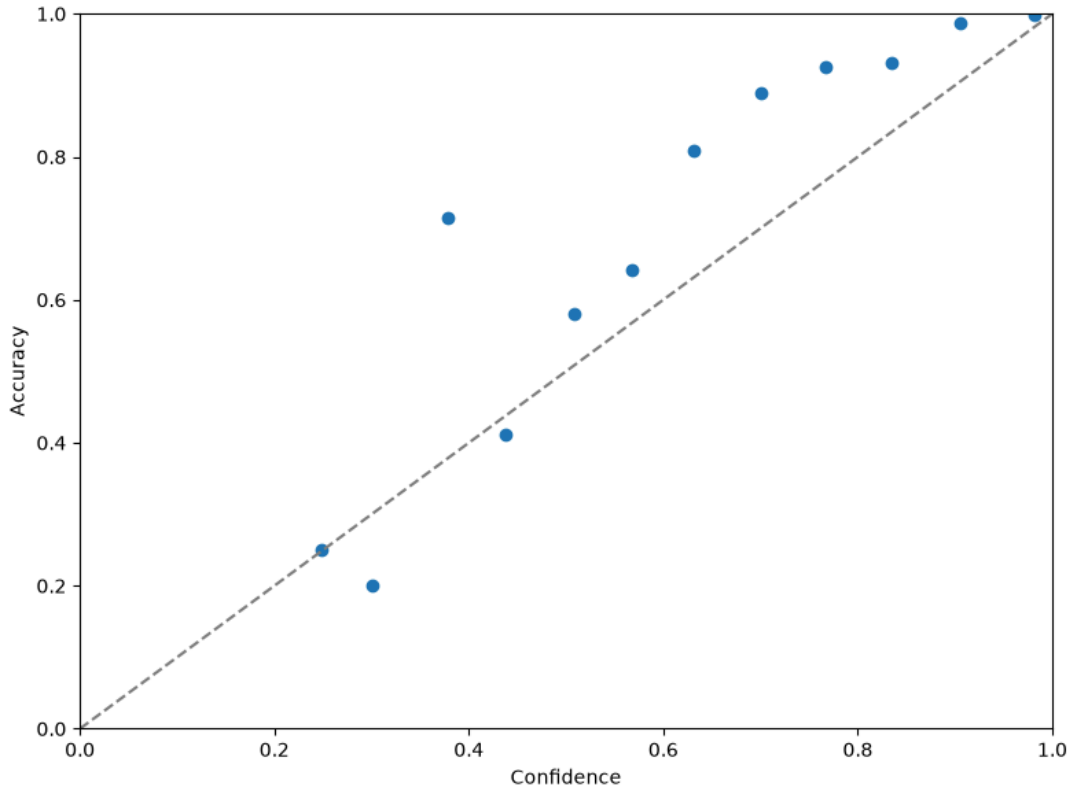
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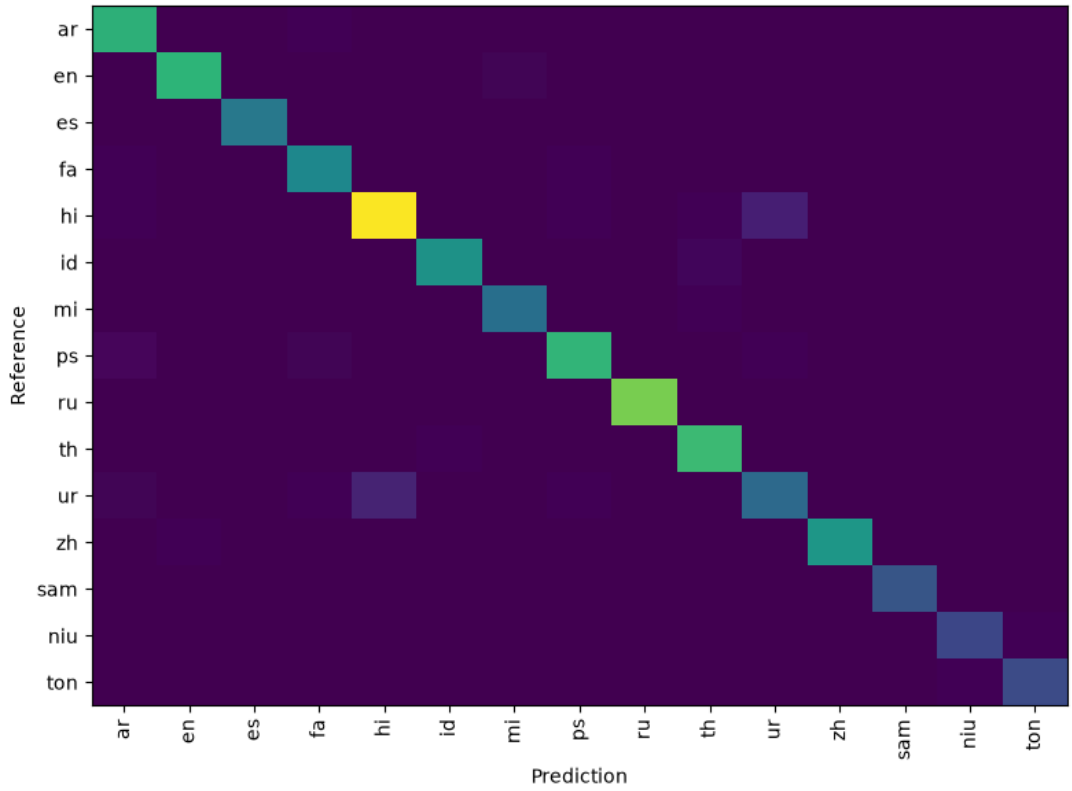
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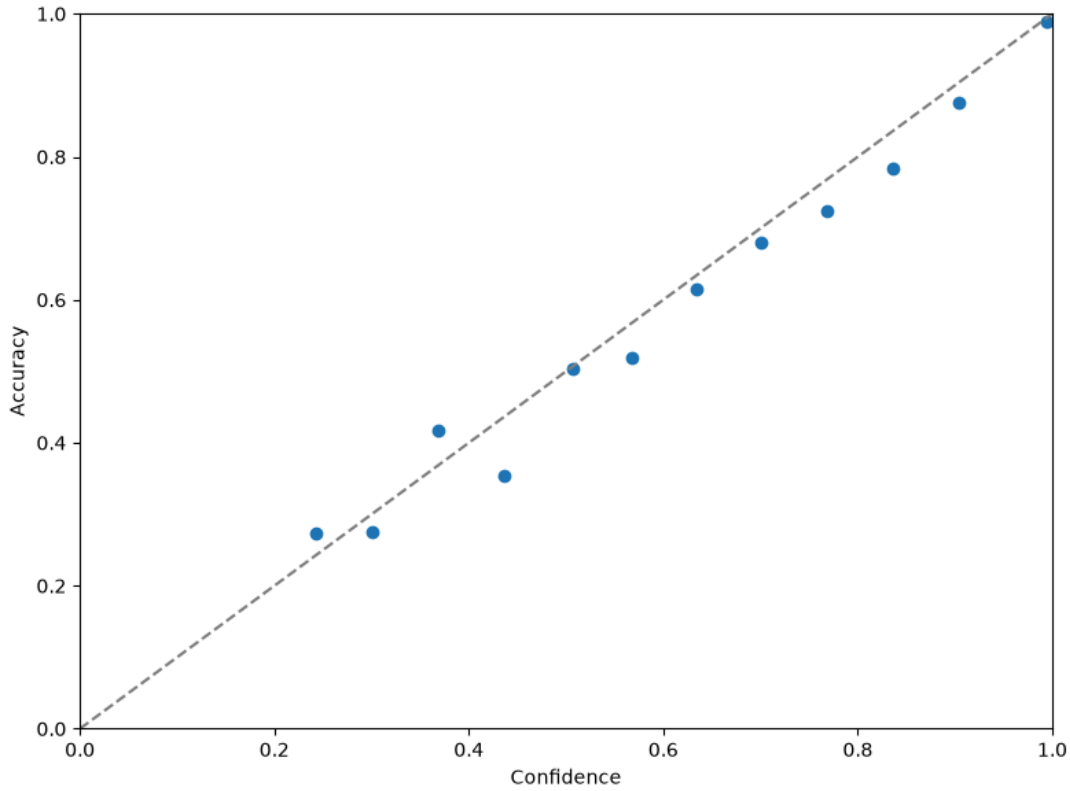
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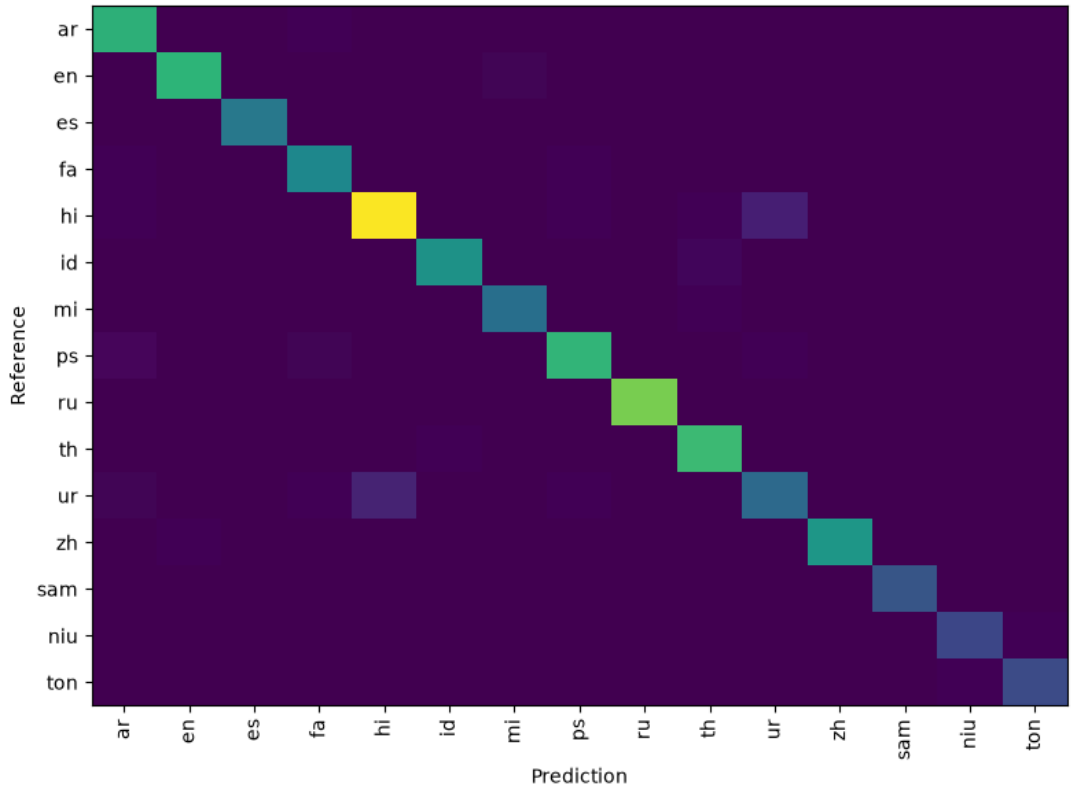
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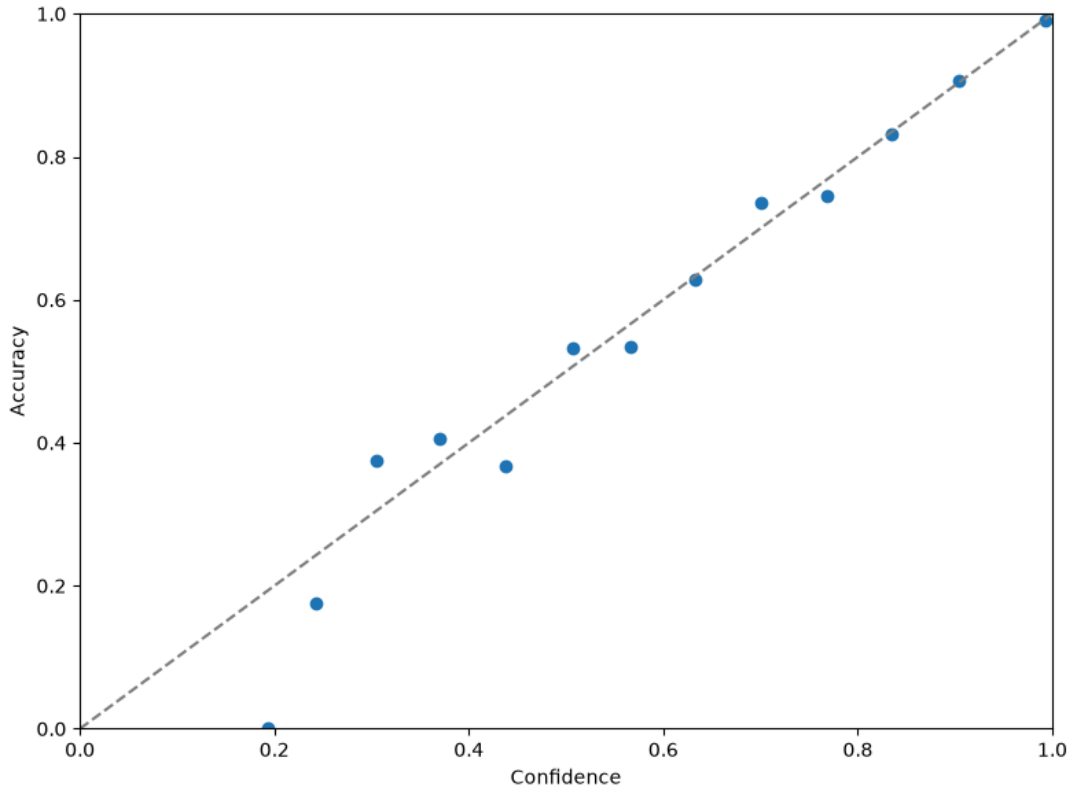
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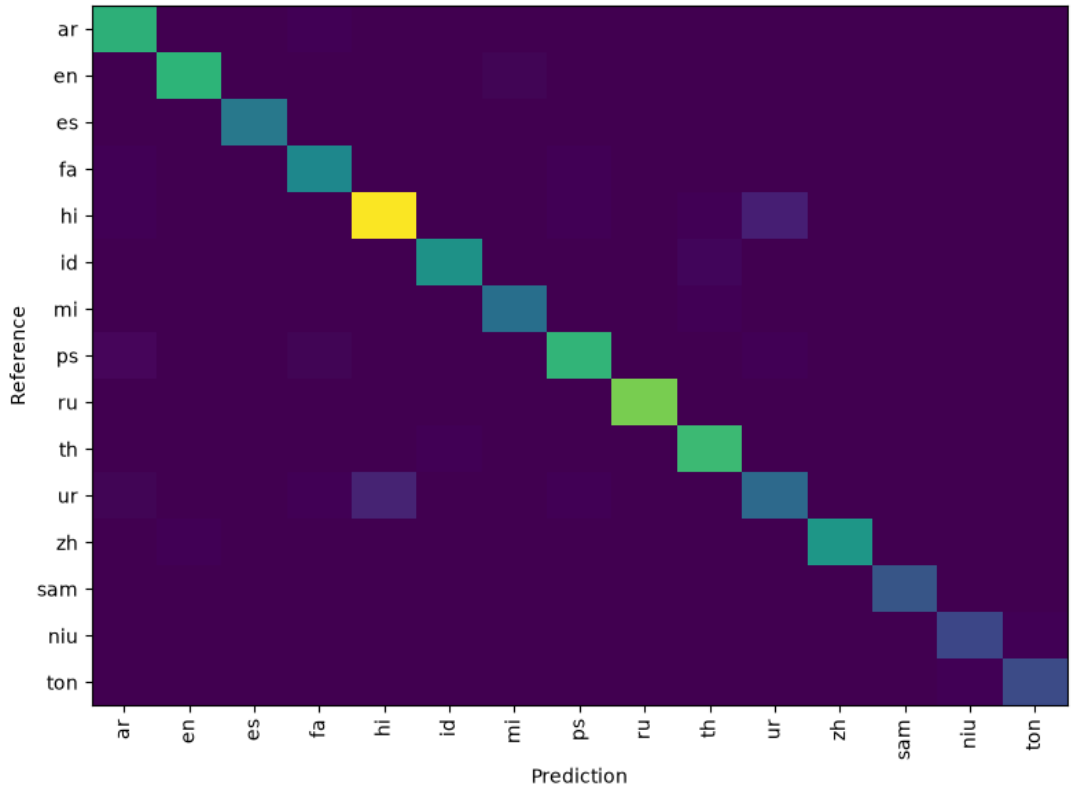
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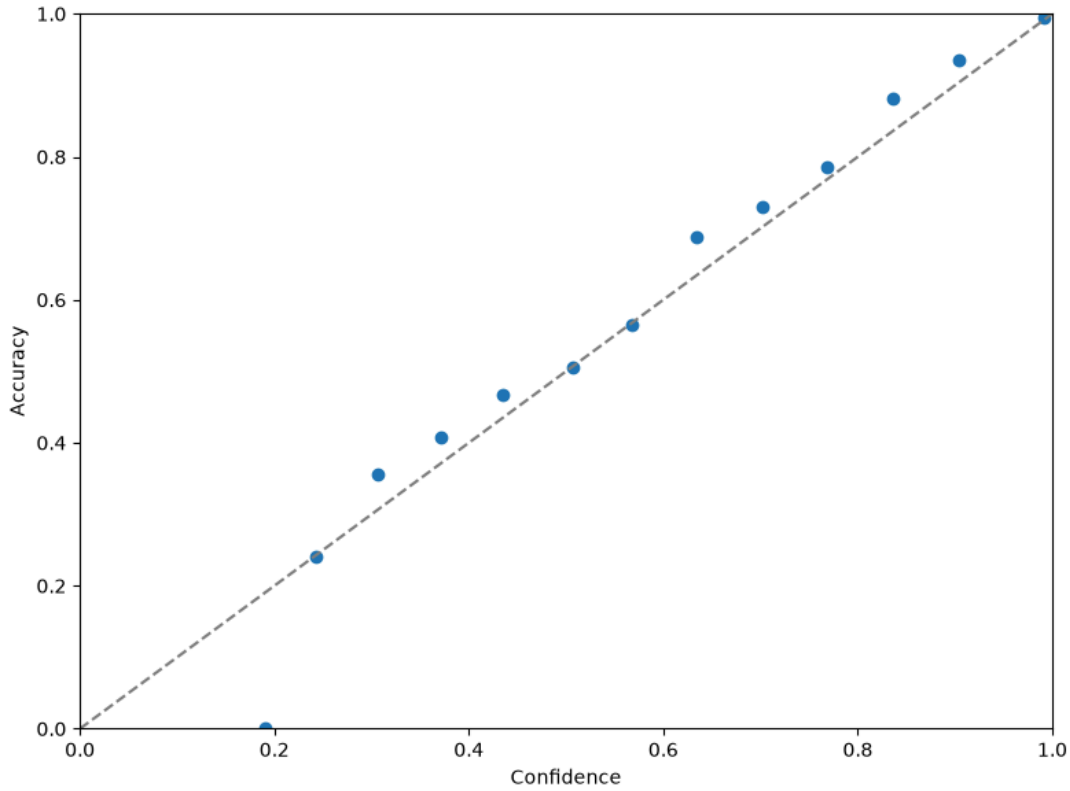
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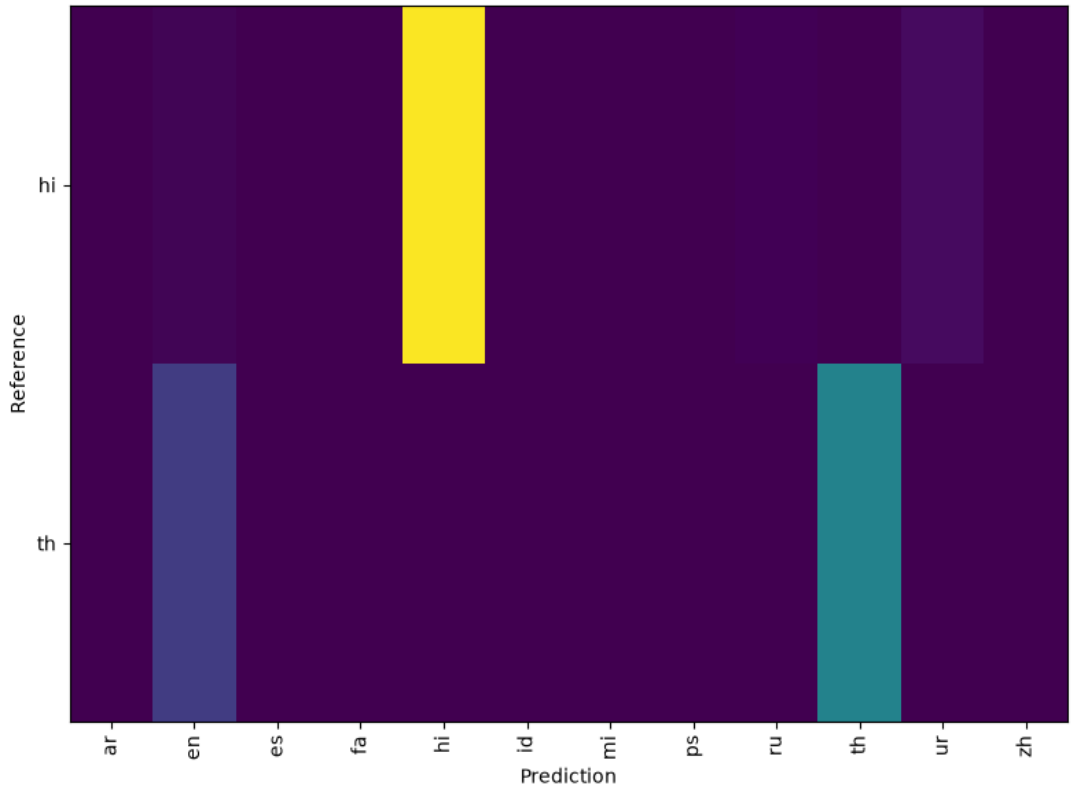
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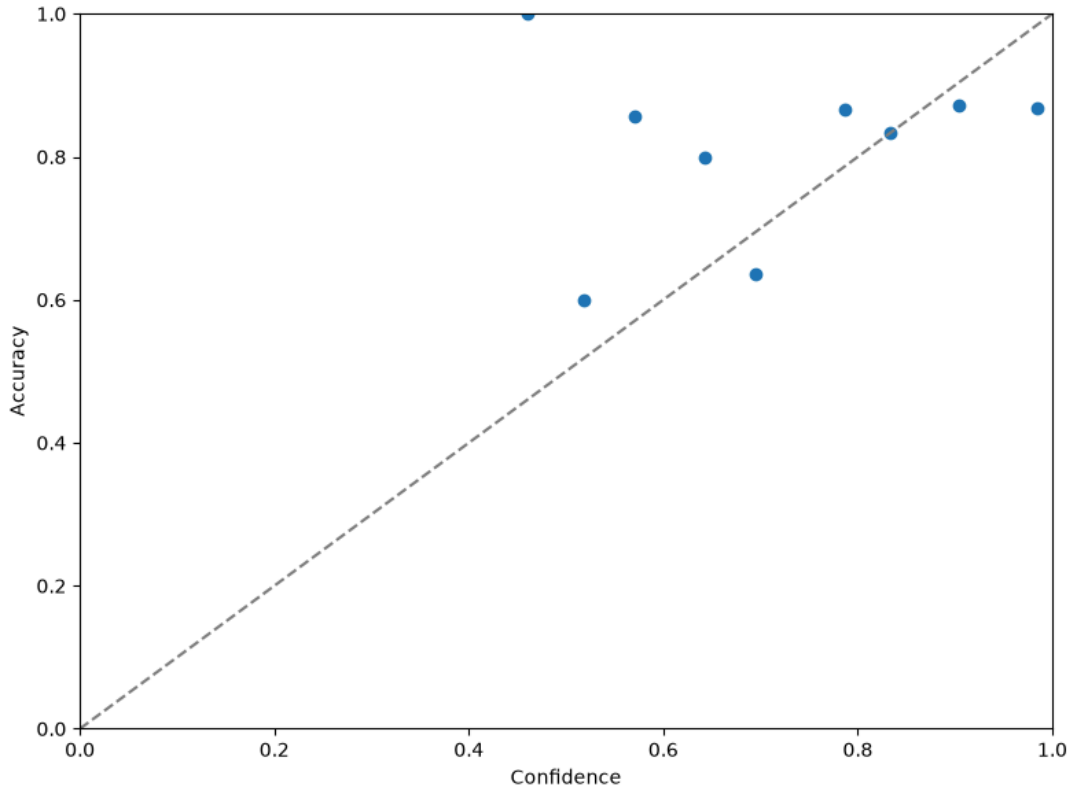
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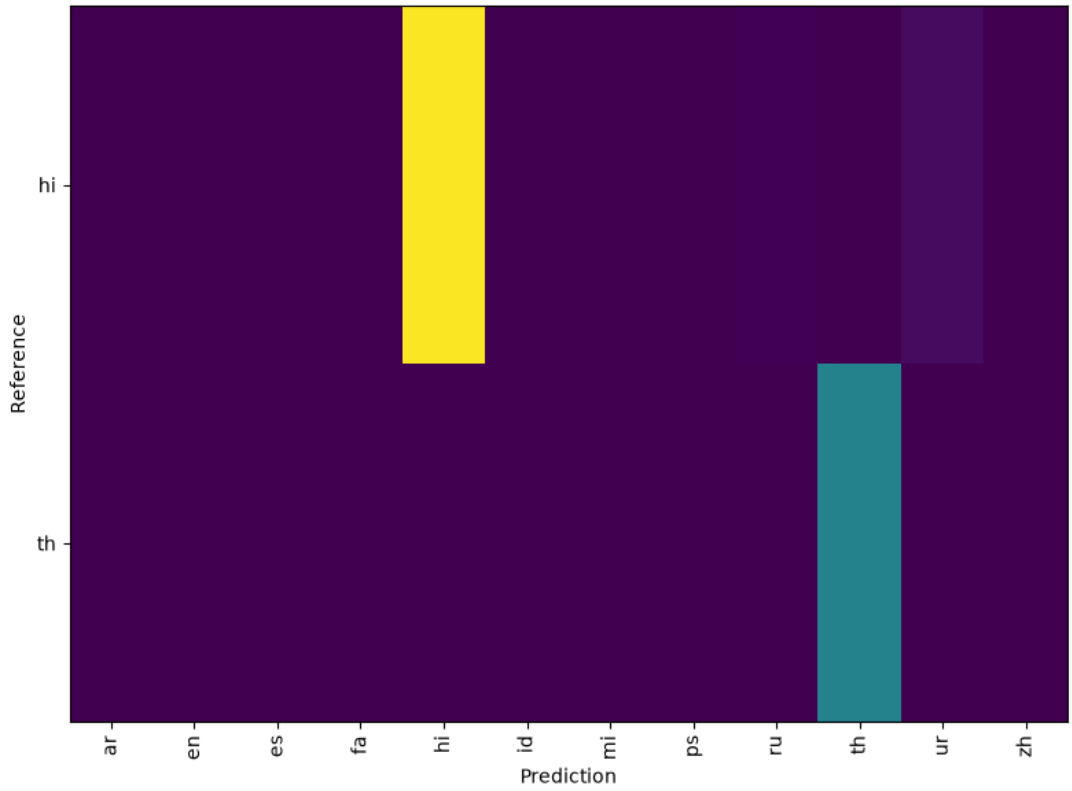
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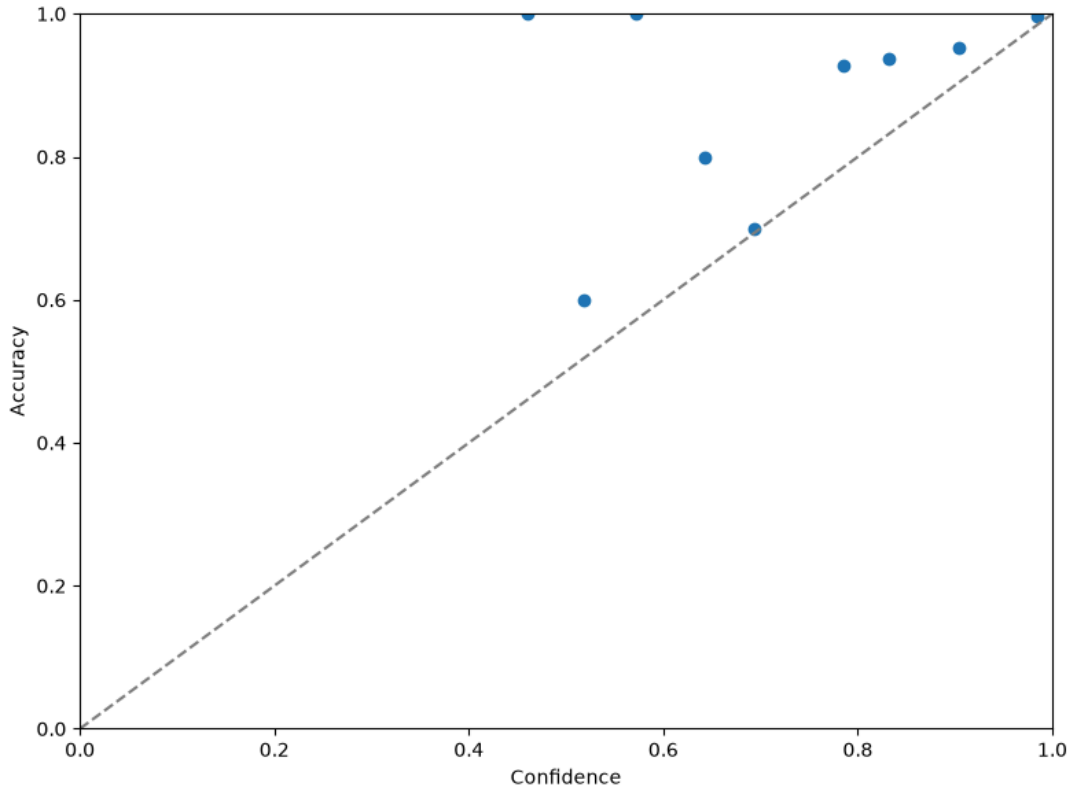
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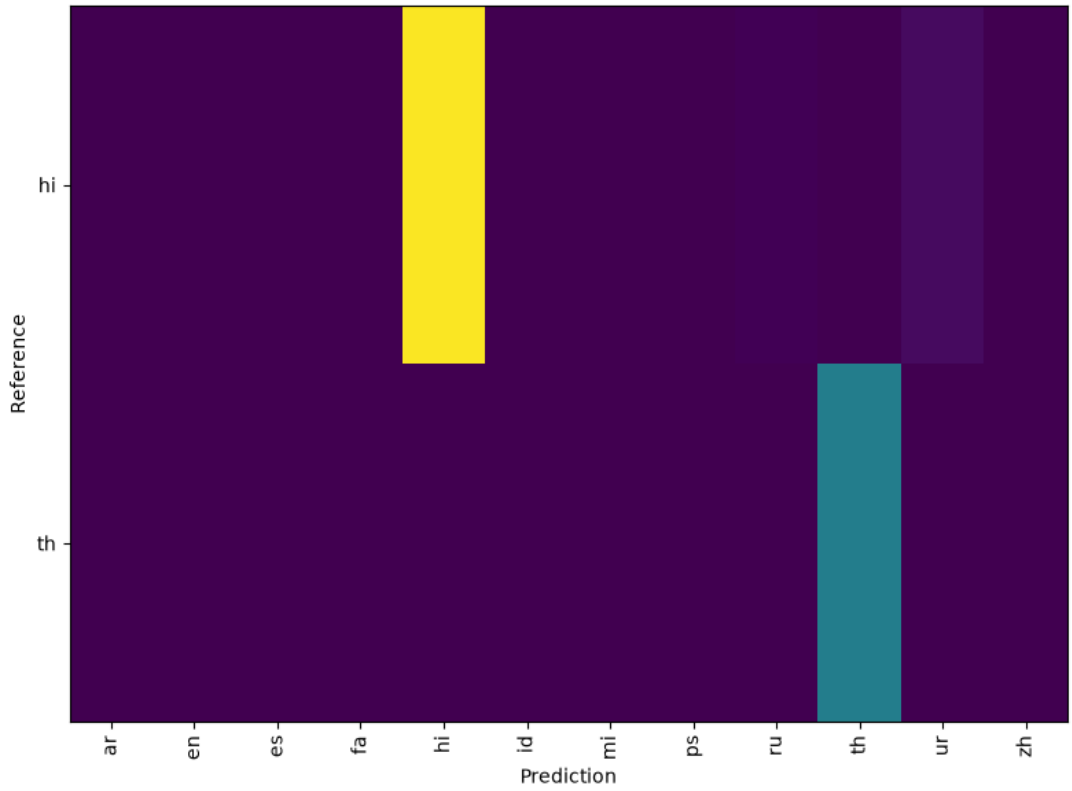
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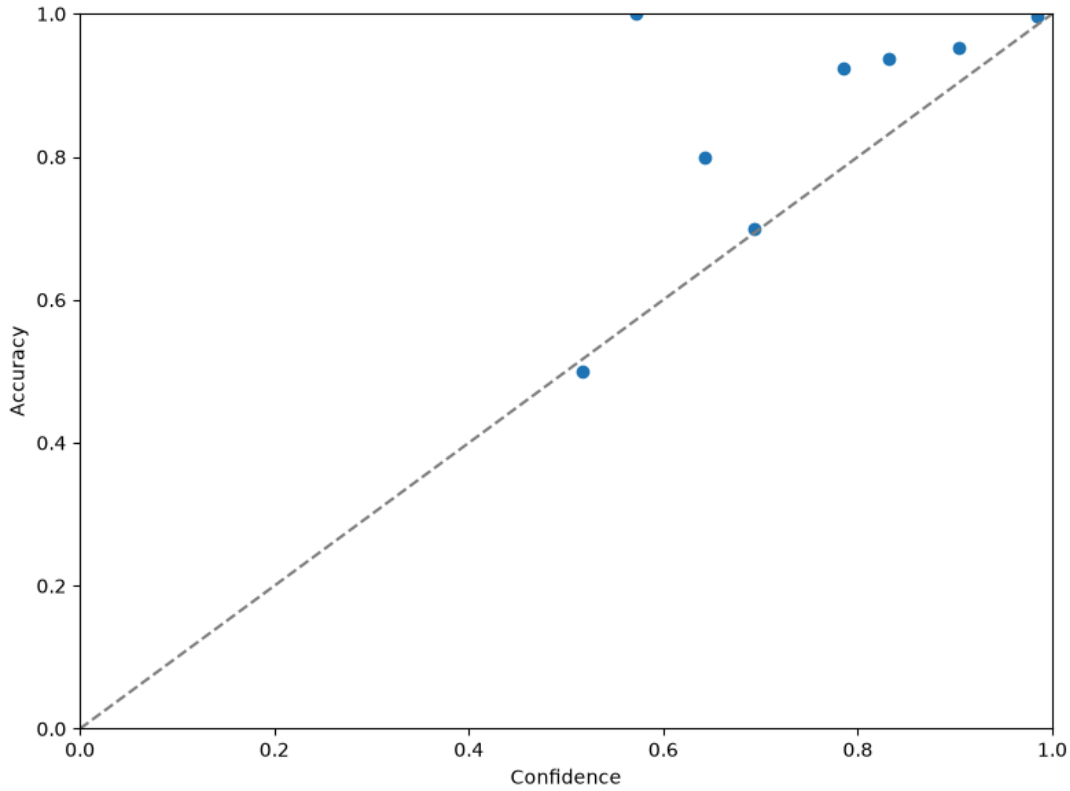
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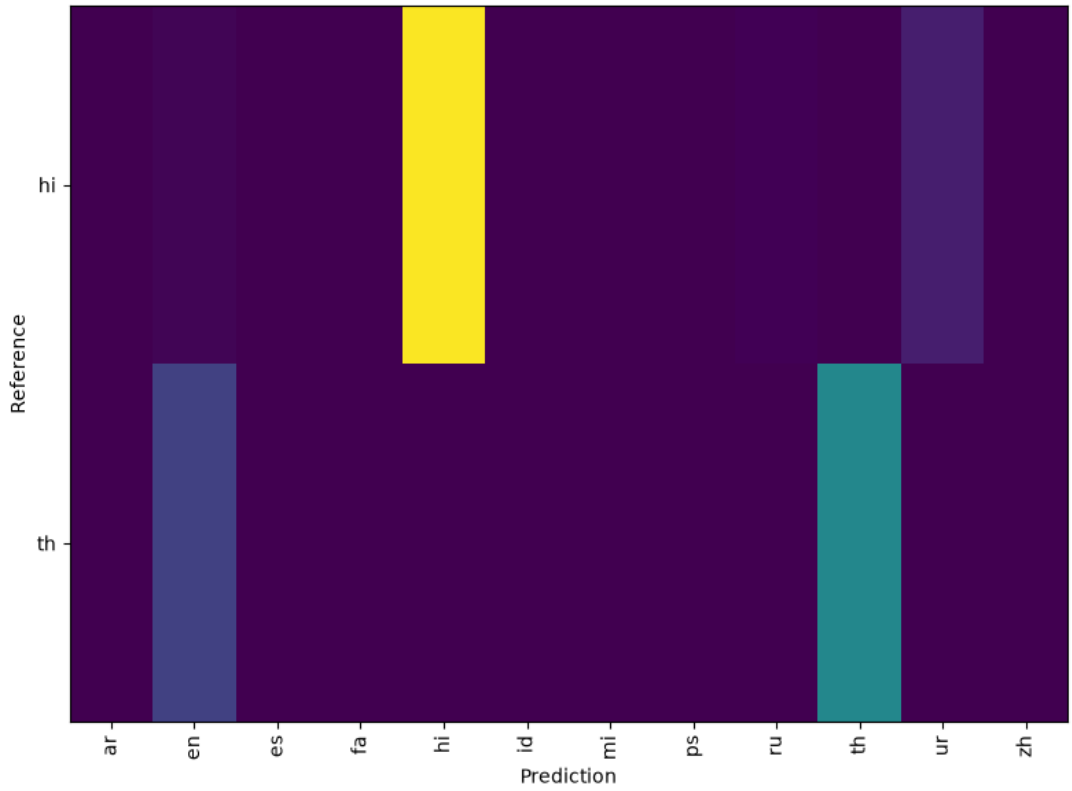
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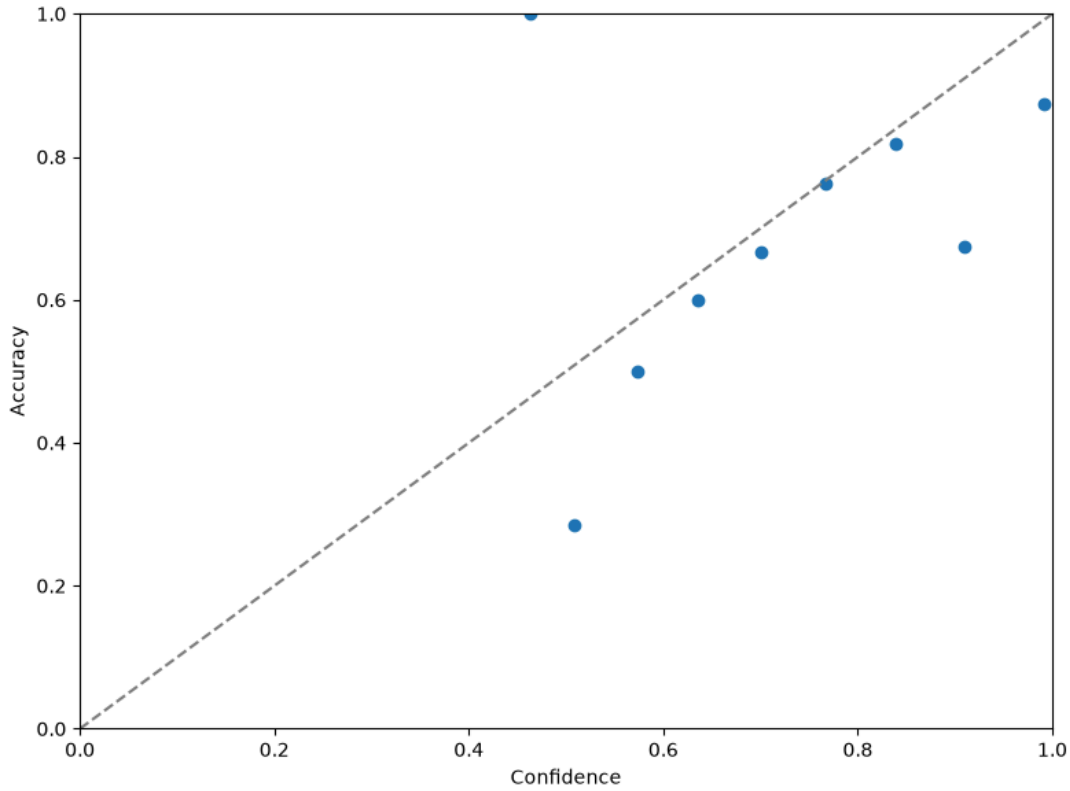
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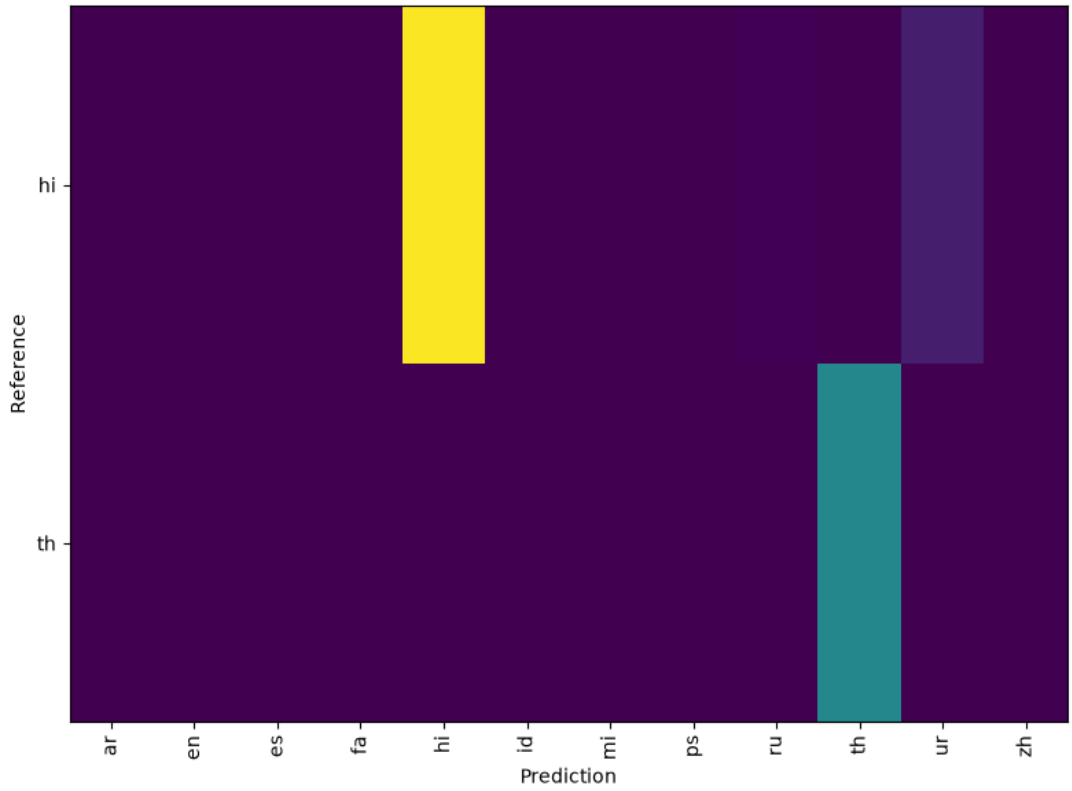
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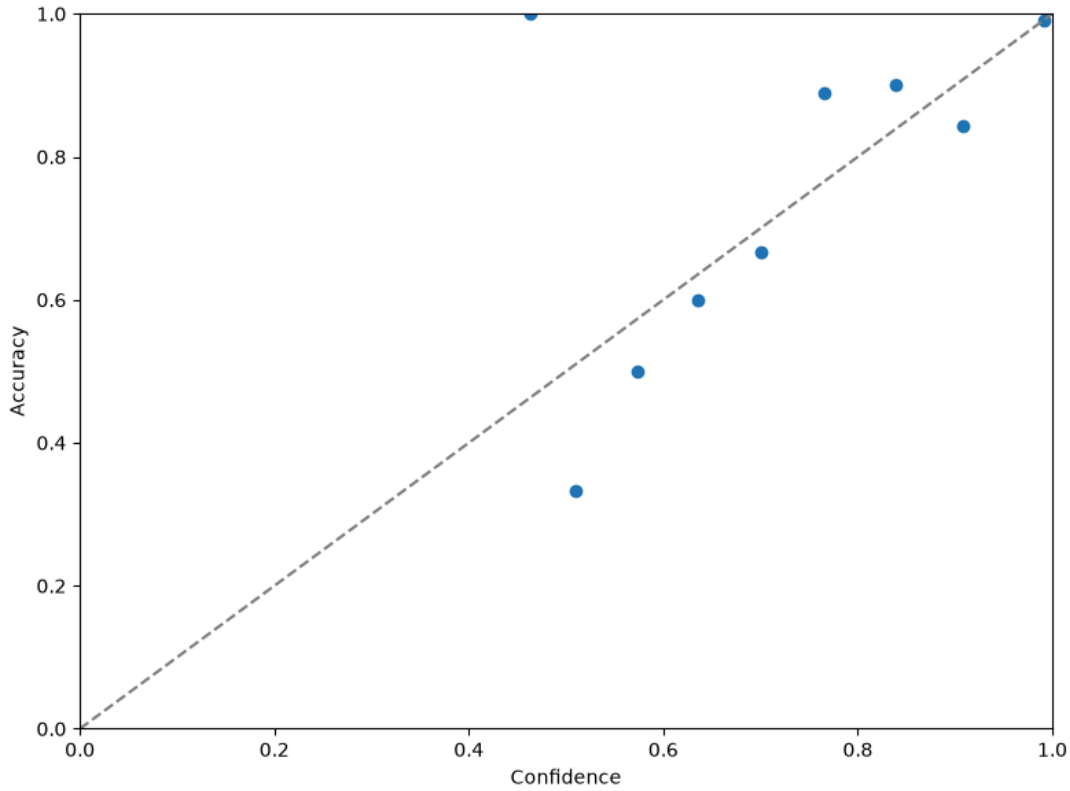
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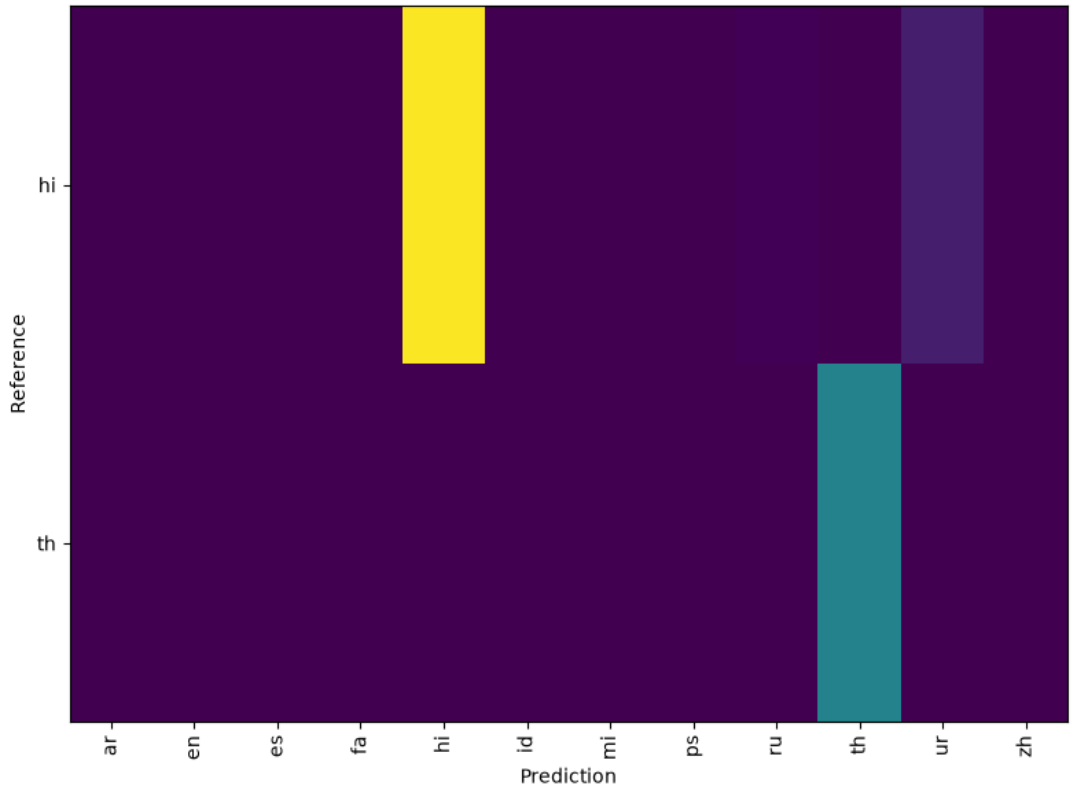
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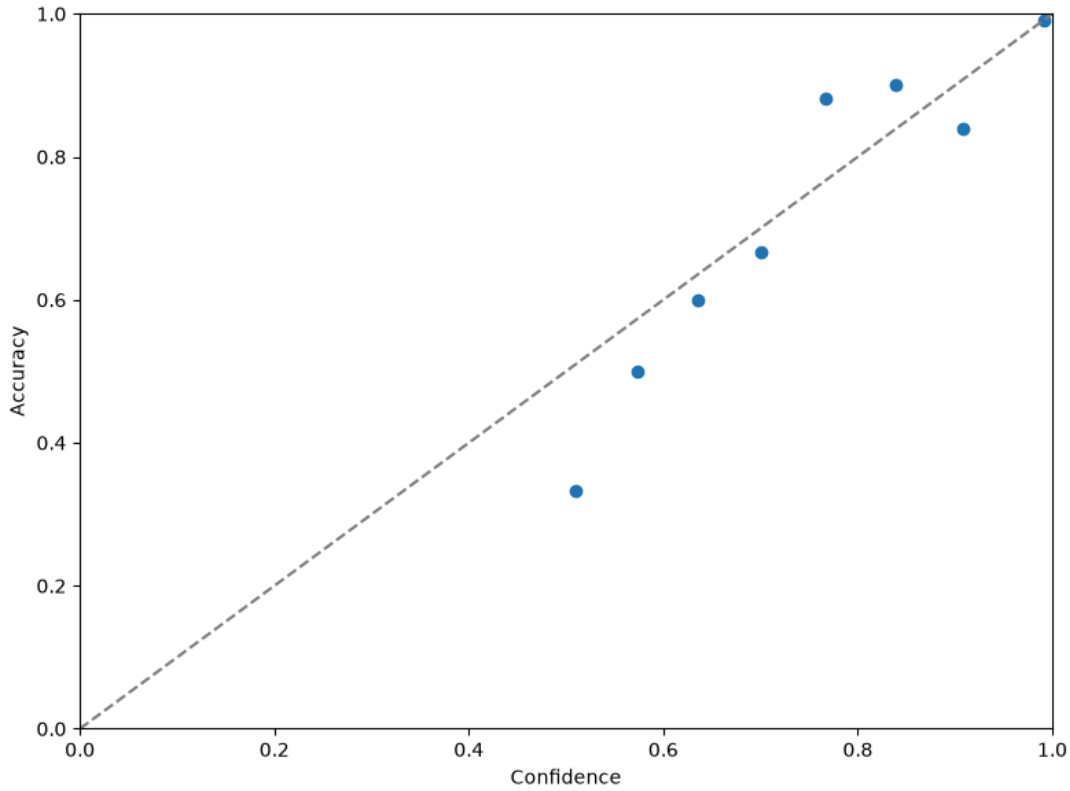
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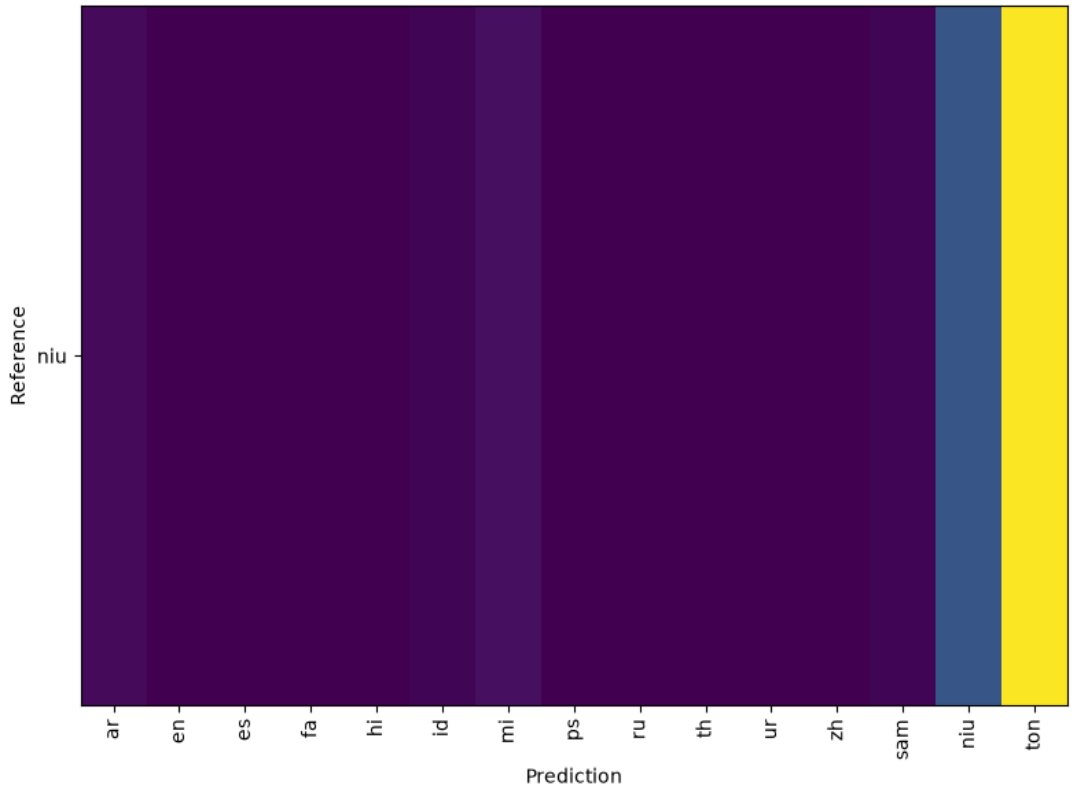
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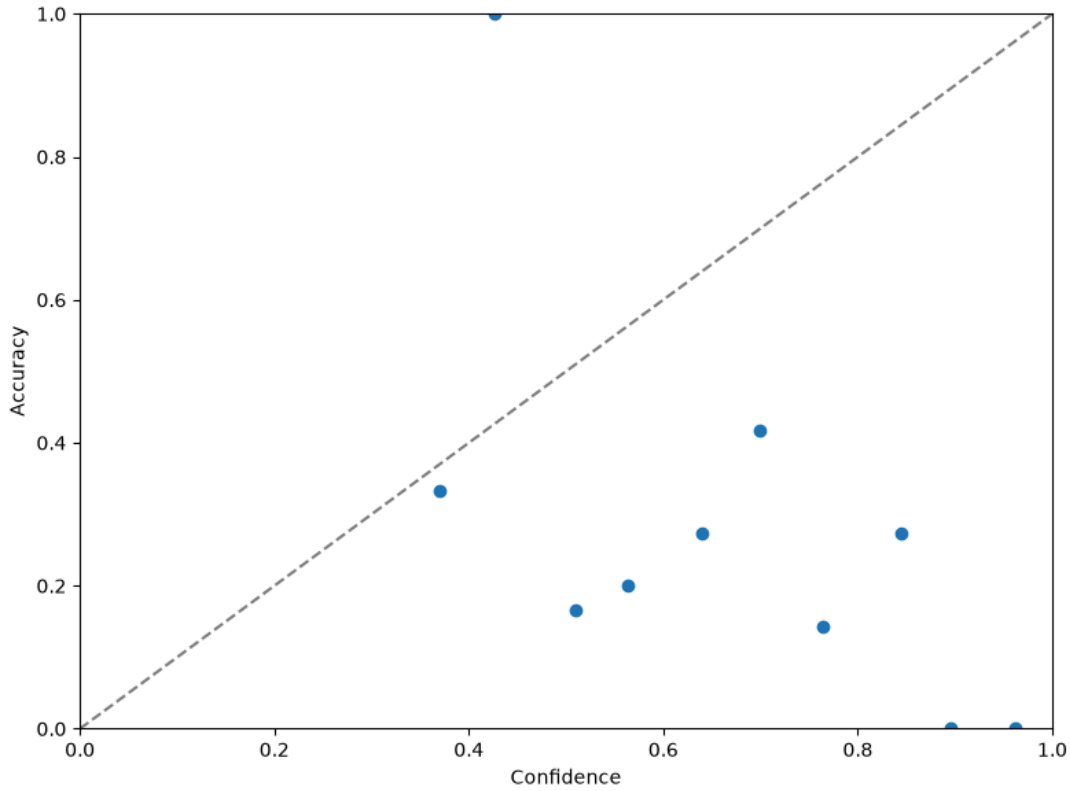
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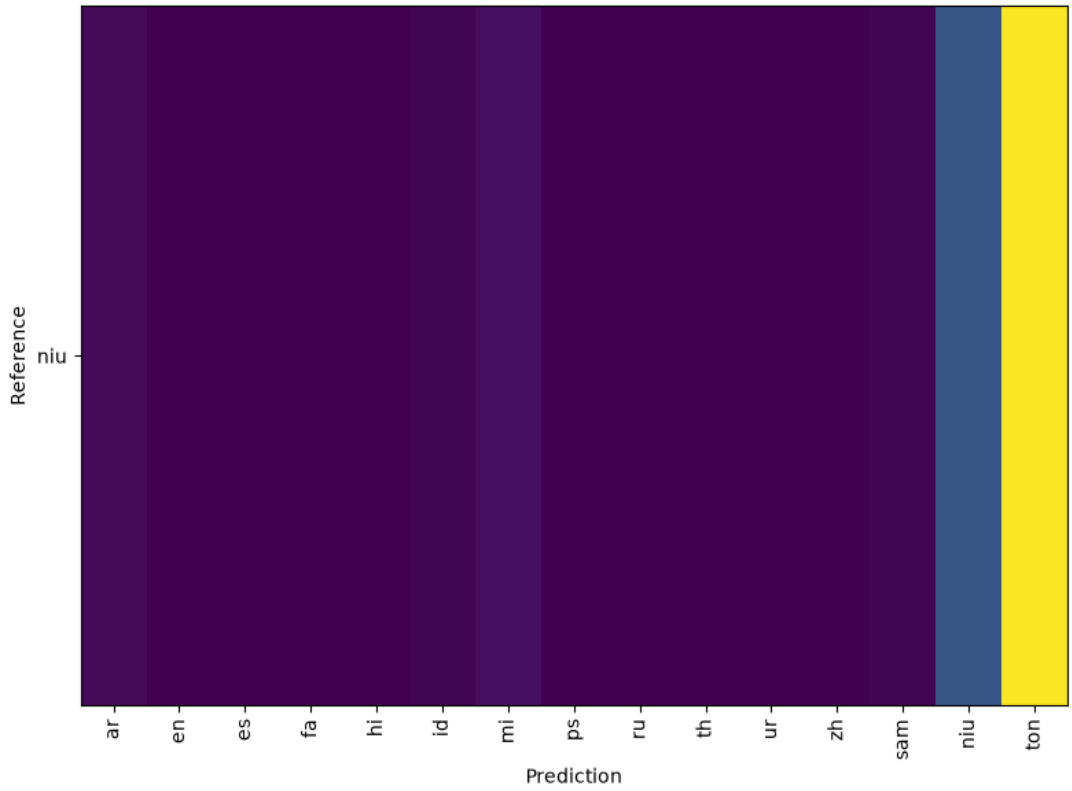
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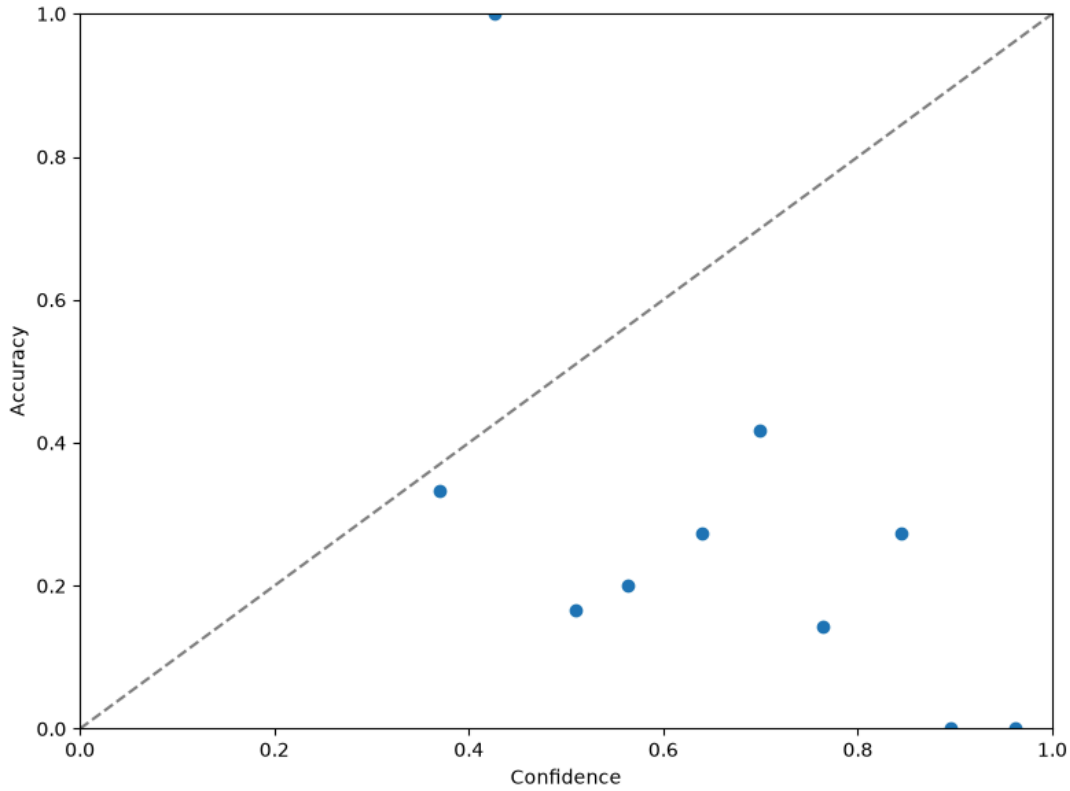
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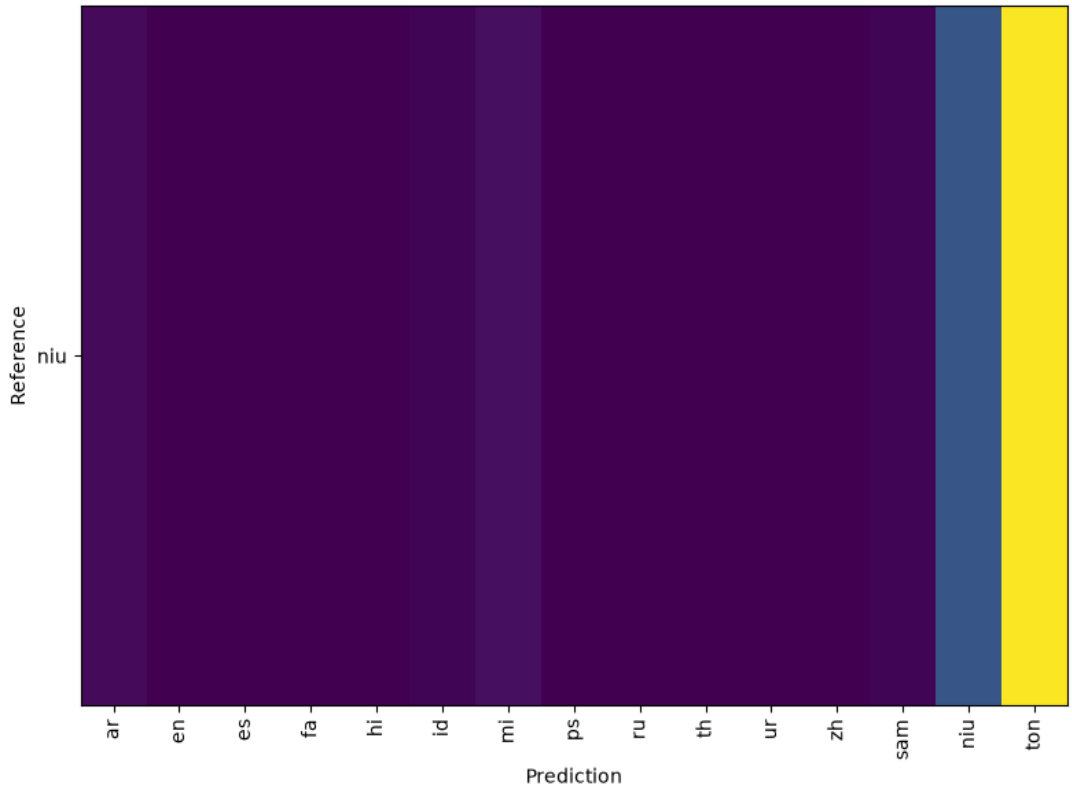
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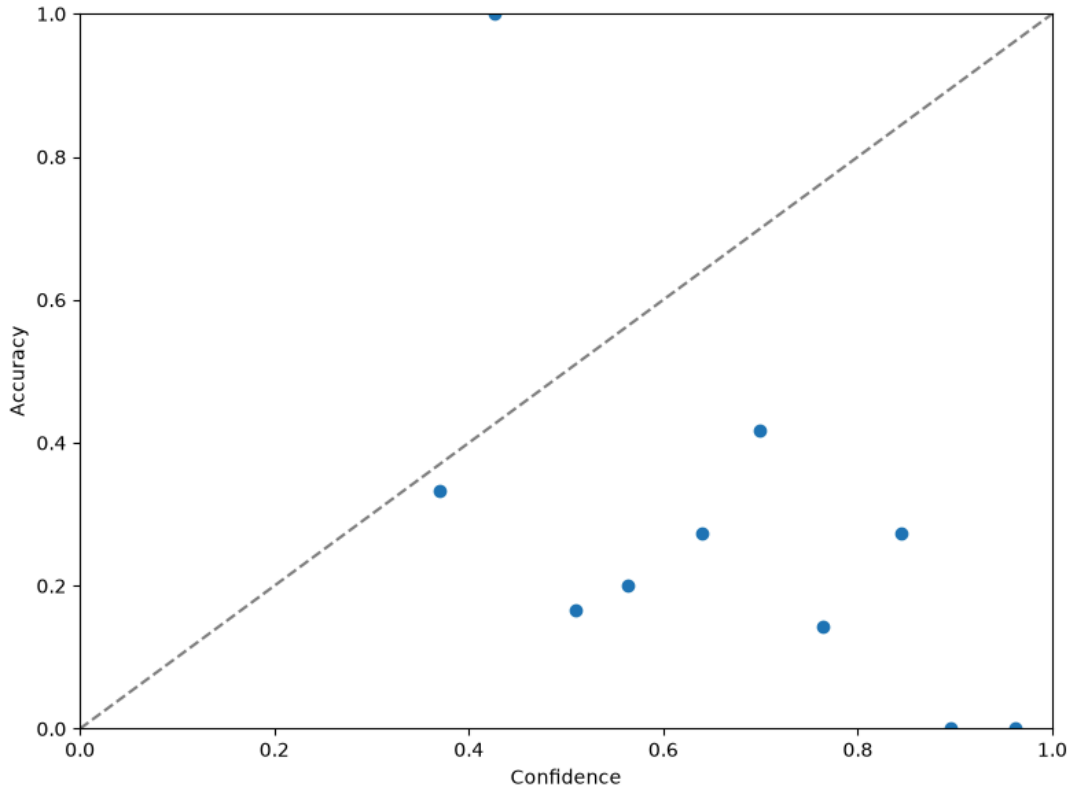
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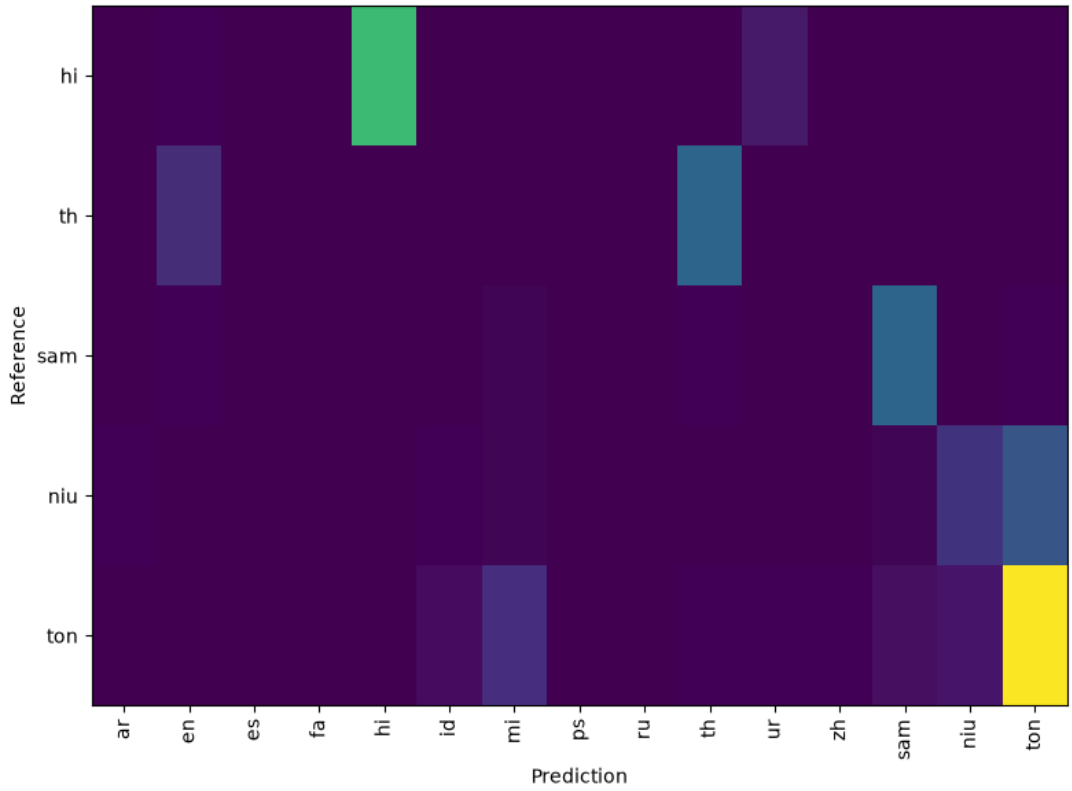
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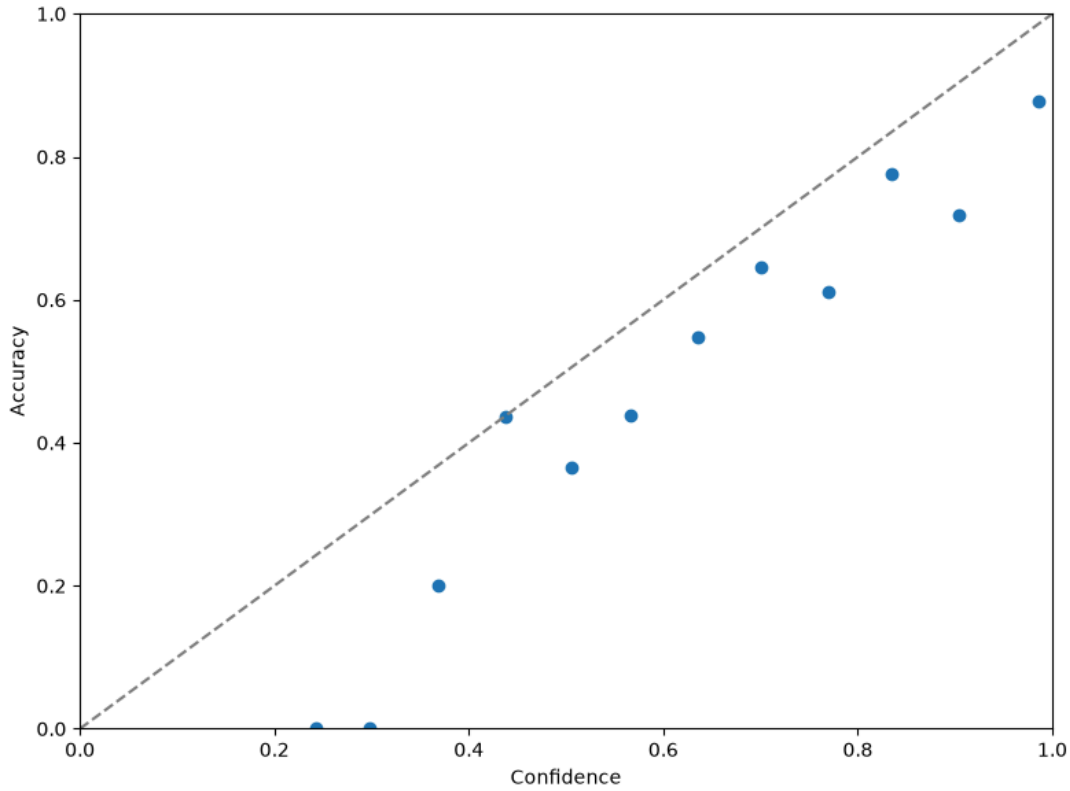
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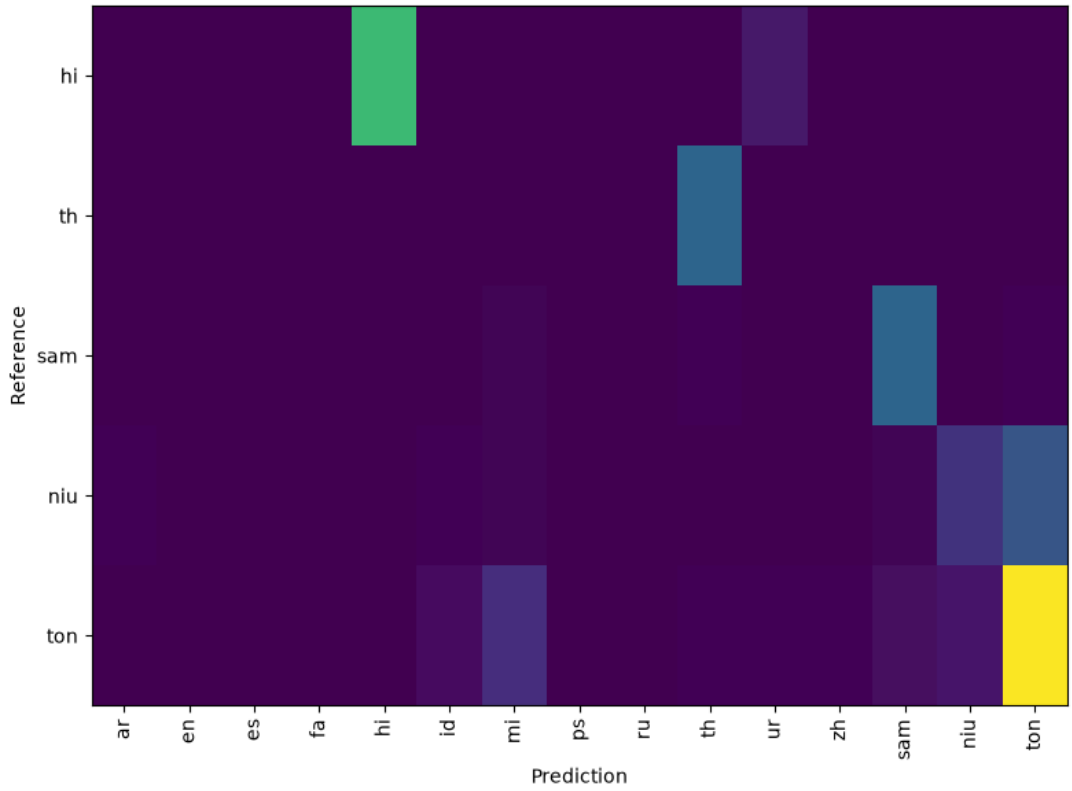
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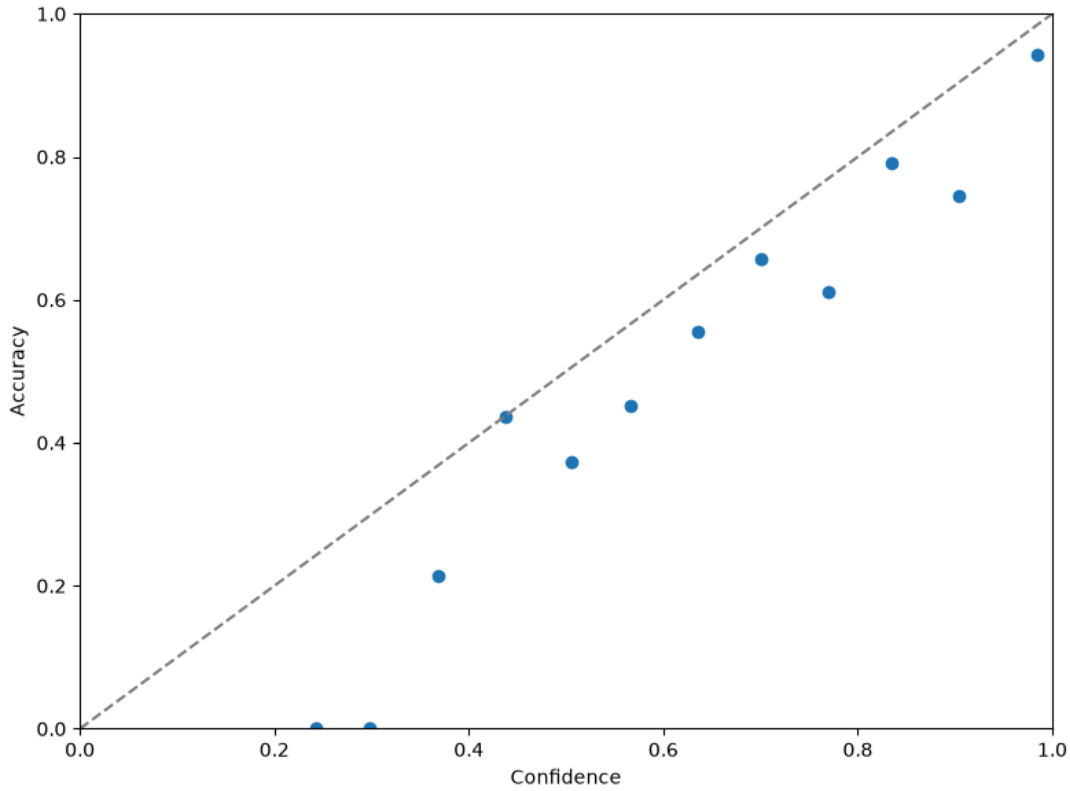
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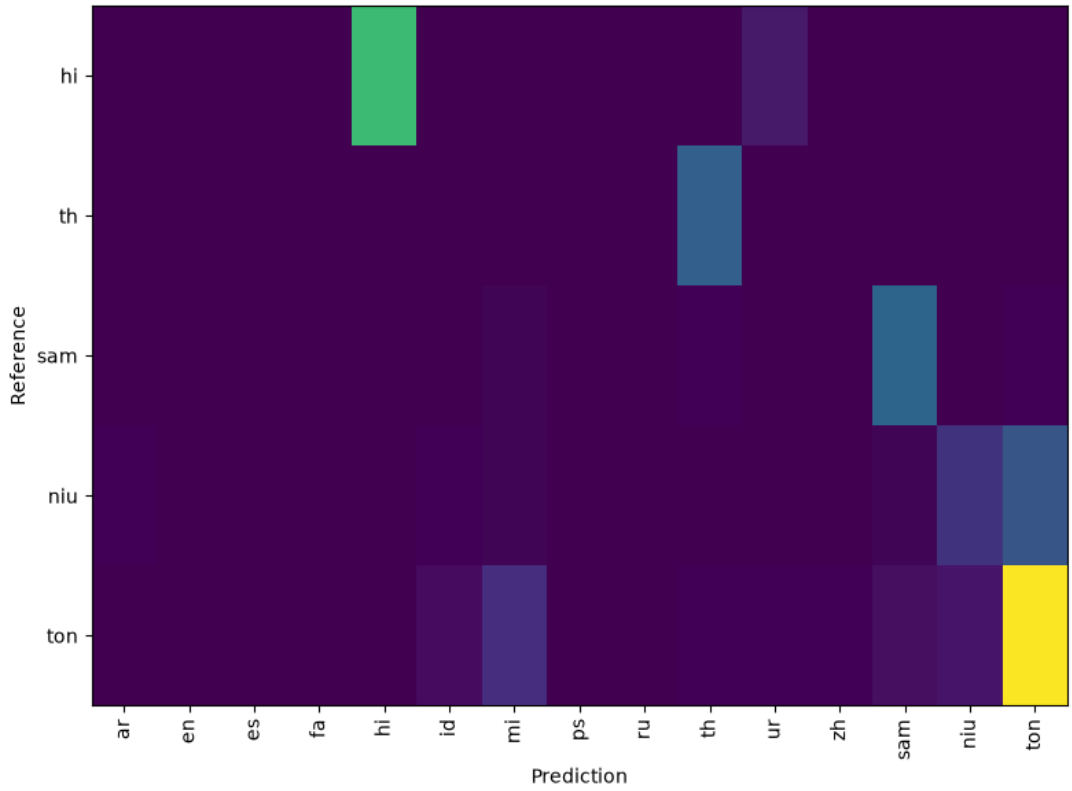
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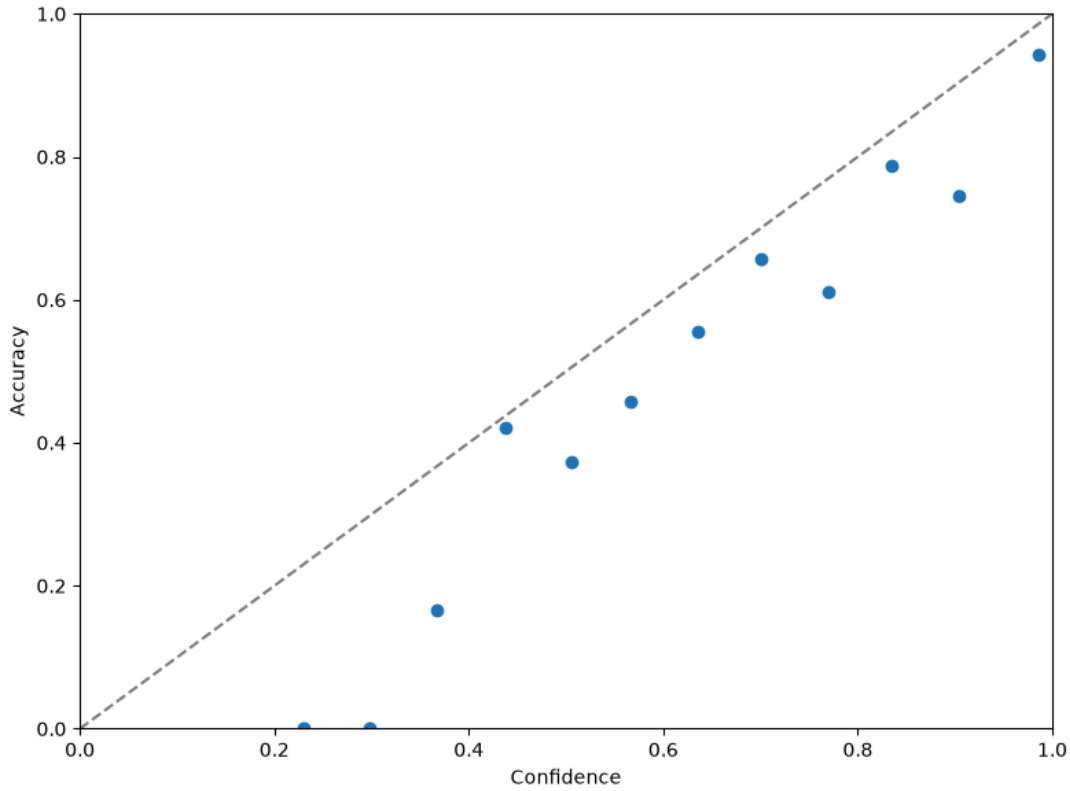
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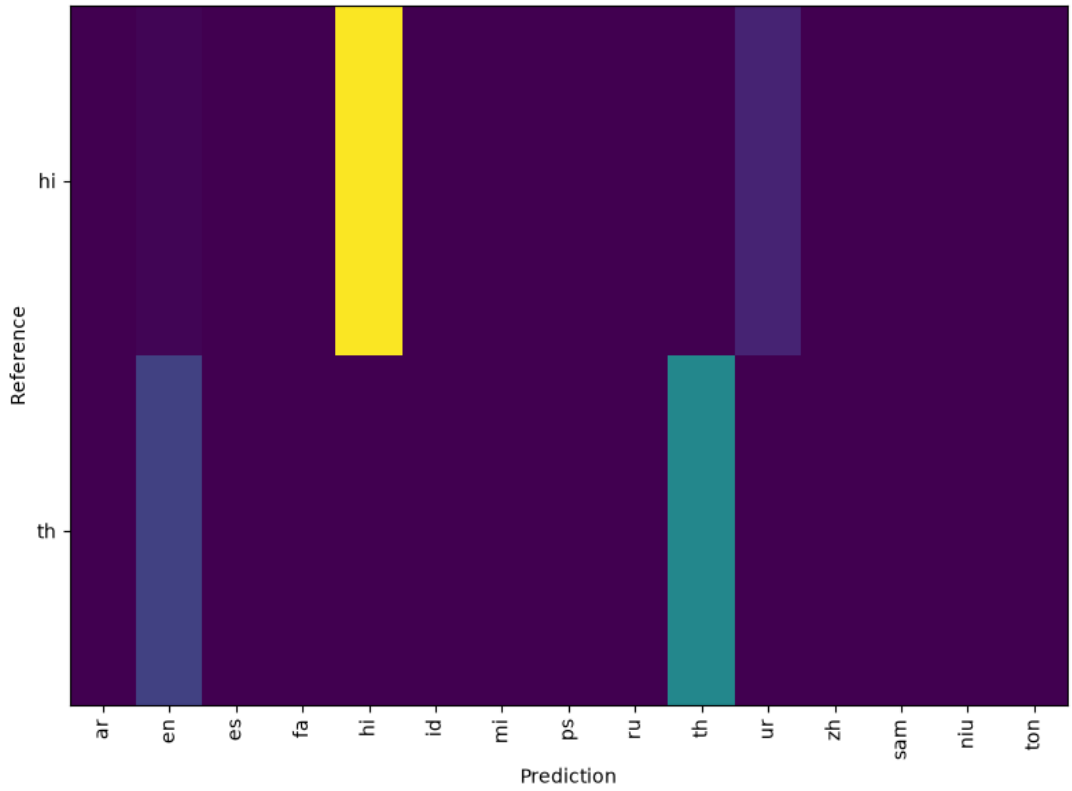
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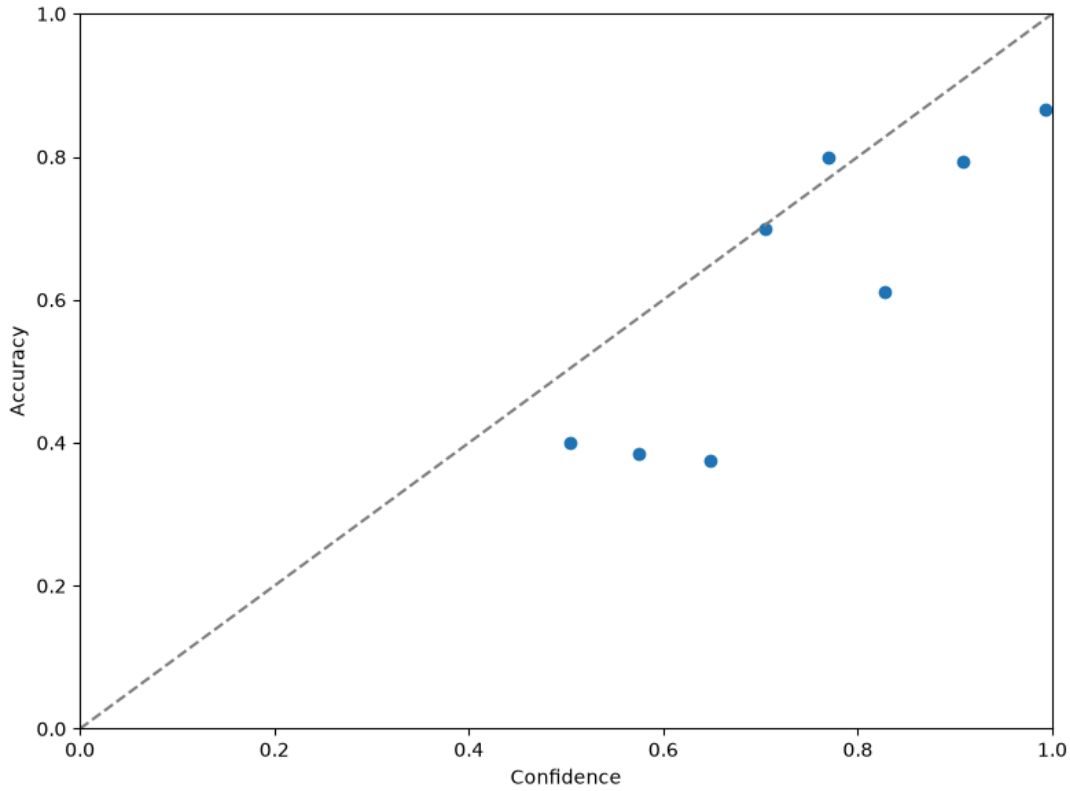
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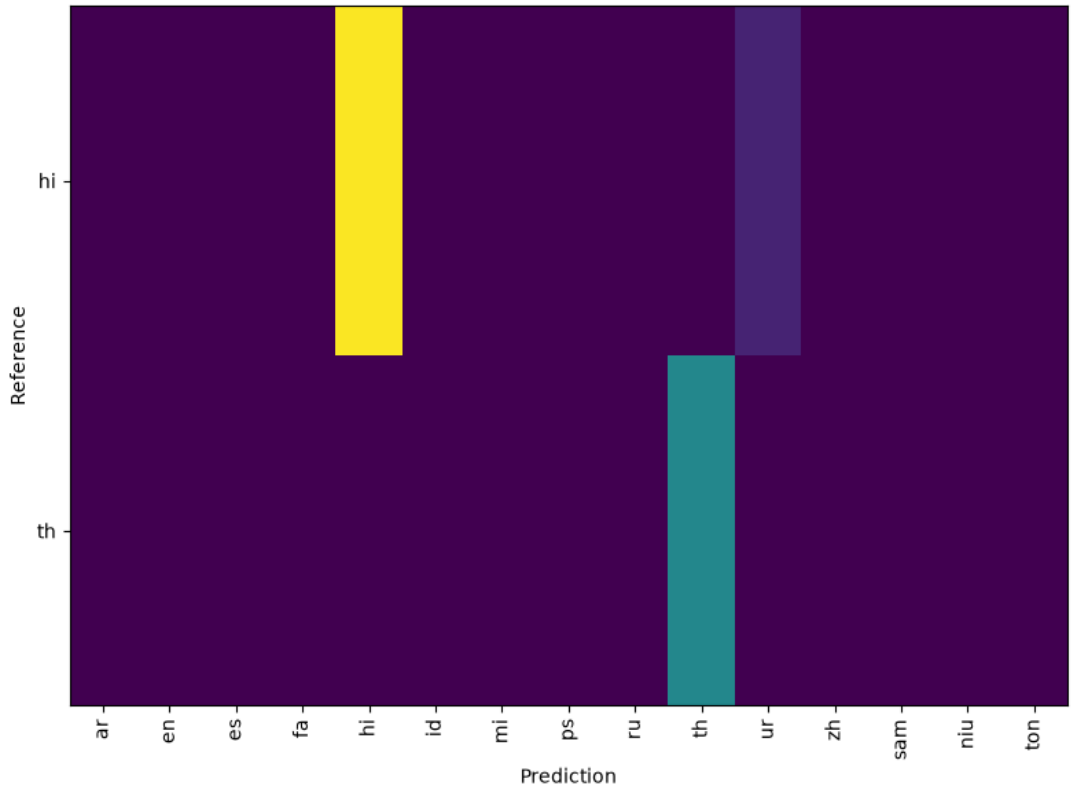
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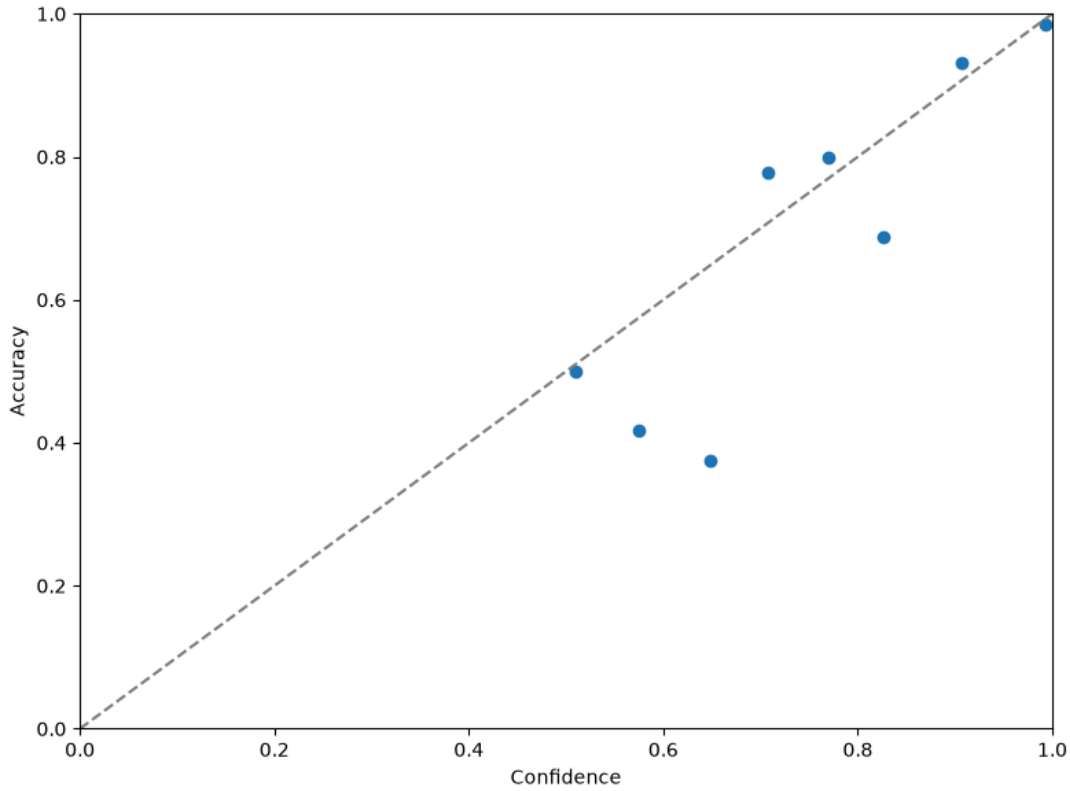
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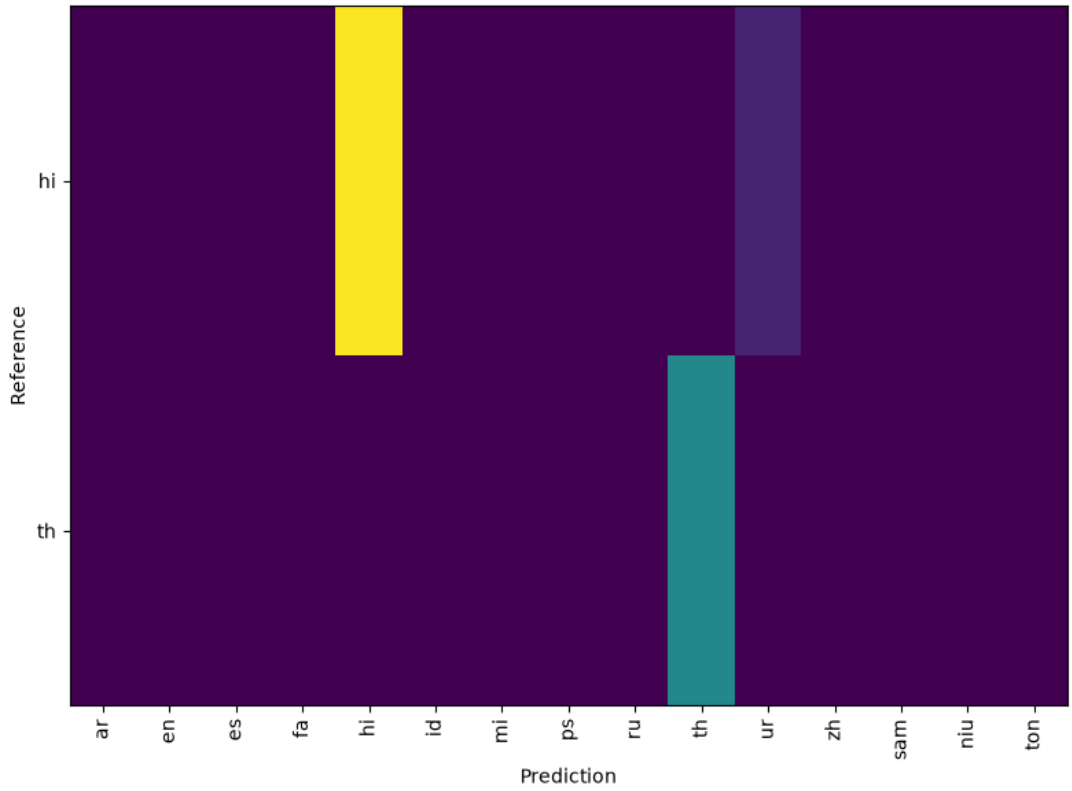
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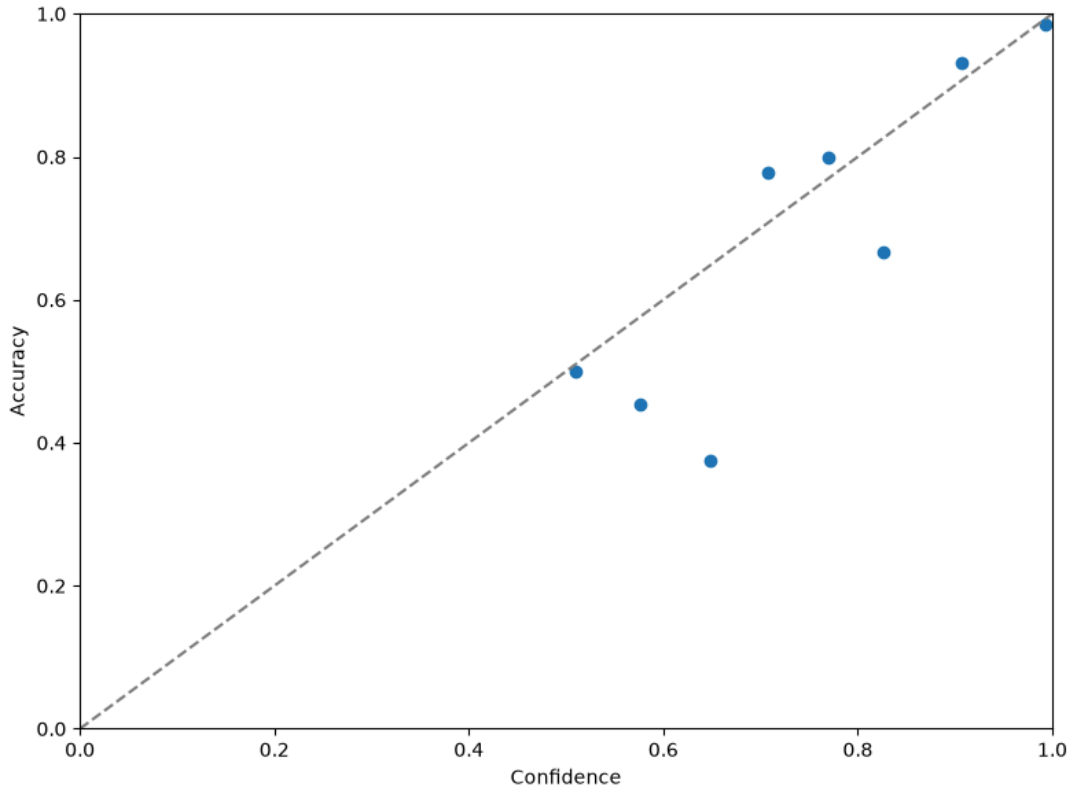
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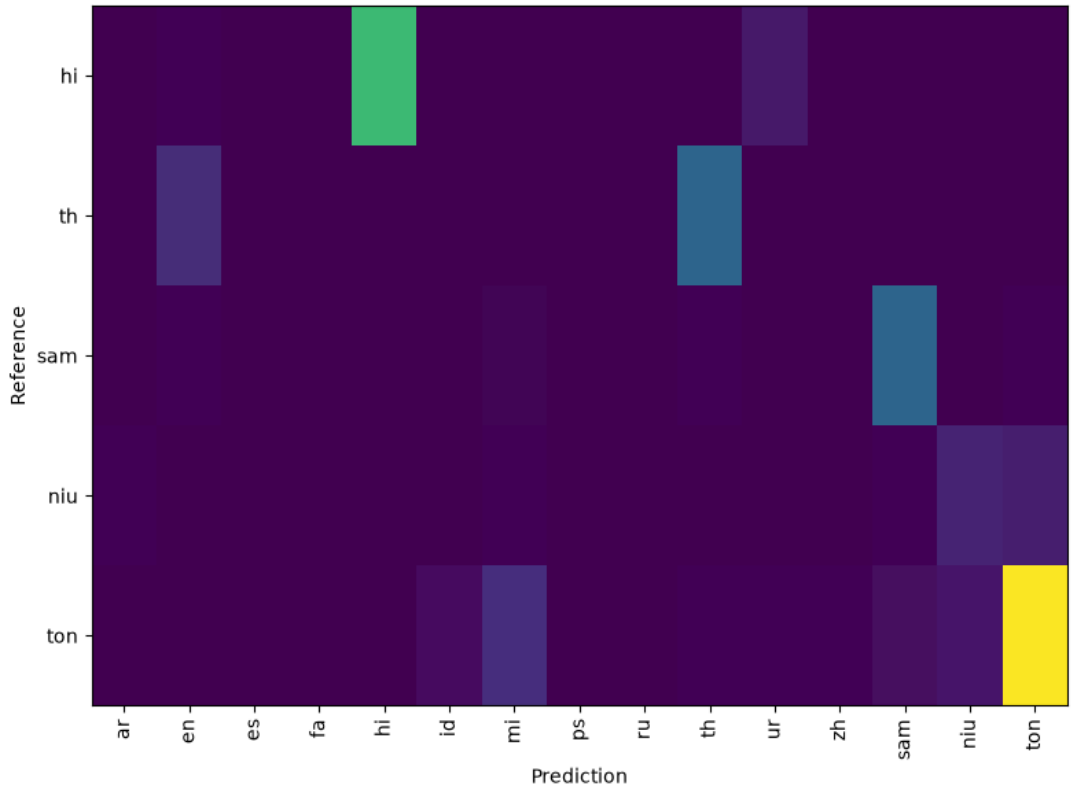
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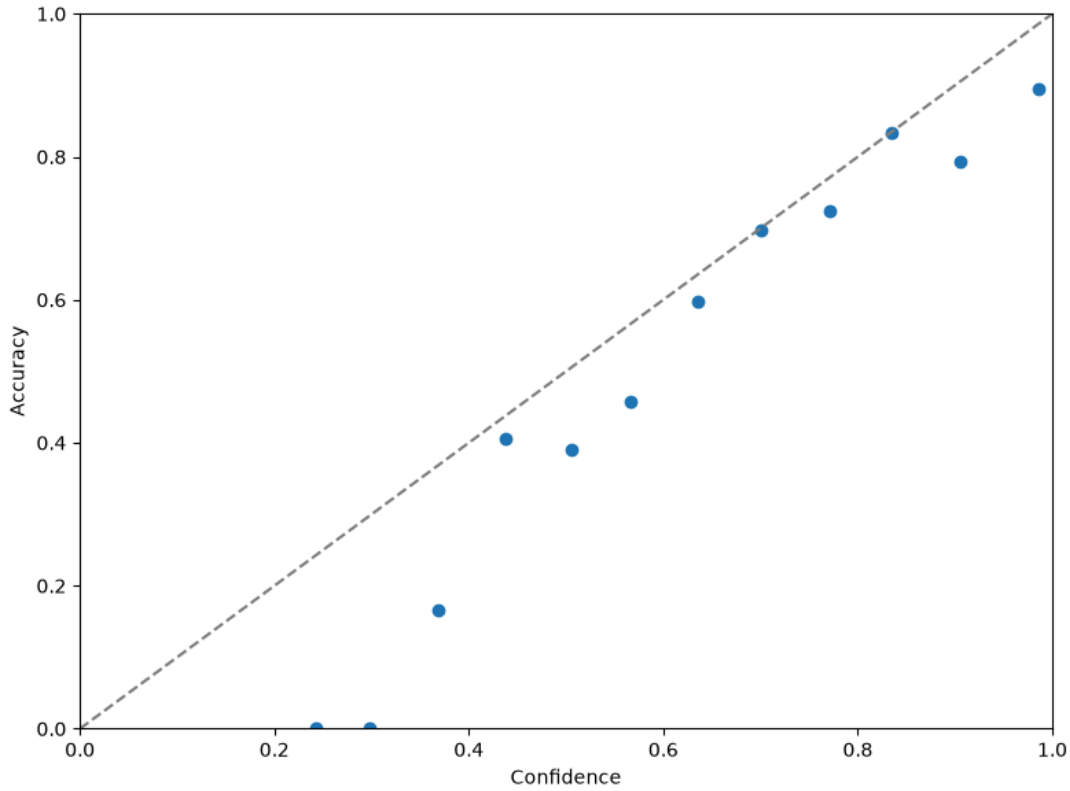
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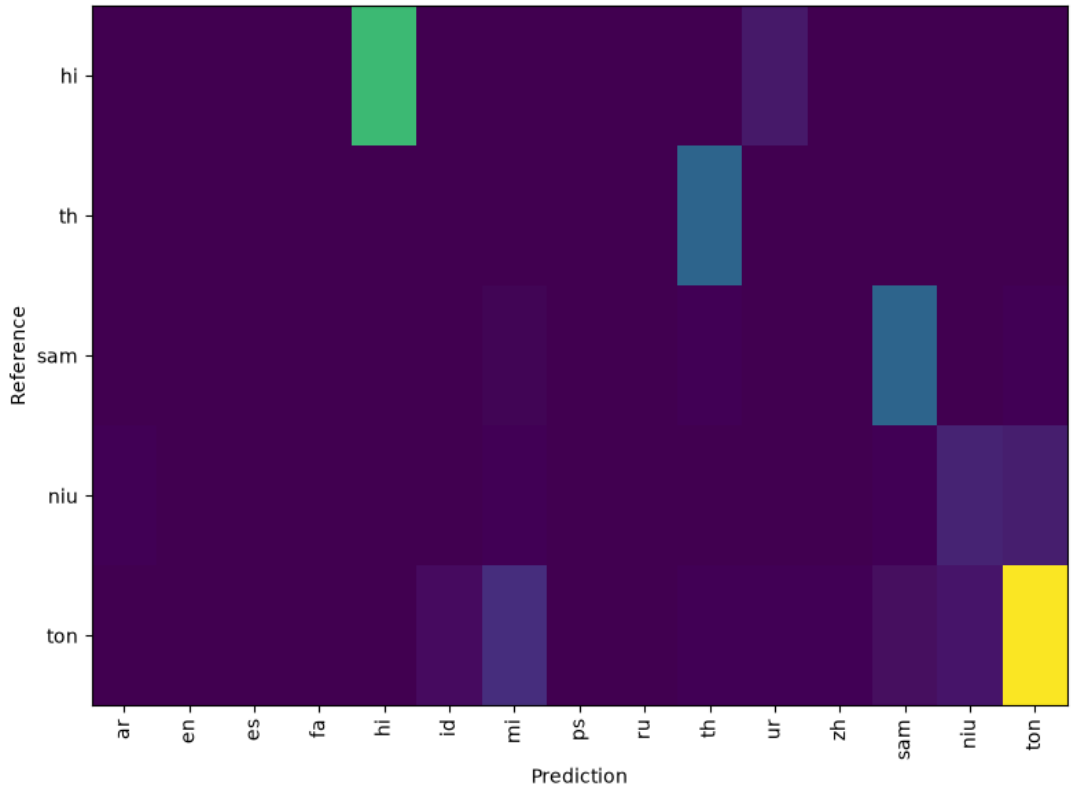
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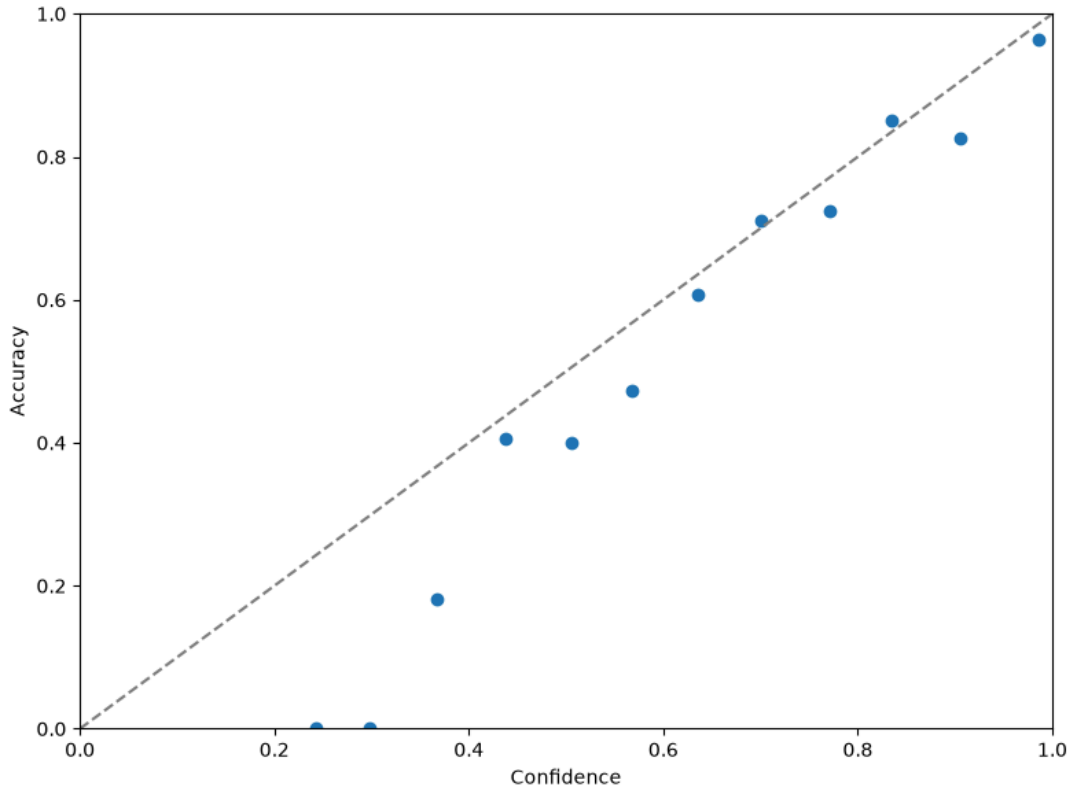
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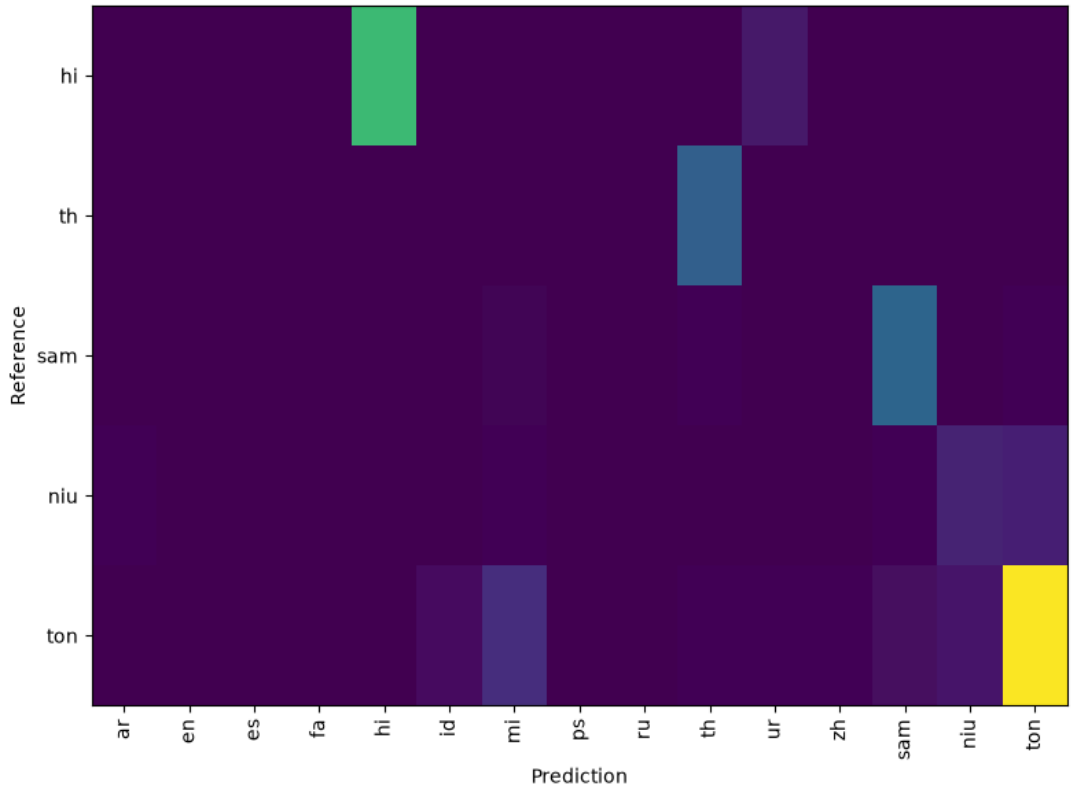
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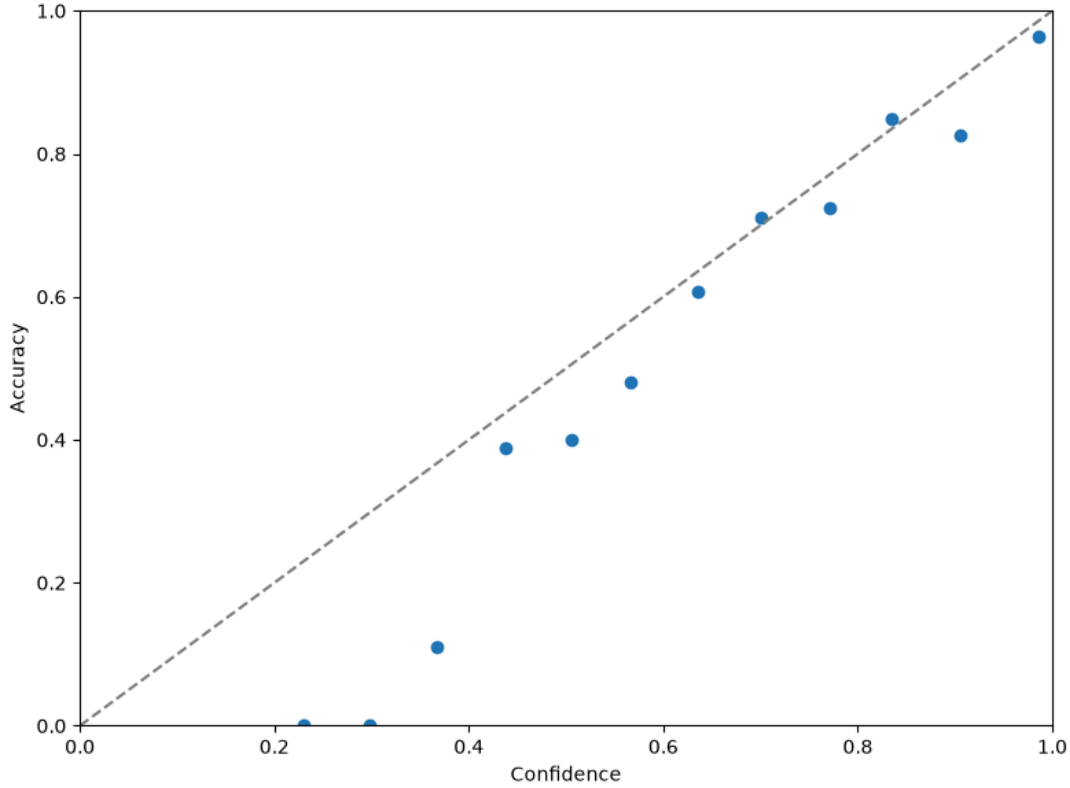
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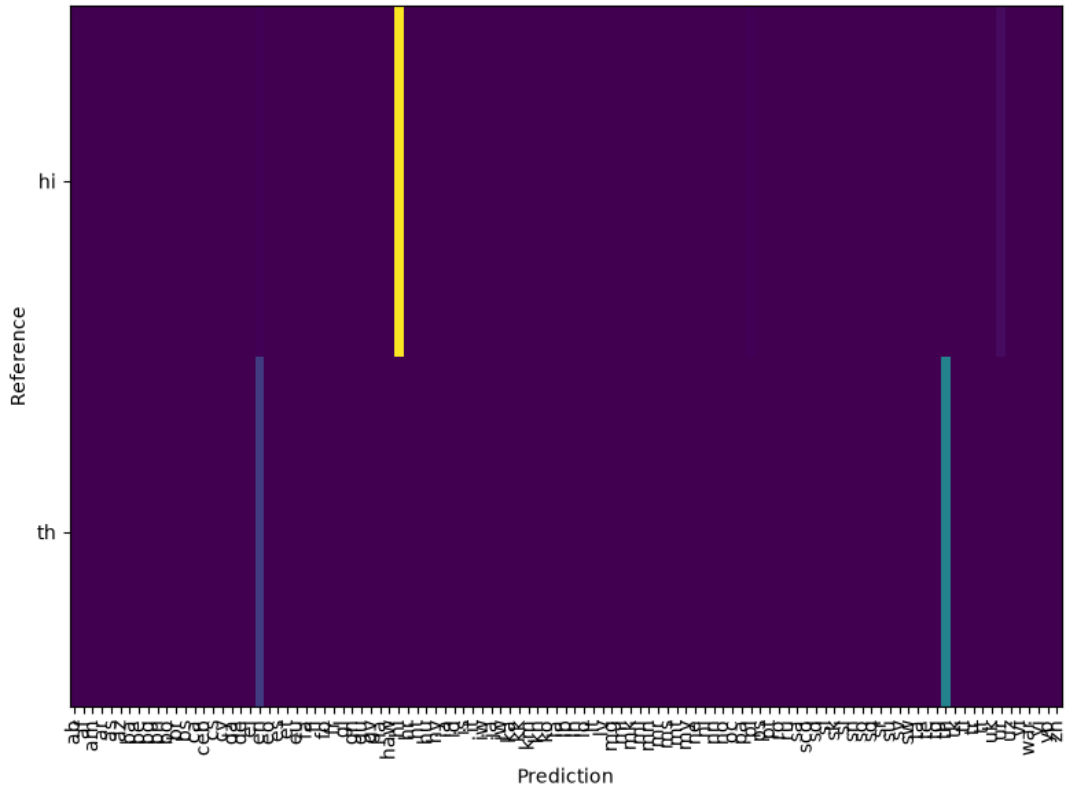
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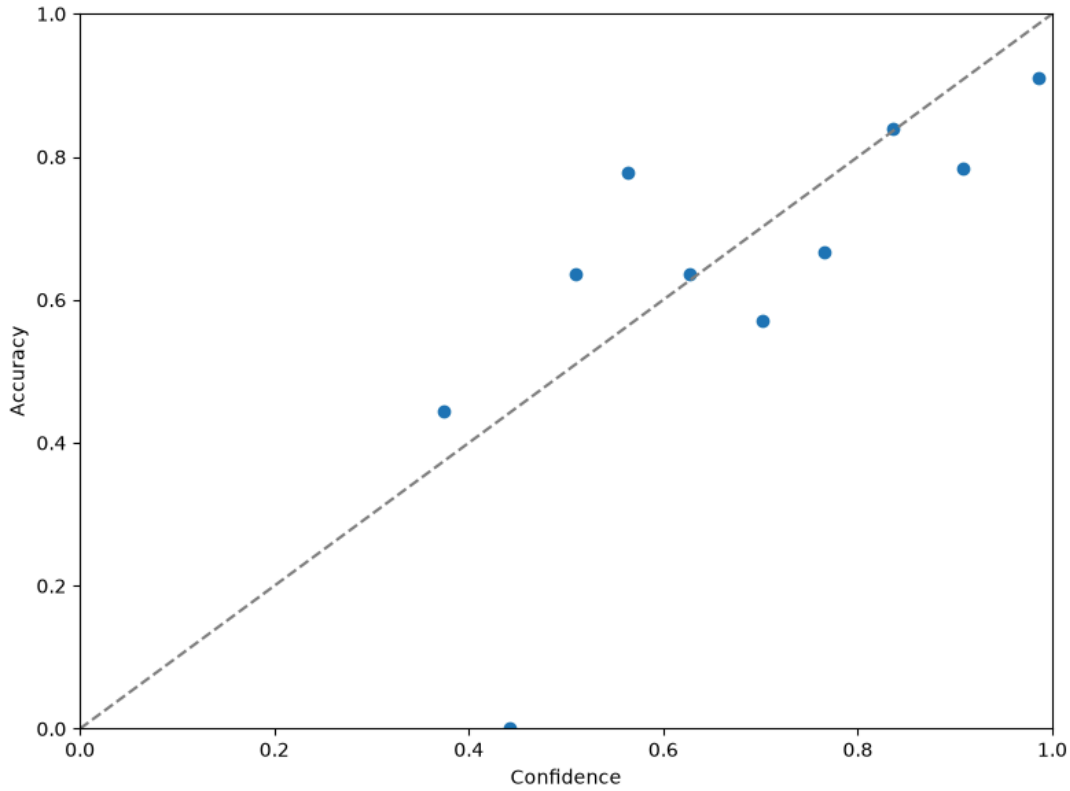
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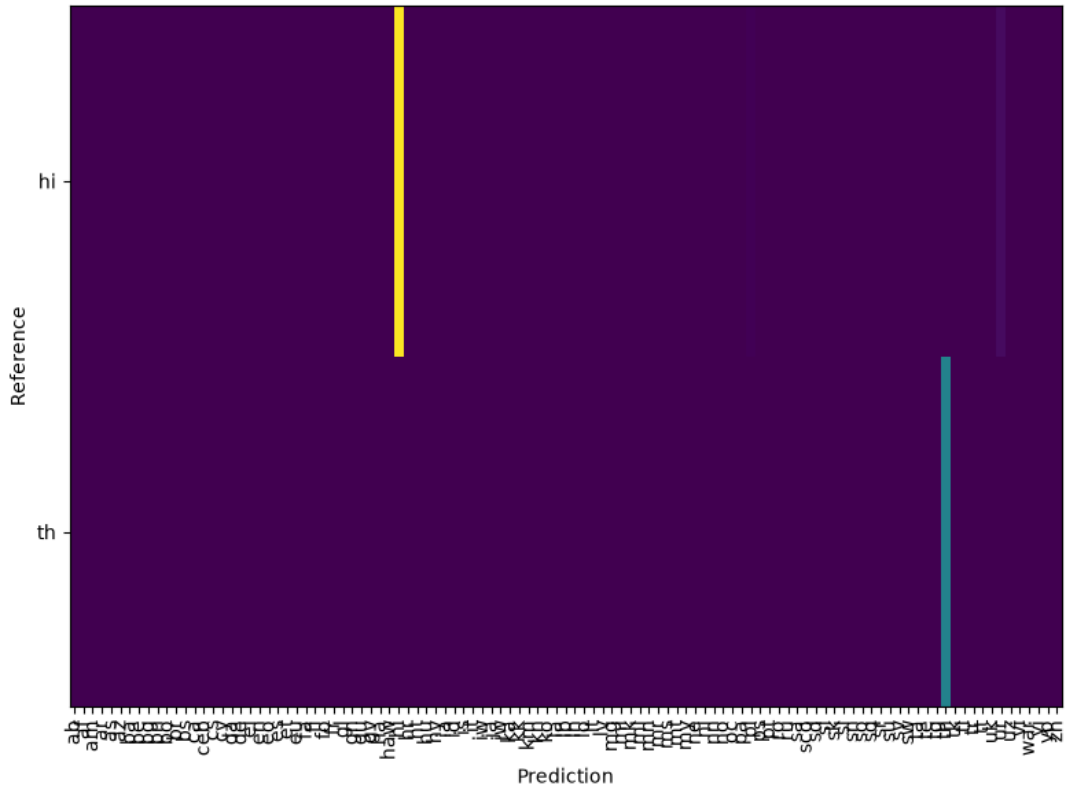
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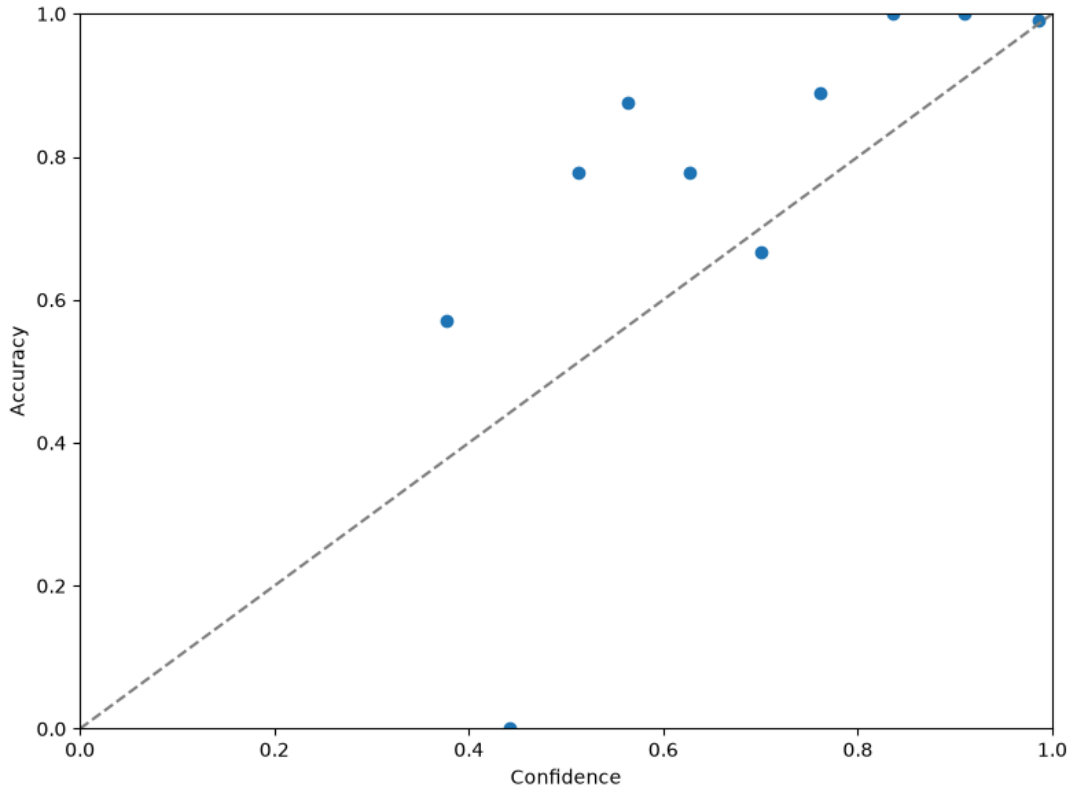
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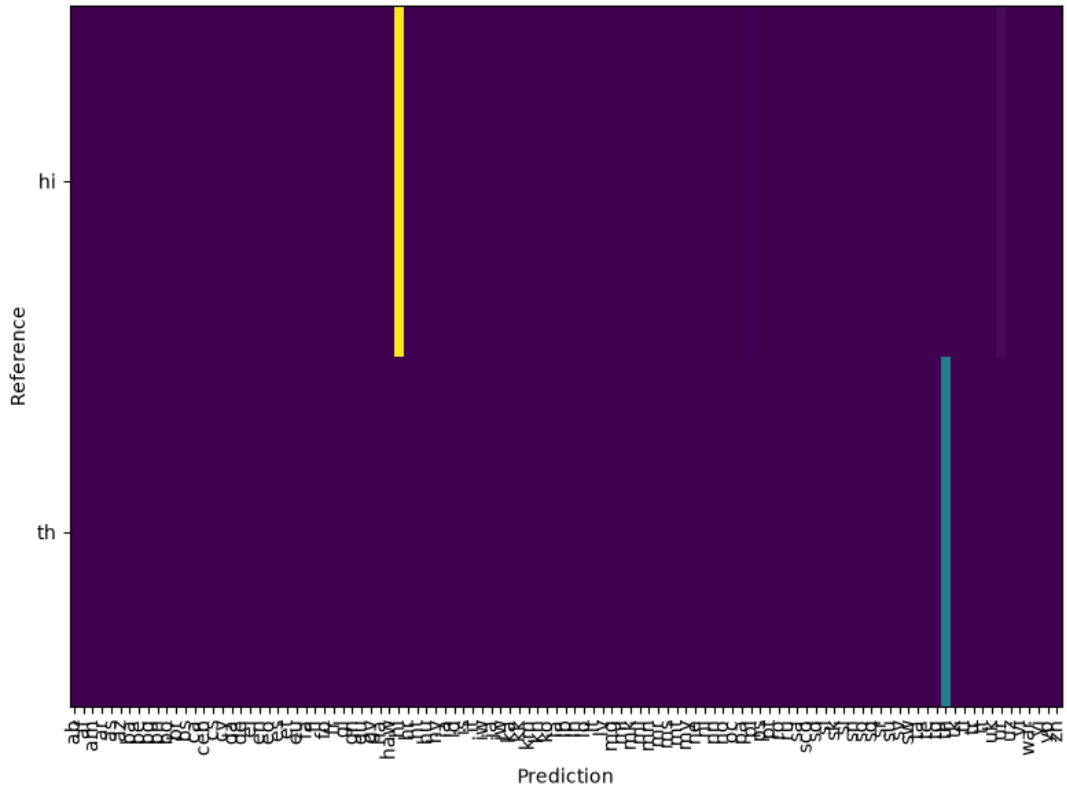
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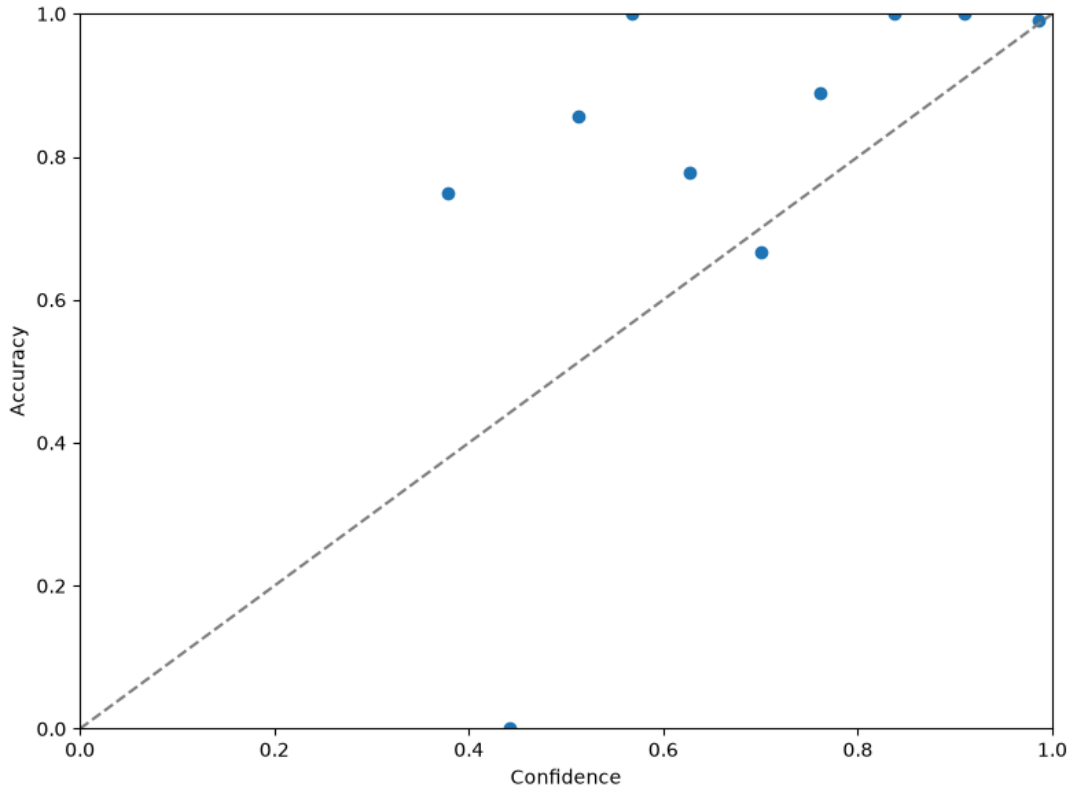
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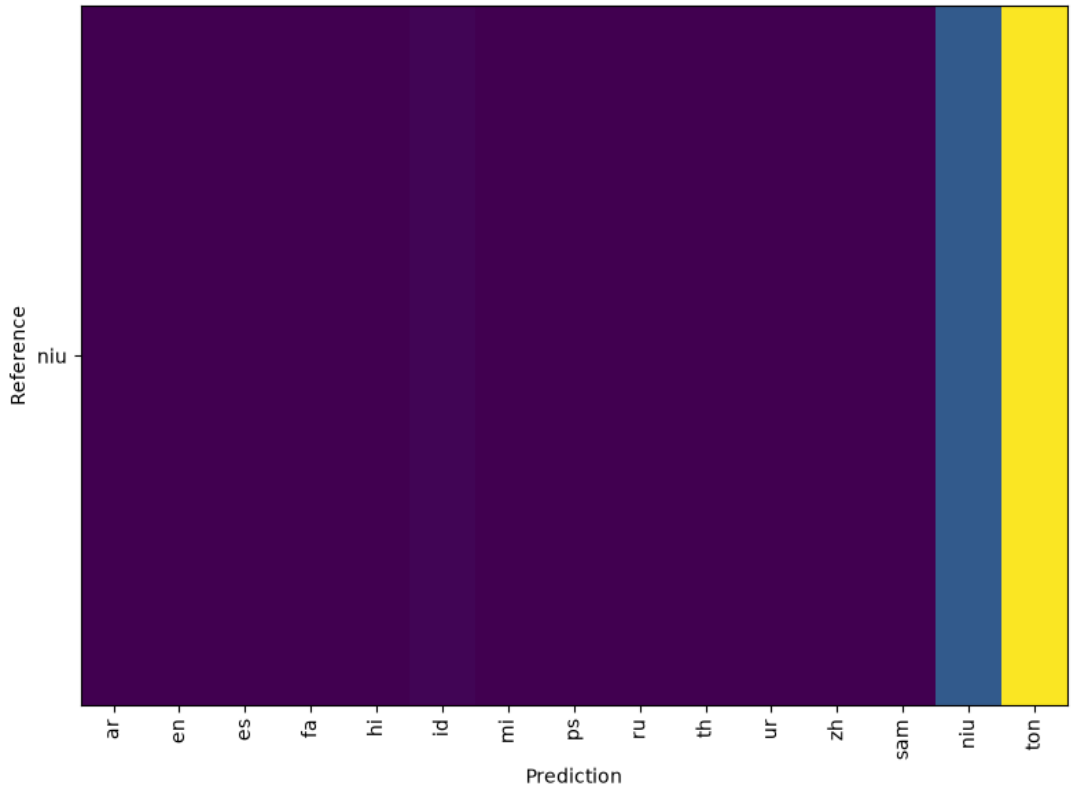
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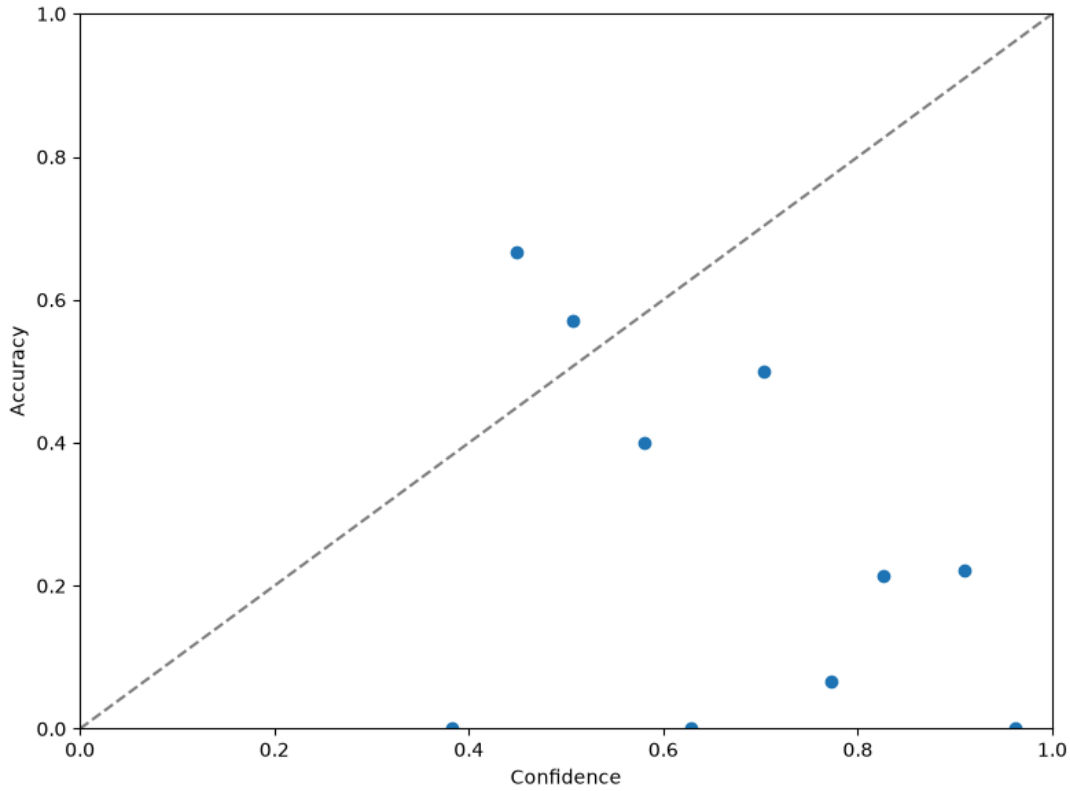
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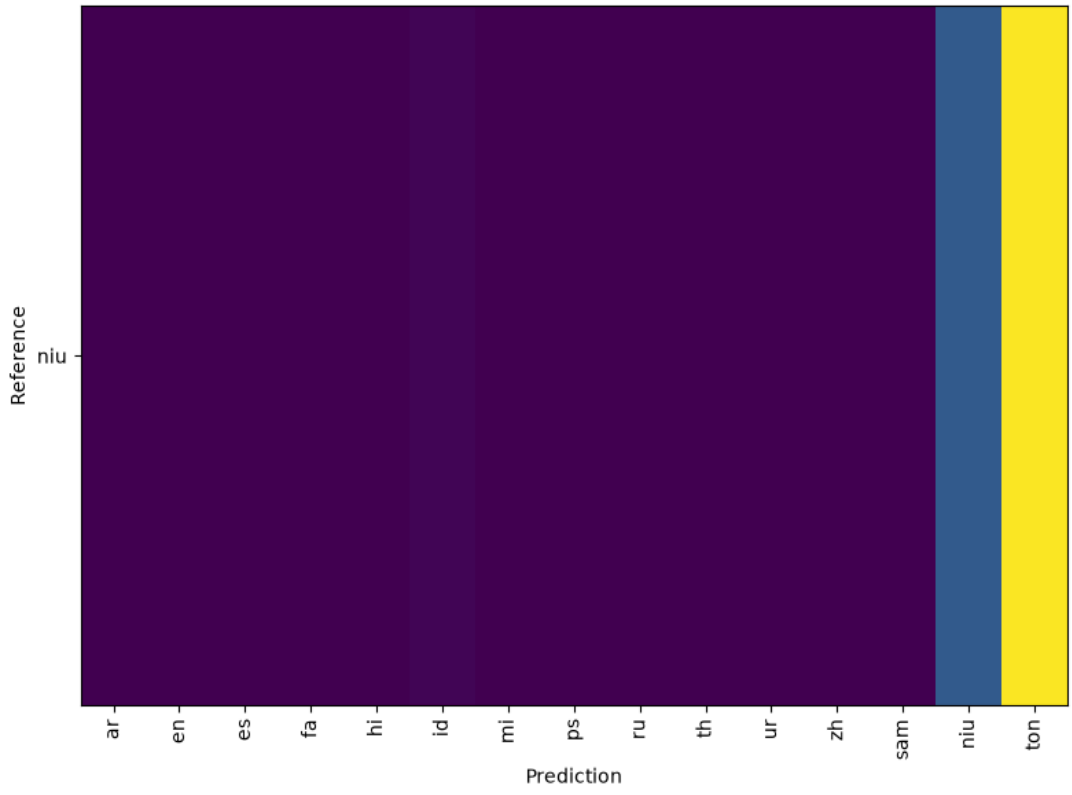
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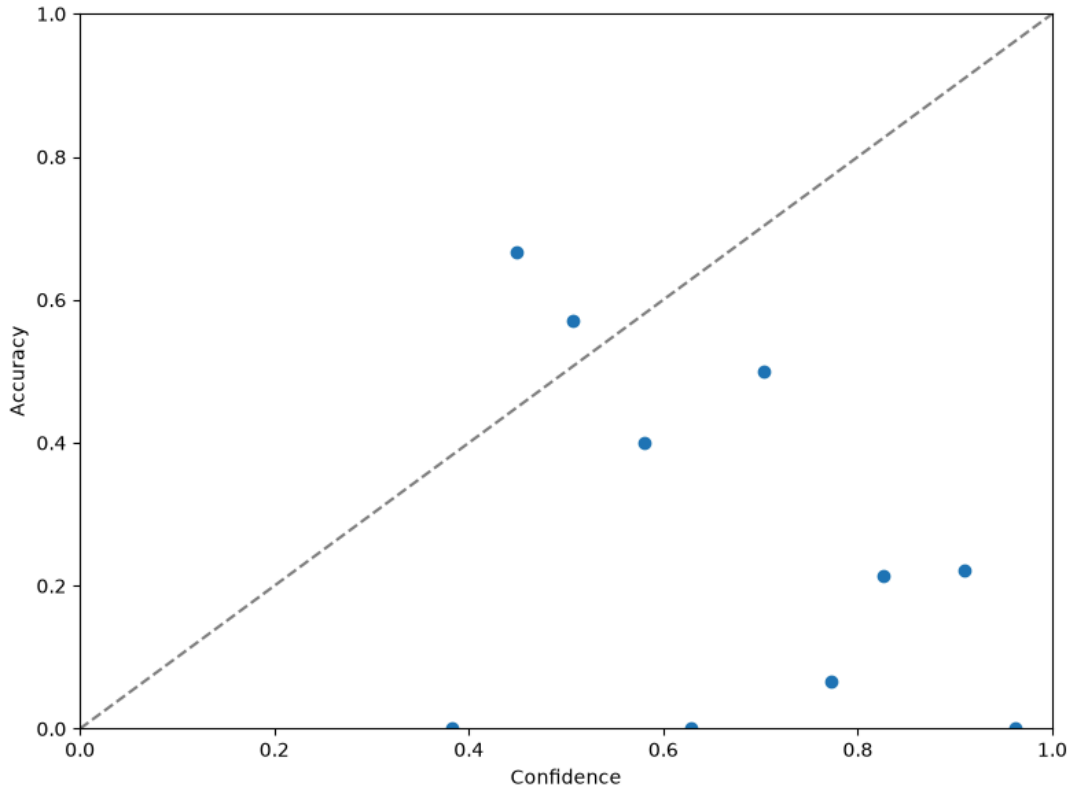
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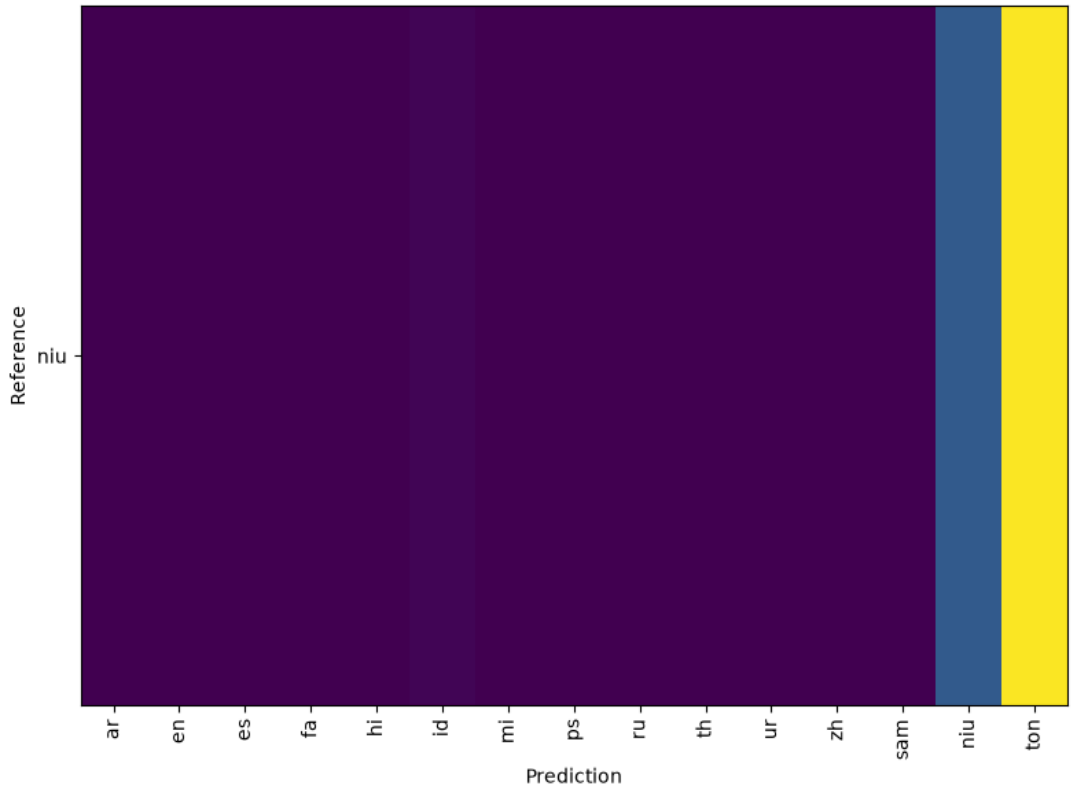
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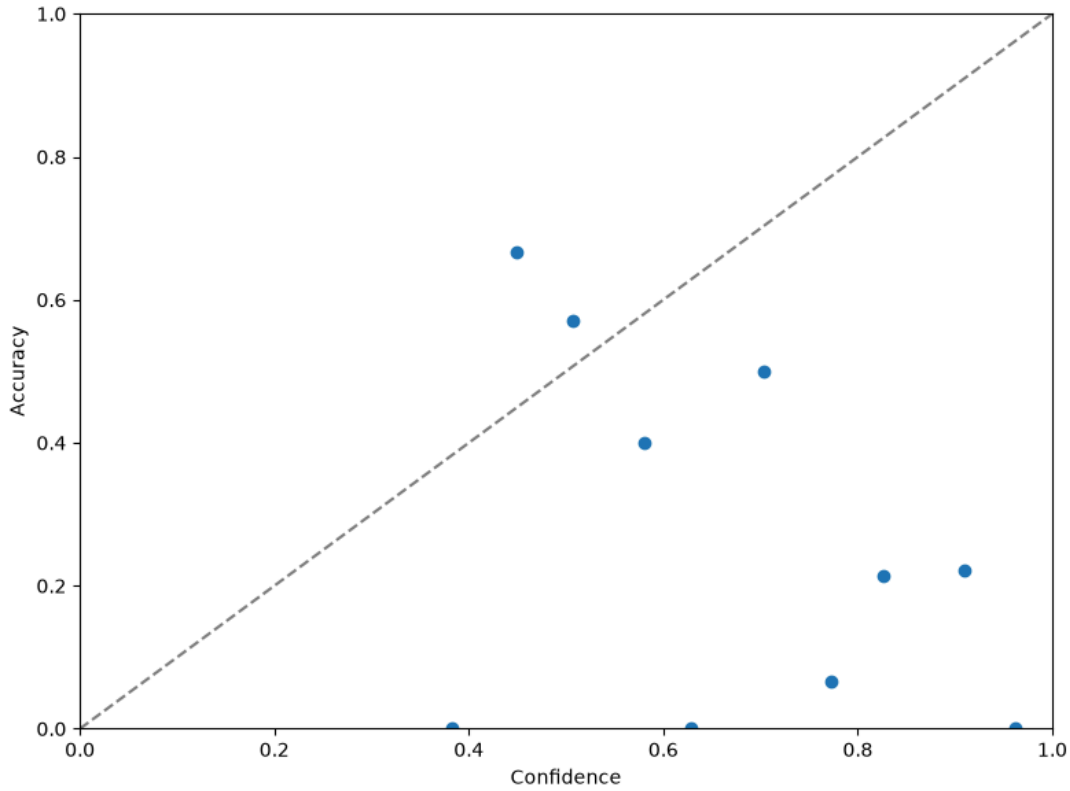
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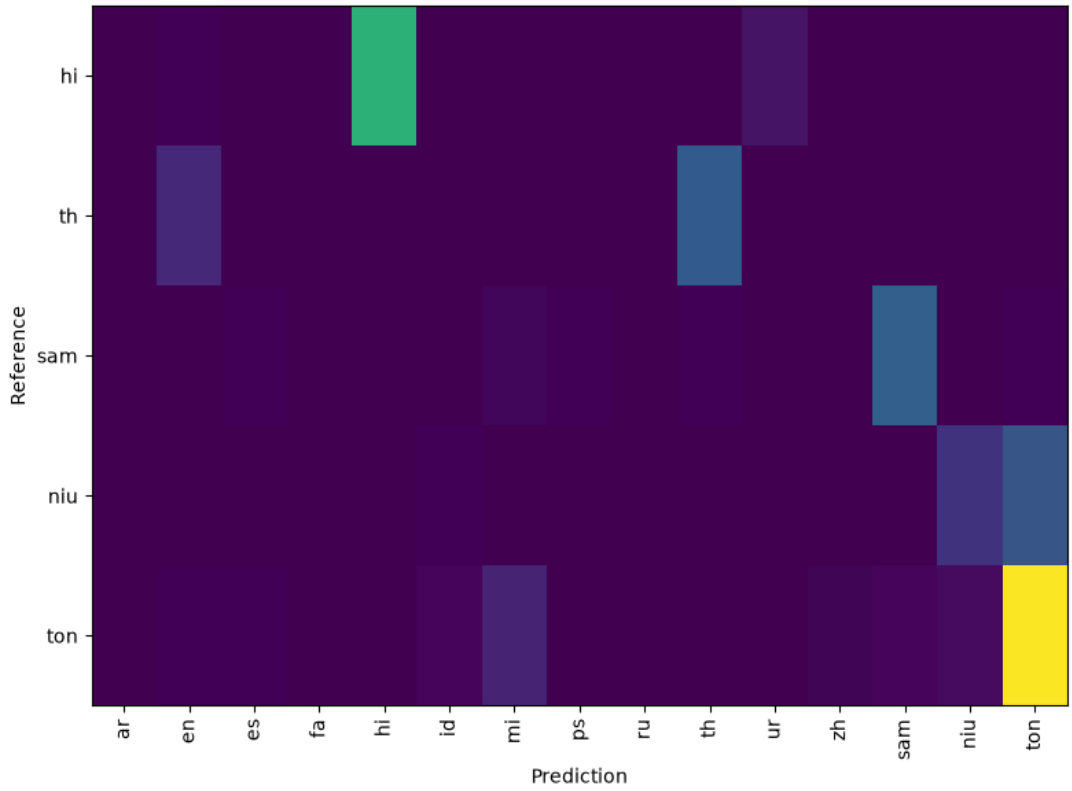
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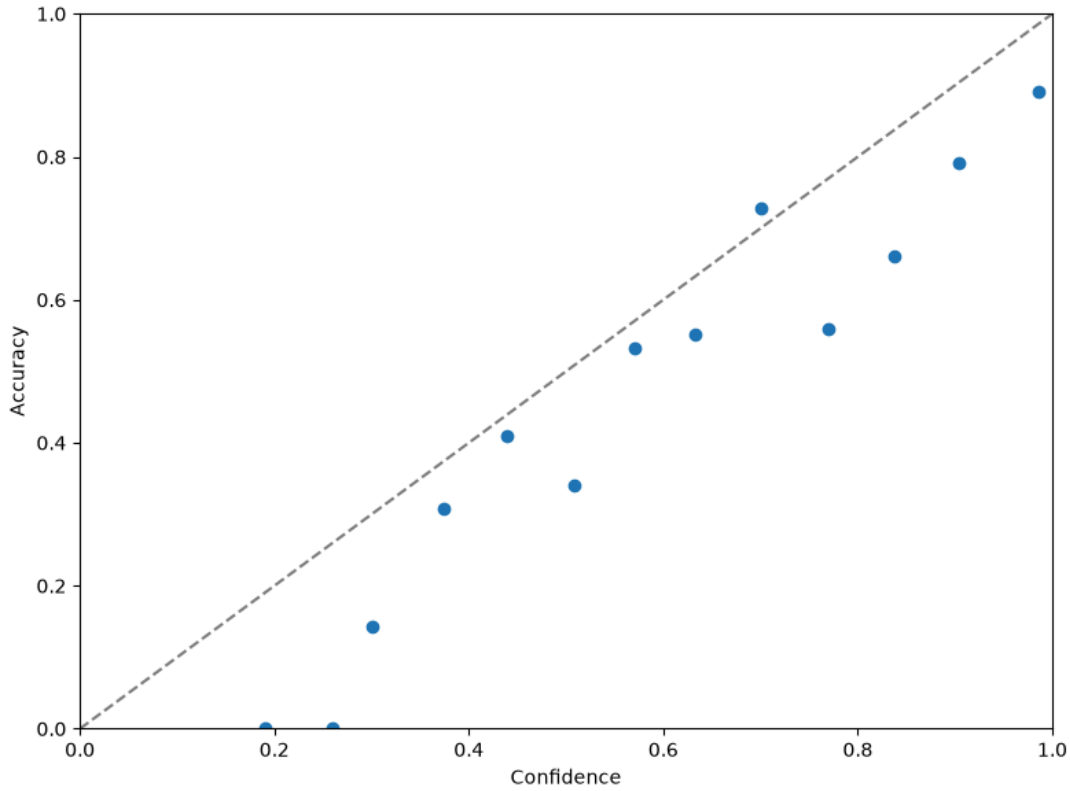
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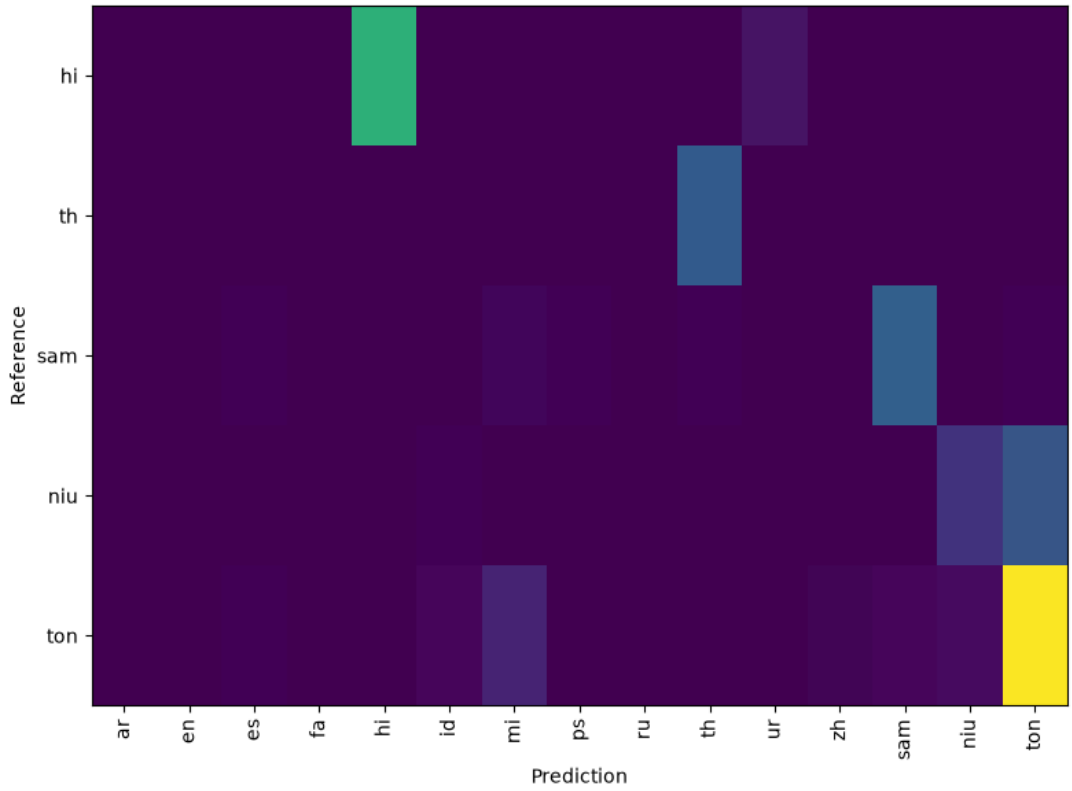
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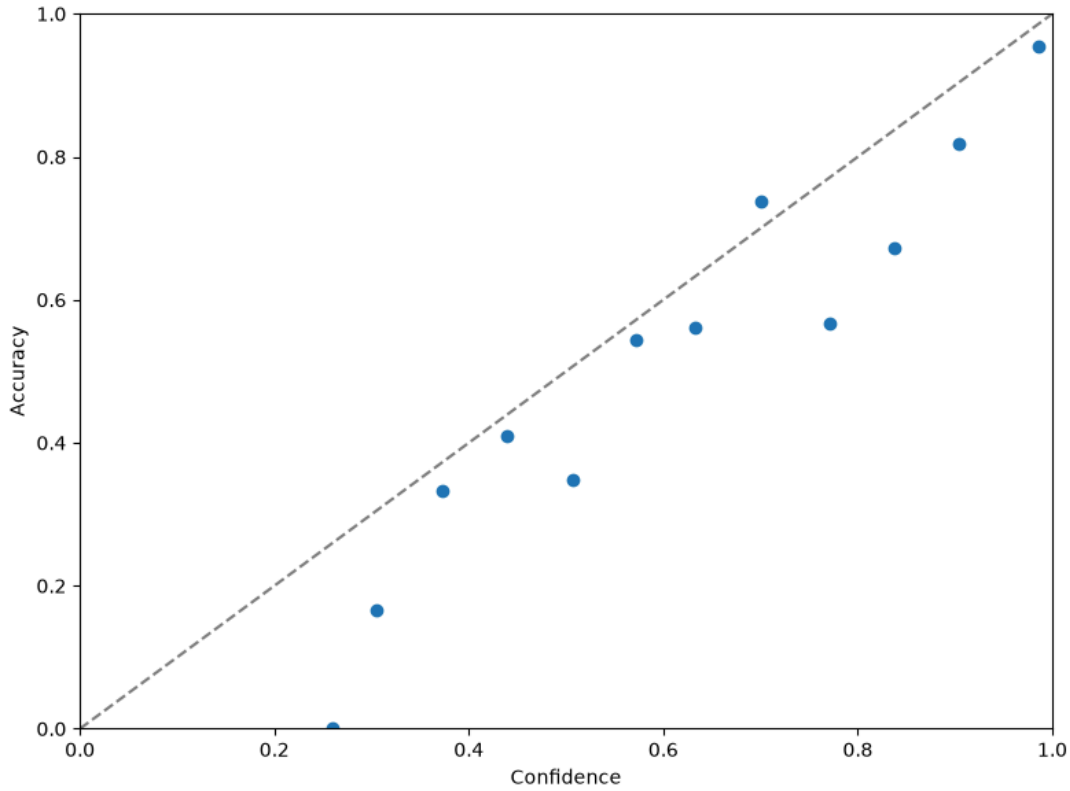
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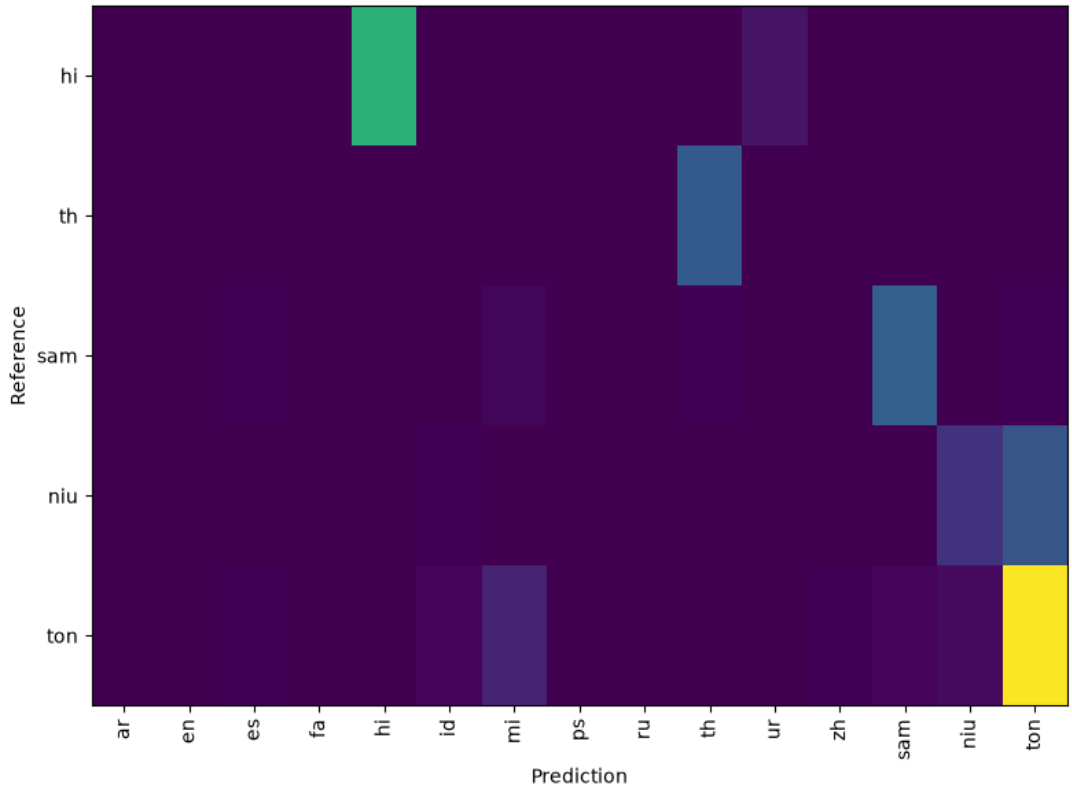
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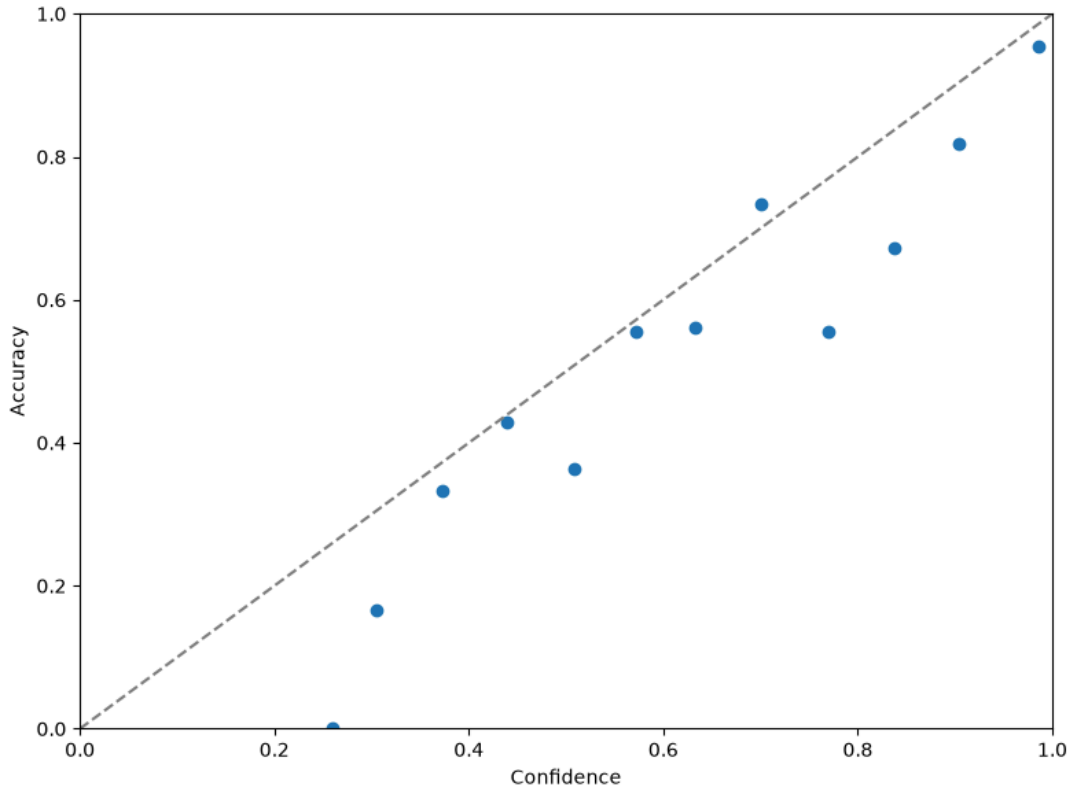
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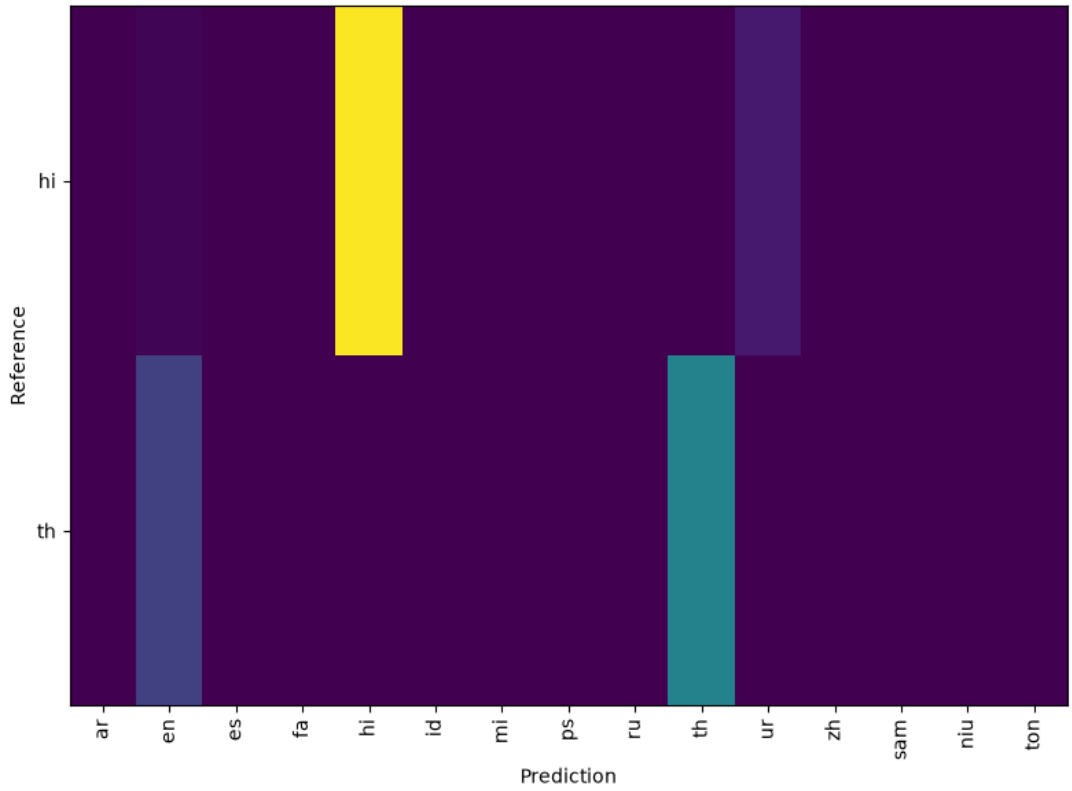
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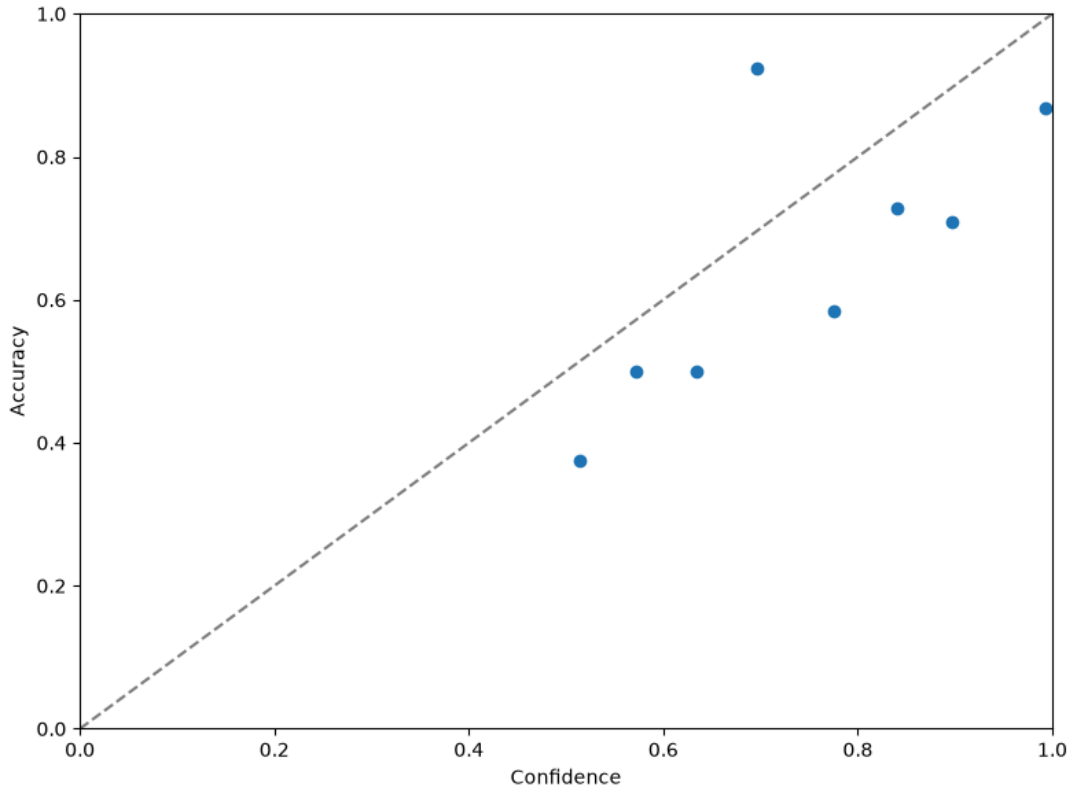
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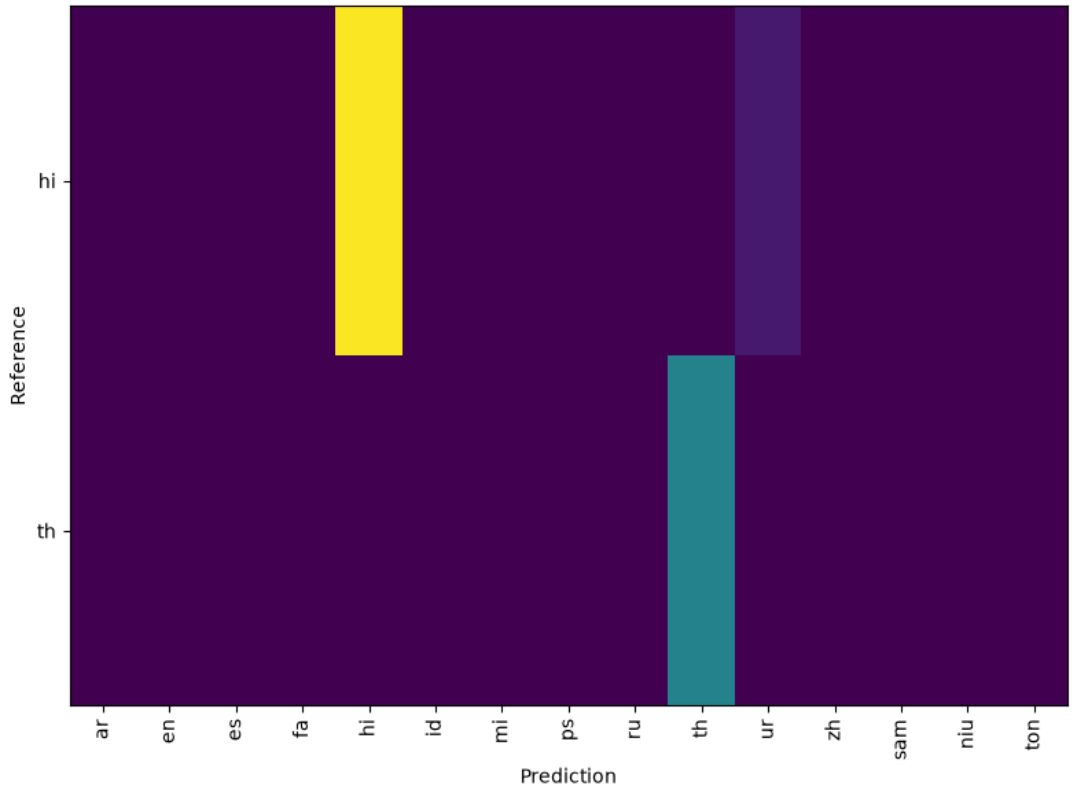
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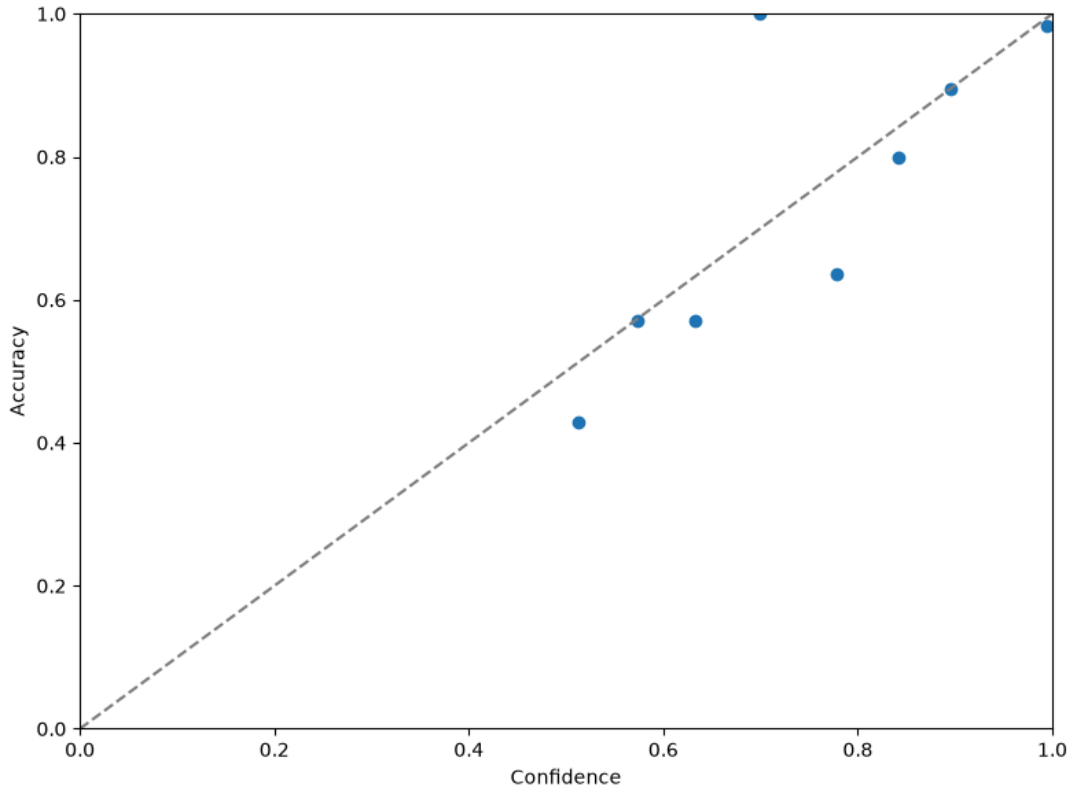
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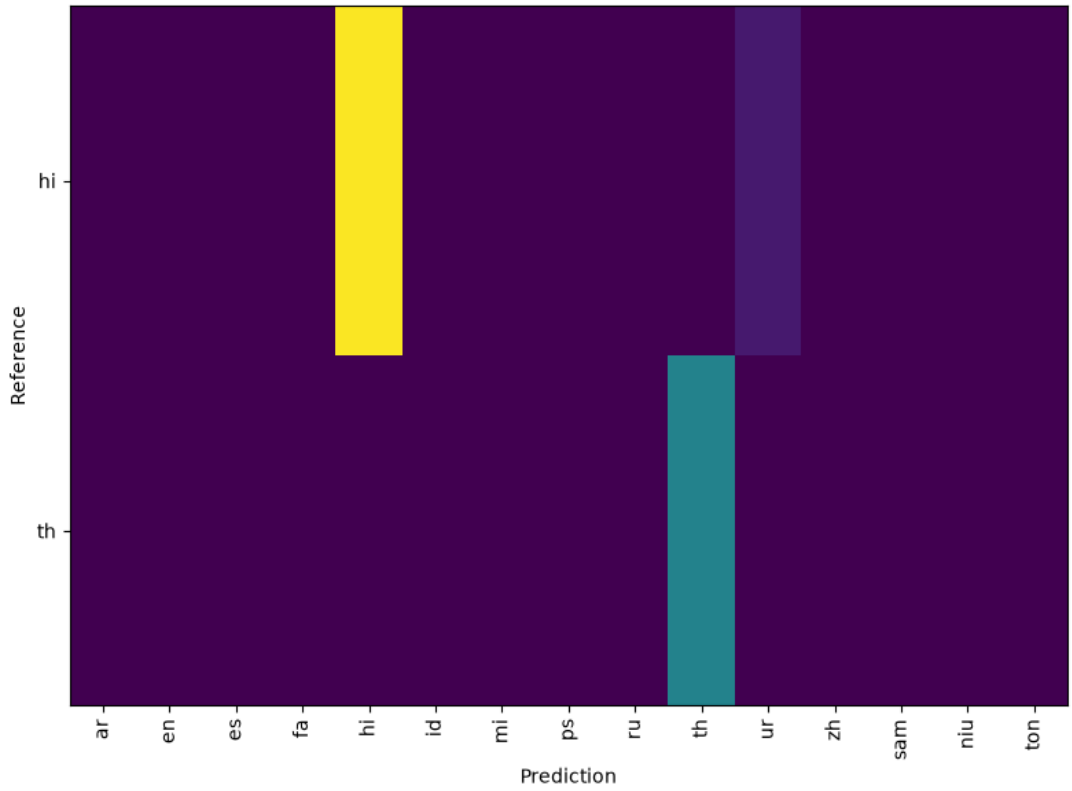
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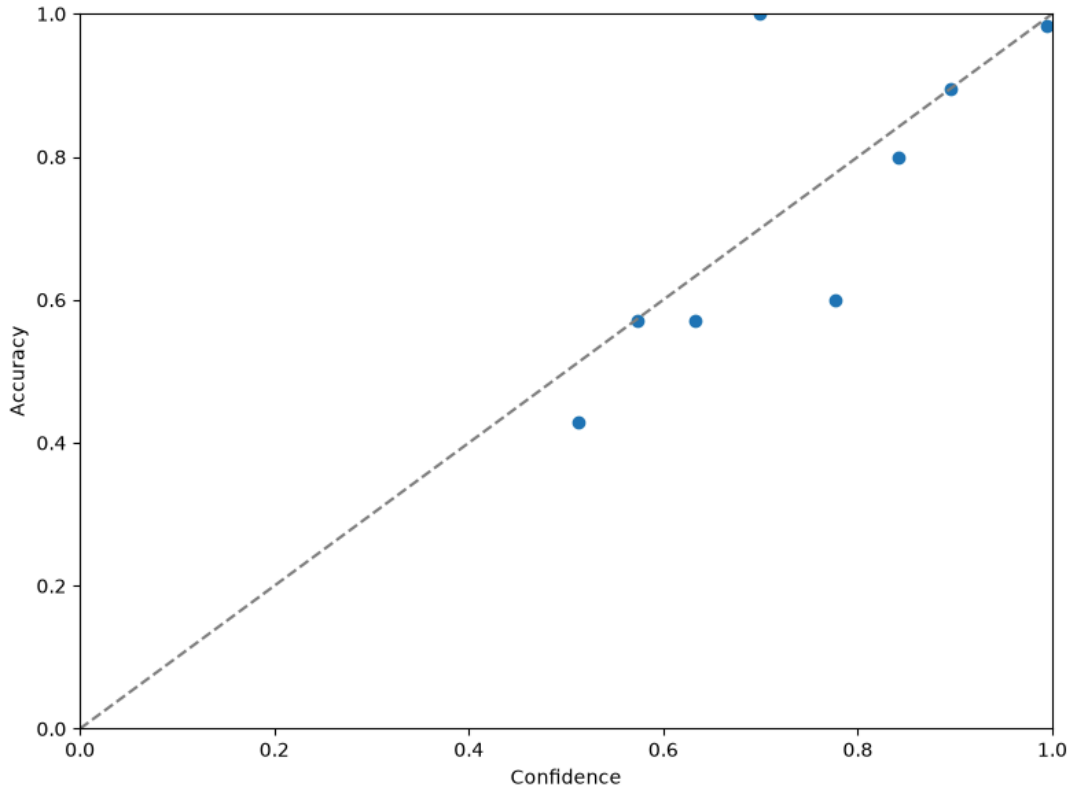
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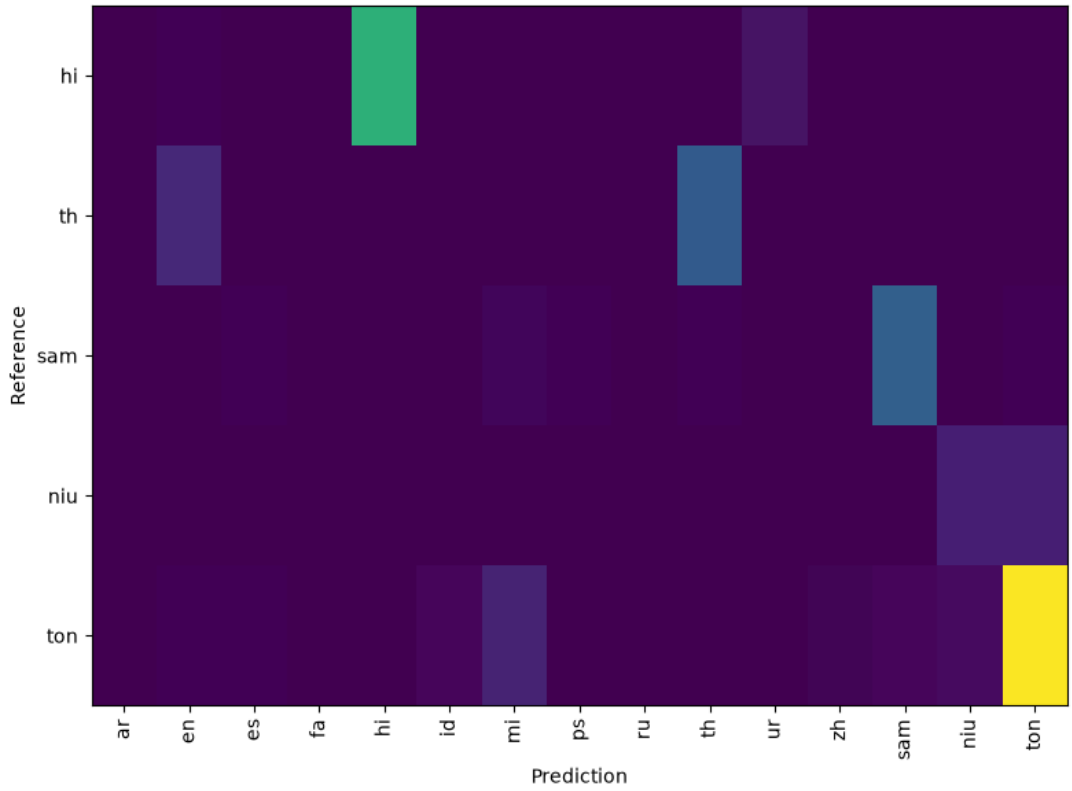
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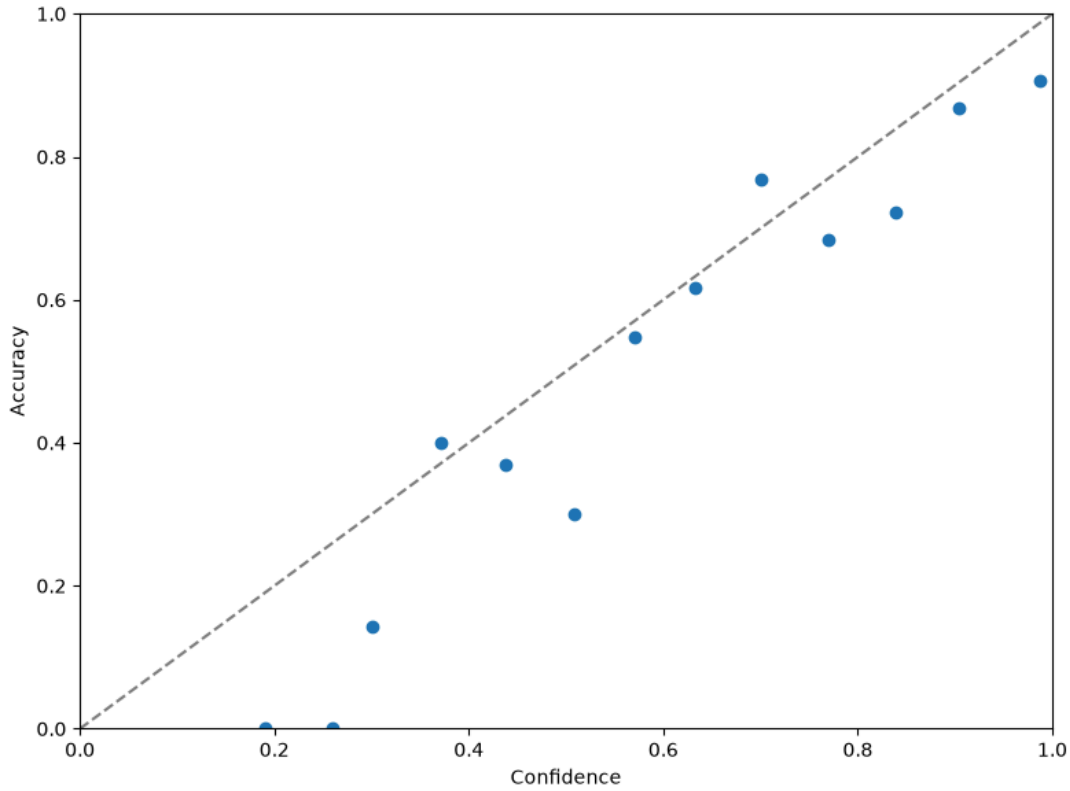
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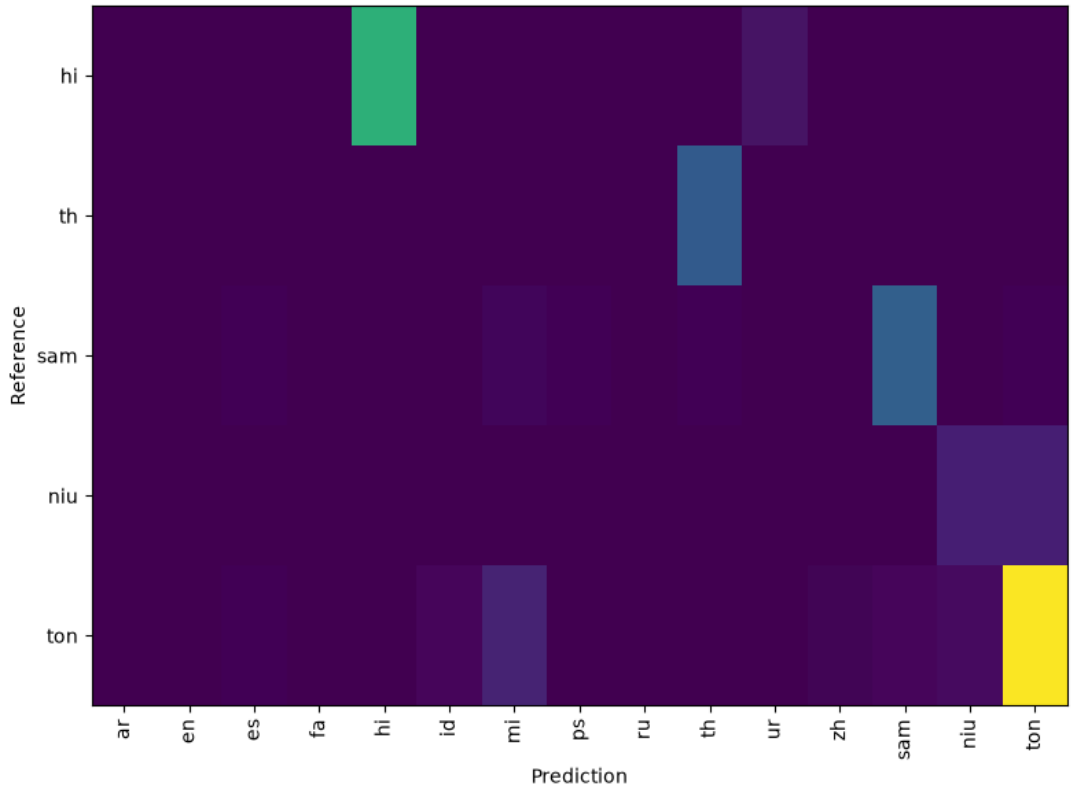
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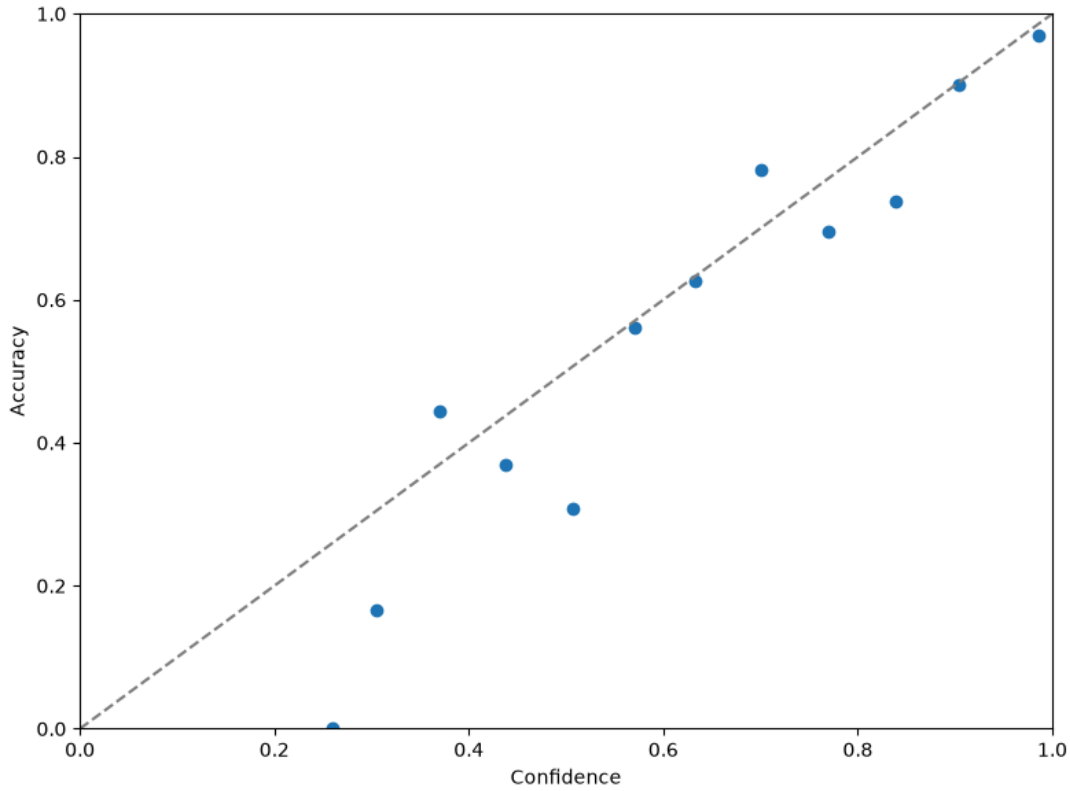
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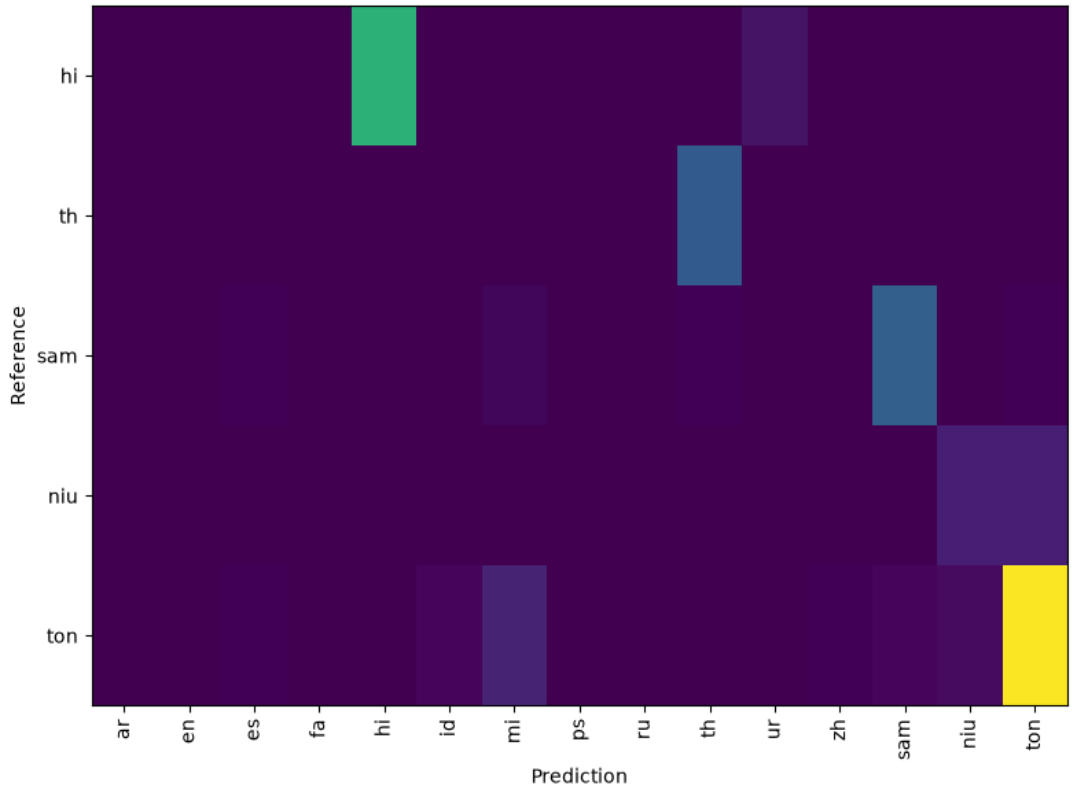
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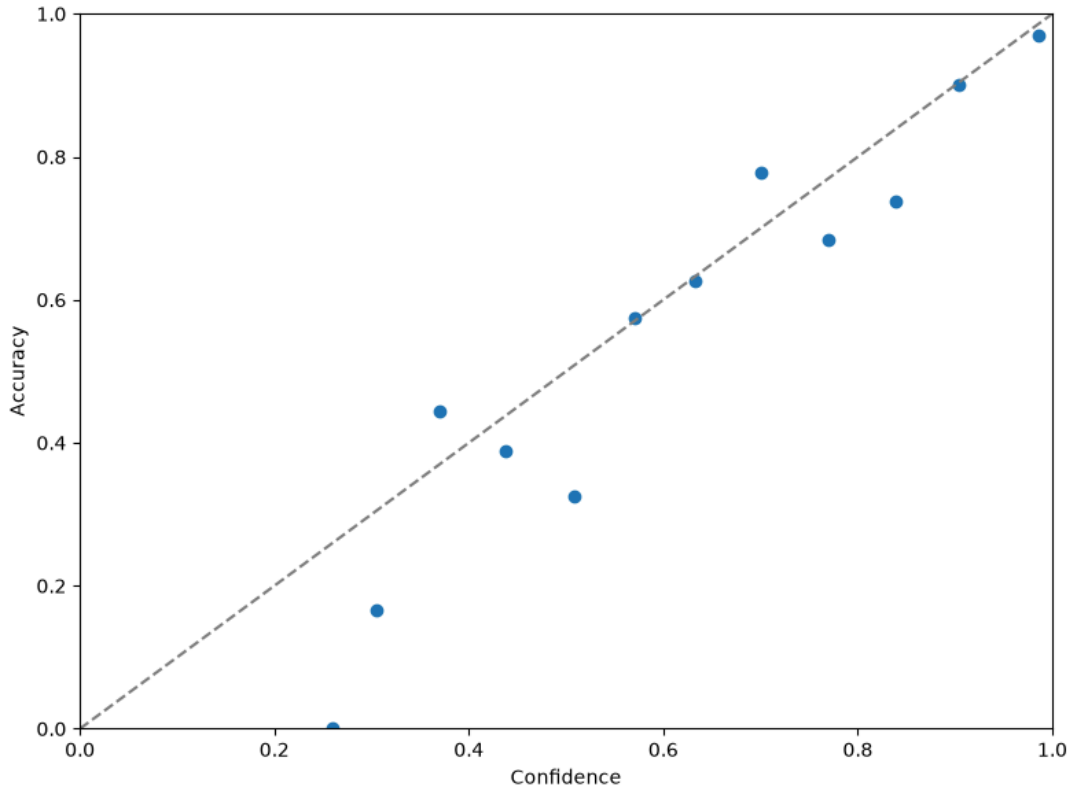
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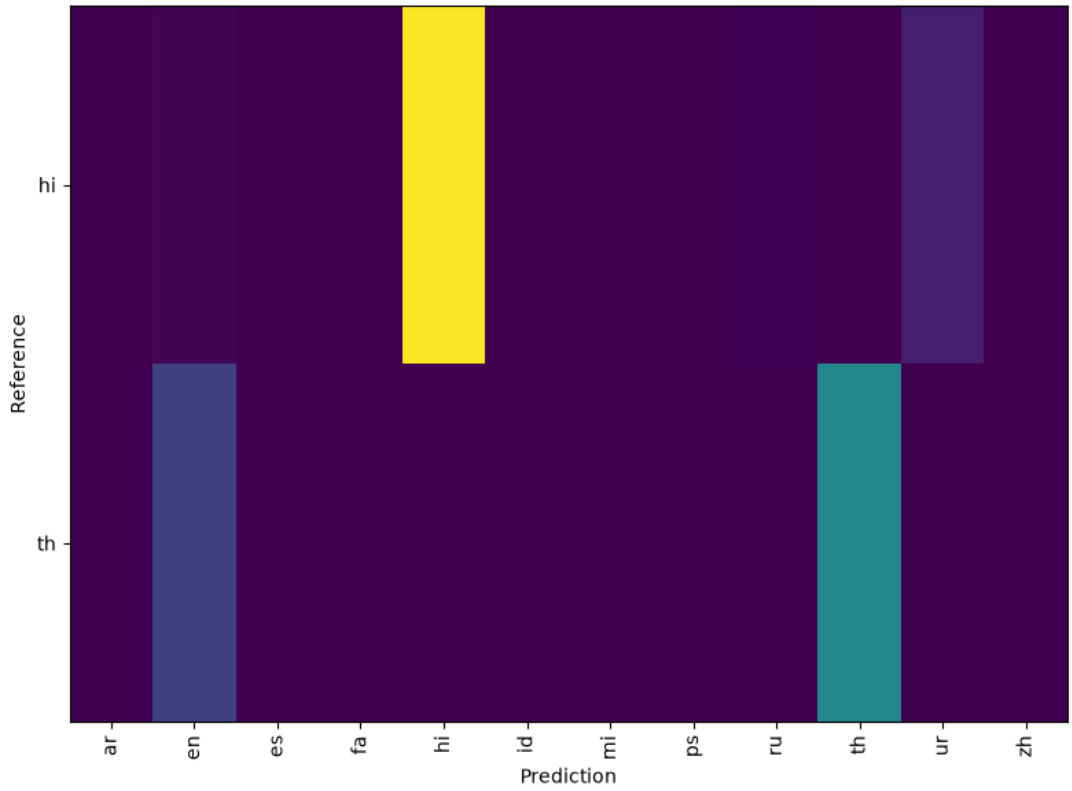
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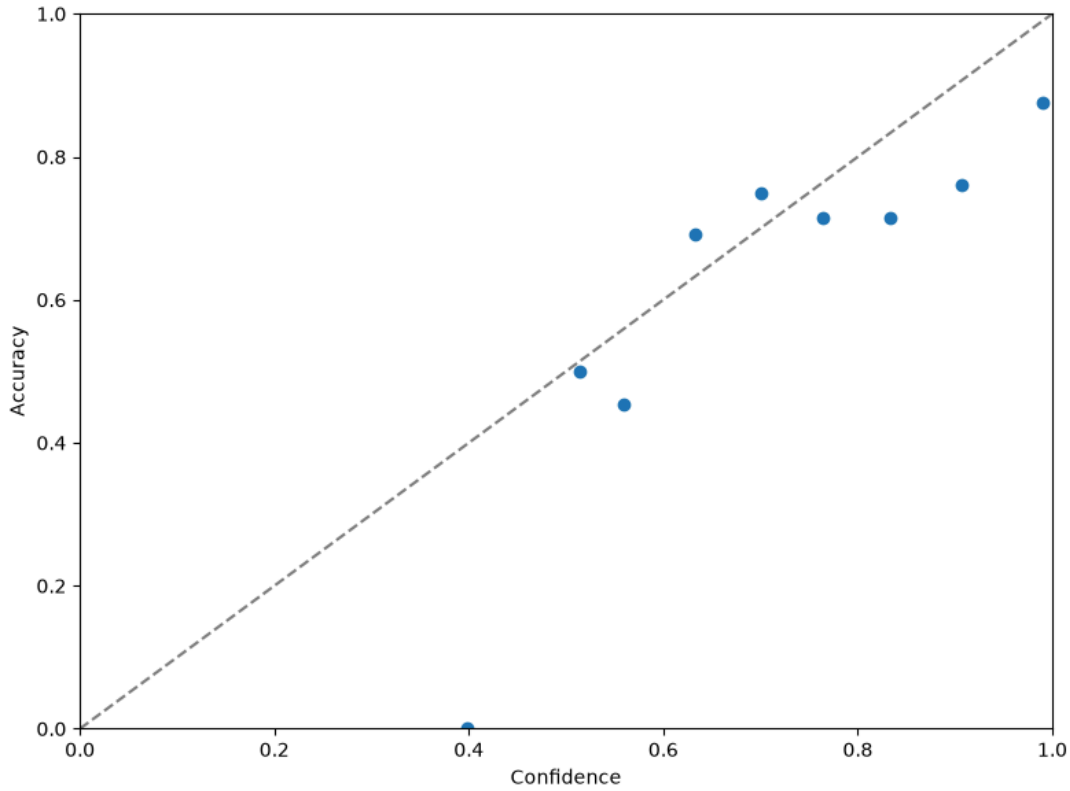
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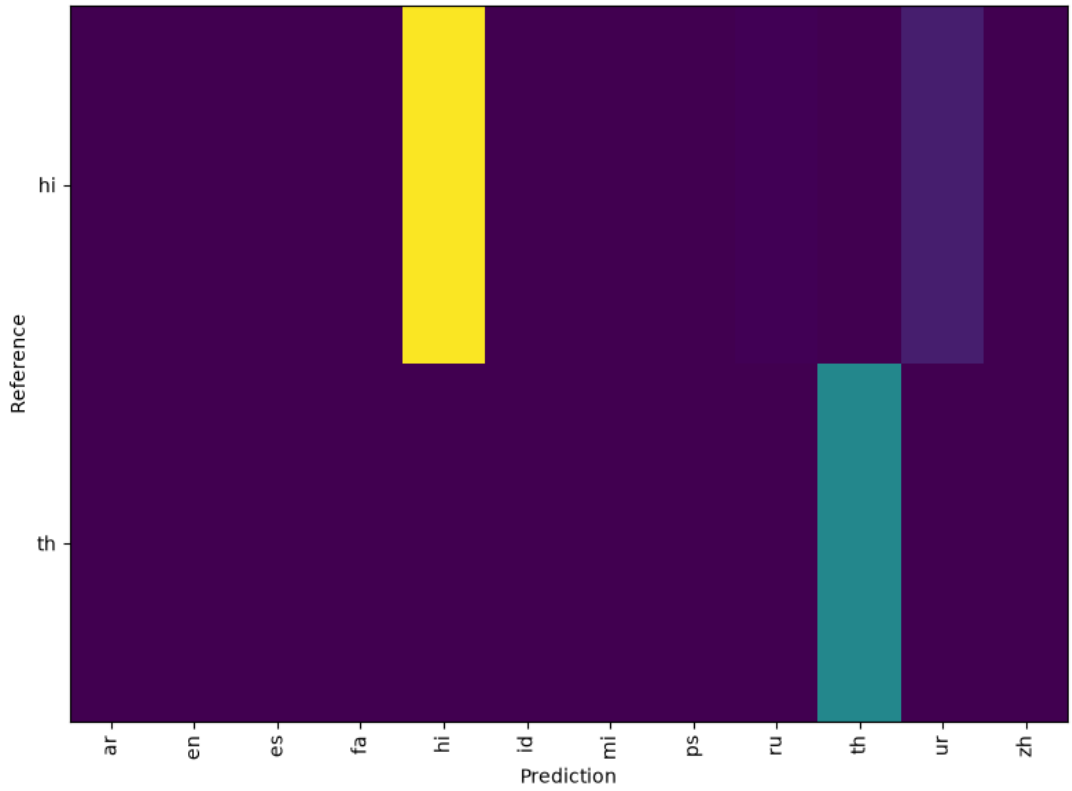
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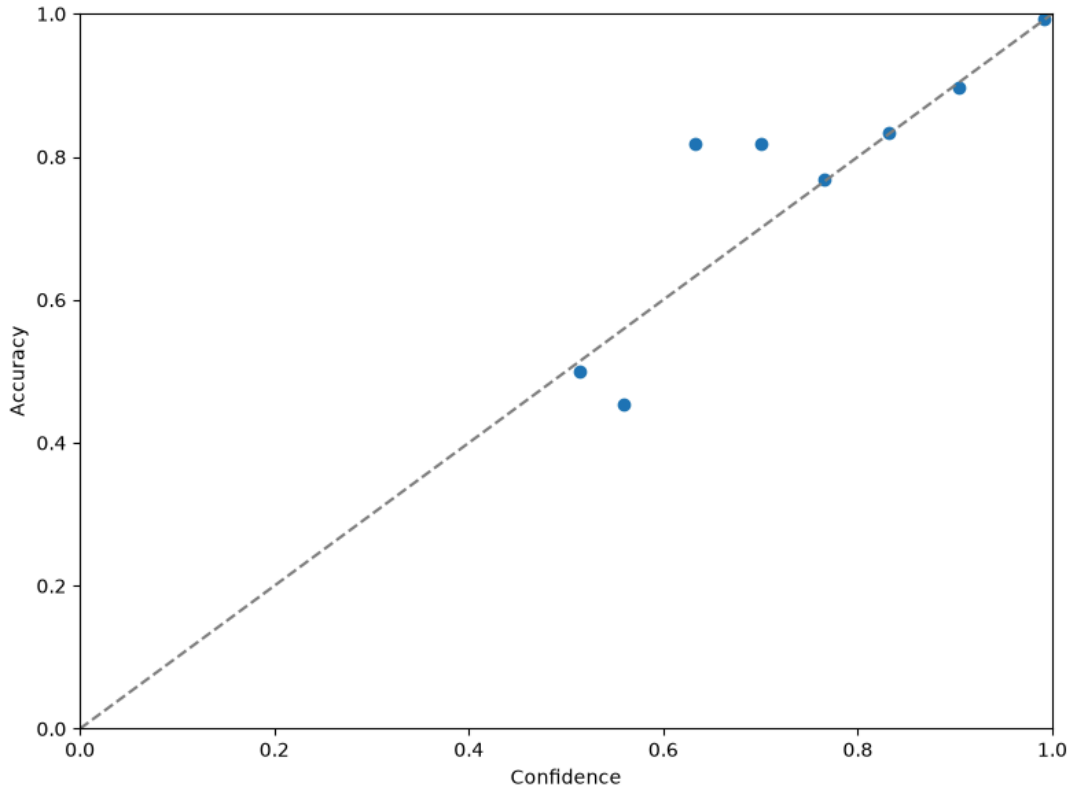
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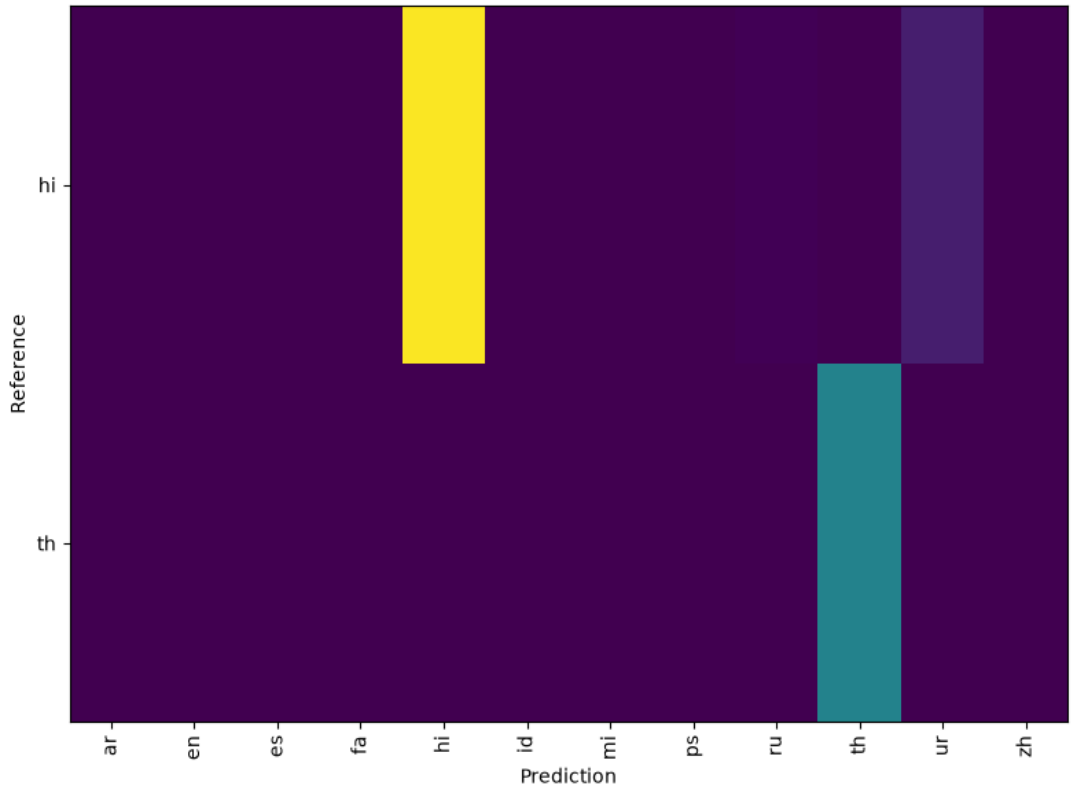
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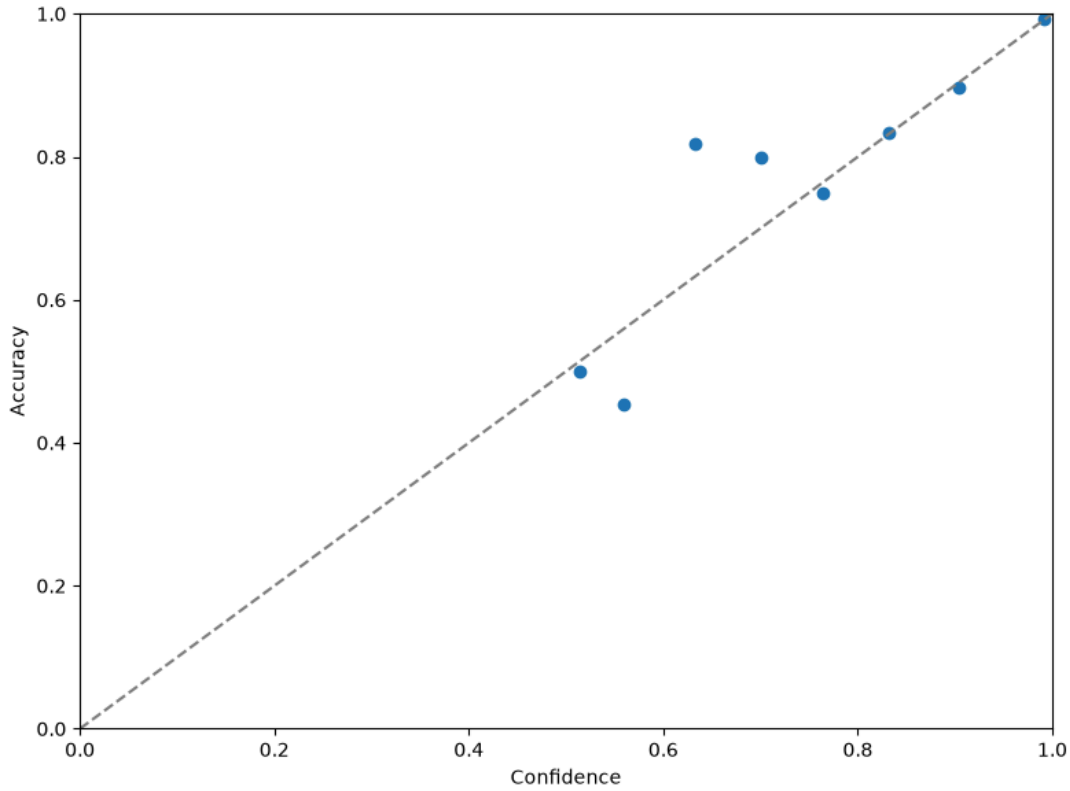
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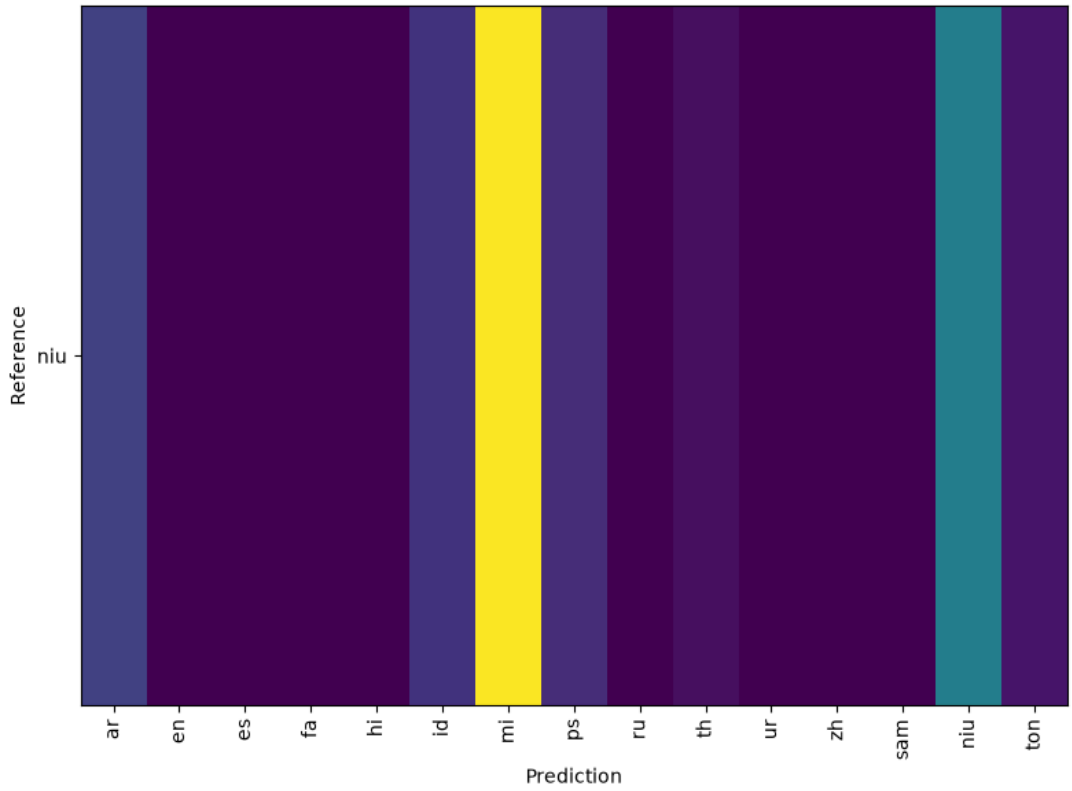
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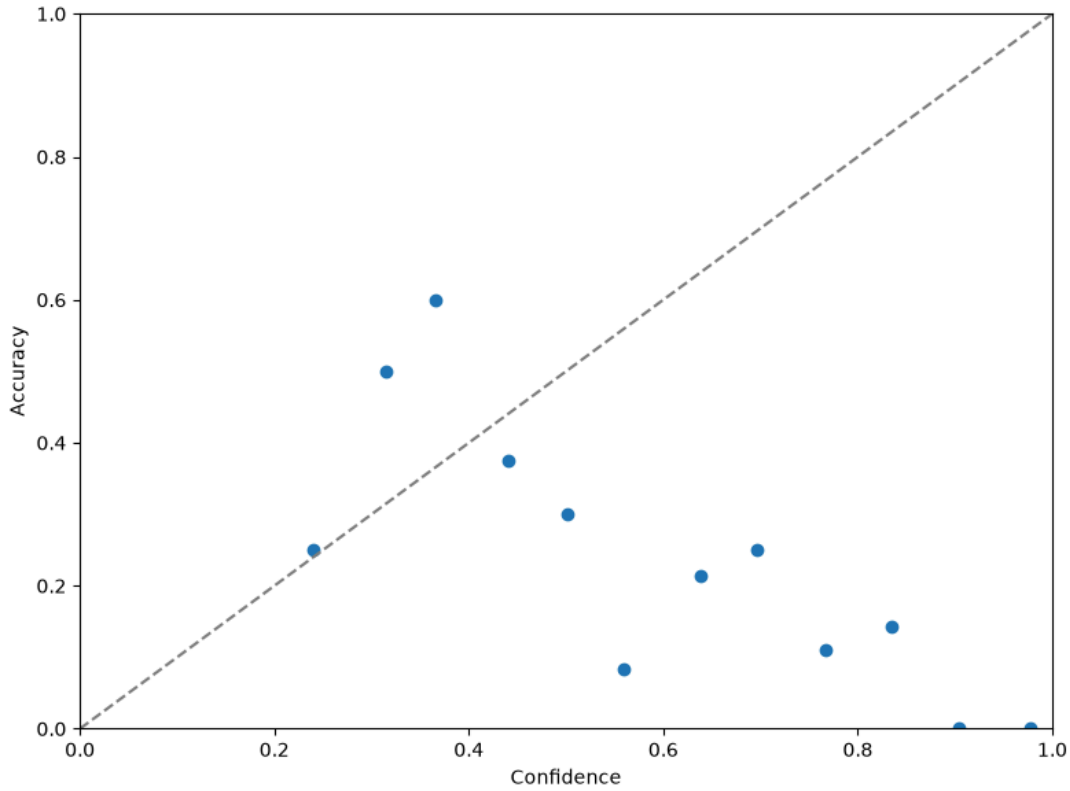
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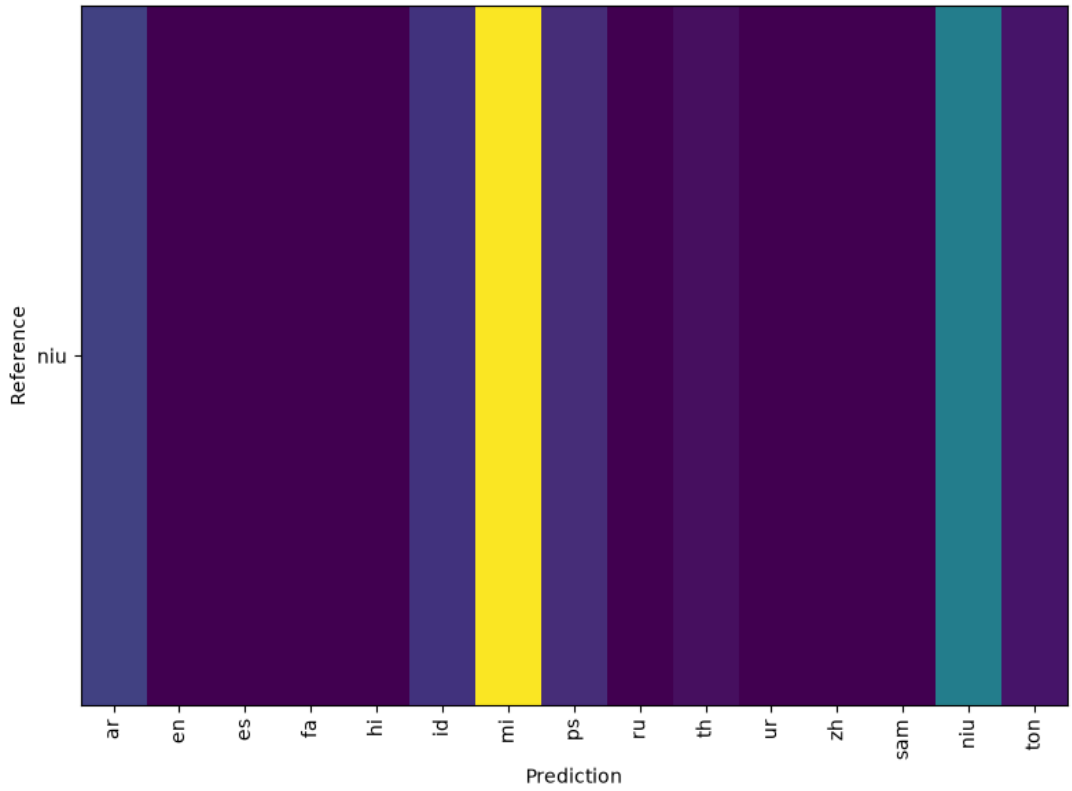
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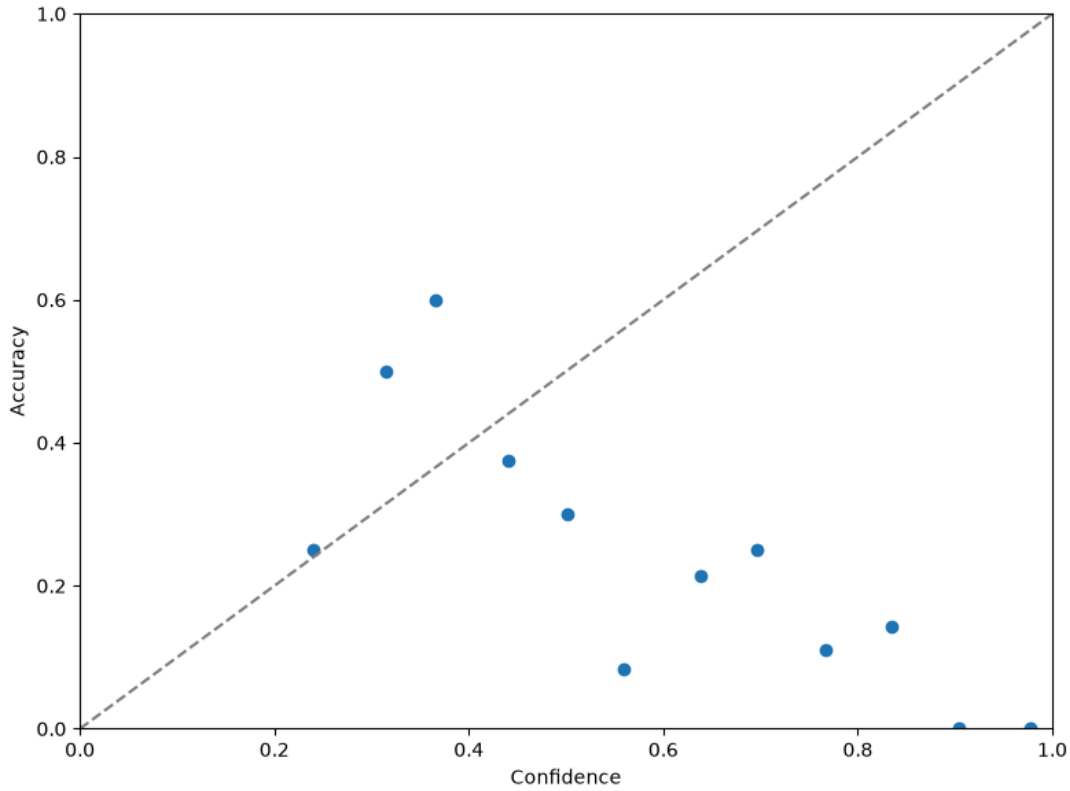
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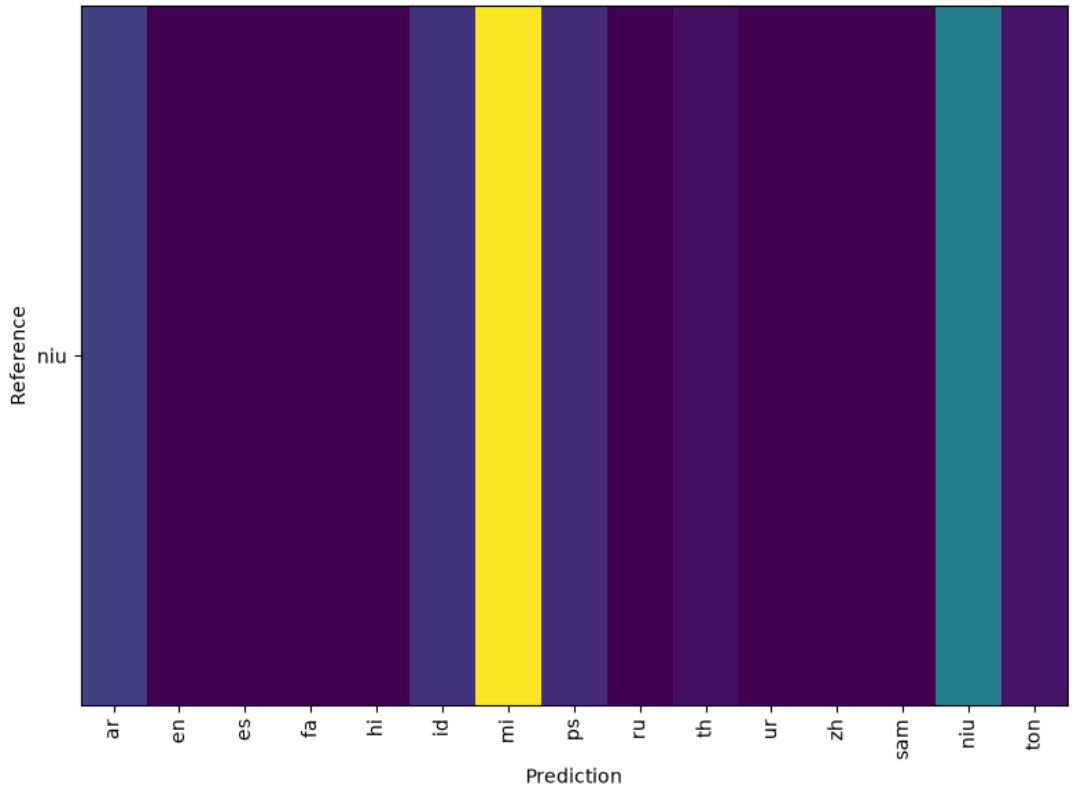
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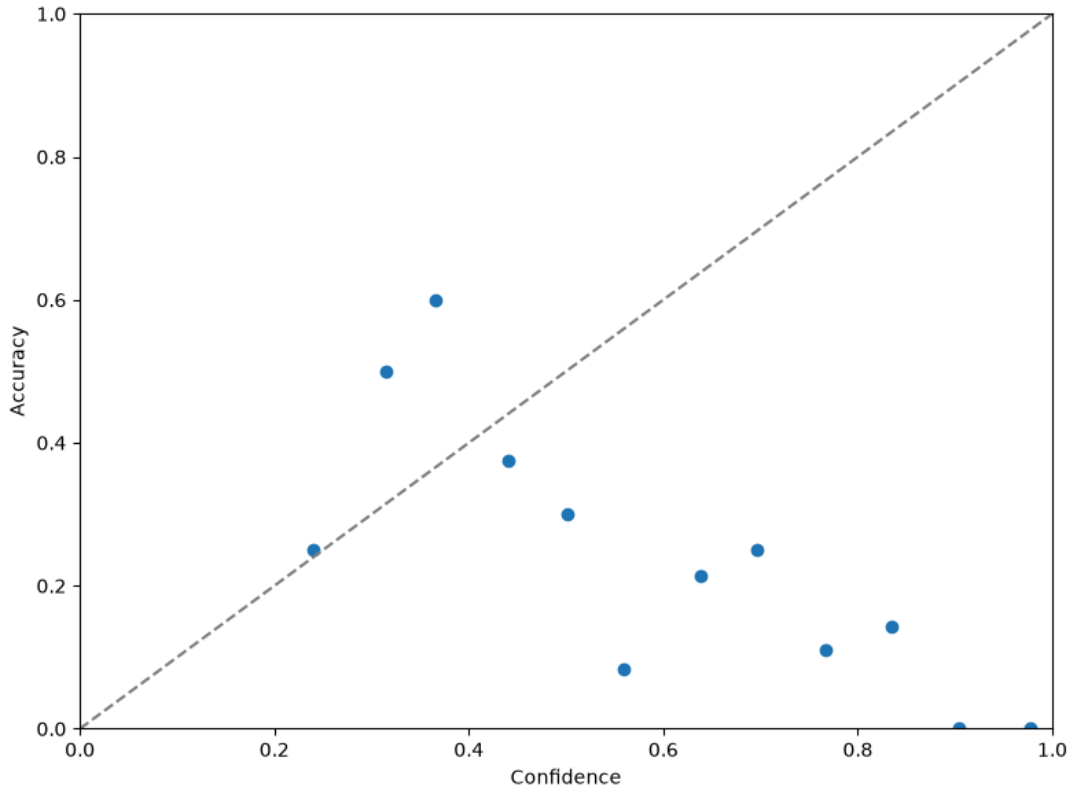
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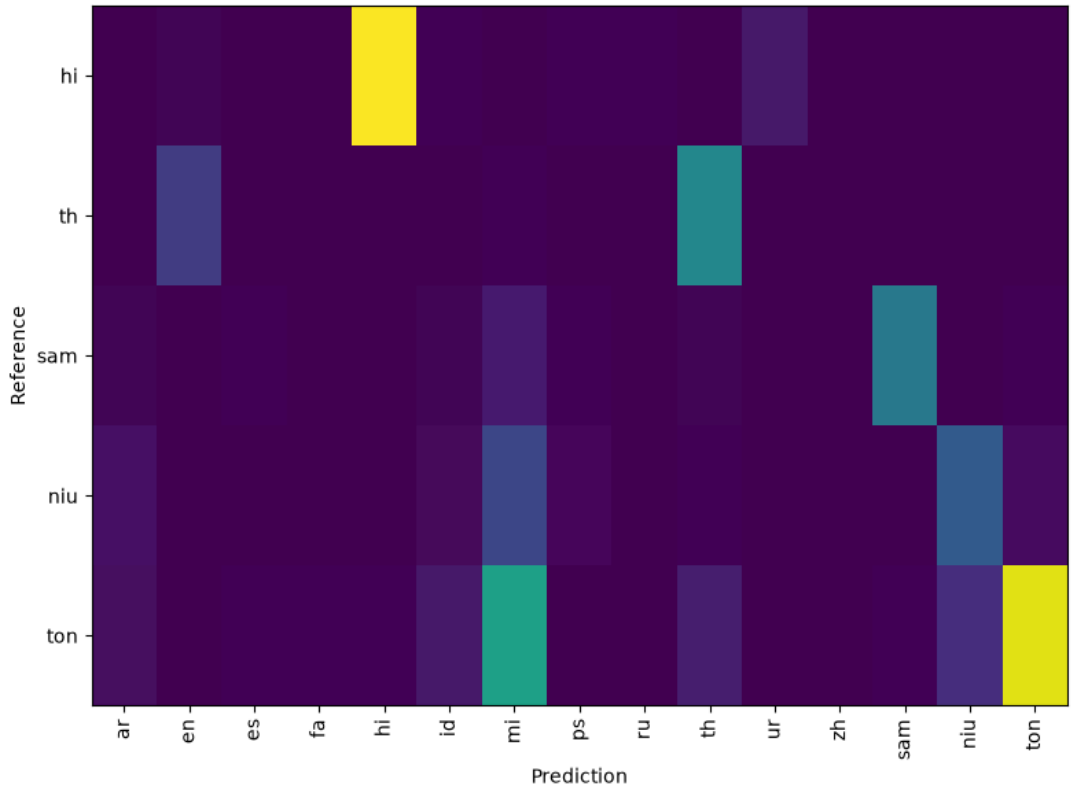
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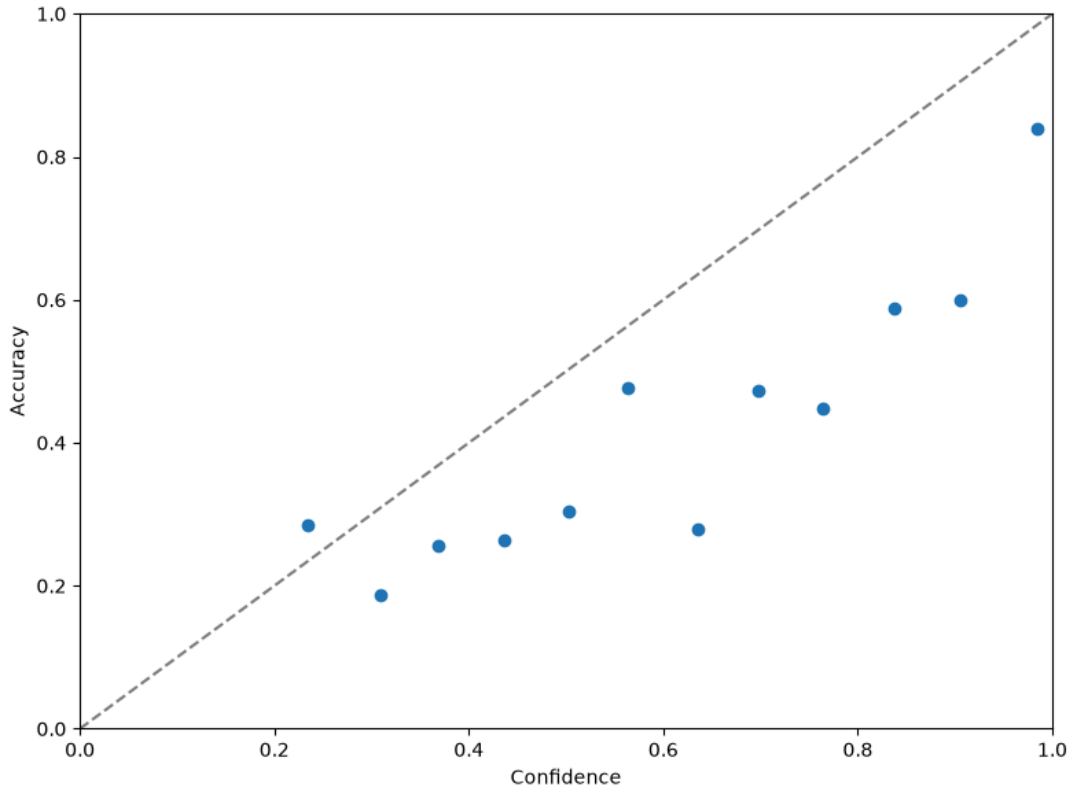
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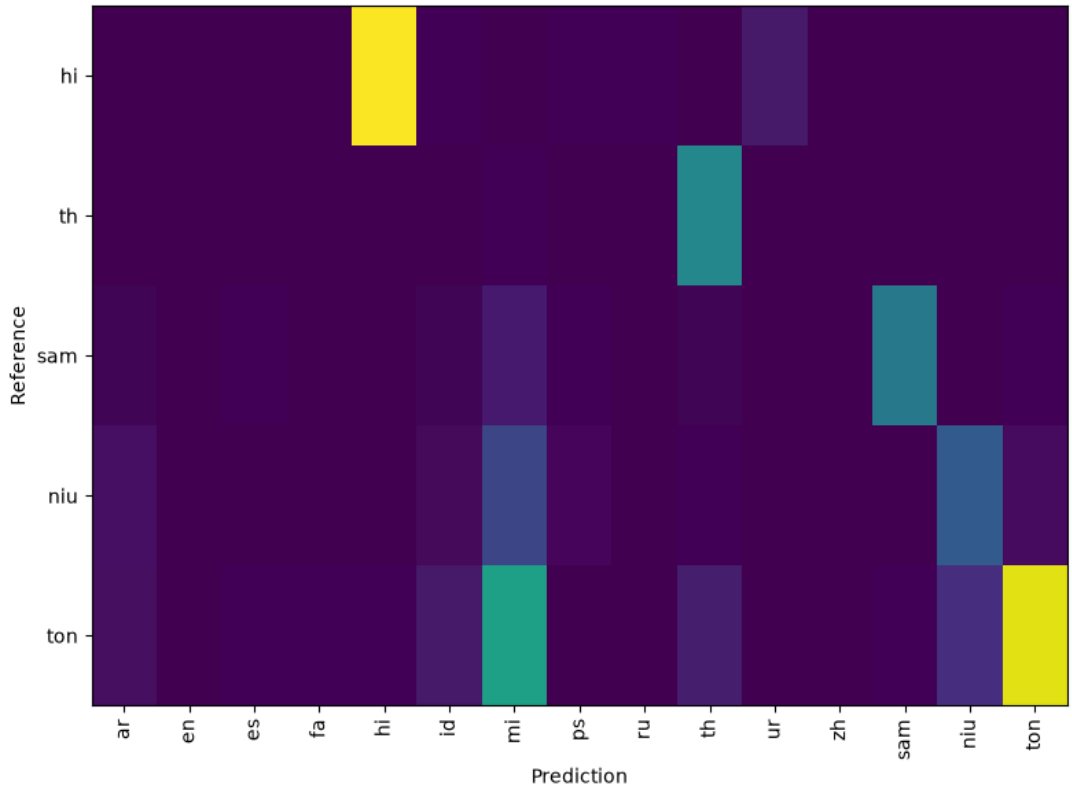
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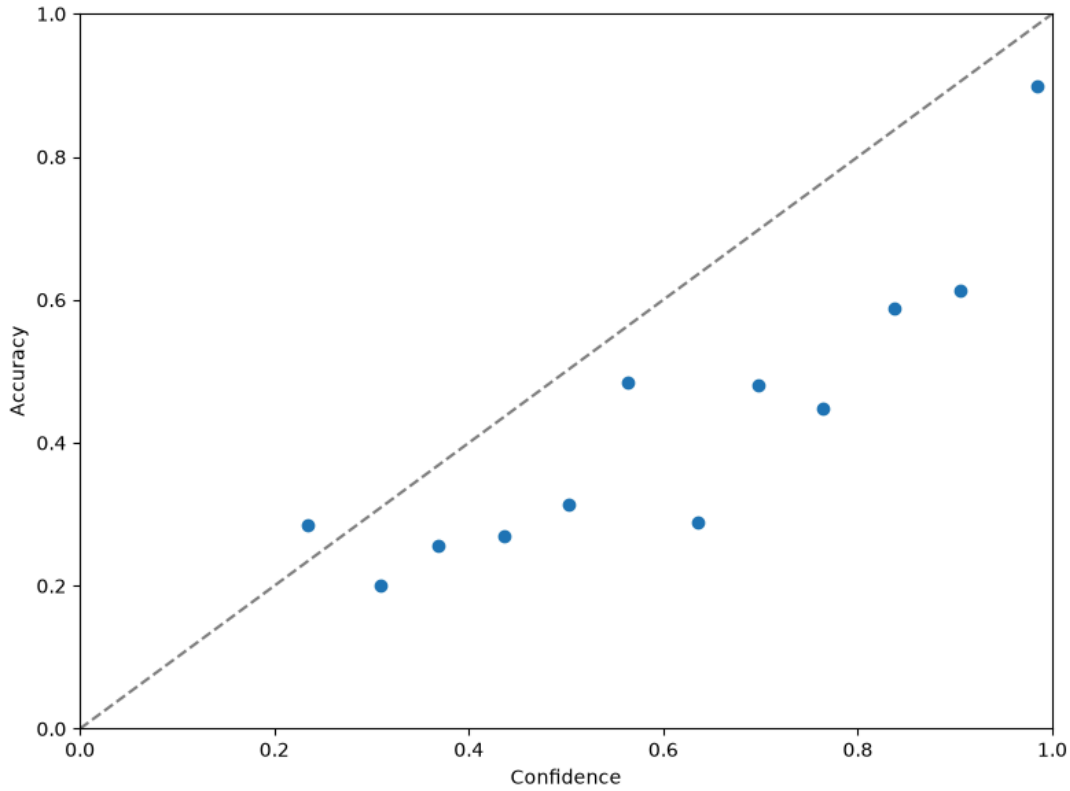
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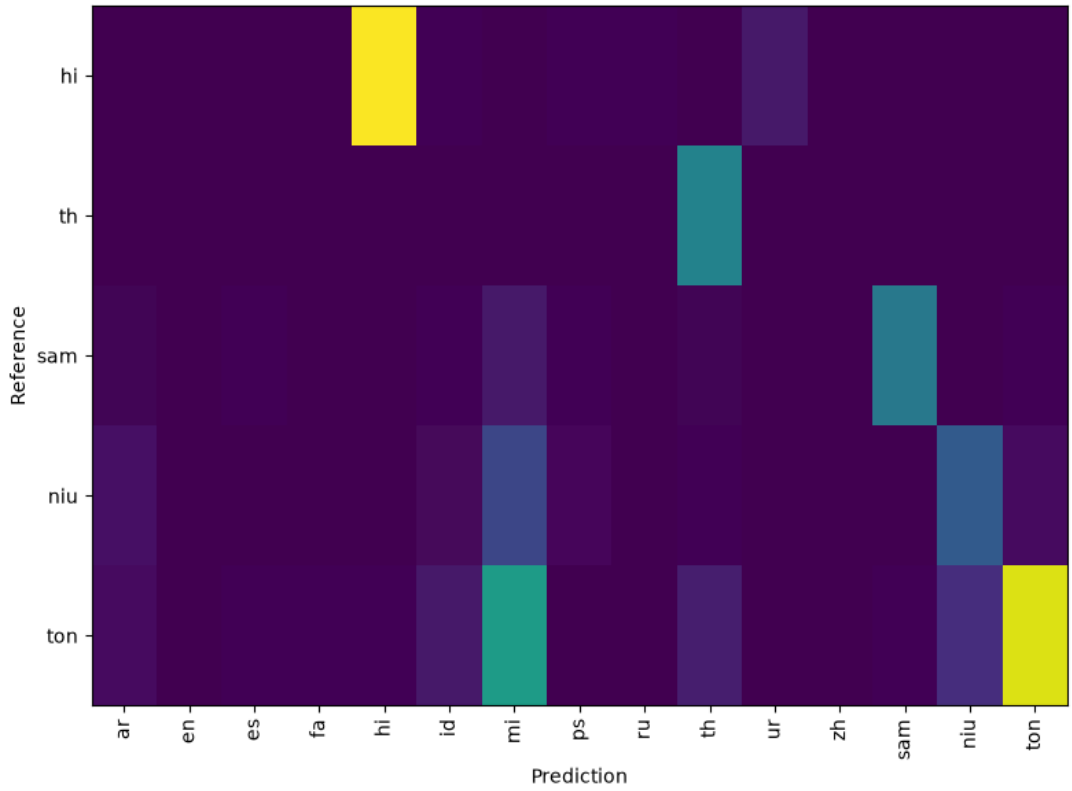
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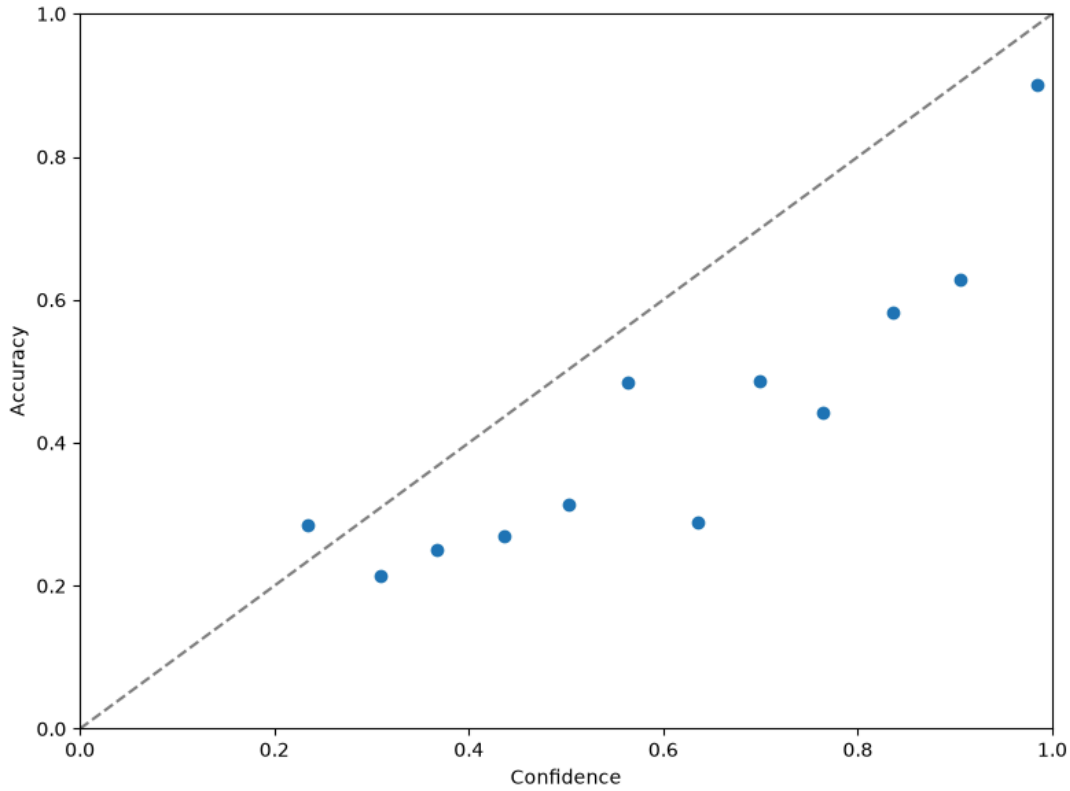
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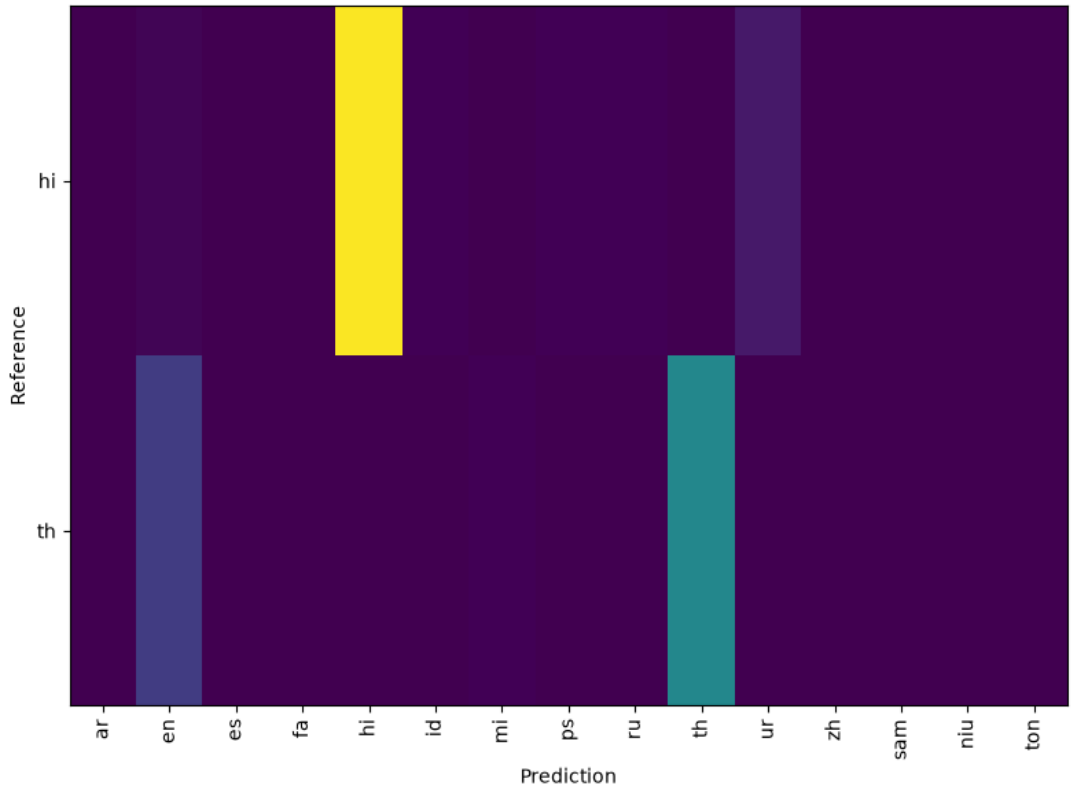
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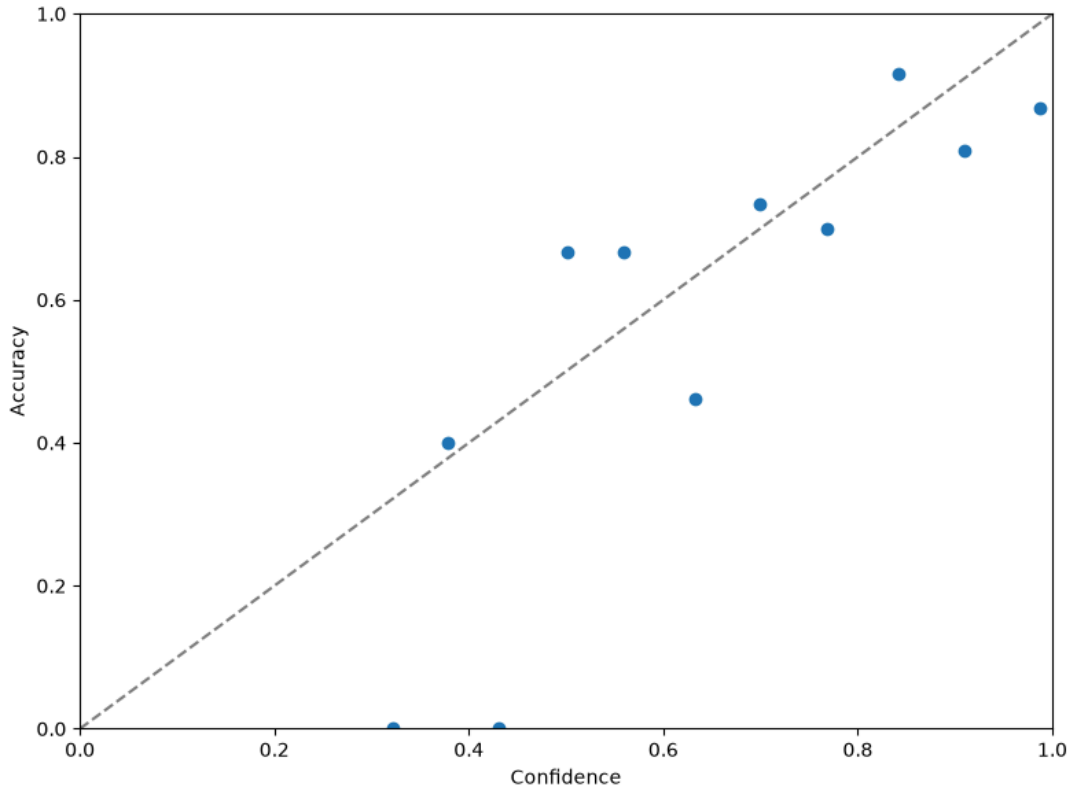
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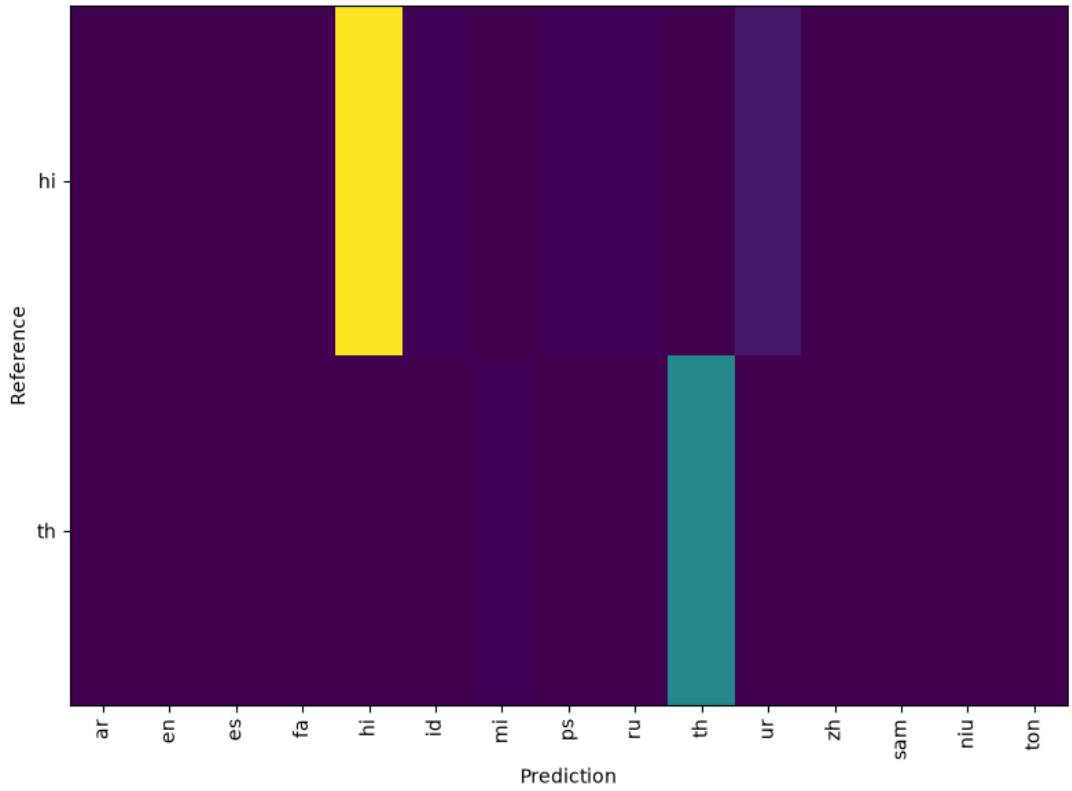
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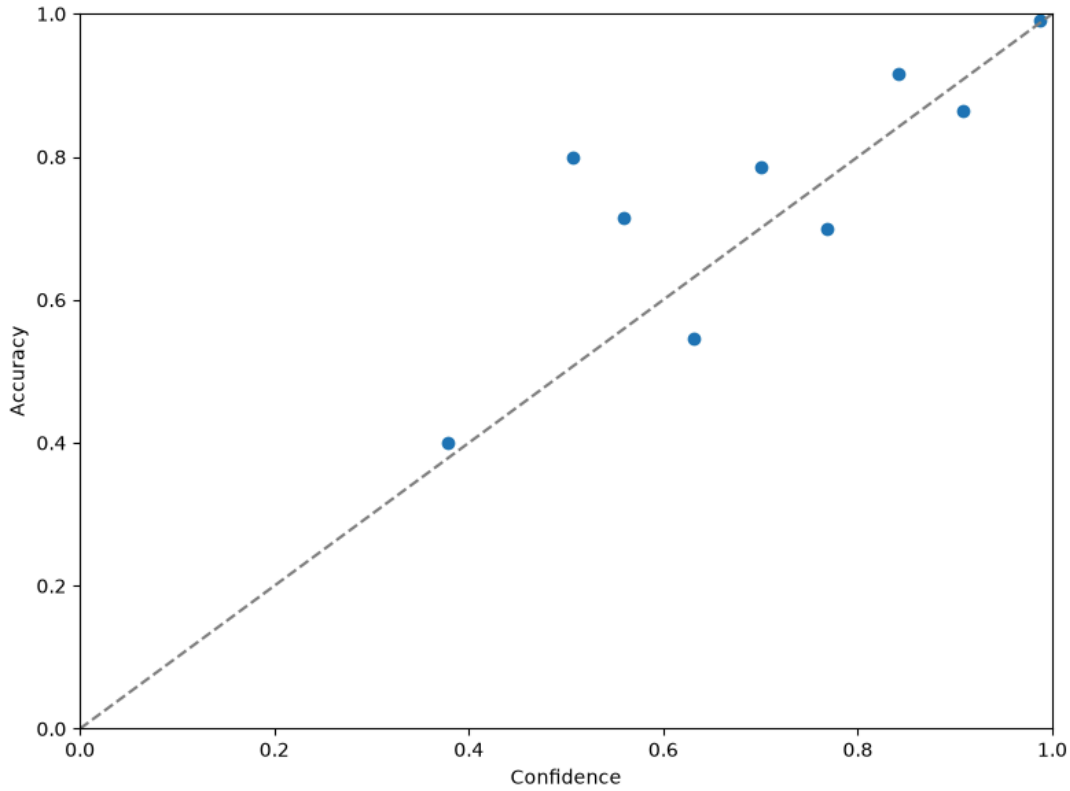
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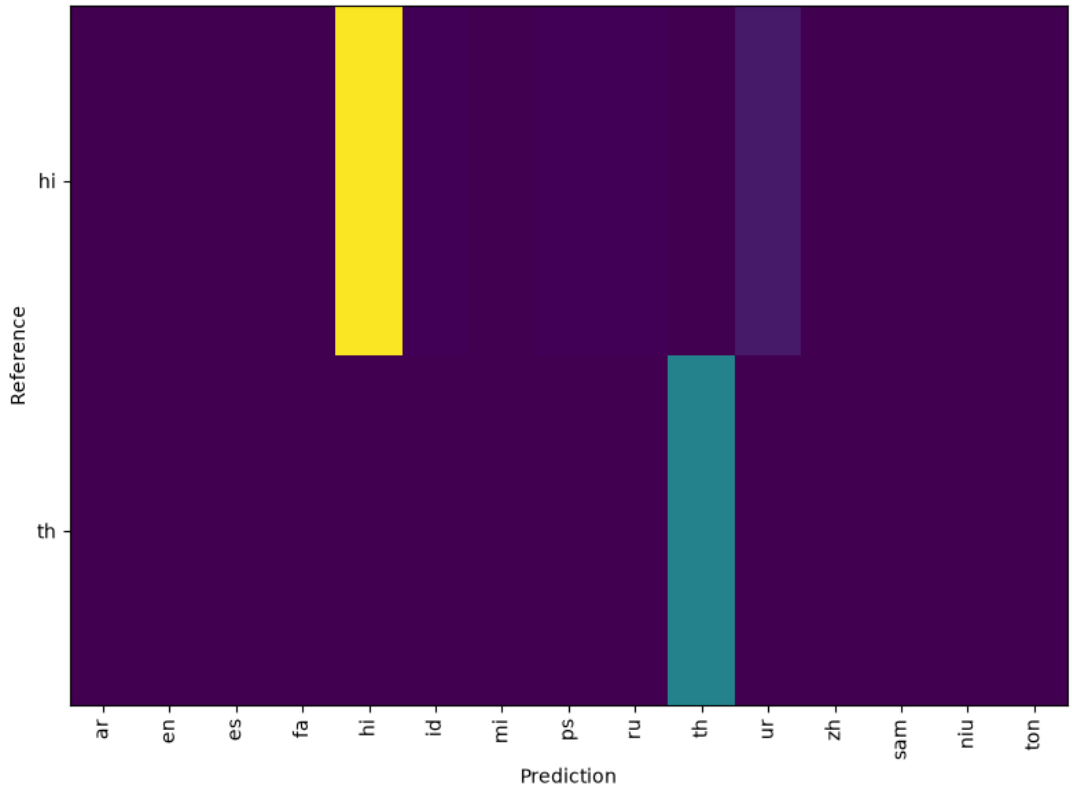
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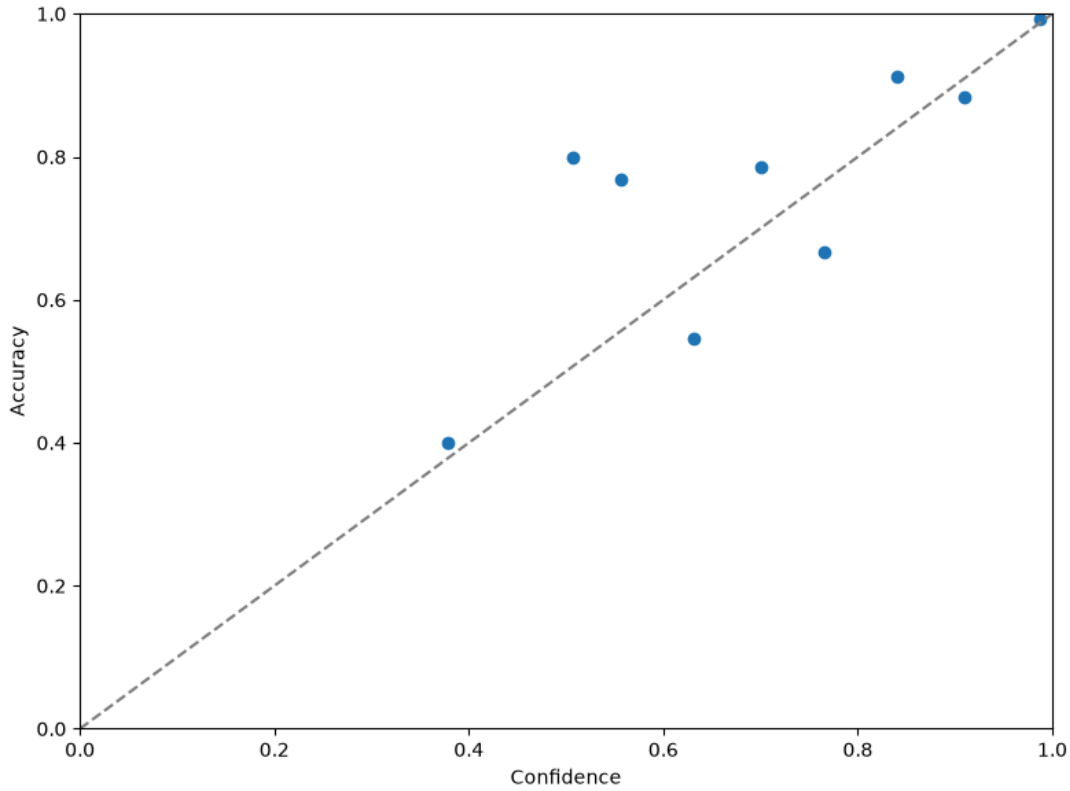
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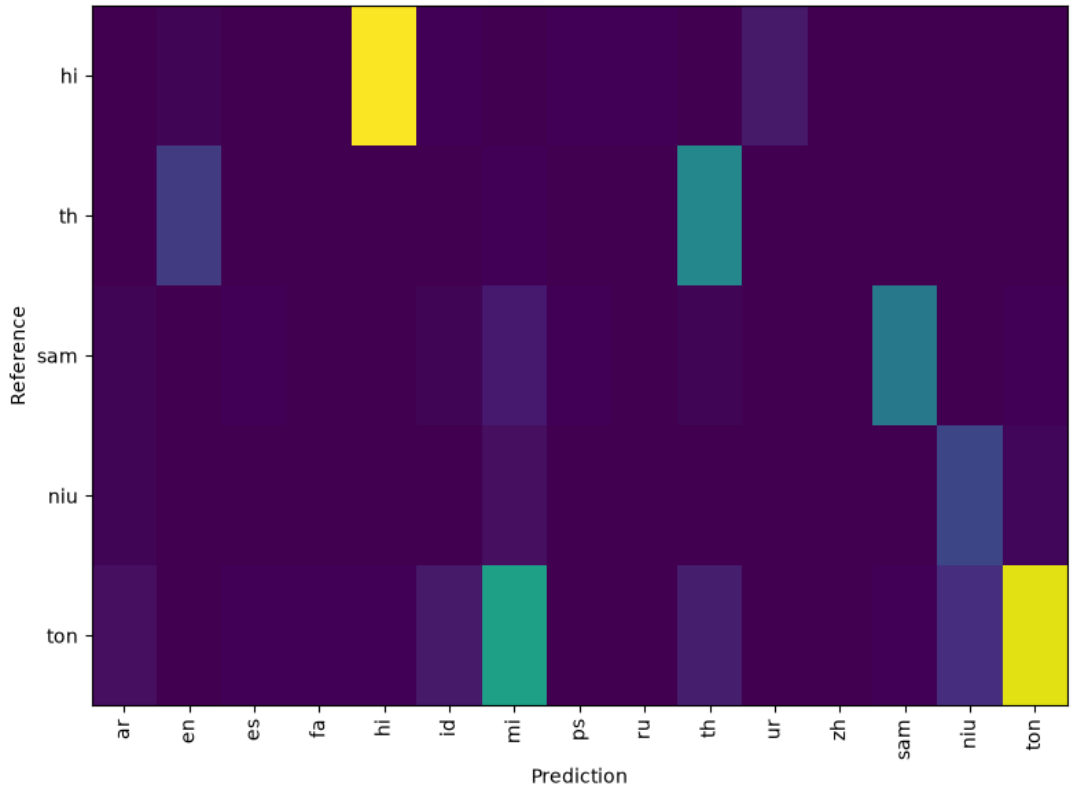
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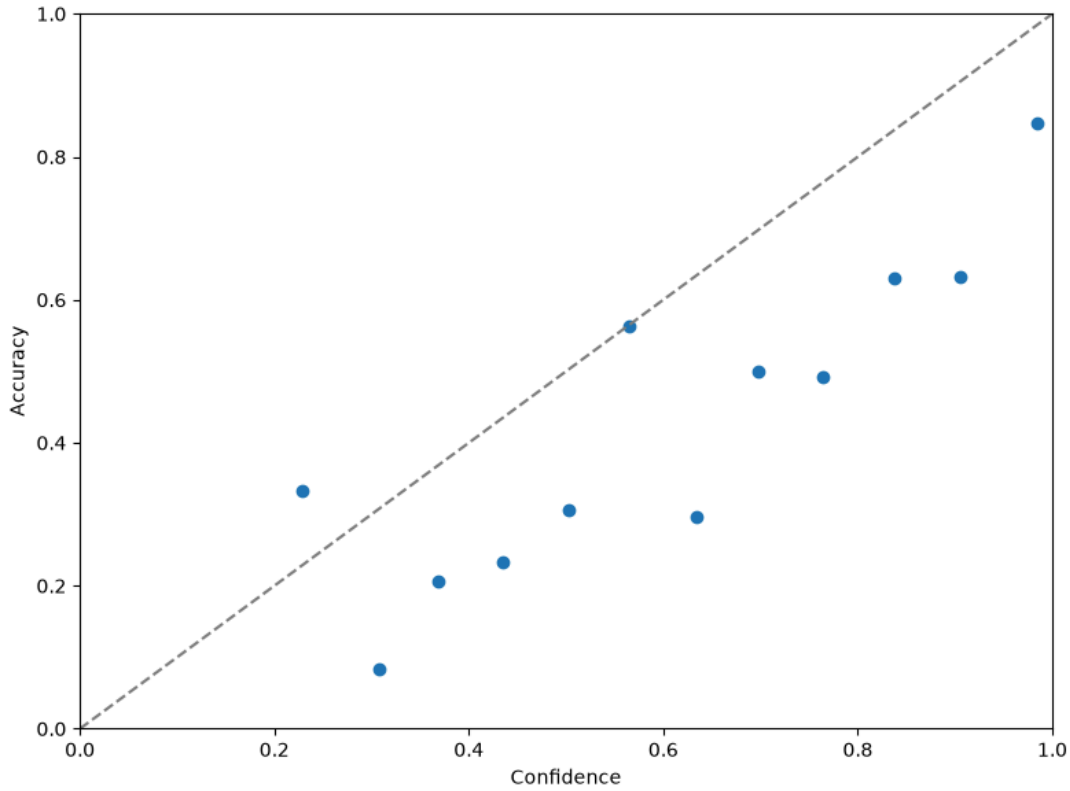
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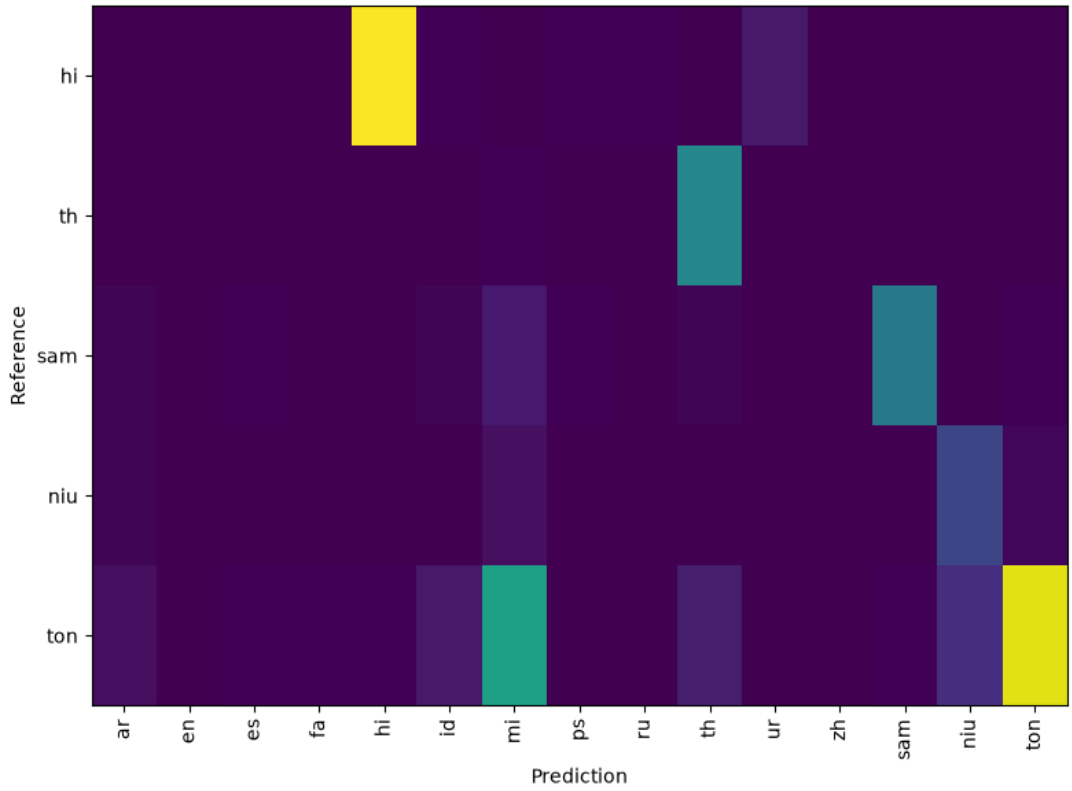
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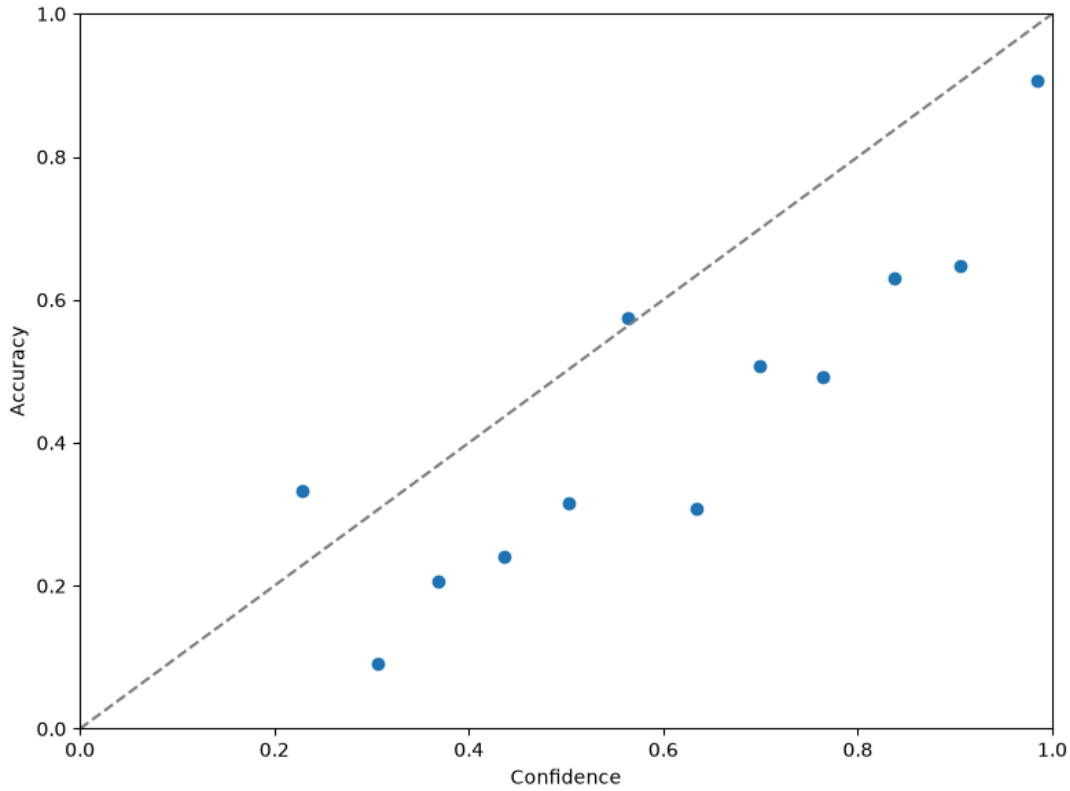
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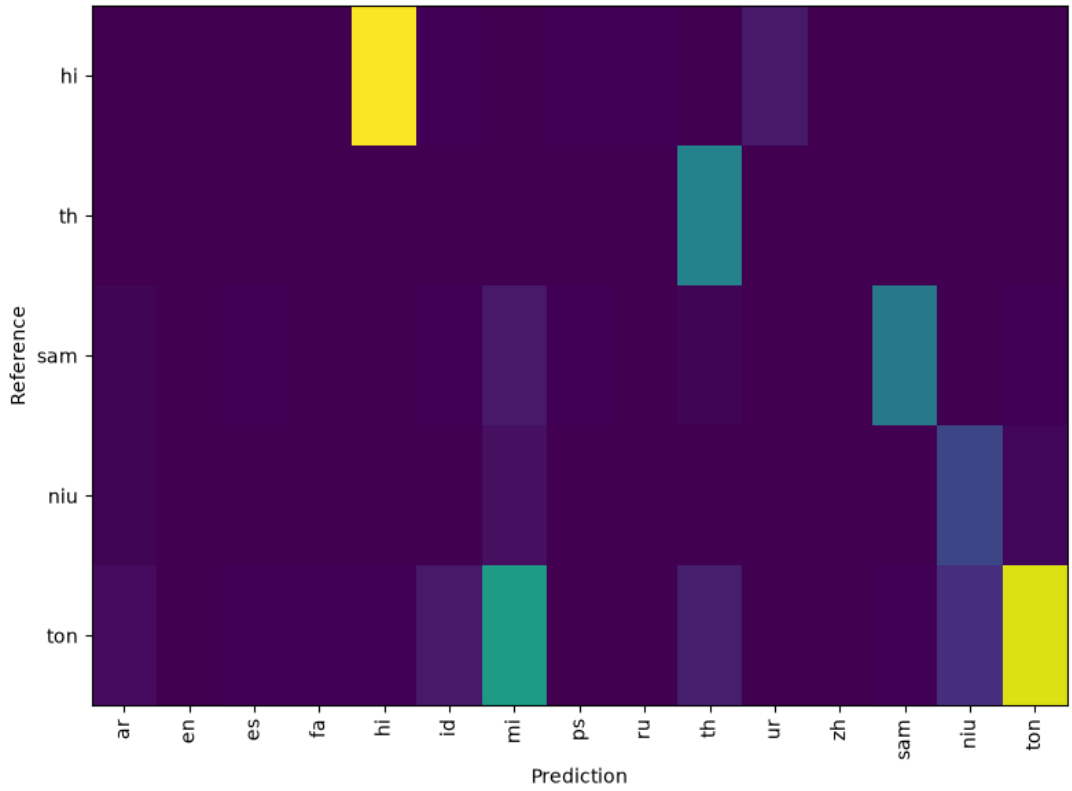
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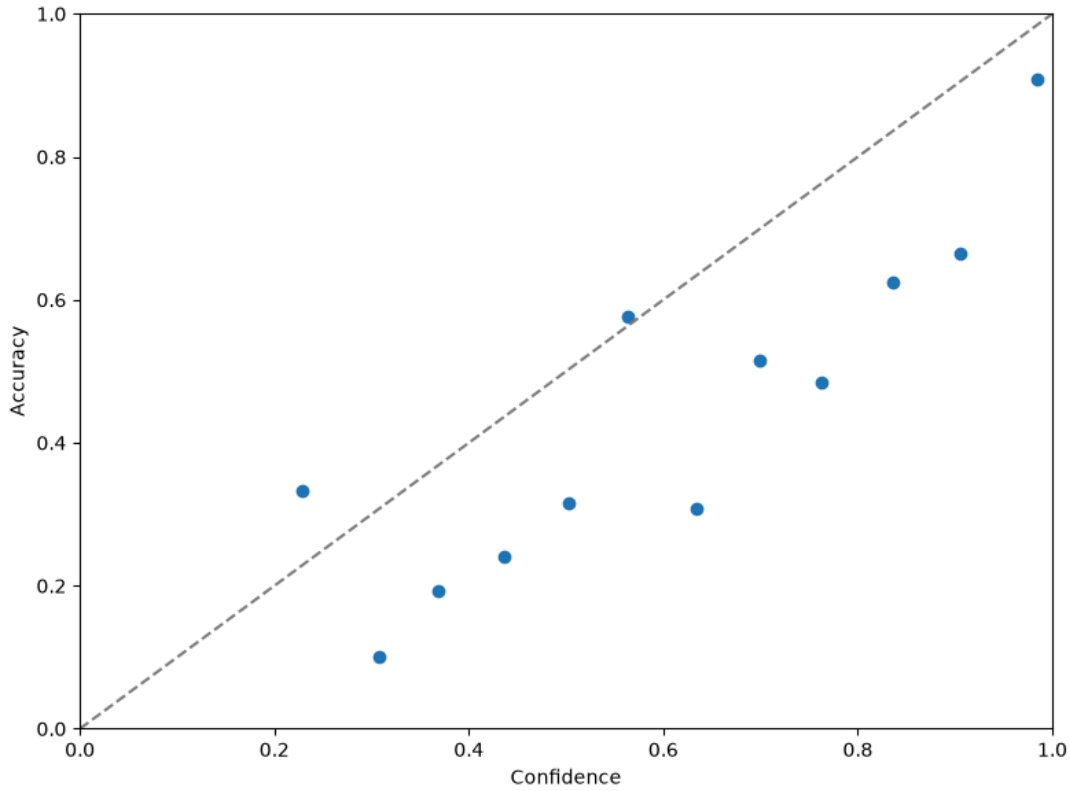
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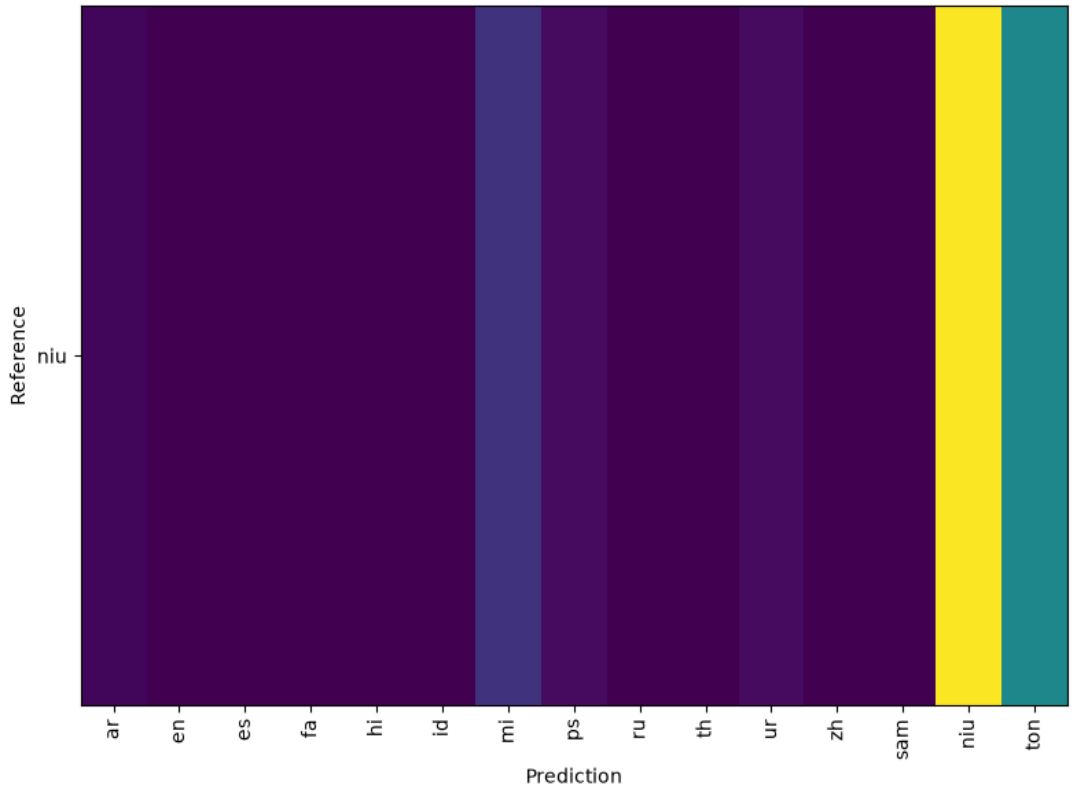
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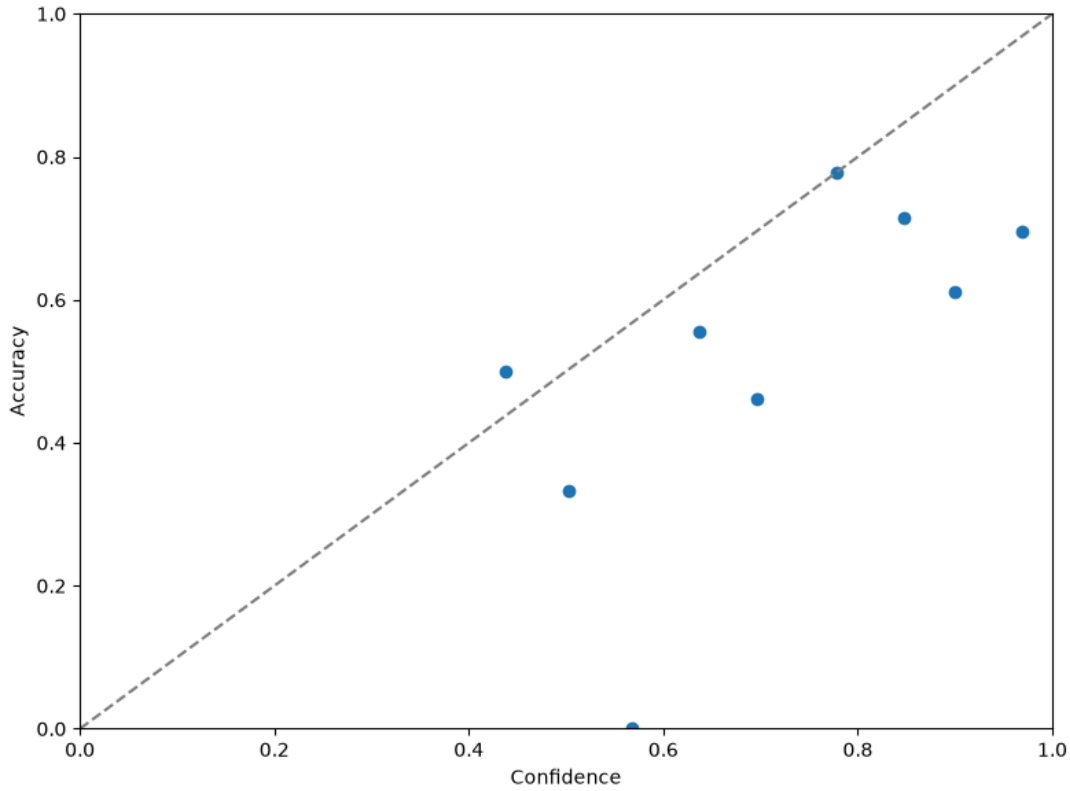
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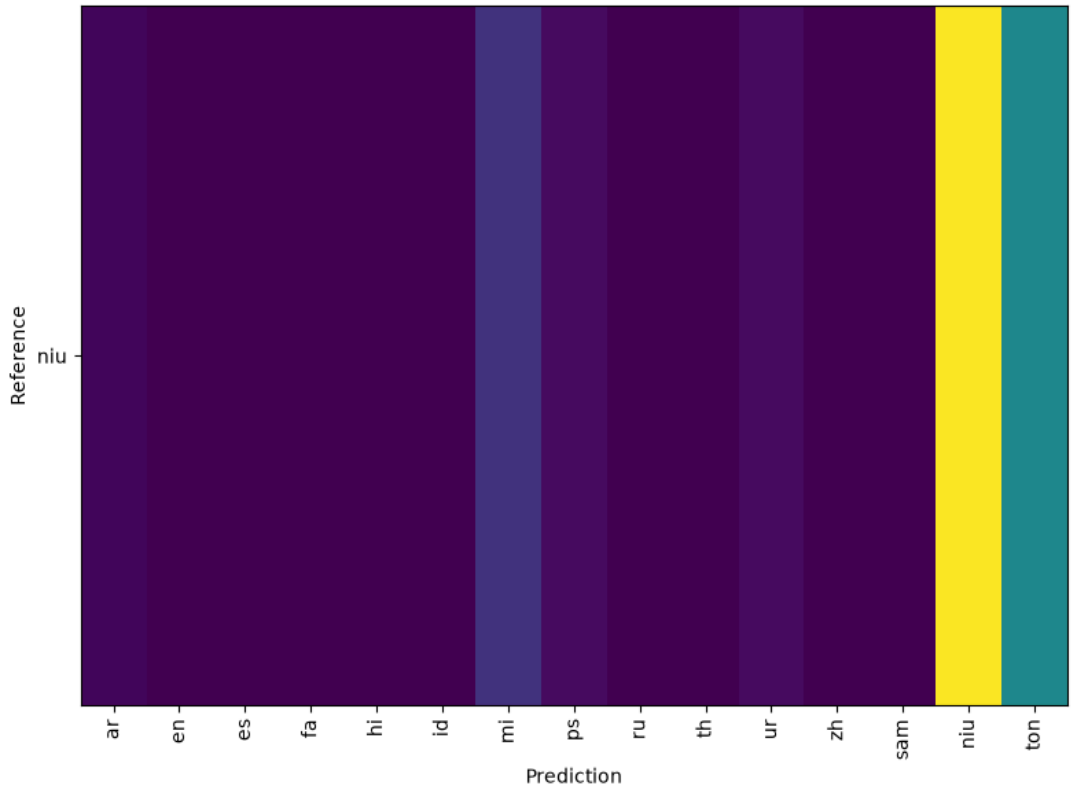
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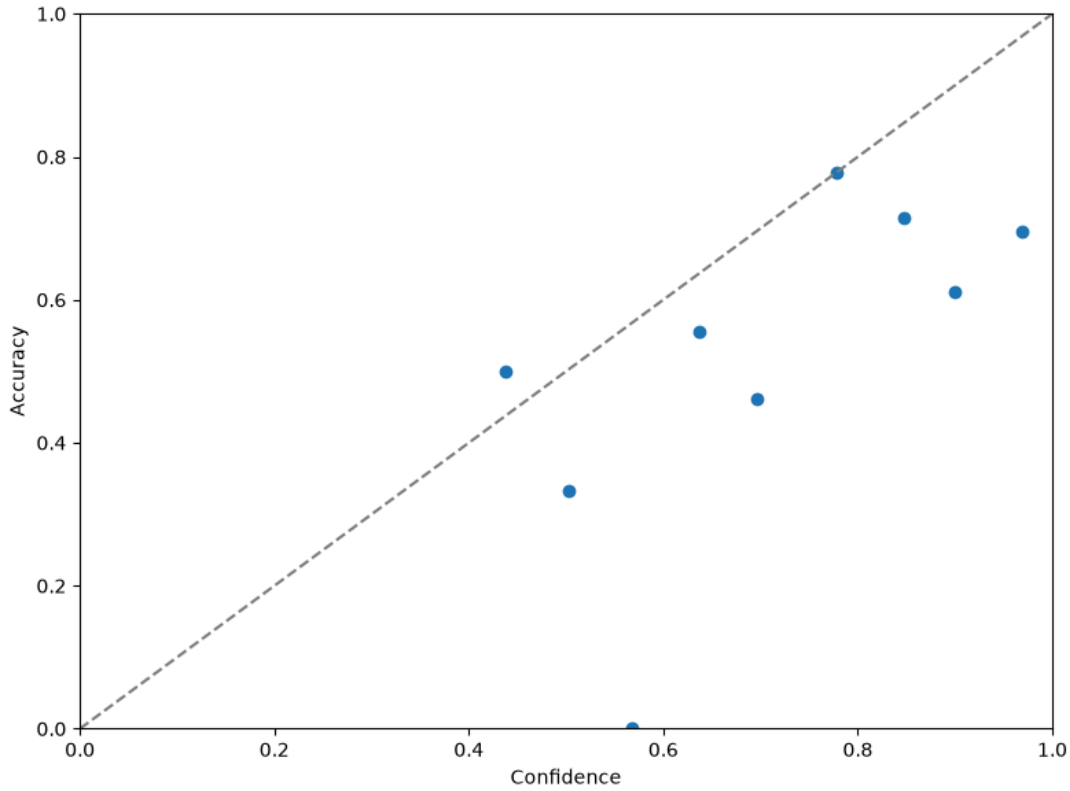
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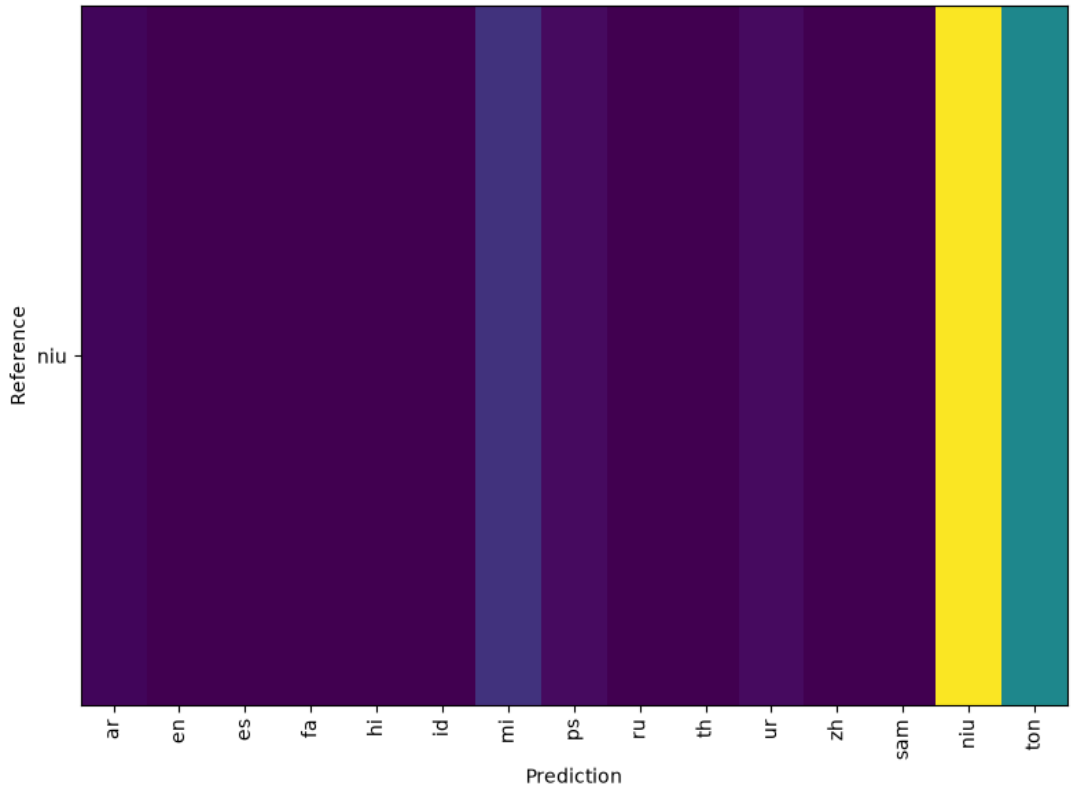
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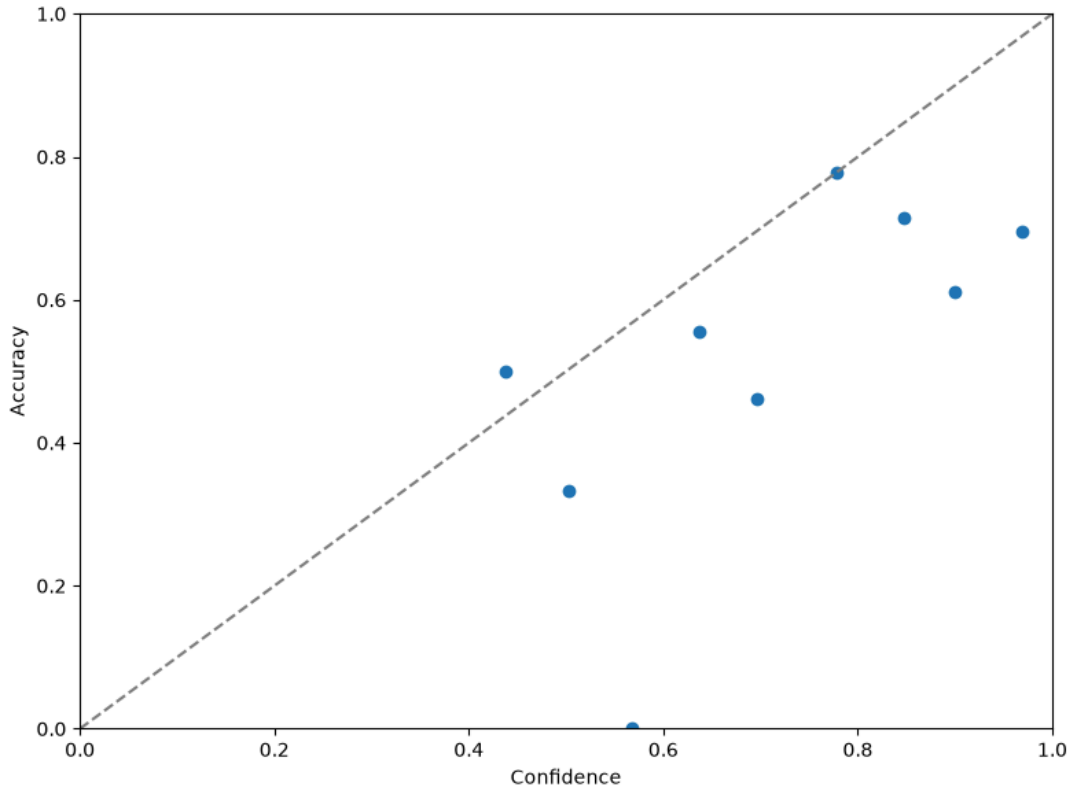
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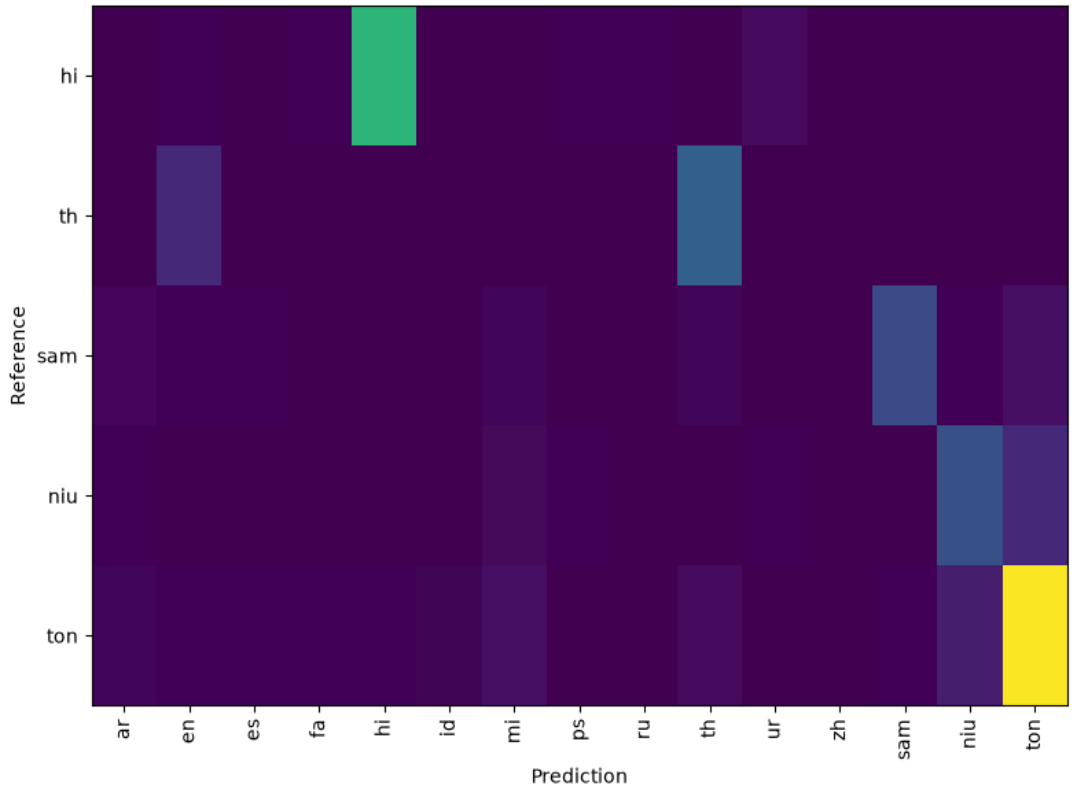
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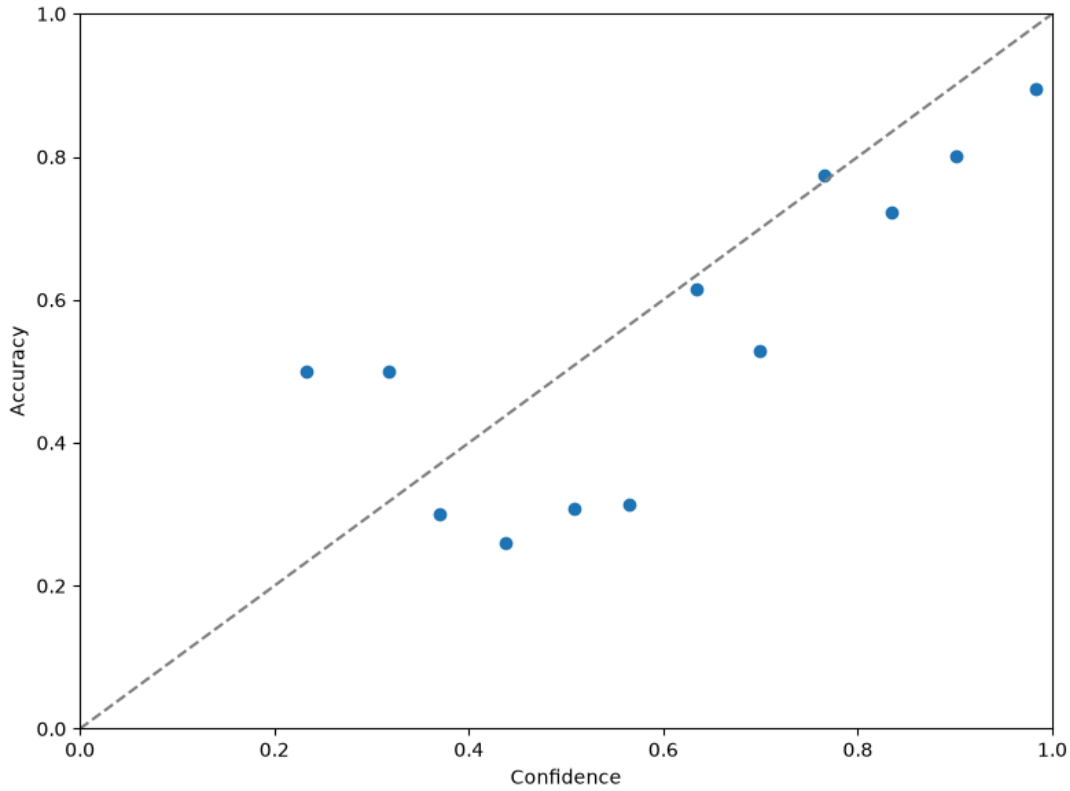
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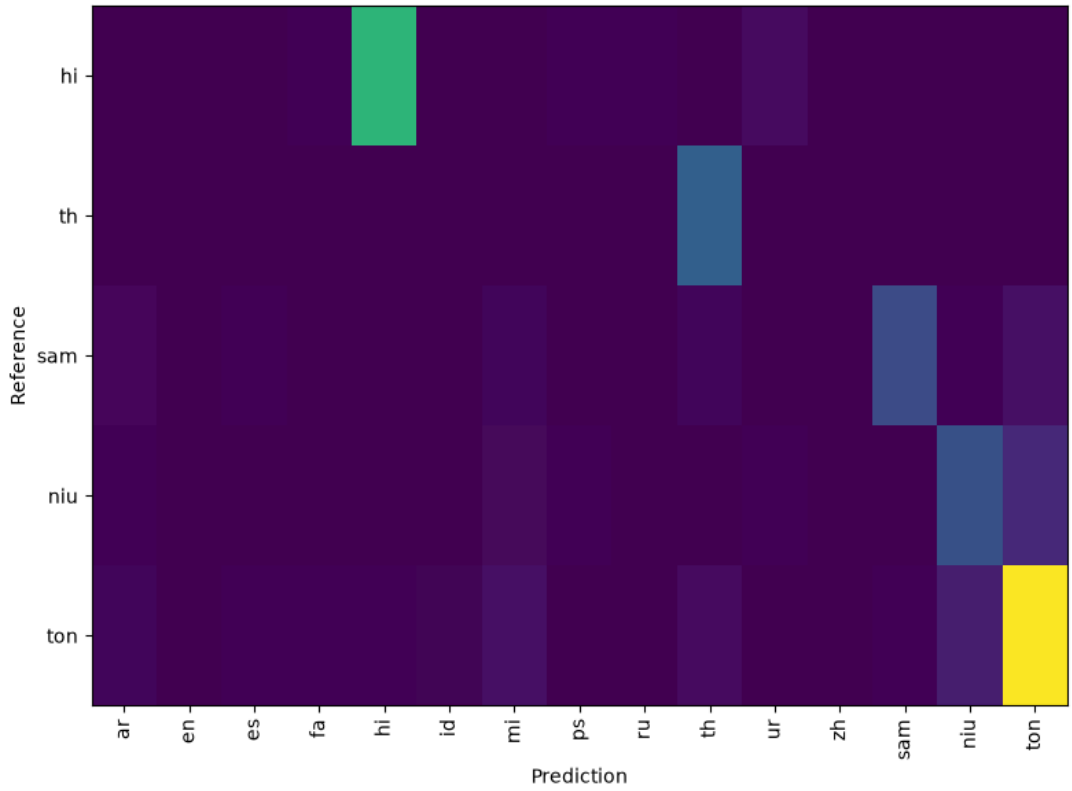
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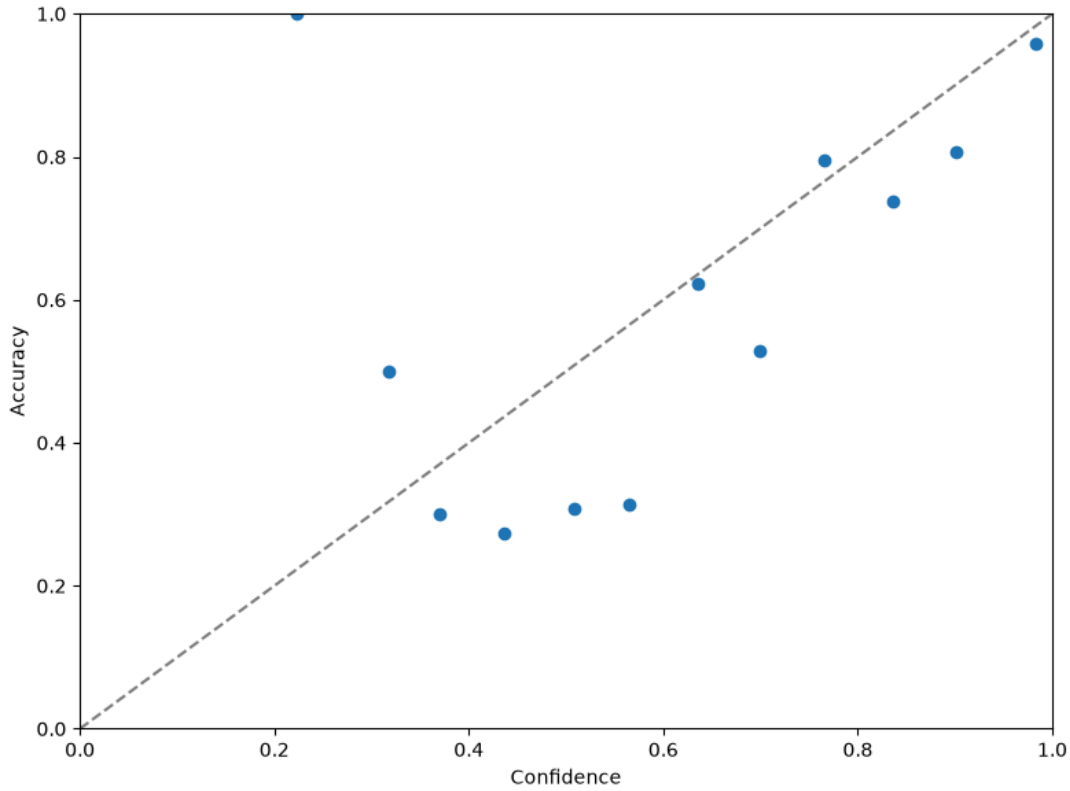
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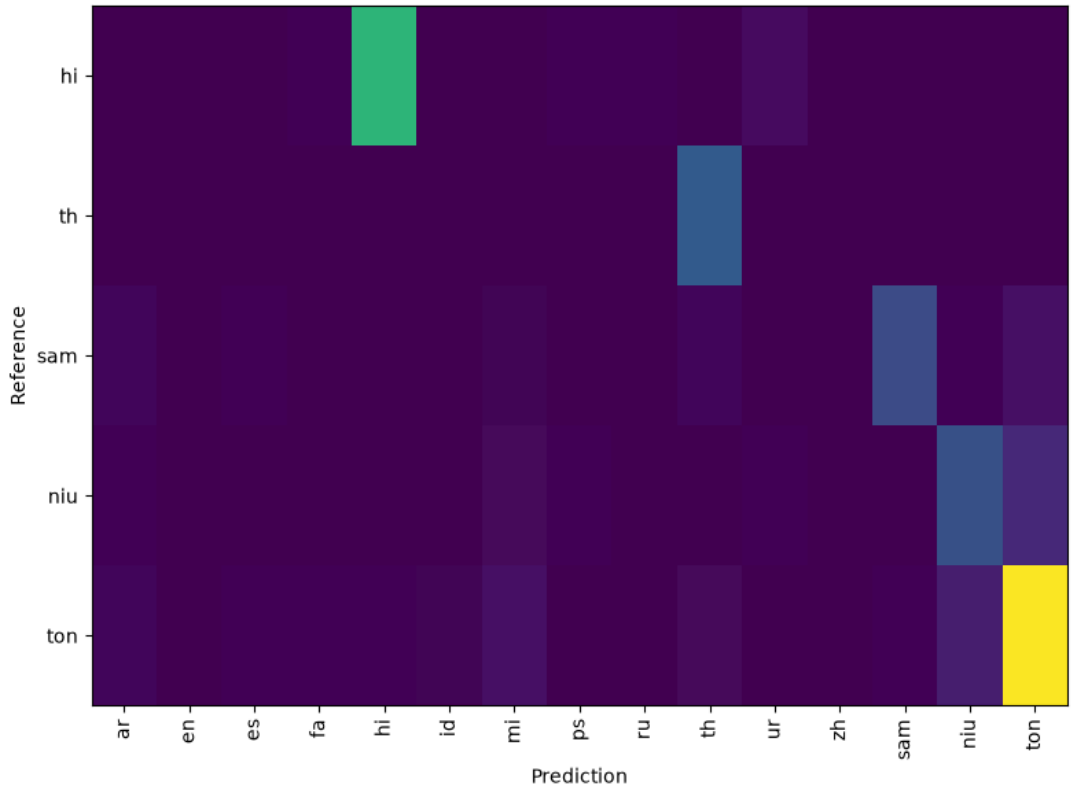
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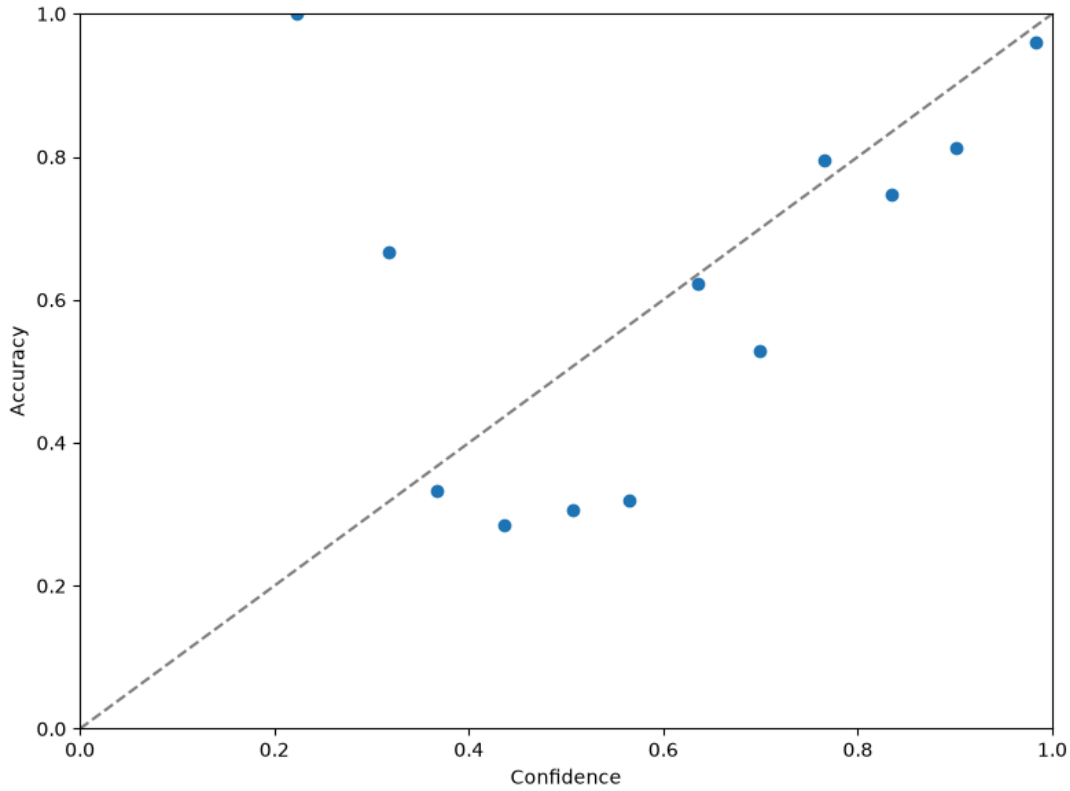
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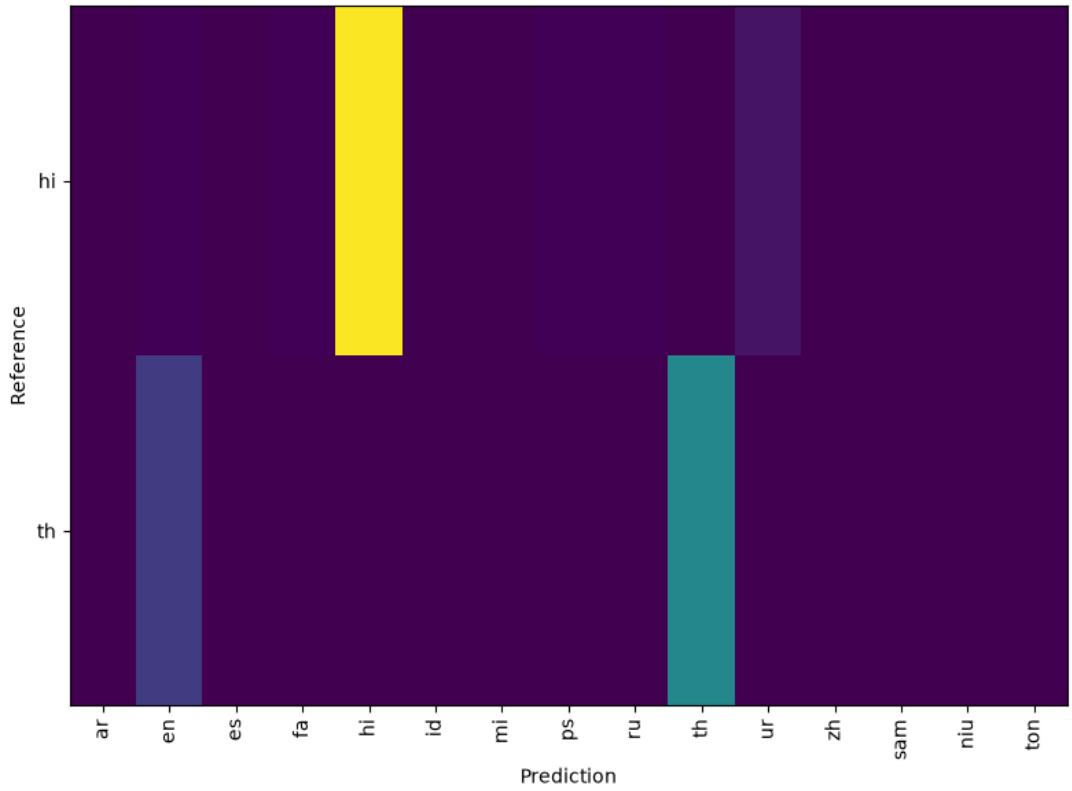
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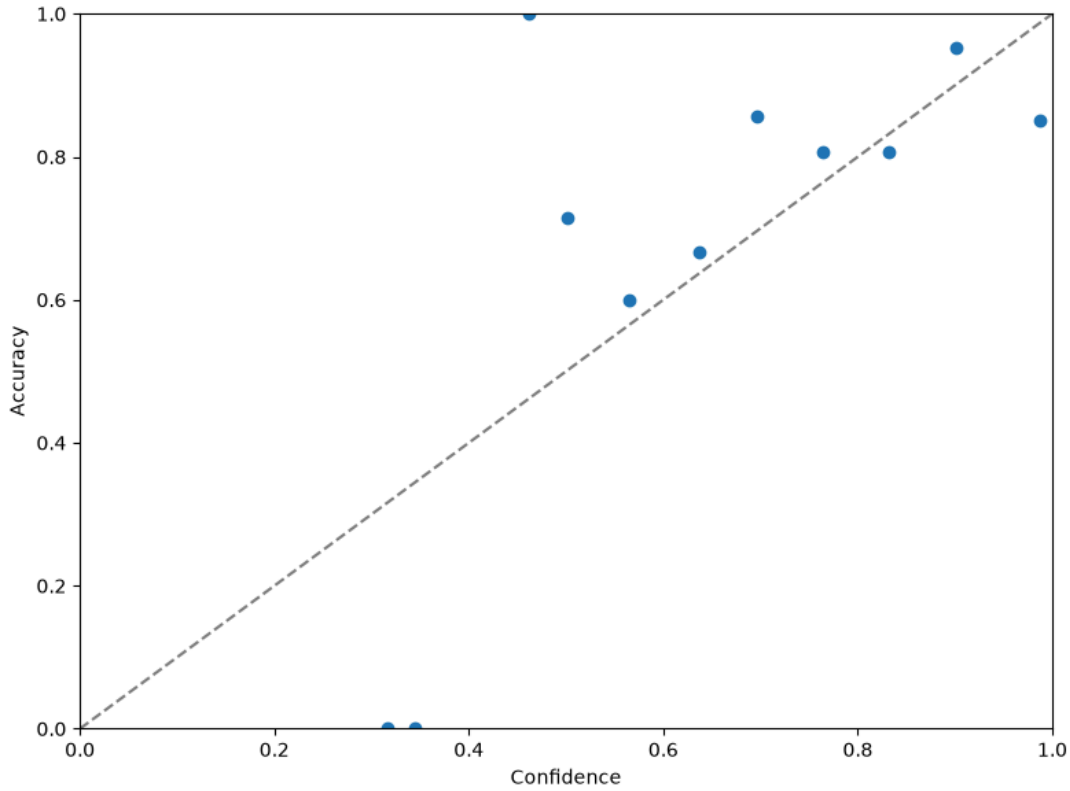
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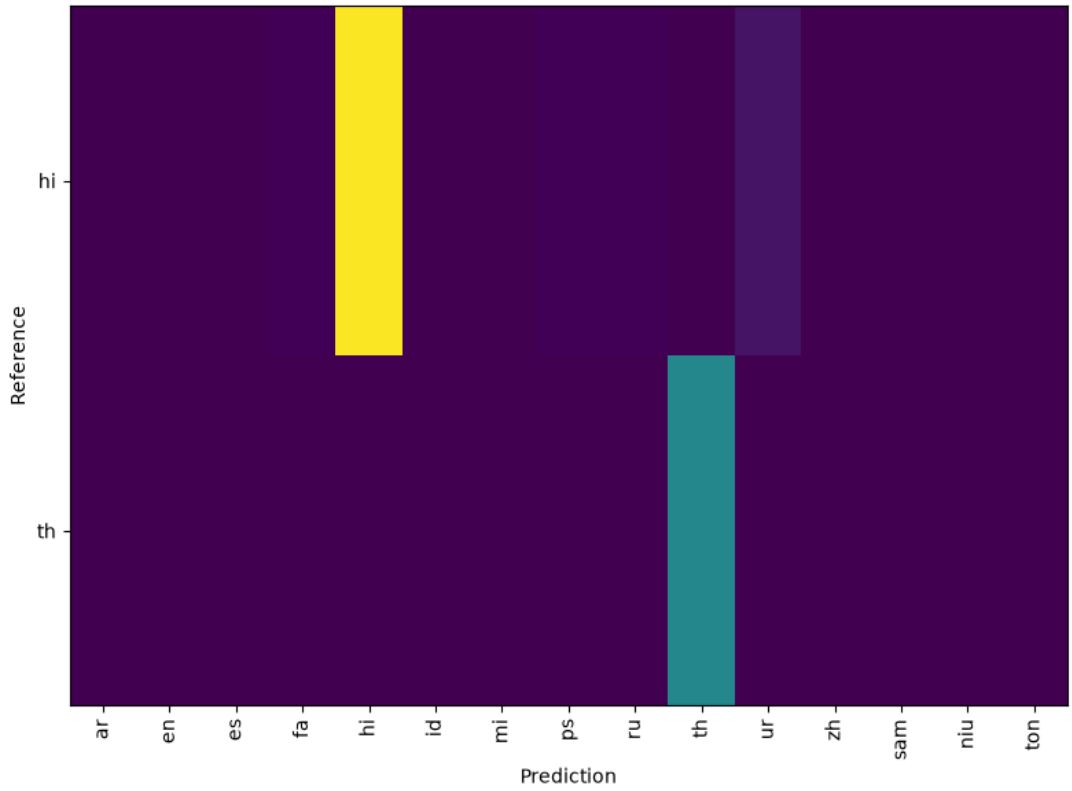
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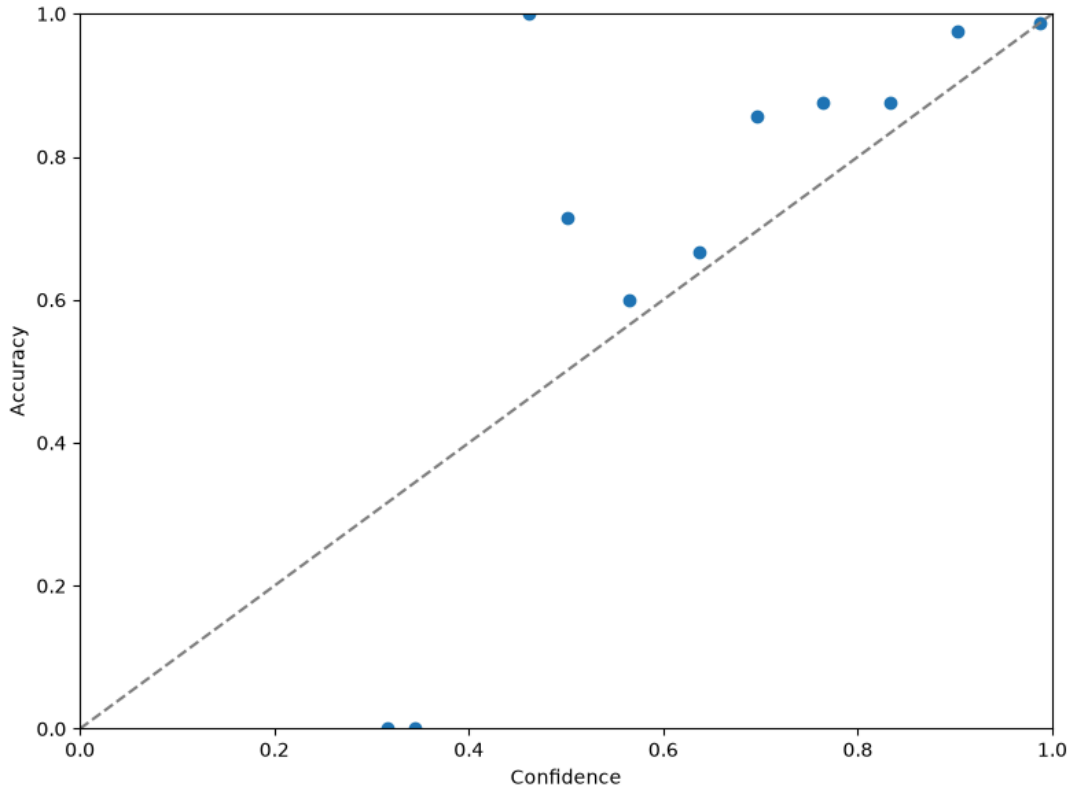
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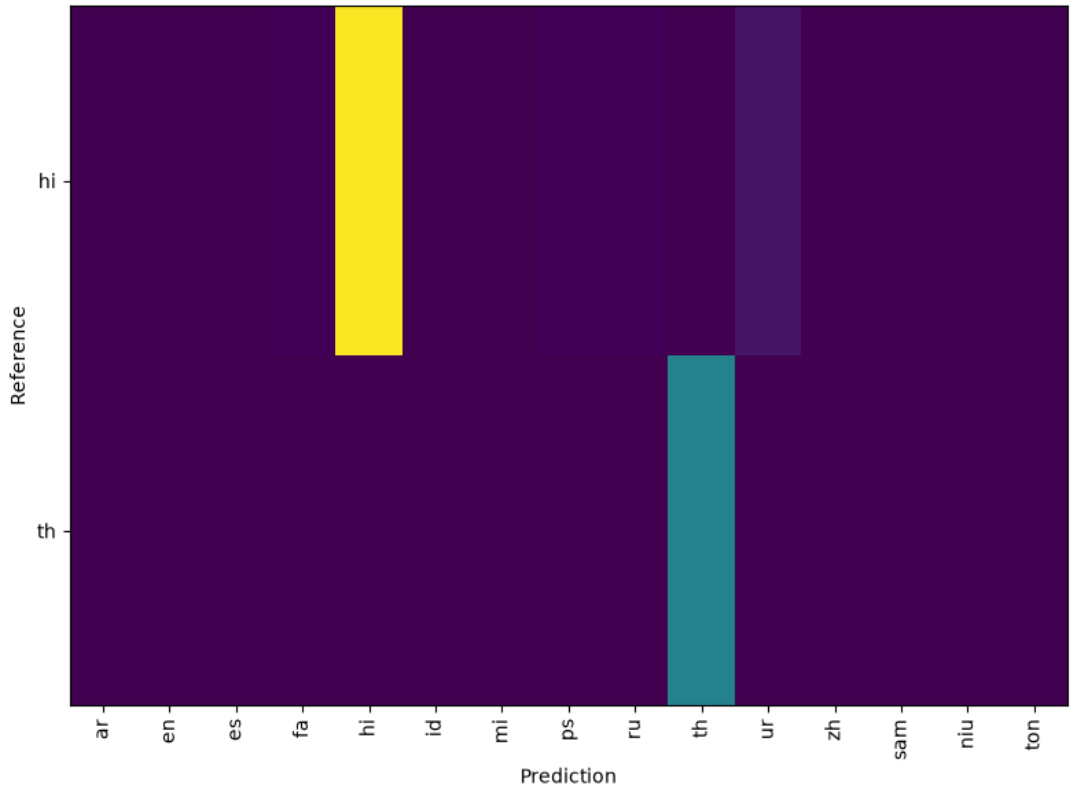
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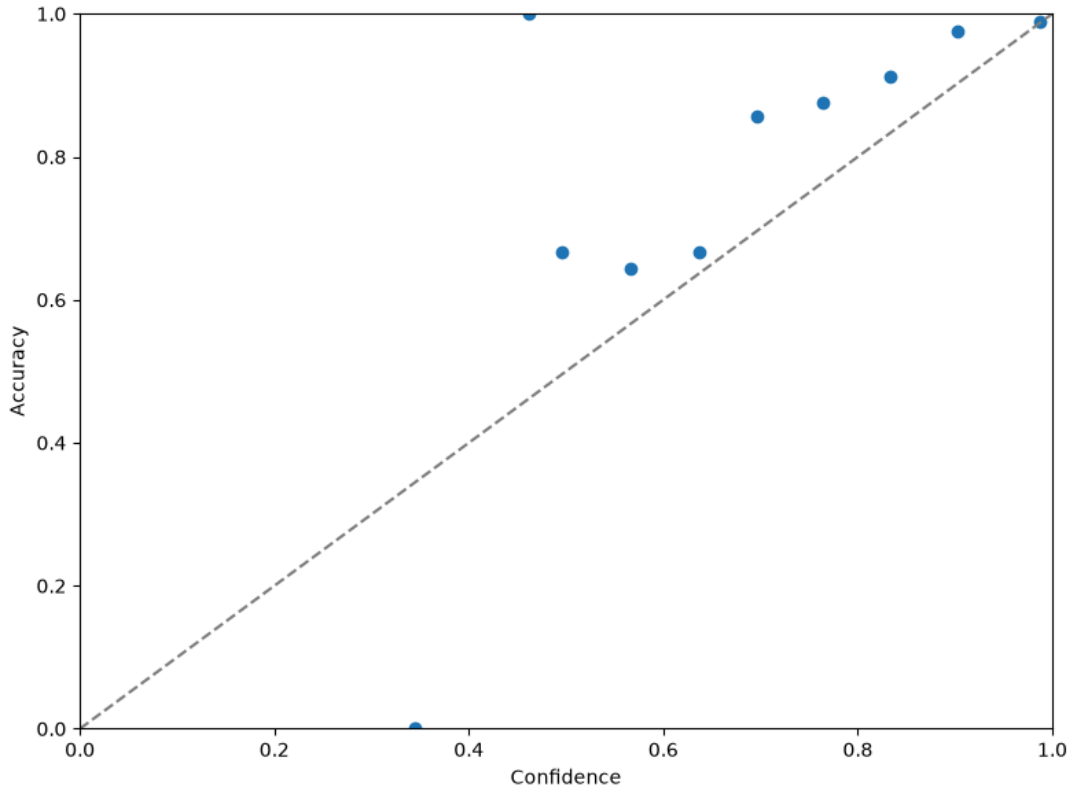
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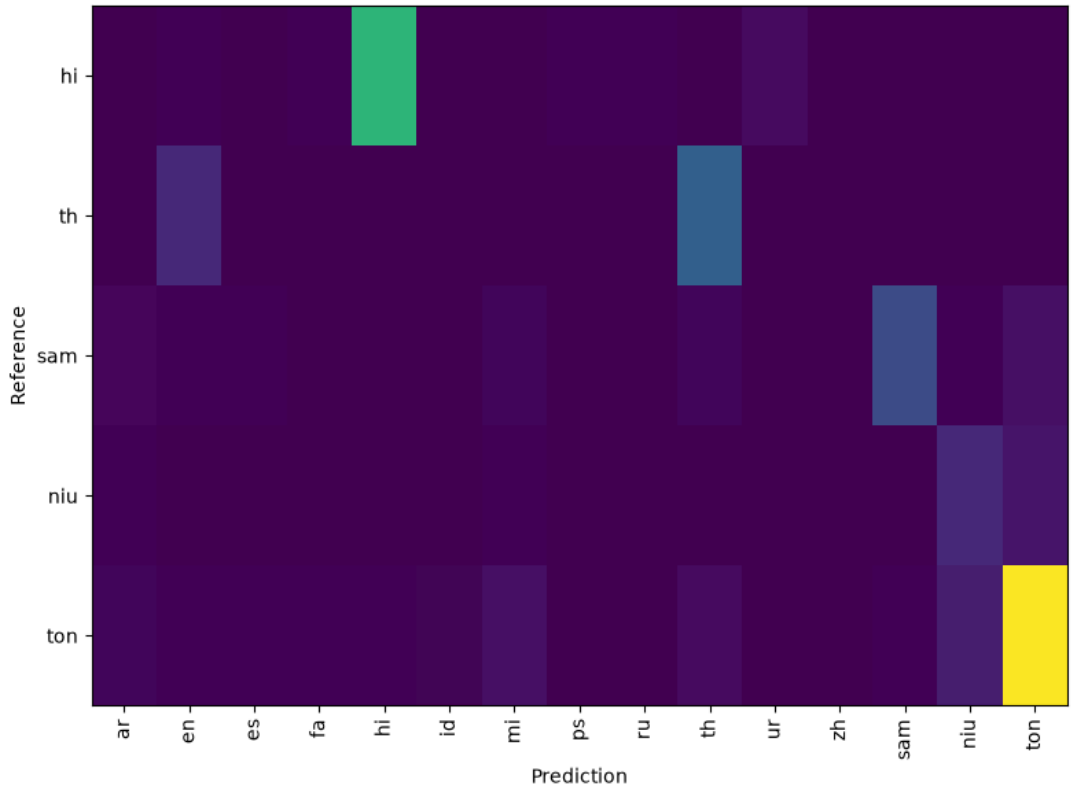
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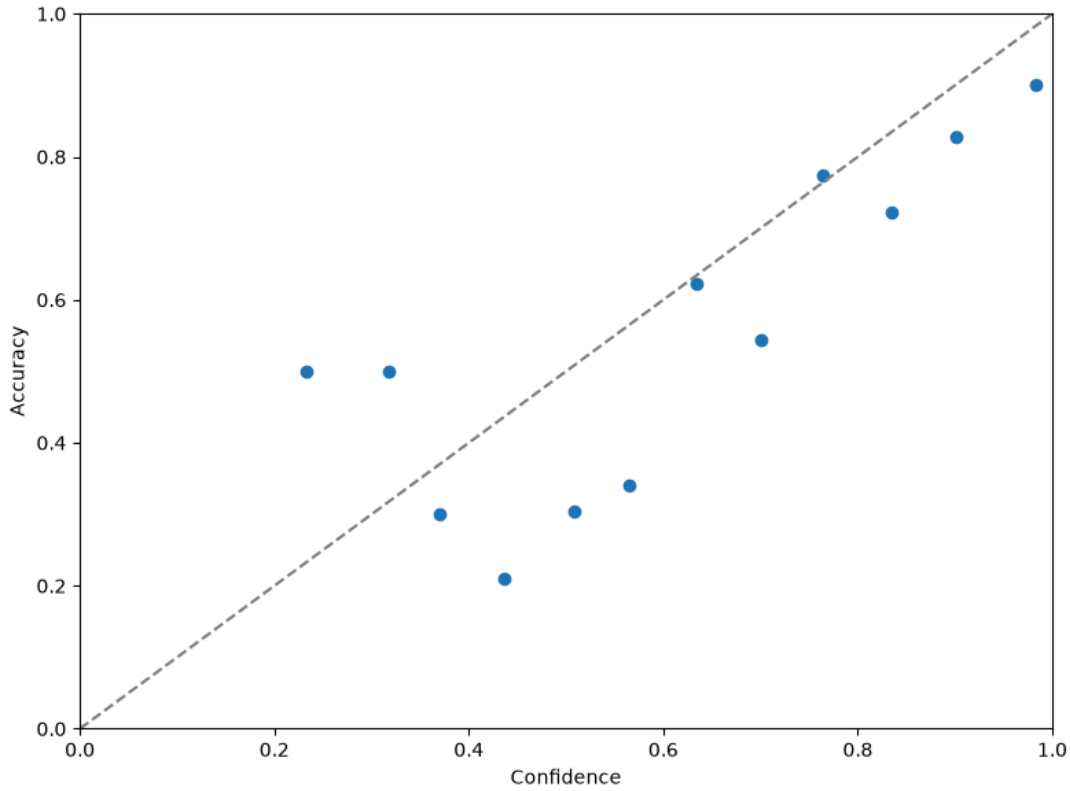
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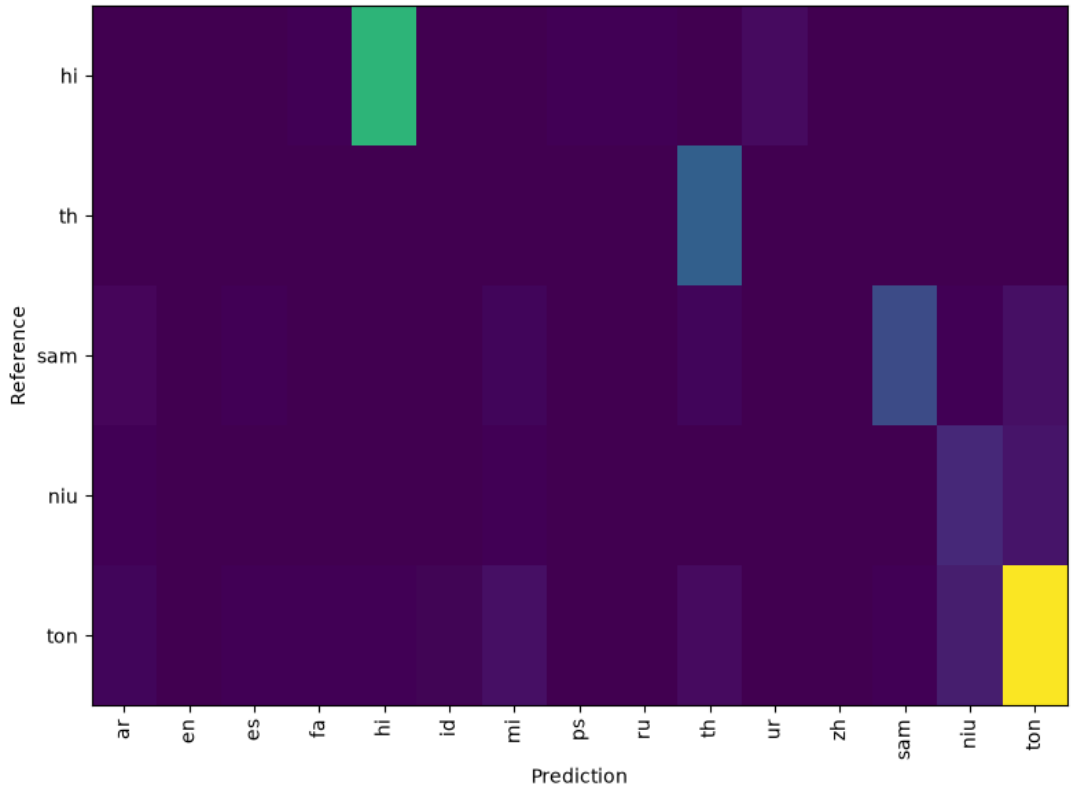
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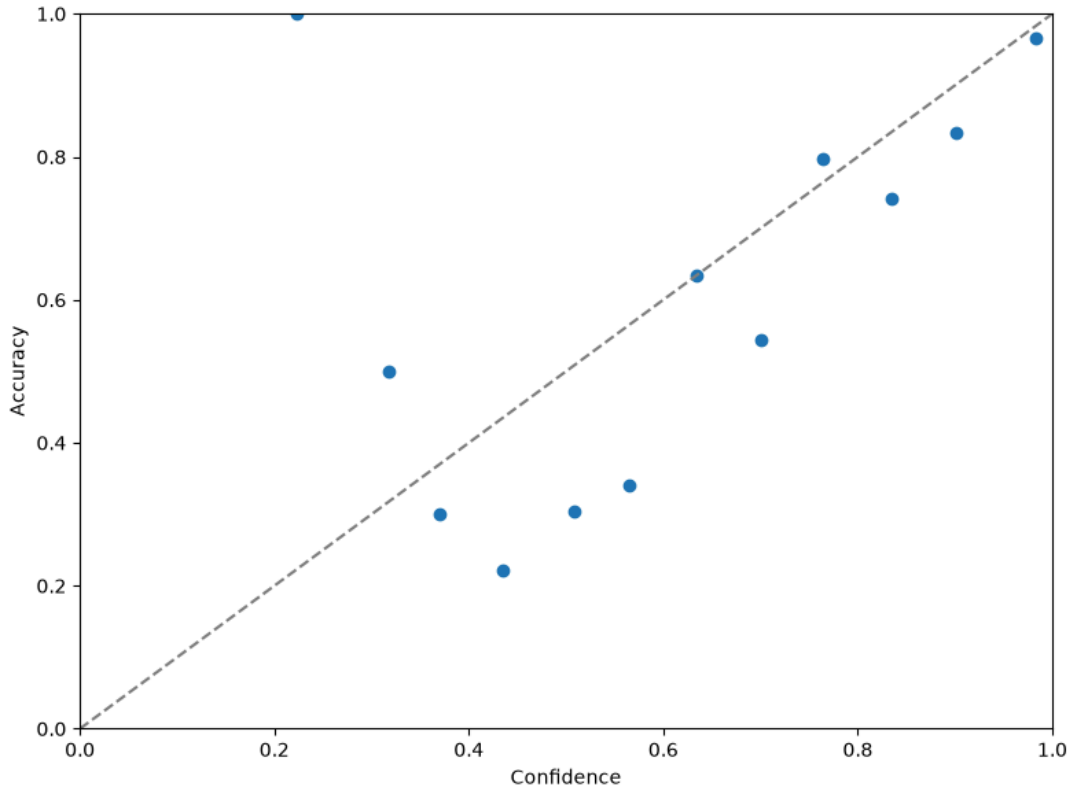
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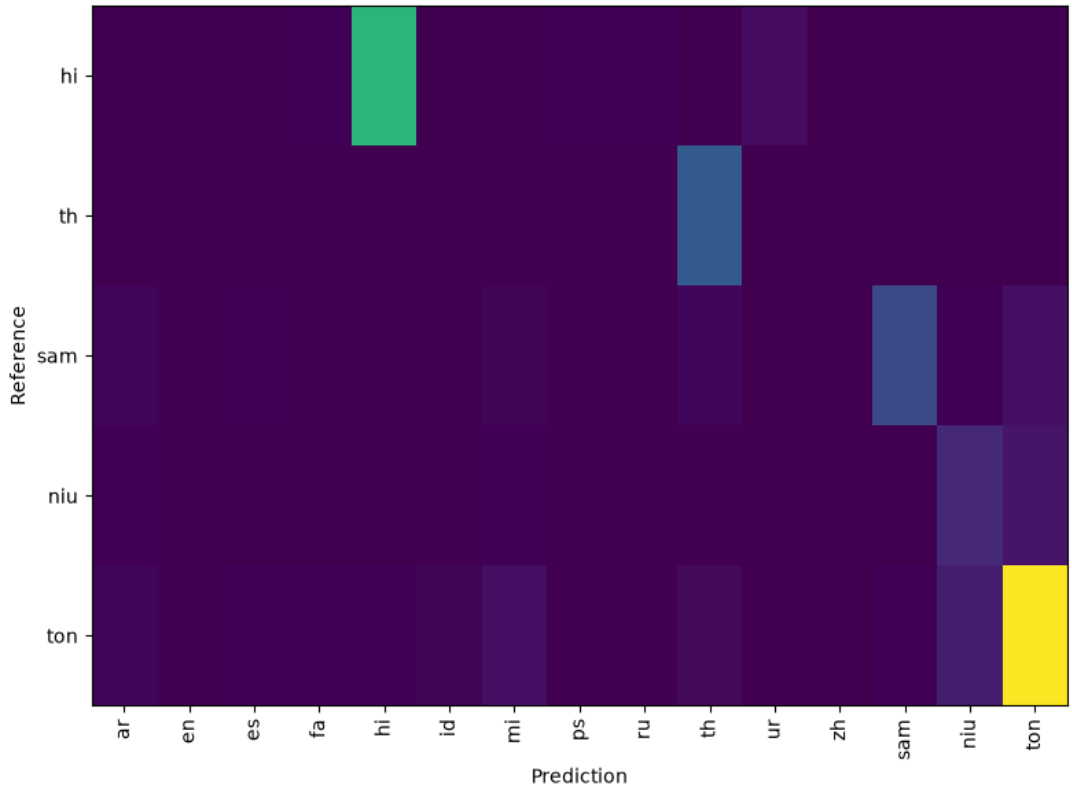
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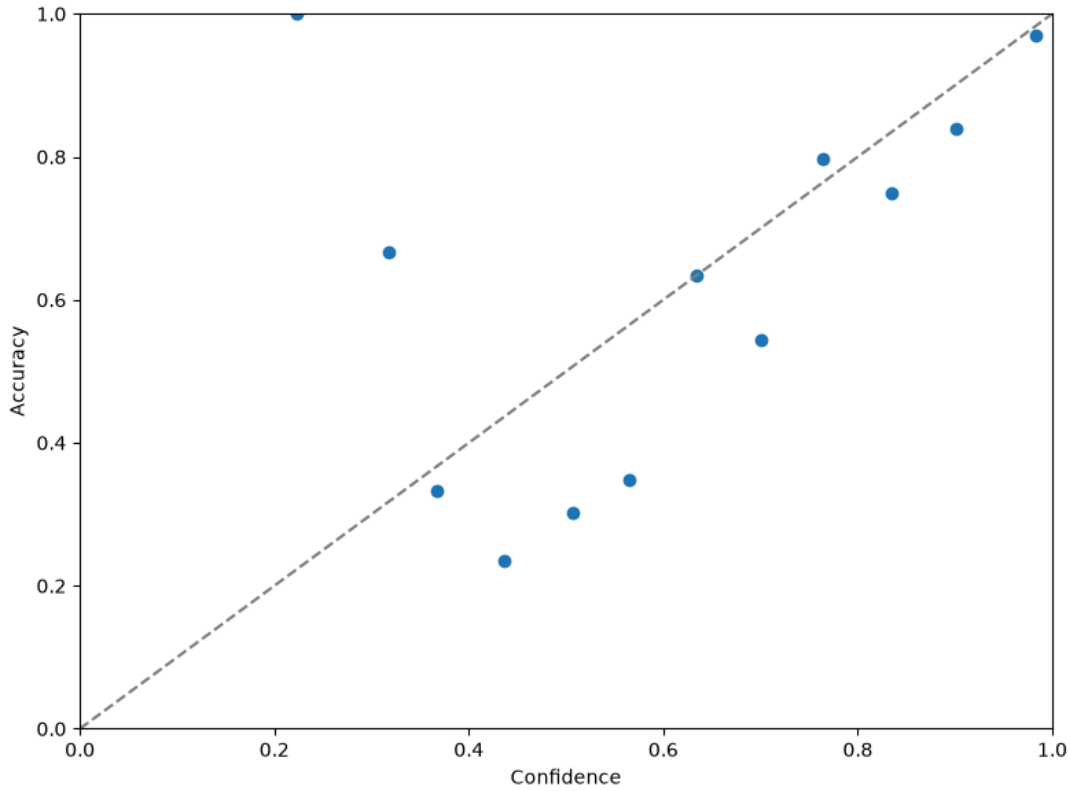
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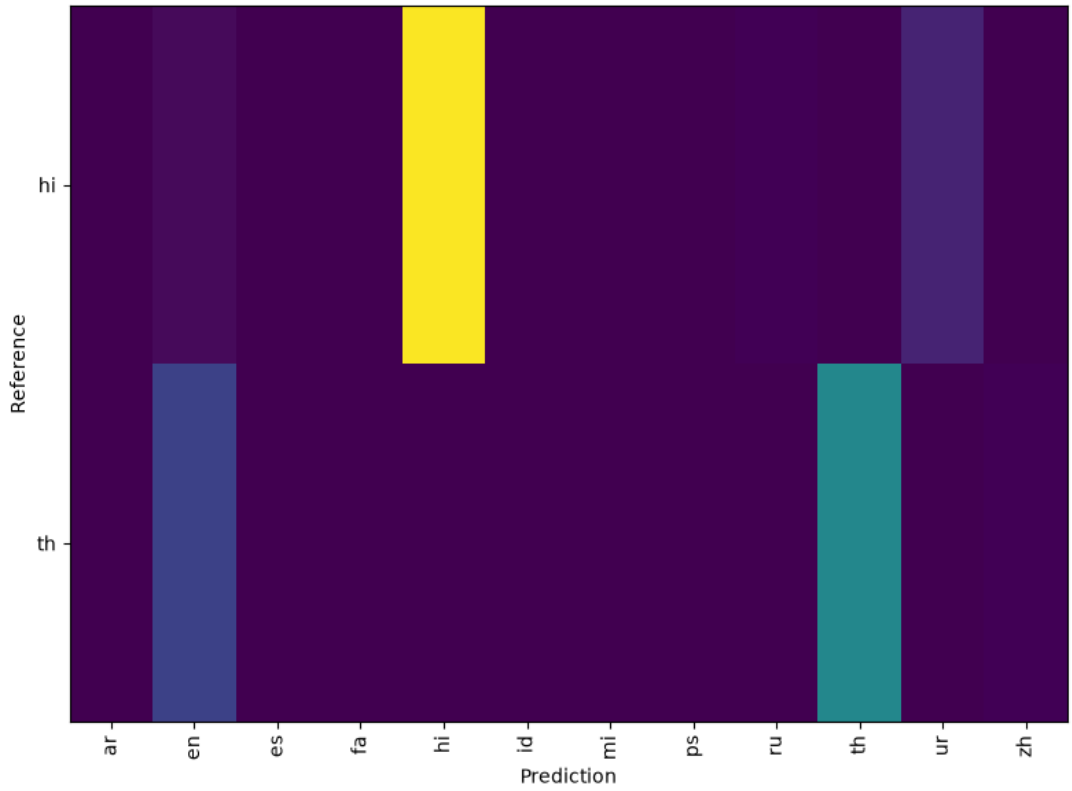
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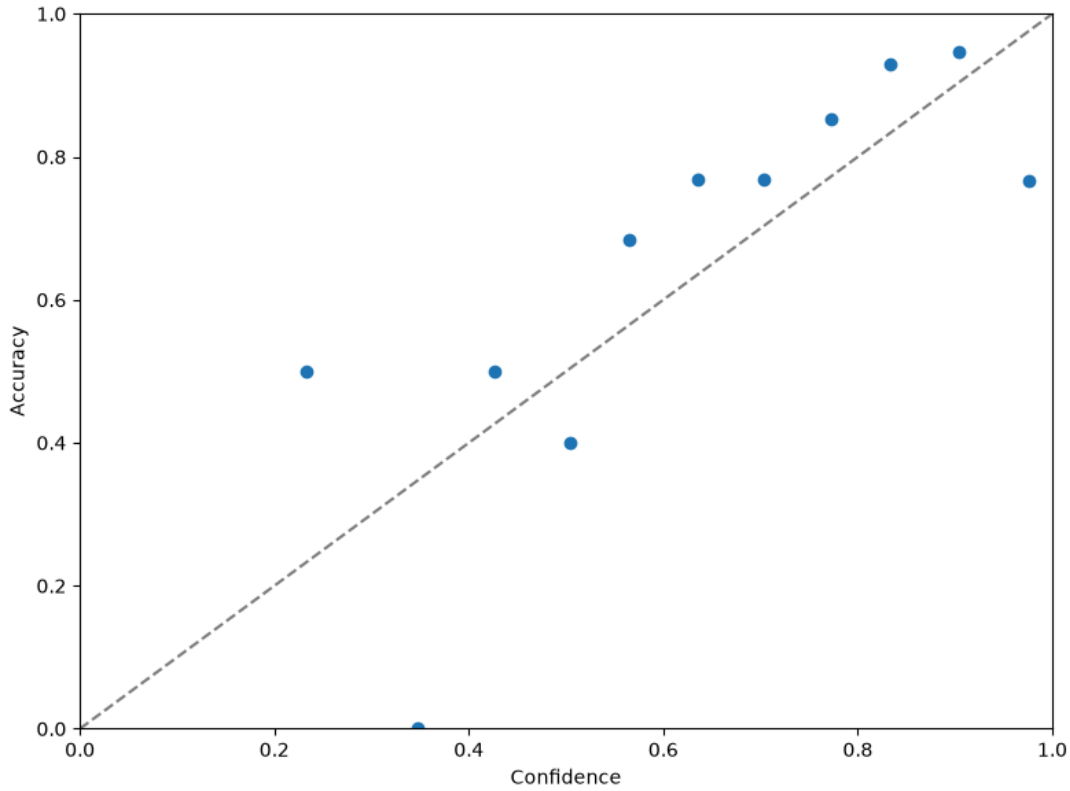
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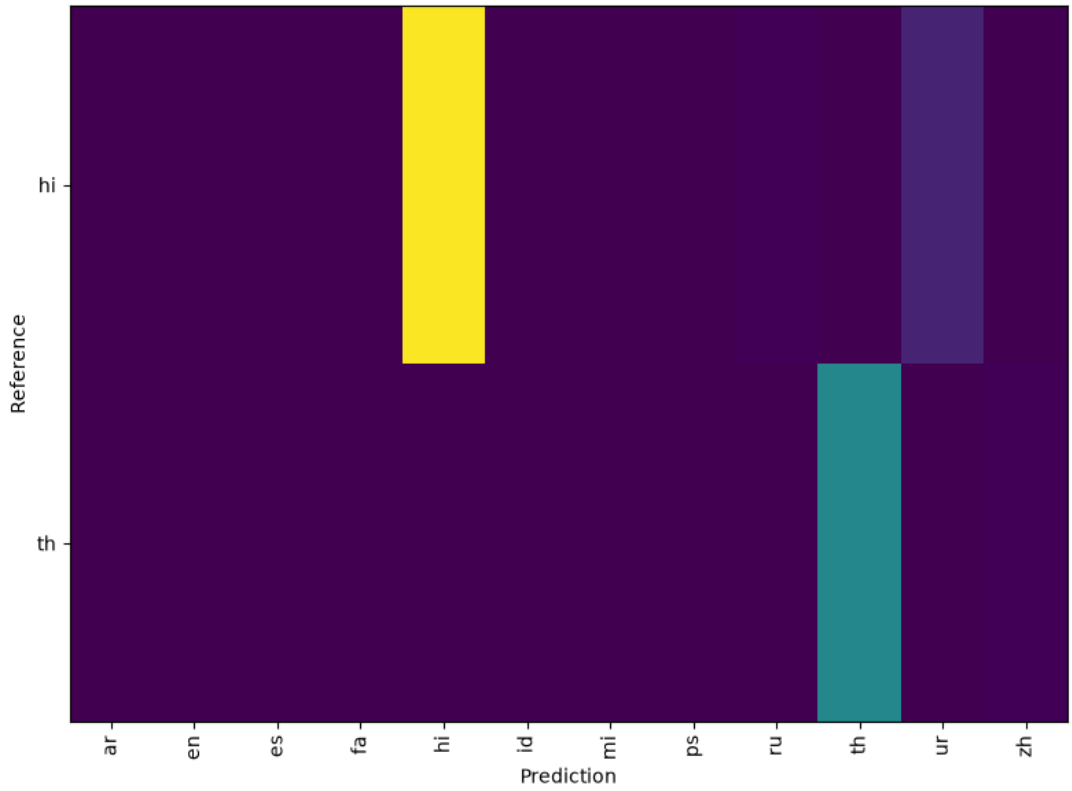
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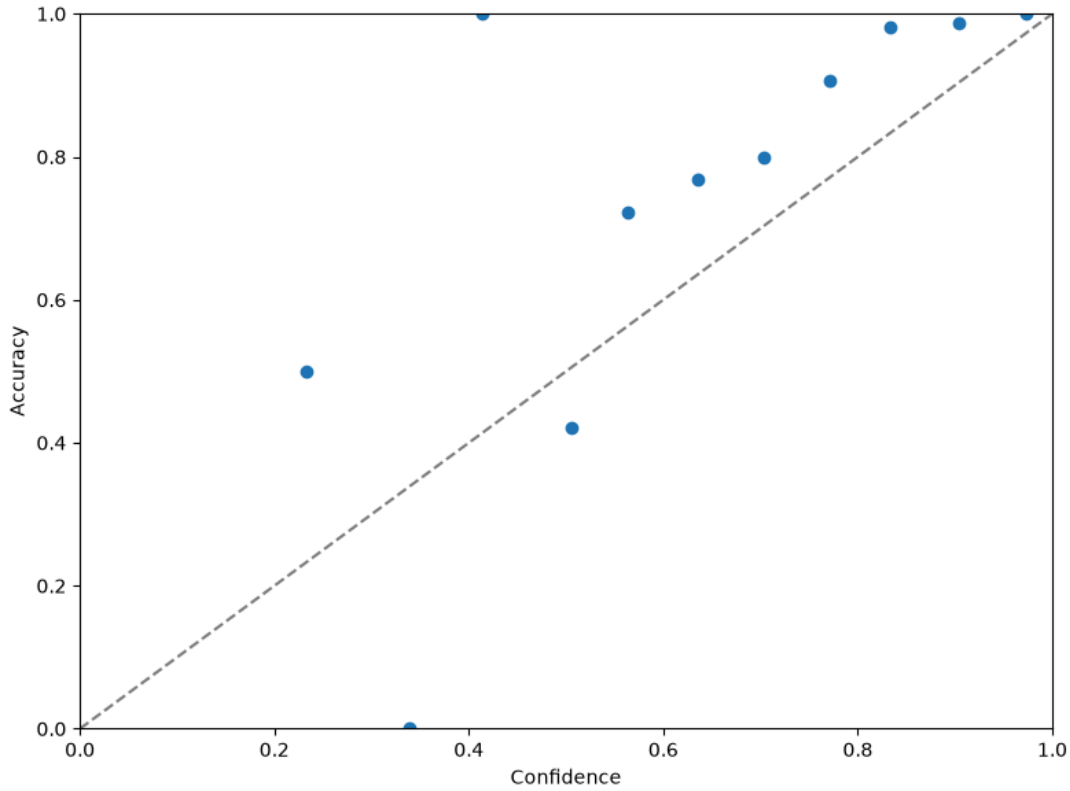
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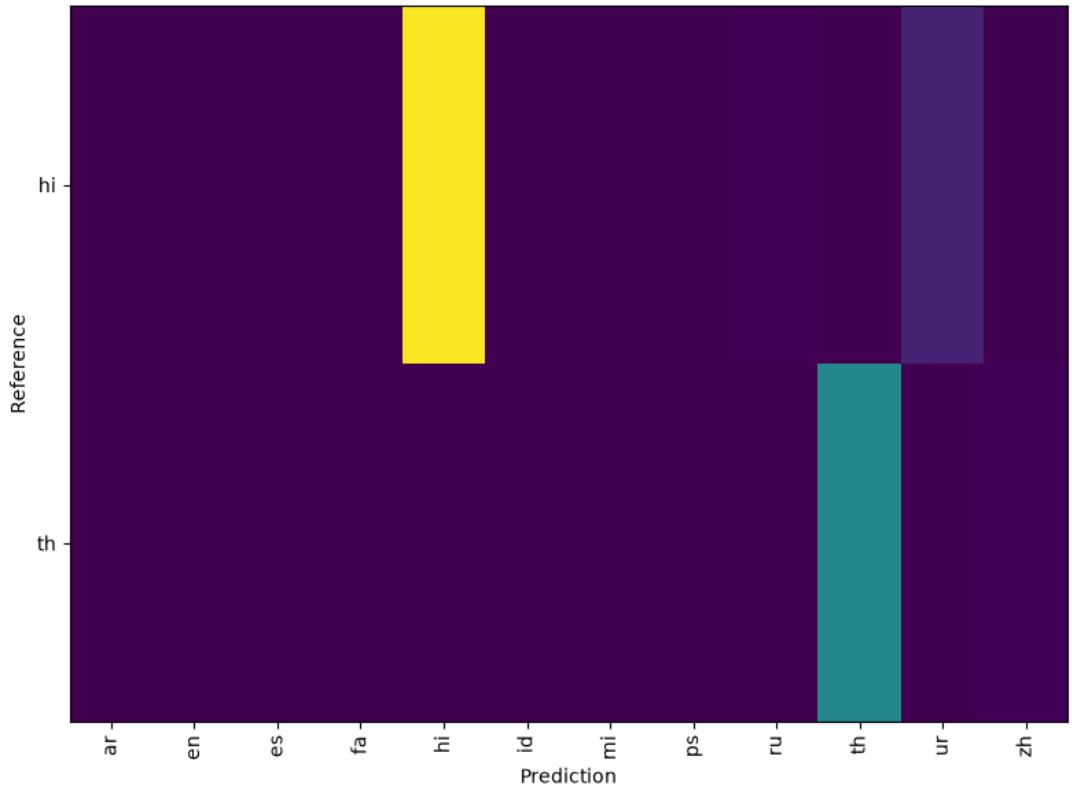
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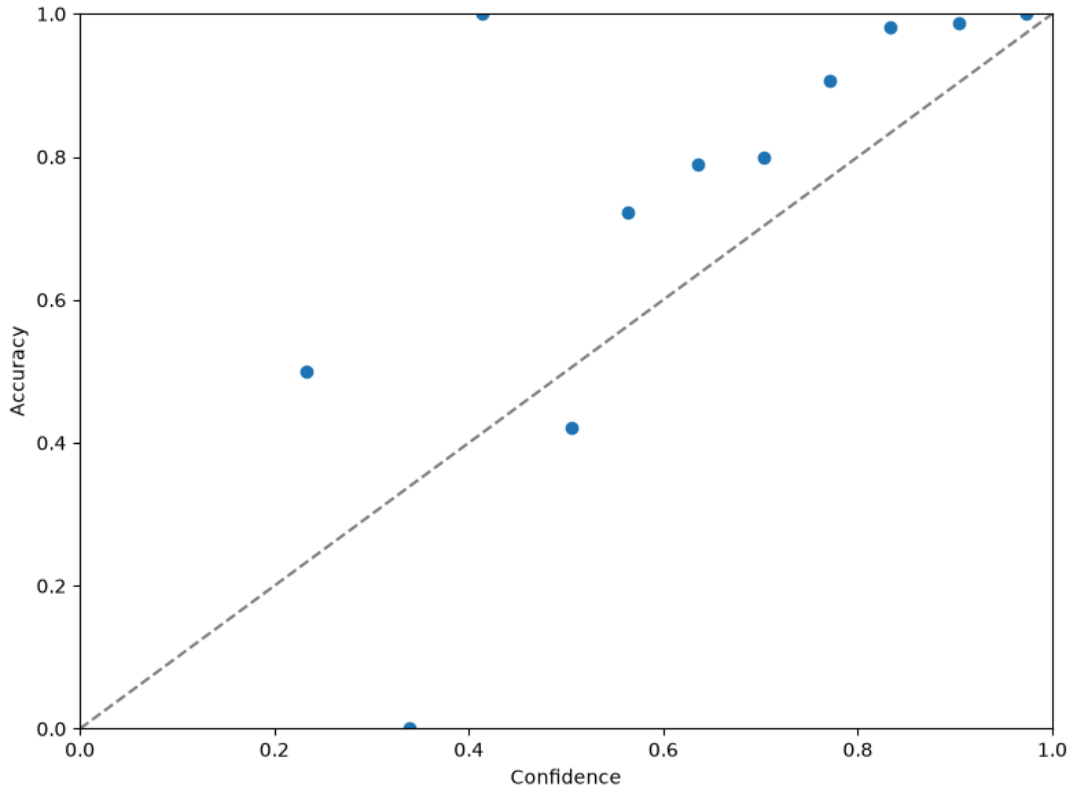
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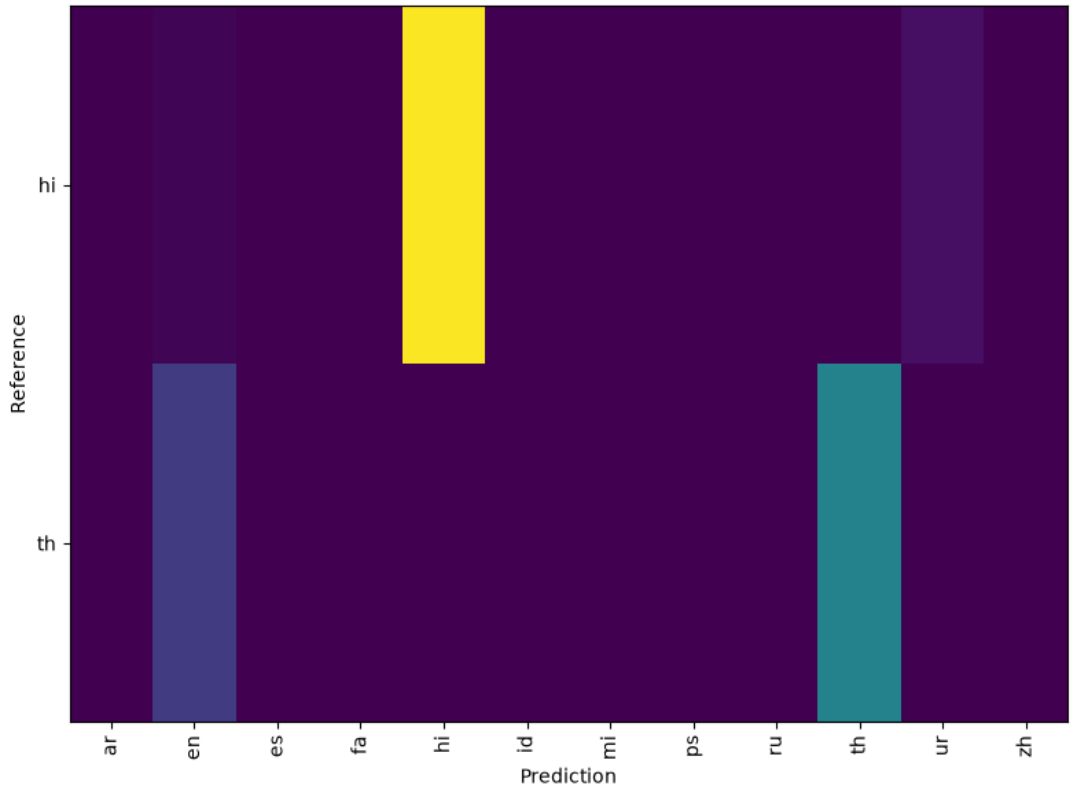
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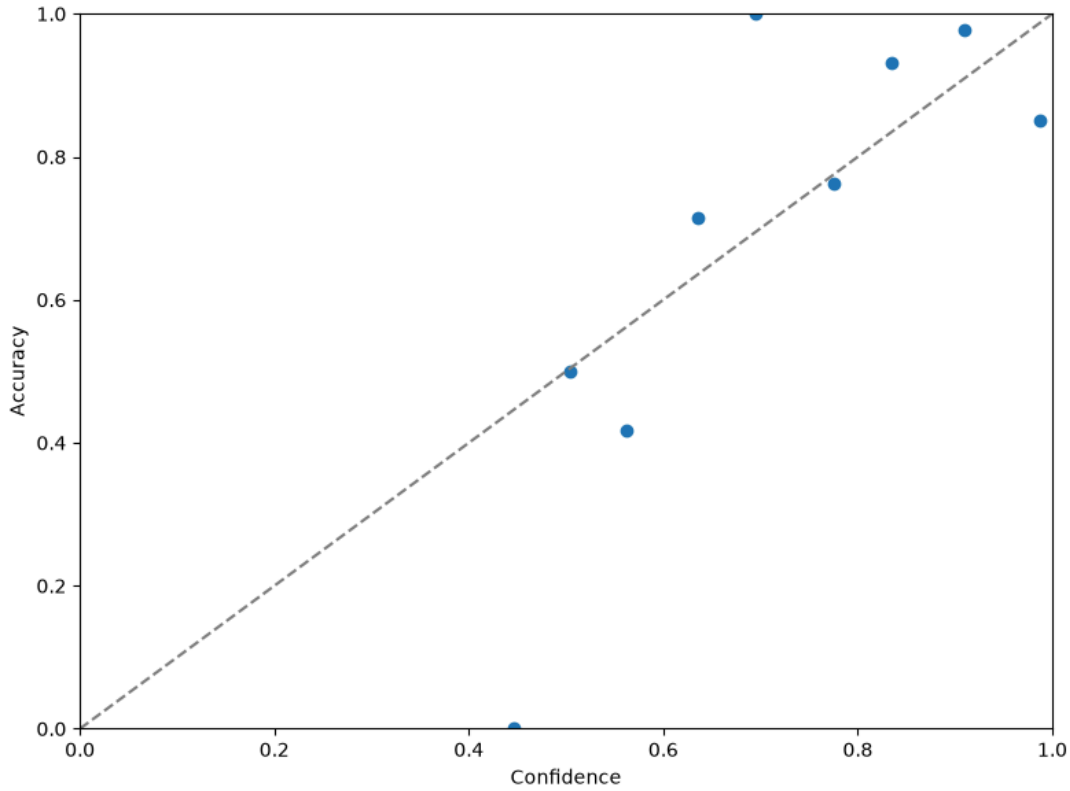
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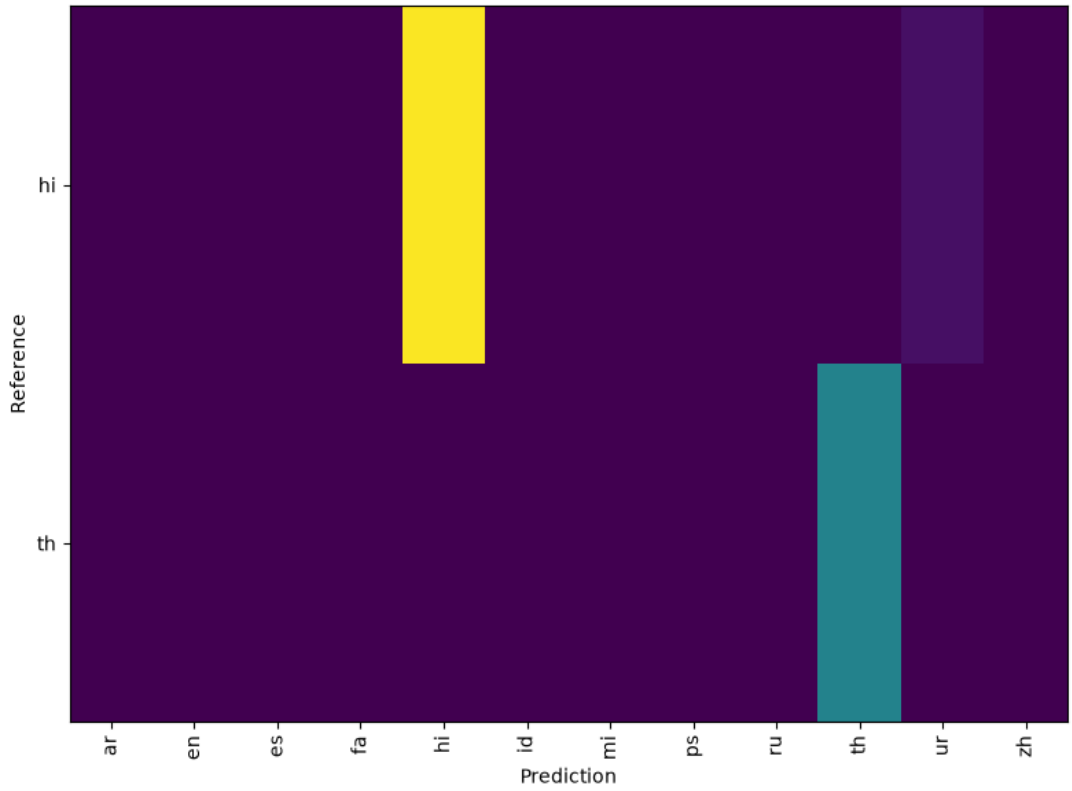
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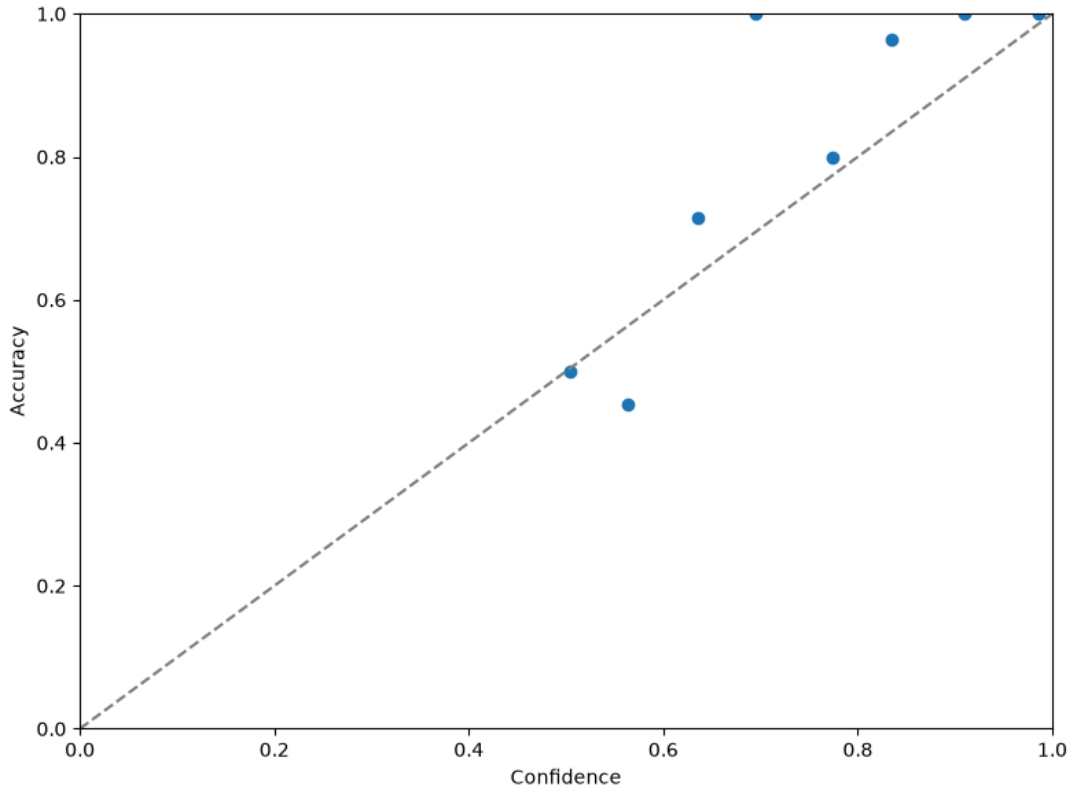
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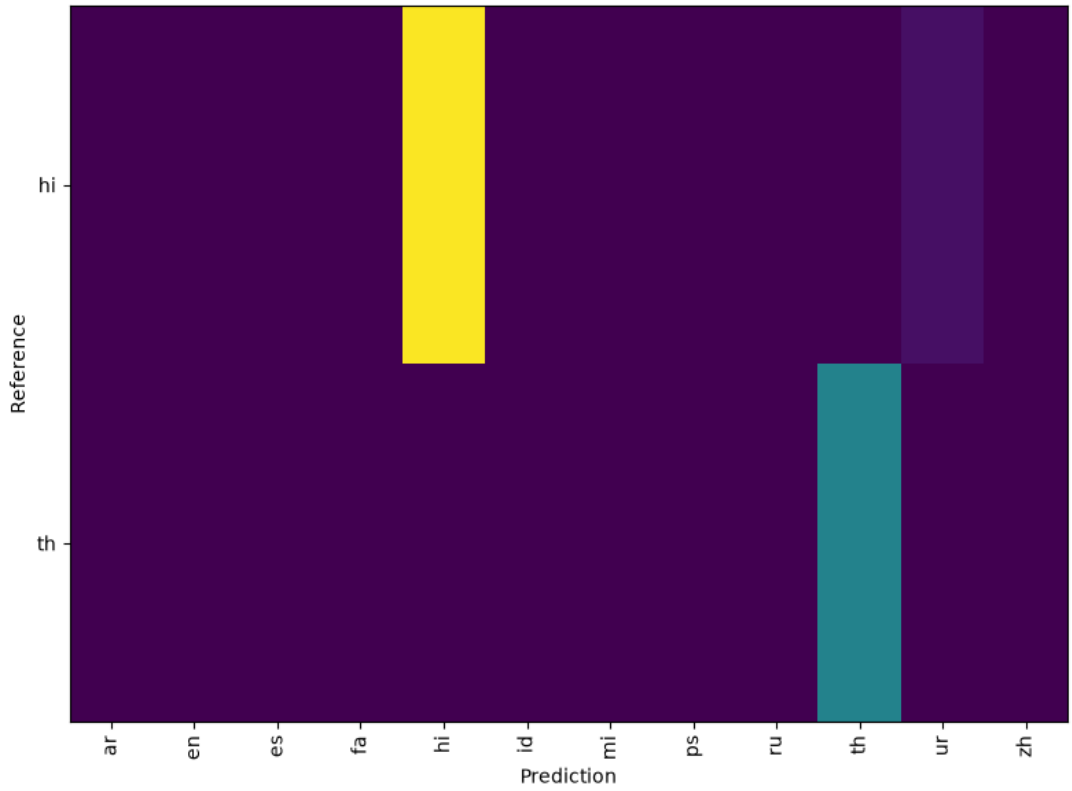
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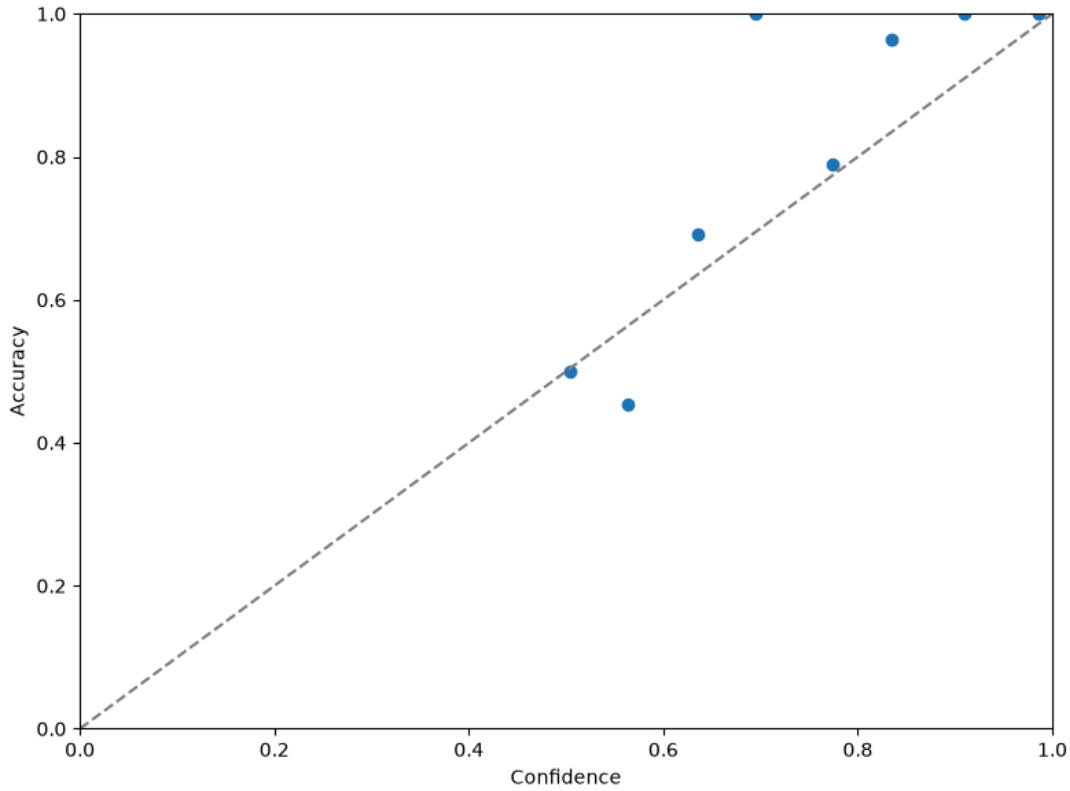
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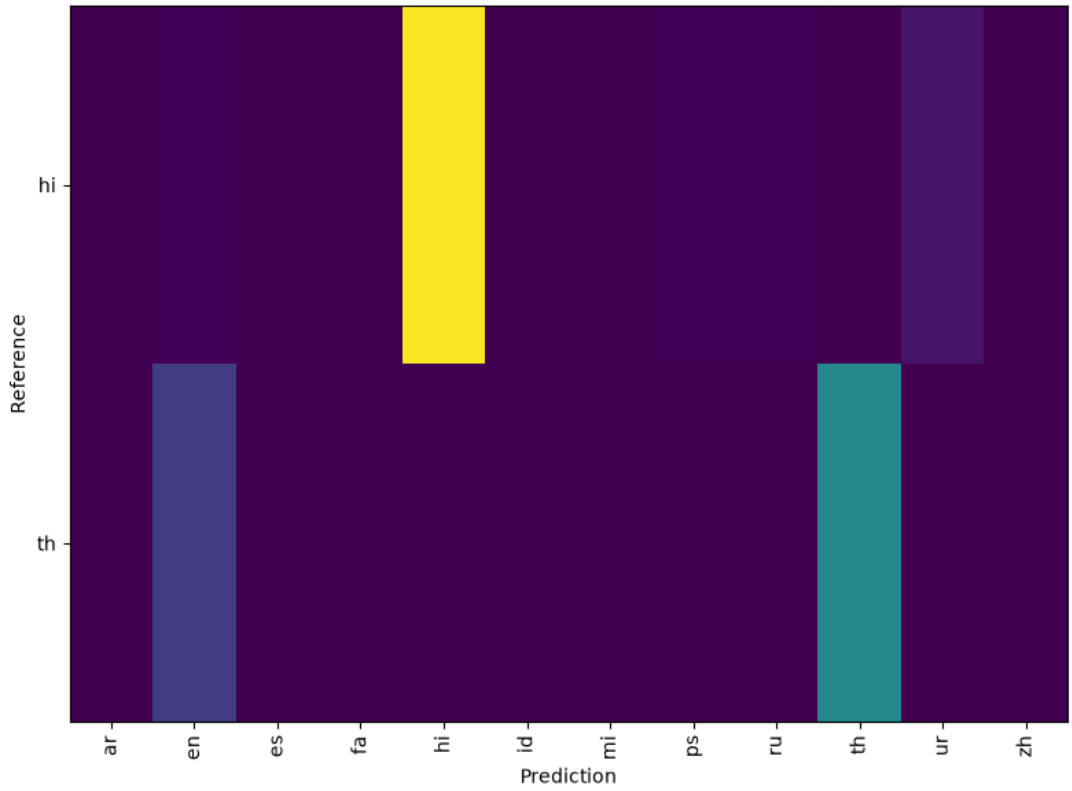
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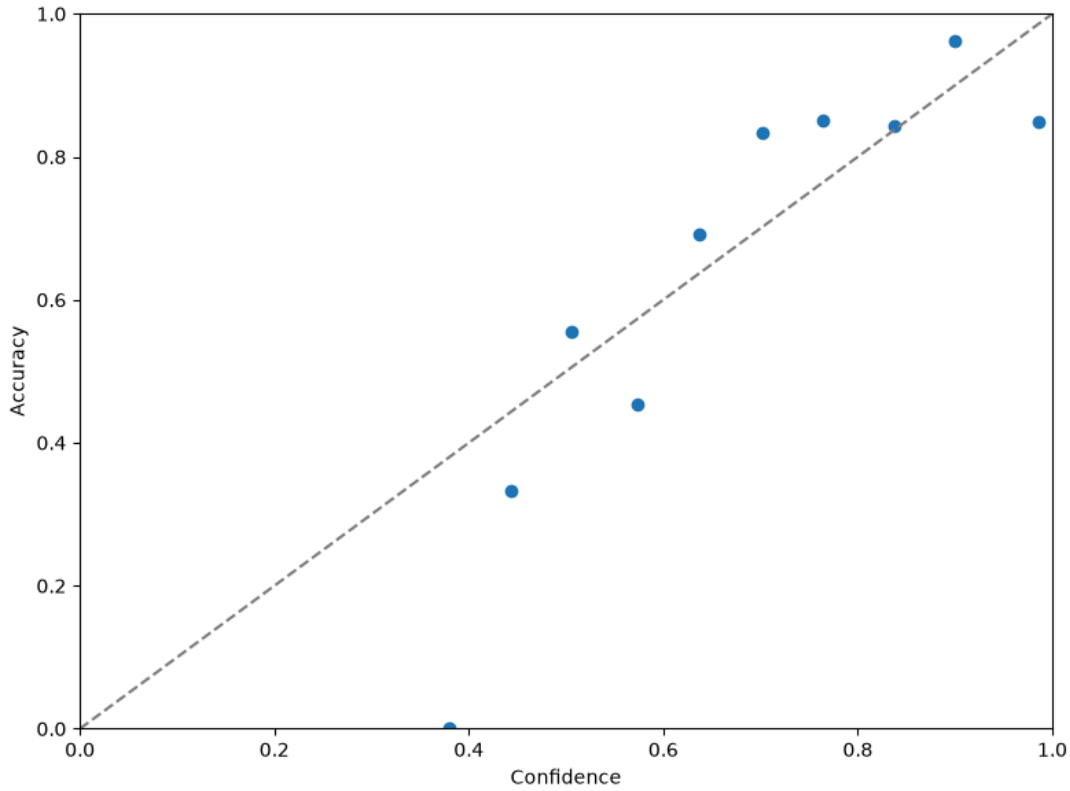
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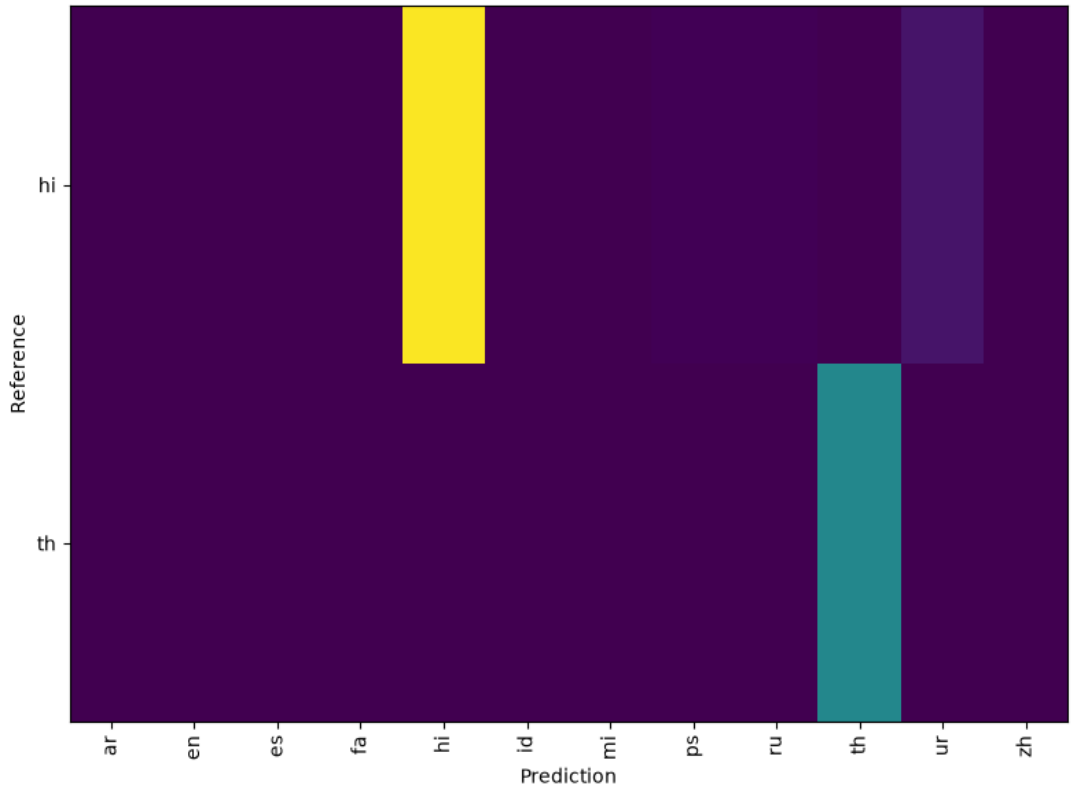
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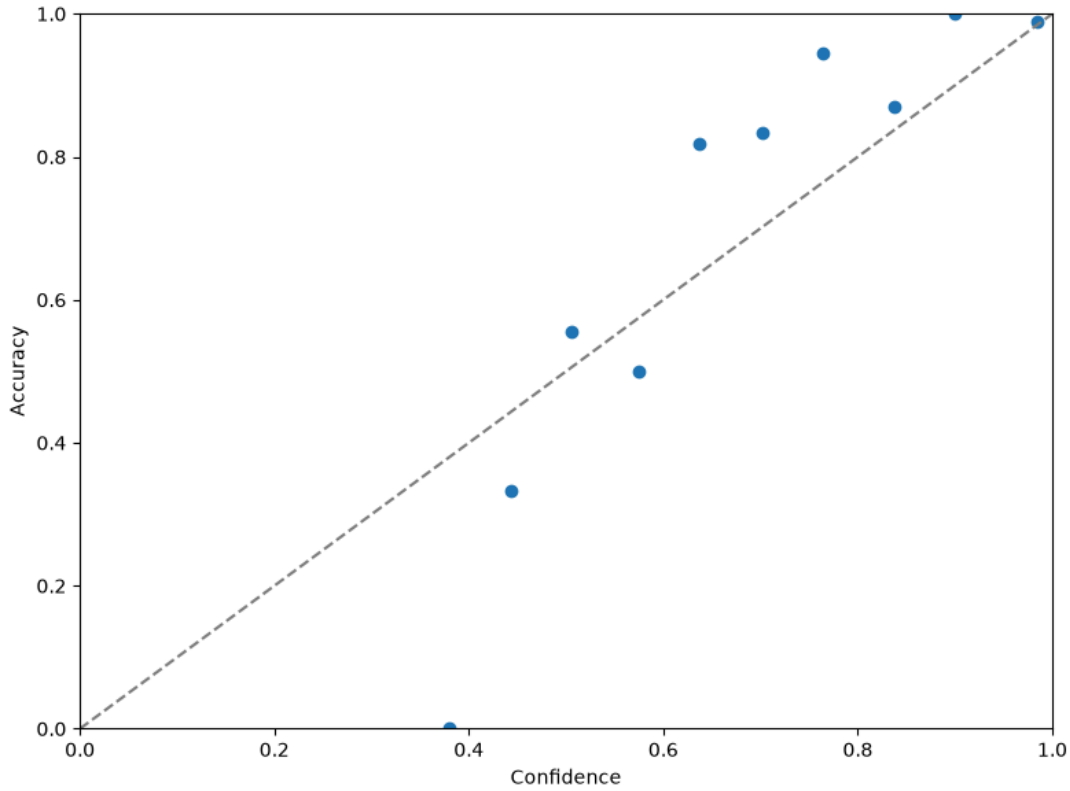
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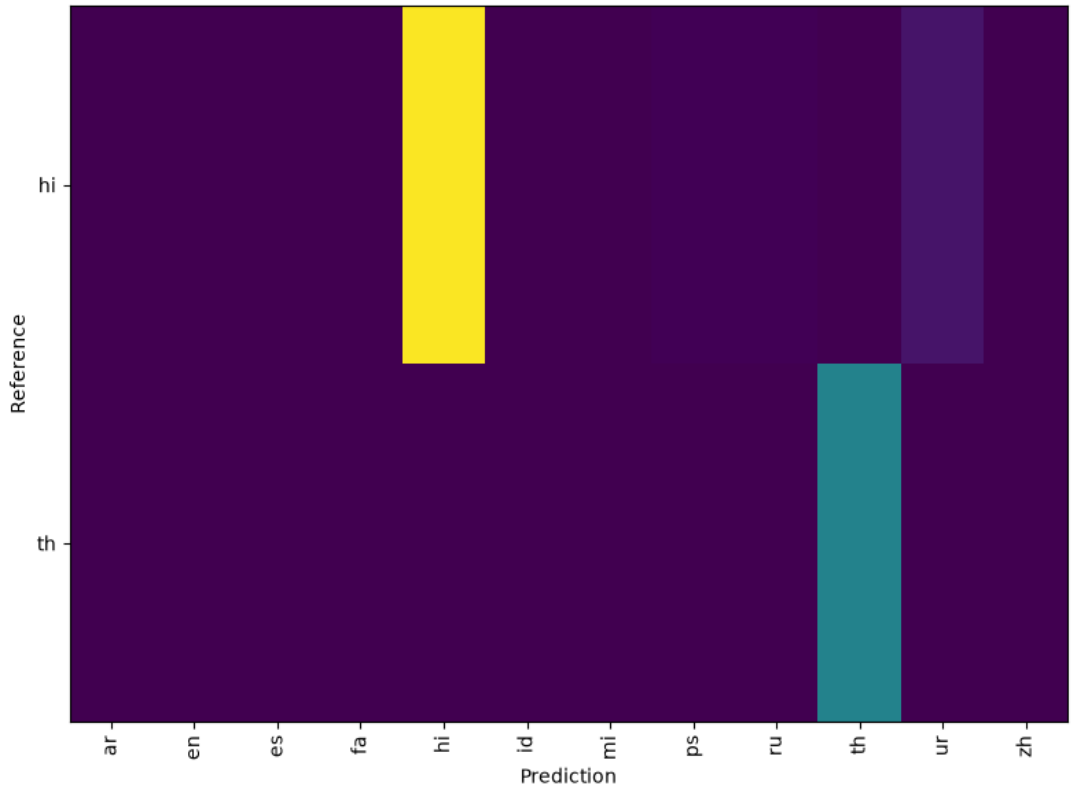
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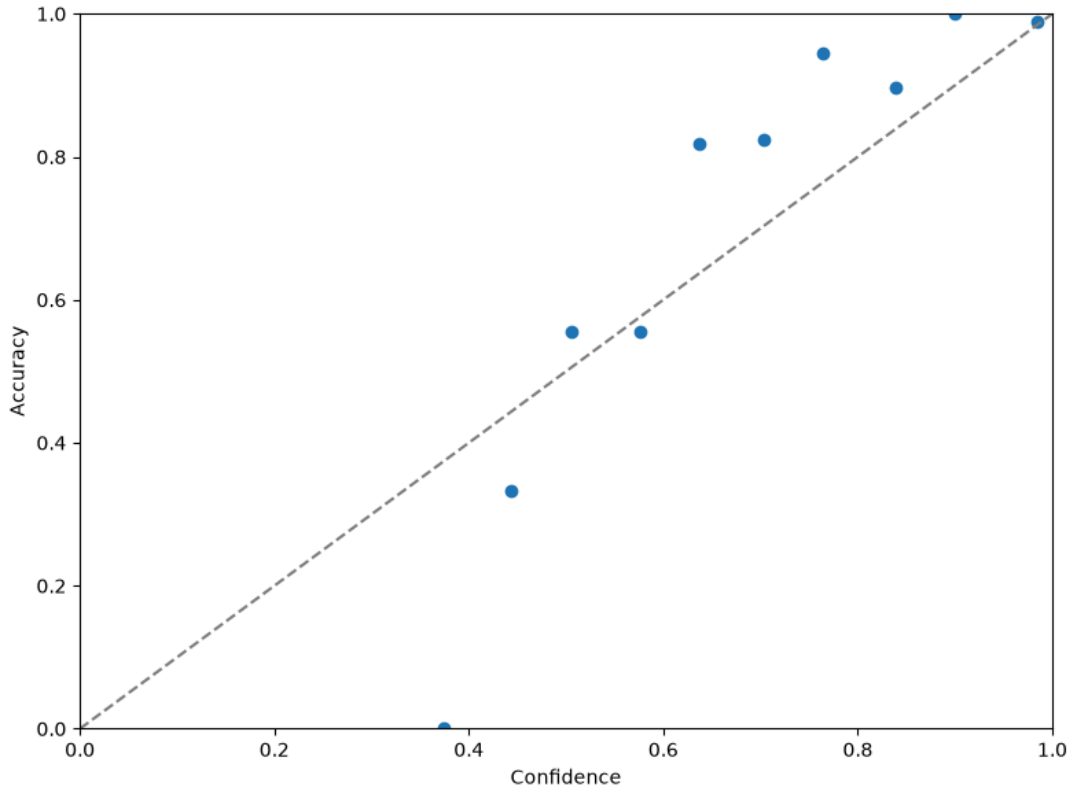
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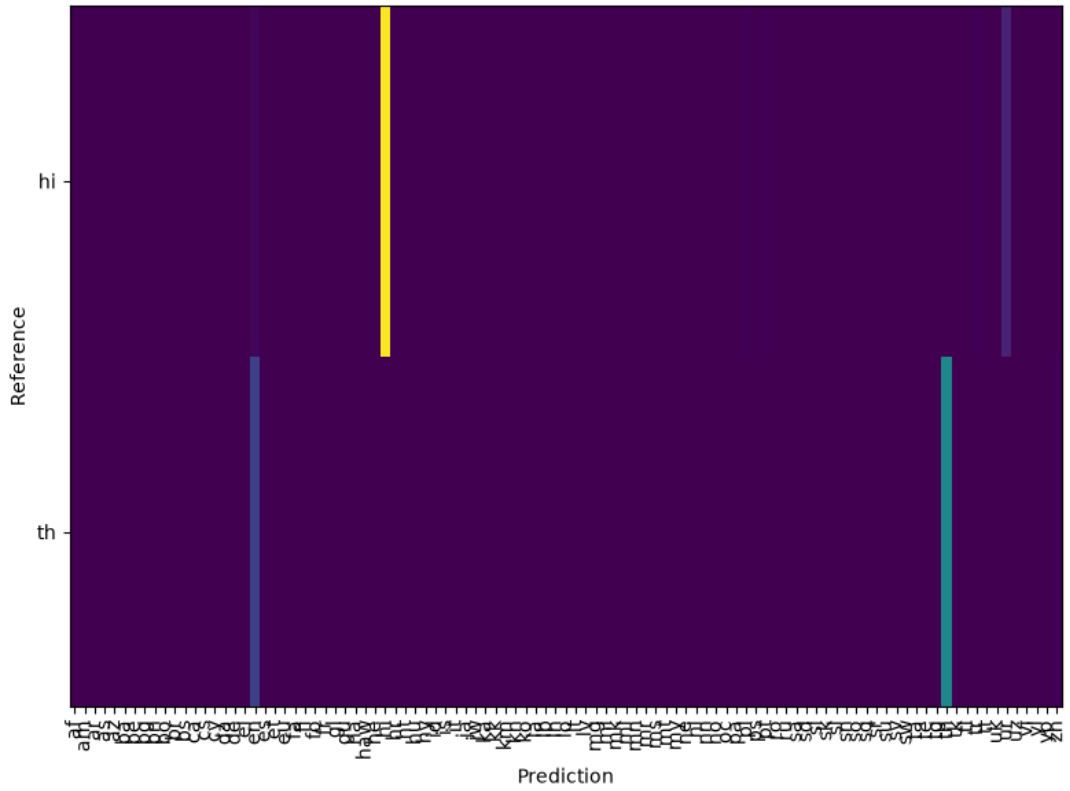
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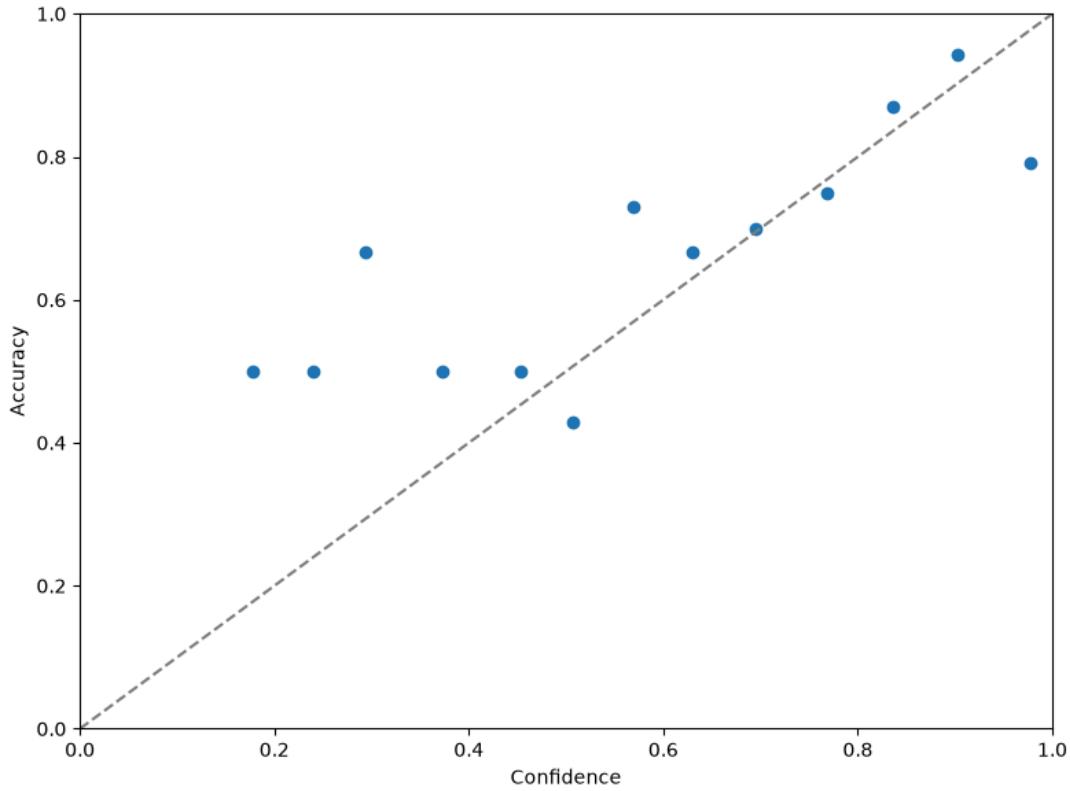
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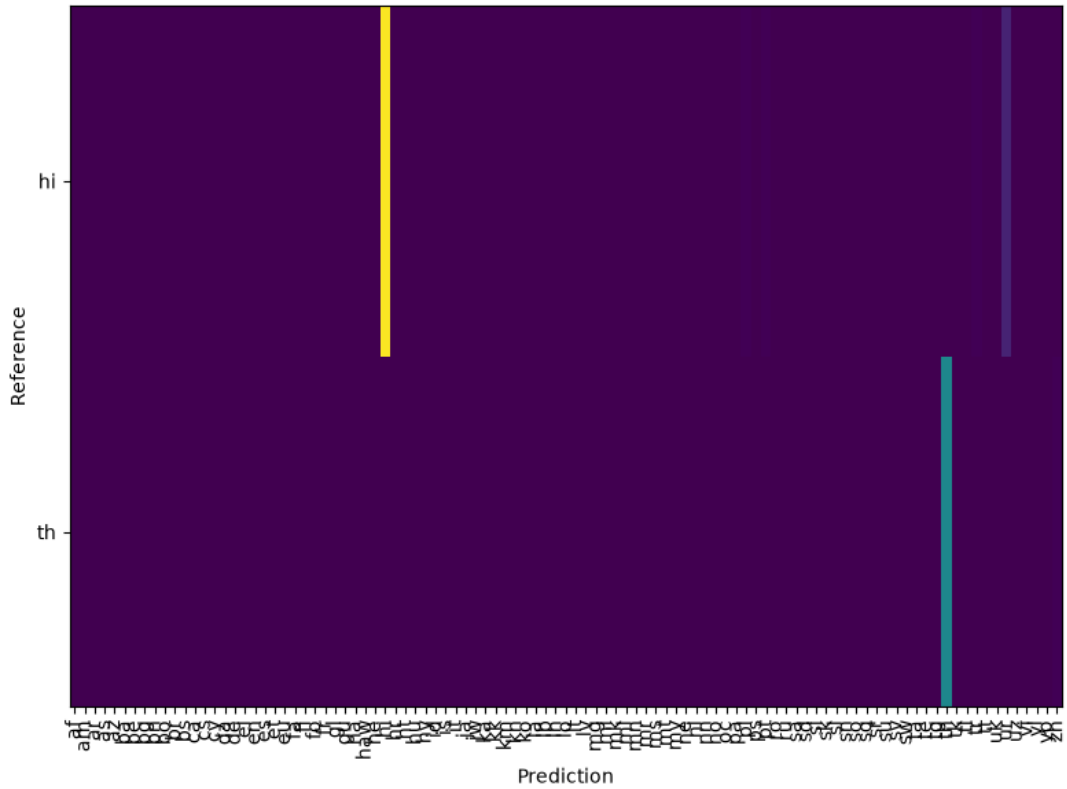
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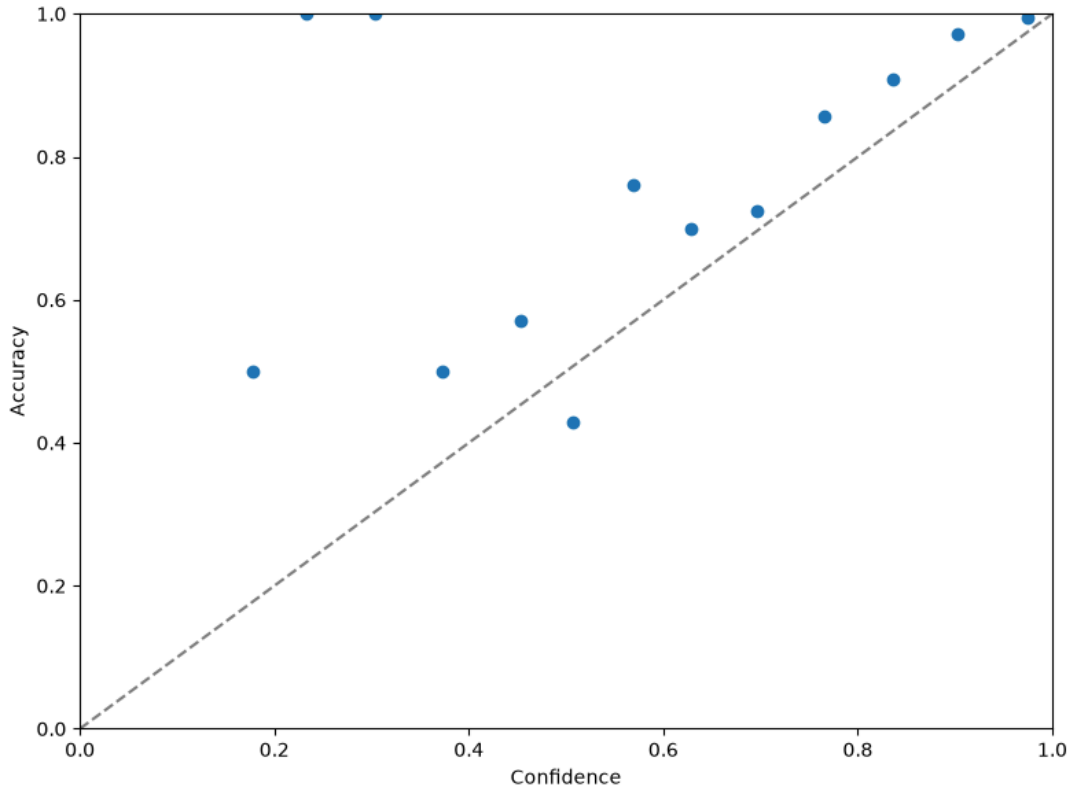
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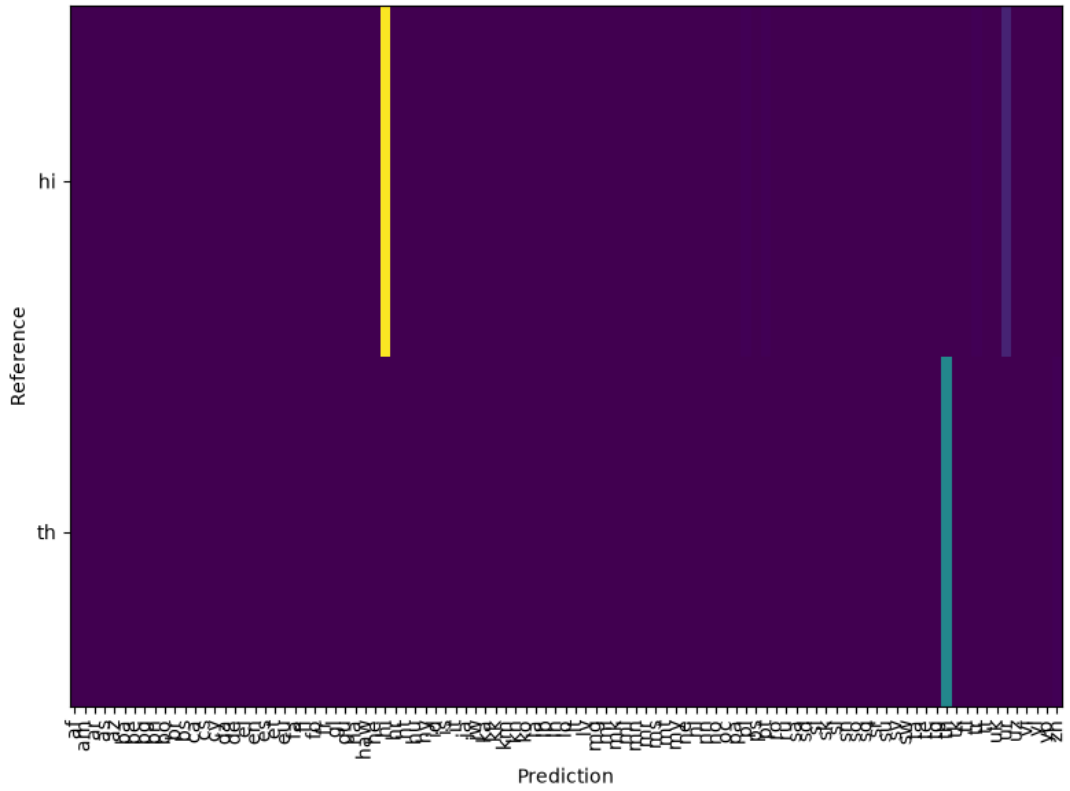
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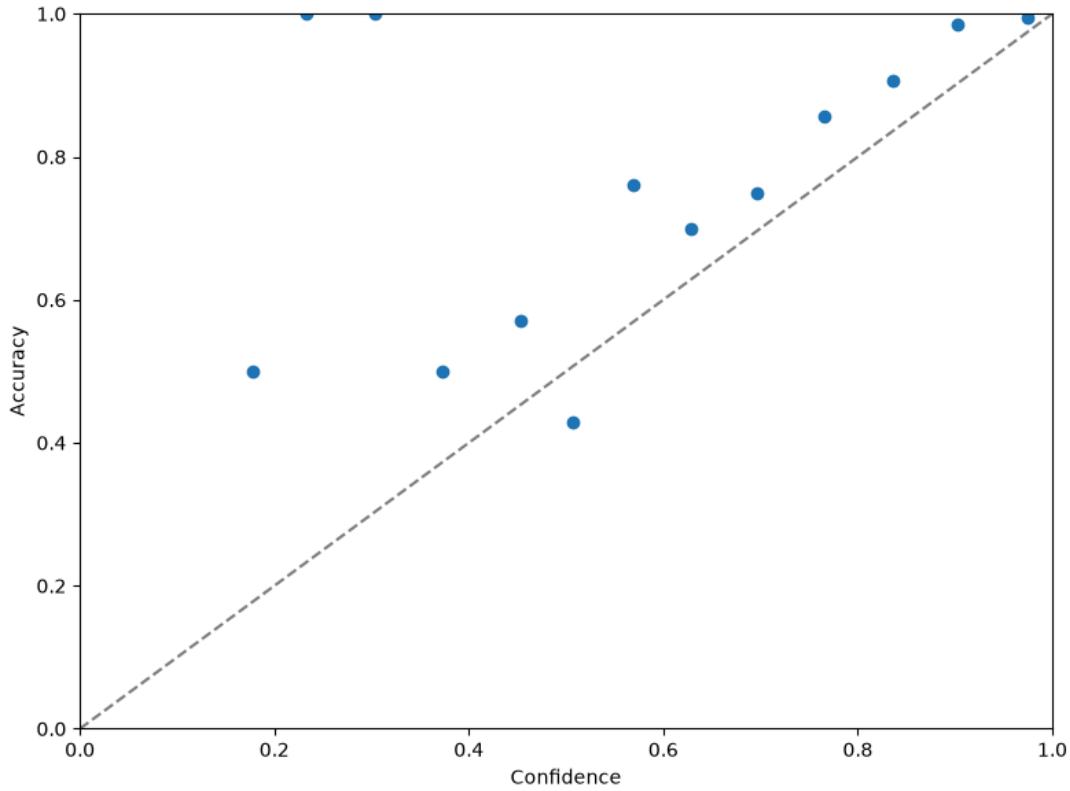
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